University of Cabuyao
(PAMANTASAN NG CABUYAO)

COLLEGE OF ENGINEERING

AUTOMATED FARE COLLECTION SYSTEM WITH SEAT ALLOCATION, TERMINAL SELECTION, AND PROFIT ANALYTICS DASHBOARDS

A Computer Engineering Research Presented to
the Faculty of the College of Engineering
Pamantasan ng Cabuyao

In Partial Fulfillment of the Requirements for the Degree
Bachelor of Science in Computer Engineering

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University of Cabuyao
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Republic of the Philippines
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ABSTRACT

Public transport fare collection remains constrained by manual payment handling, limited pre-boarding passenger information, weak seat management, and delayed operational and financial monitoring. This study designed, developed, and evaluated an automated fare collection system (AFCS) with seat allocation, terminal selection, and profit analytics dashboards. The system integrated a commuter interface; a tap-based payment process; a seat monitoring and allocation mechanism; and driver and administrator dashboards. Using the Design and Development Research approach, a working prototype was implemented using an e-bike platform and evaluated through technical testing, user evaluation, and expert evaluation based on ISO/IEC 25010:2023. Results showed that the commuter interface achieved excellent technical performance and was strongly accepted by users in terms of usability and interaction capability. The tap-based payment process performed effectively in validating cards, computing fares, allocating seats, and displaying validation feedback, although tap consistency and connectivity-status display remained areas for improvement. The seat monitoring and allocation mechanism provided acceptable seat detection and status display, with sensor consistency identified as a limitation. The driver and administrator dashboards achieved excellent technical performance and were positively evaluated for operational monitoring and financial analytics. Expert respondents consistently rated the overall system quality as excellent across functional suitability, performance efficiency, reliability, security, usability, and data quality, confirming that the developed AFCS met international software product quality standards. These findings indicate that the developed AFCS can support cashless fare processing, terminal-based fare computation, seat monitoring, and data-driven operational and financial monitoring at the prototype level.

Keywords: *automatic-fare, monitoring, occupancy, revenue, transportation*

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DEDICATION

This study is first and foremost dedicated to our Lord Jesus Christ, the author and finisher of our faith. As individuals fearfully and wonderfully made through You and for You, we offer this study as a fruit of Your gifts, for Your glory and for the good of others.

As the main beneficiary of this study, we dedicate this study to the daily commuters. Your suggestions led us to challenge the current transport system into an improvement, applying the framework provided through the 2023-2030 Research Agenda.

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CHAPTER I

THE PROBLEM AND ITS SETTING

Introduction

Public transportation is a daily lifeline for millions of Filipinos. In Philippine urban areas, public transport accounts for a very large share of trips, and studies estimate that jeepneys alone serve around 40 percent of all motorized person trips. Jeepneys therefore, remain one of the most dominant and accessible modes for everyday travel to work, school, markets, and other essential services [1], [2]. As cities grow more congested and travel demand increases, long queues at terminals, crowded vehicles, fare disputes, and uncertainty about seat availability or travel time are no longer just minor inconveniences. These issues directly affect productivity, safety, and the overall commuting experience.

Digitalization offers a way to address many of these persistent problems. Smart cities are increasingly built around data and rely on technologies such as cloud computing, the Internet of Things (IoT), artificial intelligence (AI), edge computing, and related platforms to collect, process, and act on information from the physical environment [3]. In the transport sector, these ideas are implemented through intelligent transportation systems (ITS), which apply information and communication technologies to road infrastructure, vehicles, and traffic management so that users can make safer and more coordinated use of transport networks [4]. Smart transportation systems support services such as automatic vehicle location, real-time passenger information, and electronic ticketing, all of which can improve efficiency and user experience when properly integrated into public transport operations.

One important ITS application is automated or electronic fare collection. Automatic fare collection (AFC) systems automate the process of collecting passenger fares using technologies such as smart cards, contactless bank cards, and mobile payments [5]. These systems streamline payments, verify trips, and reduce dependence on cash handling. Globally, many metropolitan areas already use AFC based on stored-value smart cards or

contactless payments. Systems such as London's Oyster [6], Singapore's EZ-Link [7], Hong Kong's Octopus [8], and Chicago's Ventra [9] show how tap-based payment can simplify boarding, reduce disputes about change, and generate detailed records of ridership and revenue. When fare payment, passenger information, and data analytics are integrated properly, operators can better match supply with demand, regulators can monitor compliance more easily, and commuters benefit from more predictable and convenient trips [10].

The Philippines is gradually moving in the same direction. The Public Utility Vehicle Modernization Program (PUVMP) is a national reform initiative that aims to transform road-based public transport through fleet modernization, industry consolidation, route rationalization, and the adoption of safer and more environmentally friendly vehicles [11], [12]. Among the proposed requirements for modernized jeepneys are global positioning system (GPS) units, closed-circuit television (CCTV) cameras, and automated fare collection system (AFCS) that can improve safety, accountability, and service quality [11], [12].

In recent years, the Department of Transportation (DOTr) and partner institutions have launched pilot projects that introduce bank card-based AFCS in selected routes. In 2022, DOTr and Land Bank of the Philippines initiated Europay-Mastercard-Visa (EMV) Contactless Pilot Production Testing for an automated fare collection system, involving around 150 vehicles and other public utility vehicles to help develop national AFCS standards [13]. In 2024, Visa and its partners launched a transit project that enables contactless Visa card payments in vehicles in Cebu and Mandaue City, initially covering around 140 units [14]. These pilots show that bank card-based AFCS is technically and operationally feasible for Philippine road-based public transport.

Alongside these pilots, the Beep™ card system has become a widely recognized stored-value medium for rail and bus services. AF Payments Inc. reports that as of July 2025, the Beep system has processed about 2.9 billion transactions and issued approximately 11.6 million cards across thousands of acceptance points [15]. This suggests that many commuters are already familiar with tap-based payment for transport and other services, and

that large-scale AFCS platforms can be deployed and maintained in the Philippine context.

Despite these advances, several gaps remain, especially in the operations of public transportation vehicles. Many routes still rely on cash or partially cashless setups, and even when an AFCS is present, it often focuses narrowly on collecting payment. In typical operations, commuters usually board without any digital information about real-time seat availability and rely on verbal confirmation of their destination. Fares are often collected in cramped conditions where cash handling can be slow, error-prone, and vulnerable to pilferage. For operators and cooperatives, manual or semi-manual head counting and paper-based expense tracking make it difficult to reconcile revenue with actual passenger loads, to compute profit per trip or per terminal, or to detect anomalies such as missing fares and unauthorized stops.

The province of Laguna, particularly the City of Cabuyao, offers a concrete setting where these challenges can be observed. Public utility vehicles serving Cabuyao and nearby cities typically operate under a flexible boarding and alighting scheme. Commuters may board or alight at many points along the route instead of only at fixed stations, while fare structures defined by the Land Transportation Franchising and Regulatory Board (LTFRB) are based on distance travelled or zonal rates [16], [17]. This flexibility is convenient for passengers but makes distance- or terminal-based fare computation more complex. It also complicates seat monitoring, because the number of occupied seats can change frequently between terminals, and operators still need to ensure that modern units do not carry standing passengers.

In a key informant interview conducted by the researchers with the chairman and general manager of Cabueños Transport Cooperative (CABTRANSCO) in the City of Cabuyao, the cooperative leaders reported that even with an existing cashless AFCS, they still face several operational issues. They described challenges in achieving a full shift to cashless transactions, in obtaining accurate passenger headcounts, and in monitoring individual driver performance. They also noted that their current system provides limited

visibility on expenses such as fuel, terminal fees, and allowances in relation to actual trip revenue. When asked for overall comments, they remarked that the current system still "needs improvement to perfection," indicating that payment automation alone is not enough and that richer analytics and better operational visibility are needed at the cooperative level.

From the commuter's point of view, several pain points are also evident. First, there is usually no digital way to know whether an approaching PUV still has vacant seats. Commuters often rely on visual inspection or last-minute verbal confirmation when the vehicle is already near or crowded. Second, fare computation is often opaque. Passengers may not know in advance how much will be deducted or collected for a particular terminal, and miscommunication about destinations can lead to incorrect fares. Third, when a commuter misses the intended terminal or decides during the trip to travel farther, fare adjustment is typically handled informally and may not be recorded accurately in the system. Finally, commuters rarely have access to real-time views of their stored balance, recent trips, and digital receipts, which limits transparency and trust in cashless transactions.

On the side of operators and drivers, the absence of integrated seat monitoring and trip-wise analytics has both operational and financial consequences. Drivers often track passengers mentally or by rough estimation, which is prone to error during rush hours. Without a clear record of boarding and alighting per terminal, cooperatives find it difficult to verify whether reported revenues match actual passenger volumes, to identify underperforming routes or trips, or to determine whether a particular schedule is operating at a loss once fuel, terminal fees, and allowances are taken into account. Existing AFCS deployments may capture tap events, but they do not always link these events to real-time seat allocation, extended trip scenarios, or detailed expense records. As a result, there are data gaps between what the system records and what actually happens on the road.

These gaps point to a specific research and design opportunity. There is a need for an automated fare collection system for PUVs that does not stop at collecting payment, but instead combines commuter trip support, seat availability, terminal-based fare computation,

and profit analytics in a unified platform. Such a system should allow commuters to view their card balance and see which vehicle are approaching on a selected route, including basic information about origin and destination. It should show the real-time number of available seats before boarding and provide a way for commuters to select a drop-off terminal in advance so that the fare can be computed and confirmed. It should also support an extended terminal scenario, where a commuter can change to a farther drop-off terminal during the trip and have the additional fare automatically calculated and deducted. On the back end, it should maintain an accurate, real-time model of seat occupancy and link each boarding and alighting event to specific terminals, trips, and expense records. This would enable richer analytics on revenue, cost, and profit per trip, per driver, and per route.

In response to this need, the conducted study developed an automated fare collection system (AFCS) with seat allocation, terminal selection, and profit analytics. The developed AFCS integrates a secure card-based tap-based mechanism with a commuter web interface and operational dashboards for drivers and administrators. Before boarding, a commuter can log in to the interface, view the current card balance, see incoming vehicles along a chosen route, and check the real-time number of available seats. Within the same interface, the commuter selects the current terminal and intended drop-off terminal. Upon boarding, a single tap on the vehicle reader automatically charges the correct fare based on the selected terminal, records the transaction in the system, and allocates a seat in the digital seat map. If the commuter later decides to alight at a farther terminal, an extended terminal feature in the commuter interface allows updating the destination, after which the system recalculates the fare and deducts only the additional amount required.

The developed AFCS also includes analytics and reporting functions for drivers and transport cooperative administrators. The driver dashboard provides a real-time view of seat occupancy, boarding and alighting per terminal, and basic service status. This information helps the driver avoid overloading and understand where passengers are expected to disembark. The administrator dashboard aggregates transaction and operations data into

analytics on revenue, encoded expenses such as fuel, terminal fees, and maintenance, and resulting profit per service run, day, route, or vehicle. By linking automated payment, seat allocation, terminal selection, and structured financial reporting, the developed AFCS enables transport operators to identify profitable and unprofitable runs, monitor compliance with policies, and support evidence-based decisions on scheduling and resource allocation.

The developed system was implemented as a prototype using contactless smart cards, an on-board fare validation terminal, a centralized database, and web-based interfaces for commuters, drivers, and administrators. For safety and practicality, the study used simulated commuter balances and fare transactions that followed LTFRB fare structures instead of handling real money [16], [17]. The prototype was tested in terms of its functional behavior, such as successful tap-in transactions, correct seat updates, and accurate terminal-based fares. Moreover, the developed system was evaluated using selected product quality characteristics from ISO/IEC 25010:2023 [18]. These characteristics include functional suitability, performance efficiency, reliability, security, usability, and data quality.

Research Objectives

This study aimed to design, develop, and evaluate an automated fare collection system (AFCS) with seat allocation, terminal selection, and profit analytics dashboards. Specifically, the study sought to:

1. Design and develop a commuter interface that displays card balance details, seat availability, and terminal selection before boarding the incoming vehicle.
2. Design and develop a tap-based payment process that validates commuter cards, computes terminal-based fares, and automatically allocates seats after successful validation.
3. Design and develop a seat monitoring and allocation mechanism that provides accurate seat status indicators for commuters and the driver.

4. Design and develop the driver and administrator dashboards for operational monitoring and financial analytics.
5. Evaluate the overall system using ISO/IEC 25010:2023 in terms of functional suitability, performance efficiency, reliability, security, usability, and data quality.

Theoretical Framework

This study adopted two theoretical frameworks relevant to the use of innovative technologies in public transportation: Intelligent Public Transportation Systems (IPTS) and Dynamic Data-Driven Application Systems (DDDAS).

Intelligent Public Transportation Systems (IPTS)

The Intelligent Public Transportation Systems (IPTS) aim to improve the accessibility, efficiency, sustainability, and safety of transportation systems by integrating digital technologies. Within the context of public utility vehicles (PUVs) in the Philippines, specifically the vehicles, by integrating the IPTS framework in the form of cashless and contactless transactions and trip monitoring systems, it optimizes the commuter experience, fare collection system, route management, and efficient fuel consumption, and turns the traditional operations into data-driven systems [19].

Dynamic Data-Driven Application Systems (DDDAS)

The model of Dynamic Data-Driven Application Systems (DDDAS) provides an adaptive feedback loop, which enables real-time data to inform and control computational and analytical system models. It lies in transforming traditional static systems into dynamic and responsive applications operated by real-time data inputs, which enables it to make predictions and decision-making processes [20]. These data are well capable of integrating into near-field communication (NFC)-based transactions, fuel usage and revenue analytics, and dashboard visualization in a cashless and contactless transportation system.

Conceptual Framework

This section presented the conceptual framework that guided the design, development, and evaluation of the developed AFCS. Using the Input, Process, and Output (IPO) model, it illustrated how user data, hardware, and system resources, and configuration rules are transformed through the developed AFCS processes into operational and research outputs that addressed the study objectives.

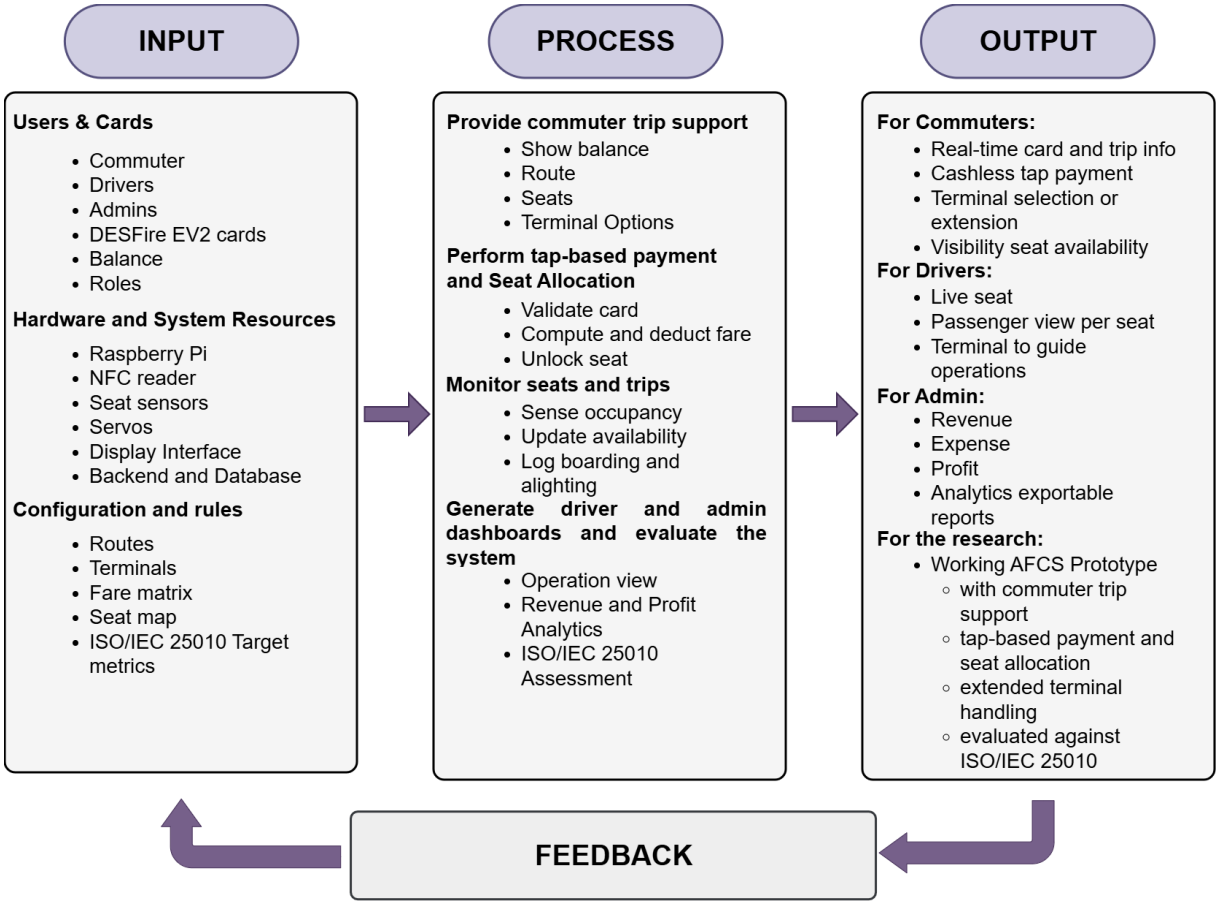


Fig. 1. Conceptual Framework of the Developed AFCS

Figure 1 specified the conceptual framework of the developed AFCS system using an input, process, and output structure with feedback. It showed how user information, hardware and system resources, and configuration rules were transformed by the system processes into operational and research outputs that addressed the objectives of the study.

It represented the resources and parameters required by the developed AFCS. Users and cards included commuters with MIFARE DESFire EV2 cards, stored balances, and selected drop-off terminals, as well as driver accounts and administrator-maintained data on routes, terminals, fare matrices, and expense categories. Hardware and system resources included the Raspberry Pi 4 controller, NFC card reader, commuter display or web interface, driver display, seat sensors and indicators, backend server with database, and the network or Internet connection. Configuration and rules consisted of the route and terminal list, the fare matrix by terminal or distance, the seat map and capacity, rule sets for tap-in validation, fare deduction, seat allocation and release, and extended terminal handling, together with ISO/IEC 25010:2023 target metrics for functional suitability, performance, reliability, security, usability, and data quality.

Furthermore, it described the major functional components implemented in the developed AFCS. First, the commuter trip support interface retrieved card balance, route information, incoming vehicle status, and seat availability, and displayed these on the commuter interface so that the commuter can choose a drop-off terminal. Second, the tap-based payment and seat allocation function validated the card at the NFC reader, deducted the fare based on the selected terminal, confirmed the transaction, and marked a seat as allocated. Third, the seat monitoring and allocation mechanism coordinated card validation with the seat sensors and actuators so that a valid tap results in an available seat being opened and recorded as occupied. Fourth, the extended terminal mechanism allowed the commuter during the trip to request a change to a farther terminal; the system computed the additional fare and checked that the remaining balance was sufficient. Fifth, the driver and administrator dashboards collected logs and presented real time views tailored to drivers and

cooperative administrators. Sixth, the system evaluation component measured the developed AFCS prototype against ISO/IEC 25010 quality characteristics and compared the results with predefined thresholds.

It summarized the expected outcomes of these processes. For commuters, the developed AFCS provided real-time views of card balance, incoming vehicle and route or destination, available seats, and supports fast tap-based payment with clear feedback and automatic fare recomputation for terminal changes. For drivers, the system offered live seat status and passenger distribution per seat or terminal, together with a clear view of boarding and alighting per terminal. For administrators, the developed AFCS generated real-time and historical views of revenue, expenses, profit, and ridership per route, terminal, and time period, along with exportable reports. For the research, the framework yielded a working AFCS prototype with commuter trip support, tap-based payment, and seat allocation, extended terminal handling, and real-time driver and administrator dashboards, evaluated using ISO/IEC 25010.

Lastly, the results from system operation and evaluation were used to refine configurations, rules, and design decisions in subsequent development iterations.

Assumptions

This study was conducted under a set of assumptions that clarified the conditions under which the developed automated fare collection system (AFCS) prototype was designed, tested, and evaluated. These assumptions specified the expected behavior of the test commuters, the e-bike–mounted hardware, the configured fare and route data, and the evaluators, so that the results could be interpreted within a clear and consistent context. The following items list these assumptions, which the researchers accepted as true for the purpose of this research.

1. The e-bike prototype with a two-seat capacity was assumed to be a reasonable small-

scale representation of the incoming vehicle for testing fare collection, seat availability, and terminal selection.

2. The route, terminals, and fare matrix used in the prototype were assumed to be clearly defined, fixed for the duration of the tests, and sufficient to simulate actual public transport operations.
3. It was assumed that the evaluators would follow the prescribed procedures, including selecting a terminal before boarding, tapping their card upon boarding, and requesting any terminal extension only as instructed in the scenarios.
4. All NFC cards used in the study were assumed to be properly registered, uniquely linked to individual testing UIDs.
5. The hardware components of the prototype were assumed to be correctly installed, calibrated, and functioning within their specified ratings during the tests.
6. The system and network connectivity used during experiments were assumed to be stable enough to allow reliable logging of transactions and periodic updating of dashboards, with any brief disconnections handled by local data logging as designed.
7. The data displayed on the commuter, driver, and administrator interfaces (such as card balance, route, seat availability, boarding and alighting, revenue, expenses, and profit) were assumed to accurately reflect the records stored in the prototype system at the time of viewing.
8. Each prototype seat was assumed to correspond to only one commuter at any given time, and evaluators were assumed not to tamper with or bypass the seat monitoring and locking mechanisms.
9. The fare matrix configured in the system was assumed to be internally consistent and complete so that the system can always compute the correct fare and any additional amount required for extended terminals.
10. The ISO/IEC 25010:2023 quality characteristics selected for this study (functional suitability, performance efficiency, reliability, security, usability, and data quality) were

assumed to be appropriate and sufficient for evaluating the developed prototype system.

11. Evaluators were assumed to give honest, thoughtful, and unbiased feedback based on their actual interaction with the prototype and to understand that the tests were conducted under simulated, prototype-level conditions.

Significance of the Study

The developed AFCS with seat allocation, terminal selection, and profit analytics dashboards was envisioned to benefit key stakeholders in public transport operations, even at the prototype level using an e-bike platform. By integrating cashless fare payment, pre-boarding terminal selection, seat availability monitoring, and operational and financial dashboards, the system demonstrated how automated and data-driven processes could improve commuter convenience, support safer driver operations, and provide organized records for system monitoring and decision-making.

Commuters. Commuting would become more convenient by allowing commuters to see their balance, view incoming vehicles, check available seats, and choose their drop-off terminal before boarding. Tap-based, cashless payment enabled fast transactions at the door, reduced boarding delays, and helped avoid fare discrepancies since the system automatically computes the correct fare.

Transport Operators. Transport operators, including drivers and system administrators, would benefit from the developed AFCS by using the driver and administrator dashboards to monitor seat status, passenger activity, fare transactions, and profit analytics. The system could support safer vehicle operation, reduce manual fare-related tasks, and provide organized records for operational and financial monitoring.

Future Researchers. This study would serve as a reference for future research on smart public transport systems, especially in exploring autonomous or driverless vehicles, and advanced analytics for operations planning.

Scope and Limitations

This study focused on the design, development, and evaluation of an automated fare collection system with seat allocation, terminal selection, and profit analytics dashboards for vehicles, implemented at the prototype level on an e-bike with a two-seat passenger capacity.

Scope of the Study

1. The study covered the development of a commuter interface that allows users to login, view their card balance, see incoming vehicles represented by the e-bike prototype, check available seats, and select a drop-off terminal before boarding.
2. The system implemented a tap-based payment process using MIFARE DESFire EV2 near-field communication (NFC) smart cards, in which the commuter taps the card on the reader upon boarding, and the prototype automatically deducted the fare based on the terminal selected in the commuter interface.
3. The prototype included a seat allocation and monitoring mechanism for two passenger seats on the e-bike, excluding the driver, where each seat has a servo-controlled lock that opens only after a valid tap for that trip, and an infrared (IR) sensor that detects whether the commuter is seated or has already vacated the seat at the chosen drop-off terminal, resulting in a total of two servos and two IR sensors for the two seats.
4. The study modeled an extended terminal feature, where a commuter who has already selected a drop-off terminal may request to extend to a farther terminal during the simulated trip, and the system recalculates and deducts any additional fare required.
5. The system included web-based driver and administrator dashboards that present operational information such as current seat availability, boarding and alighting per terminal, and financial analytics that summarize revenue, expenses, and profit at the prototype scale.
6. The backend infrastructure of the developed AFCS, including database storage, GPS-based tracking of the e-bike route, and basic security controls for card

transactions and user access, is within the scope to the extent necessary to support tap-based payment, terminal selection, seat monitoring, and dashboard analytics for the prototype.

7. The prototype was designed to require an active internet connection for card validation, transaction processing, and real-time database updates. When connectivity is temporarily lost or the device enters a dead spot, the system displays a system unavailable status on the TFT screen, denies card validation, and resumes normal operation once connectivity is restored.
8. The study evaluated the developed system using selected product quality characteristics from ISO/IEC 25010:2023, specifically: functional suitability, performance efficiency, reliability, security, usability, and data quality. Evaluation procedures were designed to suit a prototype mounted on an e-bike platform and were conducted through structured assessments by 12 experts, and user evaluation of 100 end-users for Objectives 1 and 2 and another 100 end-users for Objectives 3 and 4, through a questionnaire based on usability in terms of interaction capability.
9. The operational scenarios in this study were simulation-based, using the e-bike platform, a limited set of test routes and terminals, two passenger seats, and a controlled set of test commuters and trips that were considered sufficient to demonstrate the key features and gather evaluation data.
10. The web interface designs were initial versions and may be improved by future developers. The core functionalities specified in the objectives would remain unchanged.

Limitations of the Study

1. The system was implemented only at the prototype level using an e-bike with two passenger seats; therefore, it did not capture the full operational complexity, passenger volume, and traffic conditions of an actual fleet of vehicles.
2. The study used 10 test commuter cards, two seats, four terminals, and routes, and

the findings were interpreted only within this constrained setting; large-scale deployment behavior, peak-hour loads, and multi-route operations were not covered.

3. All fare transactions in the prototype utilized simulated monetary values. Real banking integration, actual electronic payment processing, and cash-in or cash-out operations involving real currency were not implemented or evaluated within the scope of this study.
4. Testers' and evaluators' misbehavior, such as intentional card sharing, refusal to tap, bypassing seat mechanisms, or deliberately entering wrong information, was considered beyond the control of the researchers and was not fully modeled or addressed in this study.
5. The prototype depends on internet connectivity for online card validation and real-time database communication. As a result, when connectivity is unavailable, the system cannot process validations or update records until the connection is restored. In addition, the study did not provide a comprehensive evaluation of large-scale network reliability, coverage, or all possible communication failures that may occur in real city-wide deployments.

Definition of Terms

This section listed the terminologies used in this study and provided their operational definitions as they were applied within the context of the developed AFCS.

ACR1252U NFC Reader. A USB-powered contactless smart card reader and writer that supports ISO/IEC 14443 and NFC standards, used in this study to read and communicate with MIFARE DESFire EV2 cards for tap-based fare transactions.

Admin/Administrator dashboard. The web-based interface used by cooperative administrators or system managers to view operational data, configure fares and terminals, record expenses, and monitor revenue and profit generated by the developed AFCS prototype.

Automated Fare Collection System (AFCS). A system that uses electronic devices, smart cards, communication networks, and software to automate fare payment, validation, recording, and reporting in public transportation; in this study, the developed AFCS functions as an AFCS for vehicles at the prototype level.

Boarding and alighting. Boarding refers to the act of a commuter entering the vehicle at a terminal or stop, while alighting refers to the act of a commuter leaving the vehicle at the chosen drop-off terminal.

Commuter. A person who uses the transportation service and interacts with the developed AFCS by selecting terminals in the commuter interface and tapping an NFC card to pay fares and access a seat.

Commuter interface. The web-based application provided to commuters allows them to login, view and top up their card balance, see incoming vehicles, check available seats, and select or extend a drop-off terminal before or during a trip.

Data analytics. The process of examining transactional and operational data using statistical and computational techniques in order to generate meaningful insights; in this study, data analytics was applied to metrics such as ridership, revenue, expenses, and profit.

Descriptive analytics. A type of data analytics that summarizes historical data to describe what has happened, typically through counts, totals, averages, and visualizations; in this study, examples include daily revenue, boarding and alighting per terminal, and seat utilization.

Diagnostic analytics. A type of data analytics that examines data to understand why certain patterns occur, such as identifying which terminals are frequently used, which periods show higher boarding counts, or which trips generate more profit in the simulated operations.

Driver. The person responsible for operating the e-bike prototype as a model of incoming vehicle and for viewing the driver dashboard that indicates boarding and alighting events and seat availability.

E-bike prototype. An electric bicycle used as a physical platform in this study to mount the

developed AFCS hardware and simulate the movement of the incoming vehicle, configured with two passenger seats and the necessary sensors and controllers.

Extended terminal. A farther terminal chosen by a commuter while already on board, beyond the original drop-off terminal, requires the system to recalculate the fare and deduct an additional amount from the commuter's balance.

Fare matrix. A structured representation of fare values between defined origin and destination terminals along a route, used by the developed AFCS to determine how much to charge the commuter based on the selected or extended drop-off terminal.

Global Positioning System (GPS). A satellite-based navigation system will be used in this study to determine the position and movement of the e-bike prototype along a simulated route, and to support the display of incoming vehicles in the commuter interface.

Incoming Vehicle. Refers to the public utility vehicle approaching the designated terminal or boarding area where commuters may board after selecting their destination terminal and completing card validation; in this study, an e-bike prototype was used to represent the vehicle operation on a smaller scale.

ISO/IEC 25010:2023. An international standard that defines a system and software product quality model composed of quality characteristics and sub-characteristics, such as functional suitability, performance efficiency, reliability, security, usability, and data quality; this study uses selected characteristics to evaluate the developed AFCS prototype.

MIFARE® DESFire® EV2. A secure, ISO/IEC 14443 Type A compliant contactless smart card integrated circuit that supports mutual authentication, secure messaging, and multiple applications on a single card; in this study, MIFARE DESFire EV2 cards serve as the fare media for tap-based transactions in the developed AFCS.

Near Field Communication (NFC). A short-range wireless communication technology based on ISO/IEC 14443 that enables data exchange between devices placed in close proximity; in this study, NFC enabled communication between the MIFARE DESFire EV2 cards and the ACR1252U reader during tap-in transactions.

Profit analytics dashboard. The part of the administrator dashboard that summarizes revenue from fares, recorded expenses, and computed profit for the trips and operations captured by the developed AFCS prototype.

Raspberry Pi 4. A single-board computer used as the main embedded platform in the fare-validation unit; in this study, the Raspberry Pi 4 interfaces with the NFC reader, drove the servo and IR-based seat mechanisms, performed card validation, local processing, logged transactions, and communicated with the backend server.

Seat allocation. The process by which developed AFCS assigns or reserves a specific seat for a commuter after a valid tap transaction, updates the seat status to reflect that it is no longer available, and unlocks the seat mechanism to allow the commuter to sit.

Seat monitoring mechanism. The combination of infrared sensors, servo-controlled locks, and system logic was used to detect and represent the status of each seat, such as whether it is available, reserved, or occupied, and to update seat availability in the driver and administrator dashboards.

Tap-based payment. A fare payment process in which a commuter presents an NFC card to the reader at the point of boarding, after which the system validates the card, computes the fare based on the selected terminal, deducts the corresponding amount from the balance, and records the transaction.

Terminal. A designated stop or point along the route, such as an origin, intermediate, or destination location, that is used as the basis for fare computation, boarding and alighting counts, and terminal selection or extension in the developed AFCS.

Transport cooperative. A registered organization that owns, manages, and operates a fleet of vehicles or similar public utility vehicles; in this study, transport cooperative administrators were considered key stakeholders for operational monitoring and profit analytics.

Web-based application. A software application that is accessed through a web browser over a network; in this study, the commuter interface, driver dashboard, and administrator dashboard were all implemented as web-based applications.

CHAPTER II

REVIEW OF LITERATURE AND STUDIES

This chapter presented the related literature and studies that provided the conceptual and empirical foundation for the developed automated fare collection system (AFCS). It discussed works on jeepney modernization, automated fare collection, near-field communication (NFC) and MIFARE® DESFire® technologies, data analytics and computing platforms, and system quality evaluation, as well as previous implementations of cashless fare systems, transport dashboards, and smart card security in public transportation.

Review of Literature

This section reviewed conceptual and technical literature that explained the policy context of jeepney modernization, the principles and technologies behind automated fare collection using NFC and MIFARE DESFire, the role of data analytics and embedded computing in transport systems, and the use of ISO based frameworks for system quality evaluation.

Jeepney Modernization, Automated Fare Collection Systems, and Local Transport

The Public Utility Vehicle Modernization Program (PUVMP) of the Philippine government aims to replace traditional jeepneys with modern units equipped with improved safety, emissions compliance, and digital technologies such as global positioning system (GPS) tracking, closed-circuit television (CCTV), and automated fare collection systems [11]. Under this program, vehicles are envisioned to support cashless payment channels and real-time operations monitoring, in line with international best practices.

The Department of Transportation (DOTr), in coordination with other agencies, has promoted the adoption of AFCS in rail and bus services, including the use of stored value cards, QR codes, and bank-issued contactless cards [21], [22]. More recent initiatives include partnerships with financial institutions and e-wallet providers to enable tap-to-pay fare

systems on urban rail lines [23]. These developments show a clear policy direction toward cashless, interoperable, and data-driven fare collection in public transport.

Within this context, the developed AFCS aligns with the PUVMP by modeling how an incoming vehicle could implement automated fare collection using secure NFC cards, seat monitoring, terminal-based fare computation, and analytics dashboards, even at prototype scale. The use of an e-bike platform with two passenger seats served as a small-scale representation of the incoming vehicle suitable for laboratory-level testing and demonstration.

In parallel, local initiatives on data analytics support the trend toward evidence-based decision-making in government services. The Department of Science and Technology (DOST) has developed tools for descriptive analytics and reporting, and national repositories such as Open Data Philippines have been conceptualized to consolidate government datasets for public access and analysis [24], [25]. These efforts indicate an institutional recognition that transport operations, including fare collection, can benefit from systematic data capture and analytics, which the developed AFCS addressed through its operator and administrator dashboards.

NFC and MIFARE DESFire Technologies for Secure Cashless Fare Collection

Automated fare collection systems worldwide have increasingly adopted contactless technologies such as smart cards and NFC, replacing manual cash payment with tap-based transactions [26]. In these systems, commuters typically use contactless cards or devices at validators placed in stations or vehicles, while backend servers process transactions and generate operational and financial reports.

MIFARE® DESFire® is a family of contactless smart card integrated circuits developed by NXP Semiconductors that follows global standards for the radio frequency (RF) interface and secure cryptographic methods [27], [28]. MIFARE® DESFire® EV2 supports multiple applications on a single card, mutual authentication, secure messaging, and advanced encryption schemes such as Advanced Encryption Standard (AES). These

properties make it suitable for applications that require both security and flexibility, including transit fare collection, campus cards, closed-loop micropayments, and access management [29].

In public transportation, MIFARE-based cards have been widely adopted in well-known systems such as London's Oyster and Singapore's EZ Link [6], [7]. These implementations demonstrate how secure contactless cards, when combined with appropriate backend infrastructure, can support large-scale and high-volume fare collection.

Near Field Communication (NFC) provides the short-range wireless interface between smart cards or NFC-capable devices and readers. NFC is based on ISO/IEC 14443 standards and enables secure data exchange over a range of a few centimeters [30]. Industry organizations such as the NFC Forum define specifications and promote interoperability among devices and applications [31], [32].

In the conducted study, MIFARE DESFire EV2 cards were used as fare media and were read by an ACR1252U NFC reader, while the Raspberry Pi 4 acted as the controller that authenticates cards, computes fares, and records transactions. The choice of DESFire EV2 and NFC was consistent with current best practices for secure, contactless fare collection in modern transport systems.

Data Analytics and Computing Platforms for Smart Public Transport

Smart mobility and intelligent transport systems increasingly rely on data analytics to monitor operations, understand passenger behavior, and support planning and policy decisions [33], [34]. In public transport, data generated by fare collection devices, vehicle trackers, and sensors can be processed to produce indicators such as ridership per route, boarding and alighting patterns per terminal, revenue trends, and service reliability measures.

Descriptive analytics summarizes historical or real-time data into counts, totals, averages, and visualizations, which allow operators to see what is happening in their network. Diagnostic analytics goes further by exploring why certain patterns occur, such as

identifying causes of revenue drops or repeated delays [34]. For a prototype system such as the developed AFCS, descriptive and basic diagnostic analytics were appropriate to demonstrate how public transport operations could be monitored through simple but informative dashboards.

On the computing side, low-cost yet powerful platforms such as the Raspberry Pi have become common in transport-related applications for data processing and control [35], [36]. They can interface with sensors, card readers, and communication modules, while running embedded software that logs data and communicates with backend servers. Database management systems (DBMS) provide structured storage for transactional records, allowing reliable retrieval for reporting and analytics [37], [38]. Web-based dashboards built on top of these databases enable operators and administrators to visualize metrics in near real time [39].

The developed AFCS prototype followed this architecture by using a Raspberry Pi 4 to process NFC card transactions, monitor seat occupancy components, and perform local data handling. Transaction and operations data were stored in a centralized database and displayed through web-based dashboards for drivers and administrators. This architecture demonstrated how modest hardware and standard database technologies can support the essential functions of an AFCS, including revenue tracking, seat availability monitoring, and terminal-based boarding and alighting counts.

System Quality Evaluation Using ISO/IEC 25010 and Usability Measures

Software and system quality evaluation often relies on standardized frameworks that define quality characteristics and sub-characteristics. ISO/IEC 25010:2023 specifies a product quality model that includes characteristics such as functional suitability, performance efficiency, reliability, security, usability, and data quality [18]. Many studies selected characteristics into questionnaires or rubrics, then use these to evaluate domain-specific systems, including information systems, mobile applications, and specialized tools [40].

In transport and payment systems, usability evaluation is also critical, since

commuters, drivers, and administrators interact with interfaces that must be understandable and efficient in real operational contexts. The System Usability Scale (SUS), a 10-item Likert questionnaire that yields a single usability score from 0 to 100, has been widely used as a quick and technology-independent instrument to assess perceived usability [41], [42]. Its simplicity and robustness make it a practical choice for prototypes and deployed systems alike.

The conducted study adopted selected ISO/IEC 25010 characteristics, specifically functional suitability, performance efficiency, reliability, security, usability, and data quality, as the basis for evaluating the developed AFCS prototype.

Review of Studies

This section examined empirical studies that have implemented cashless fare collection systems, developed data analytics dashboards for transit operations, analyzed smart card security in transport payment systems, and applied standardized quality and usability instruments to transport and payment applications relevant to the design and evaluation of the developed AFCS.

Implementations of Cashless Fare Collection Systems

Multiple studies and projects worldwide have explored the implementation of cashless fare collection systems using smart cards, NFC, and related technologies [43], [44], [45]. These works typically investigate system architecture, adoption issues, and operational performance. They show that contactless payment methods can reduce boarding time, improve revenue security, and provide more accurate data for planning.

Some low-cost implementations have used RFID-based cards integrated with microcontroller platforms and backend databases to create basic AFCS prototypes with user interfaces for commuters and operators [46], [47], [48]. These prototypes demonstrate that essential functionalities such as tap-based payment, balance inquiry, and basic reporting can be achieved with affordable hardware and open-source software.

The findings from prior studies informed the design of the developed AFCS by establishing its overall system structure: a card-based payment channel, embedded vehicle controllers, a backend database, and dashboards that deliver information at both commuter and operator levels. The developed AFCS extends these foundations by explicitly incorporating seat allocation logic, terminal selection and extension mechanisms, and profit analytics tailored to the operational context of vehicles.

Data Analytics Dashboards in Transit Operations

Studies on data analytics in local public transport have shown how dashboards could be used to monitor driver behavior, route performance, and service quality [49], [50]. For example, some projects have developed dashboard systems that track driving patterns, compliance with traffic rules, and other indicators to assess the competency and behavior of public utility vehicle drivers. These dashboards translate raw sensor and trip data into visual metrics that can be used for training, evaluation, and policy formulation.

Other projects have integrated real time data analytics into broader systems for route planning, fleet management, monitoring and verification, and access to financing [51]. In these cases, dashboards serve as central tools for viewing route-level and fleet-level performance, enabling operators and regulators to make data-informed decisions about operations and investments.

These precedents supported the use of driver and administrator dashboards in the developed AFCS, where metrics such as seat availability, boarding and alighting per terminal, and revenue and profit summaries were presented in a web-based interface. By focusing on the specific case of vehicles and an e-bike-based simulation, the developed AFCS demonstrated how even small-scale systems could leverage dashboards to improve operational visibility.

Smart Card Encryption and Security in Transport Payment Systems

The security of contactless payment systems, particularly in public transport, has been examined in several studies. Researchers have analyzed authentication protocols and

encryption schemes in MIFARE DESFire cards, highlighting how mutual authentication and secure messaging help protect transactions against common attacks [51]. Other studies on older technologies, such as MIFARE Classic, have documented vulnerabilities, including weak cryptographic algorithms and susceptibility to cloning [52].

These analyses motivated the transition from less secure card families to more advanced options such as MIFARE DESFire, which provide stronger encryption, larger memory options, and better support for multi-application use cases [52], [53]. Some works have also discussed mitigation strategies and card blocking techniques for legacy systems, but generally recommended the adoption of more secure technologies for new deployments.

In addition, research on automated fare collection systems in various cities has emphasized the importance of robust cryptographic mechanisms and secure key management to protect financial transactions and personal data [54]. These findings underscored the relevance of using MIFARE DESFire EV2 and AES-based security in the developed AFCS.

Quality and Usability Evaluation in Transport and Payment Applications

Several empirical studies have applied ISO/IEC 25010-based frameworks to evaluate information systems, including transport-related applications. These studies typically create evaluation instruments that map selected quality characteristics, such as functional suitability, performance efficiency, reliability, security, and usability, to questionnaire items or rating scales [40]. They then compute descriptive statistics for each characteristic and use reliability measures such as Cronbach's alpha to assess the internal consistency of the scales.

In the transport domain, passenger information systems, ticketing applications, and mobility apps have been evaluated using standardized usability questionnaires. For example, some research has used SUS to compare prototype passenger information systems against existing commercial applications, analyzing the resulting scores and reliability coefficients to determine whether usability is acceptable [55], [56].

Design, Development, Testing, and Evaluation of the Developed AFCS

Relevant studies were reviewed to support the five modules of the present study, namely the commuter interface, tap-based payment process, seat monitoring and allocation mechanism, extended terminal feature, and driver and admin dashboards. These studies were deemed relevant because they provided the technical and conceptual foundation for the design, development, testing, and evaluation of the developed modules. As such, they strengthened the basis of the study in terms of both system implementation and evaluation.

Module 1: Commuter Interface

Recent studies on passenger information systems emphasize that commuters rely on accurate and timely information before boarding, particularly for route awareness, arrival-related information, and decision support. The guide on fare systems in [57] explained that fare platforms must calculate, verify, and present fare-related information according to the agency's fare structure and payment mechanisms, which supports the need to test whether balance-related information is correctly reflected in the commuter interface. [58] further explained that real-time transport data are already used in passenger-facing applications such as journey planning and live service information, indicating that information shown to commuters must be accurate enough to support trip decisions. [59] also show that public transport arrival-time communication is a major passenger information concern, while van [60] emphasize that passenger information systems must provide relevant and reliable information to avoid confusion and dissatisfaction among users.

Module 2: Tap-Based Payment Process

The tap-based payment process of the developed AFCS was evaluated through tap read success rate, terminal-based fare computation accuracy, and automatic seat allocation success rate, and these test areas are strongly supported by related studies on electronic ticketing and contactless transport payment. [61] proposed a reusable public transport electronic ticket system with fast validation and reported that ticket validation, get-in, and get-out use cases with real-time constraints can be completed in less than one second, showing

the importance of quick and dependable validation processes in automated transit environments. [62] also examined the emergence of contactless digital payment technology for transportation and highlighted the growing role of contactless systems in public transport innovation. In addition, the fare guide in [63] explained that technology-based fare systems must calculate, verify, and collect payment according to available fare policies and structures, which aligns with the need to test the correctness of terminal-based fare computation in the developed system. [64] likewise described an automatic bus fare collection system using RFID and GPS, where experimental testing confirmed accurate fare calculation and reduced boarding time, reinforcing the relevance of validating both correct fare processing and efficient boarding transactions. Since the developed Module 2 also included automatic seat assignment after successful validation, the study of [65] on passenger occupancy estimation and the work of [66] on bus scheduling and seat allocation are also relevant, as they show that occupancy-aware and seat-allocation mechanisms depend on correct status recognition and proper use of available capacity [65], [66].

Module 3: Seat Monitoring and Allocation Mechanism

The seat monitoring and allocation mechanism was evaluated in terms of seat occupancy detection accuracy, driver seat status view consistency rate, commuter seat status view consistency rate, and seat lock and unlock actuation accuracy. This testing approach is supported by recent studies on occupancy estimation and smart seat monitoring. [65] reviewed current methods for passenger occupancy estimation in vehicles and showed that occupancy detection systems are assessed primarily by how accurately they detect actual seat or passenger presence under operational conditions. In a more direct transport application, [67] designed an IoT-based smart occupancy monitoring system for minibus taxis that tracked seat occupancy, calculated fees in real time, and integrated sensing and location technologies into a monitoring prototype.

Implied Module: Extended Terminal Feature

The extended terminal feature was evaluated through extension success rate and

deduction correctness, and these evaluation areas are supported by studies and standards related to fare structure logic, route-dependent charging, and transaction processing. The N-CATT fare guide explains that fare systems are closely tied to fare policies and structures, and that the fare structure determines how much a passenger should be charged at a given time depending on the applicable conditions of travel [57]. This directly supports the need to test whether an updated or extended trip selection produces the correct additional fare in the developed AFCS. The PRESTO Service Design Standard provides more direct support, since it documents operational cases such as GPS-based fare switching, default trip override, tap-on and tap-off behaviors, and Fare Transaction Processors, which show that modern fare systems must correctly handle trip changes, boundary-based fare rules, and updated trip conditions during travel [68].

Module 4: Driver and Admin Dashboards

The driver and admin dashboards of the developed AFCS were evaluated through dashboard data consistency, particularly for revenue, expenses, and profit. This evaluation is supported by Firebase's documentation on Cloud Firestore real-time queries, which explains that real-time updates are propagated efficiently through the system and that the platform is designed for high availability, scalability, and support for many connected real-time queries [69].

Related studies provided concrete, reportable results that were directly usable as comparative evidence when interpreting the study's ISO/IEC 25010:2023 evaluation results in Chapter 4 for the automated fare collection system with seat allocation, terminal selection, extended terminal fare adjustment, and operational and financial dashboards.

A jeepney fare payment application evaluated using ISO-based quality criteria reported an overall mean of 3.82 (Very Good) from users and 4.29 (Excellent) from IT experts, with IT experts rating Functionality = 4.33 (Excellent), Performance Efficiency = 4.00 (Very Good), Usability = 4.22 (Excellent), and Reliability = 4.25 (Excellent). This was strongly usable, because it presented mean scores that could be aligned with the study's mean-based ISO

interpretation when reporting the ratings per quality characteristic in Chapter 4 [70]. An RFID-based automated bus fare collection evaluation also reported statistically significant results, where the computed Chi-Square was 95.06564609, exceeding the critical value 16.919 (9 df, 0.05), and the computed *t* was 2.629, exceeding the critical value 2.056 (26 df, 0.05), indicating a significant difference between traditional and automated fare systems. The same study concluded the automated system was more accurate based on the mean/average of 26.93 (automated) compared with 28.57 (traditional), which was strongly usable to support our Functional Suitability and Reliability discussion, particularly when the findings emphasized accuracy and consistency improvements after automation [71].

Performance-efficiency benchmarks that were moderately to strongly usable for the Performance Efficiency and Usability interpretation were reported in an Inter-American Development Bank technical note, which stated that reaction-time expectations for fare media range from a maximum of 300 milliseconds to 600 milliseconds and documents a gate trial where 65 to 75 users per minute could pass through while receiving visual feedback on payment success at the gate and on mobile phones. These benchmarks were usable comparison points when the study's Chapter 4 included measured tap-response time [72].

Operational outcome evidence that remained usable for discussion-level support was also provided by a World Bank report stating that smartcard ticketing could deliver savings through reduced fraud, shorter dwell times at stops, and less cash handling, which could be linked to the study's Security, Reliability, and overall operational efficiency interpretation [73]. Evidence supporting our Data Quality discussion was provided by a smart-card transaction study that processed 100,000 transactions and reported that 98,632 were error-free, while identifying an example error pattern as consecutive swiping activity occurring more than three times in the same bus trip; this supported the need for transaction validation and cleaning so that stored logs remain credible for dashboards, reports, and analytics [74]. Lastly, a rail-transit AFC paper explains that traditional AFC structures could suffer limited reliability and availability, including vulnerability to single-point failures, which was moderately usable when

the study's Chapter 4 justified reliability and availability design choices in automated fare collection systems [75].

Patent Search

The following patents focused on systems and software systems that provided foundational structures for the overall functionalities, prototype design, and software-level design of the developed automated fare collection system (AFCS).

MIFARE Push (US9607192B2)

"MIFARE Push" presented a system and method for accessing a MIFARE section within a secure memory device through an RFID reader. The invention used the RFID reader to read and write data across multiple memory sectors by transmitting the corresponding sector keys and data. It also allowed the RFID reader to write information to a predefined sector in the MIFARE partition of the secure memory device, where the stored information indicated the application identification of a specific software application [76].

Similarity

It was related to the developed AFCS because both involved the use of MIFARE-based card technology and RFID communication for secure card-related data access.

Difference

It focused on MIFARE memory access, while the developed system applied card validation to fare computation, seat allocation, terminal selection, and dashboard-based monitoring.

Method for Processing Payment of a Transportation Card and Card Reader Thereof (KR101214627B1)

"Method for Processing Payment of a Transportation Card and Card Reader Thereof" presented a method and terminal apparatus for processing transportation card payments efficiently. The invention proposed a card reader terminal that stored only positive list (PL)

information for a specific group of cardholders instead of storing and searching through the data of all members. The terminal verified the transportation card by comparing its identifier with the stored positive list data and approved the payment when a match was found. This method reduced the memory load of the terminal, improved lookup speed, and supported faster handling of valid and invalid cards [77].

Similarity

It was related to the developed AFCS because both involved transportation card validation and faster payment processing through a card reader.

Difference

It focused on positive-list-based payment approval, while the developed system integrated card validation with terminal-based fare computation, automatic seat allocation, invalid reason display, and dashboard-based operational monitoring.

Secondary Card Tap System (US10424037B2)

"Secondary Card Tap System" presented a networked fare validation architecture involving a Primary Validation Unit, a Validation Detection System, and a Secondary Validator. The Secondary Validator was positioned at a transit stop and used dual antennas, where one antenna detected nearby vehicles and the other processed short-range fare checkouts. The system enabled synchronized fare validation and data transmission to improve real-time accuracy in automated transit payment systems [78].

Similarity

It was related to the developed AFCS because both involved automated transit fare validation, card tap processing, and network-based transmission of fare-related data.

Difference

It focused on secondary tap validation at transit stops, while the developed system focused on an e-bike-based prototype that integrated card validation with terminal-based fare computation, seat allocation, seat monitoring, and profit analytics dashboards.

Interactive Dashboard Display (US10846799B2)

"*Interactive Dashboard Display*" presented a system architecture for vehicle data visualization and risk-based insurance analytics. The invention used a central computing unit with processing, memory, input/output, network, and modem components operating under a security and integration layer to coordinate data flow between local and wide area network connections. It also communicated with user and vehicular devices through internet and wireless links to collect and analyze sensor-based driving data. The system generated personalized visual dashboards that provided feedback and recommendations related to driving behavior, vehicle operation, and insurance-related performance [79].

Similarity

It was related to the developed AFCS because both used dashboard-based visualization to present vehicle-related and operational data for monitoring and decision-making.

Difference

It focused on insurance risk analytics and driver behavior assessment, while the developed system used dashboards for passenger monitoring, fare transactions, seat status, revenue, expenses, and profit analytics.

Research Gaps

After reviewing local and foreign literature and studies on automated fare collection, NFC-enabled MIFARE DESFire technology, seat and capacity management, and transport data analytics, the following research gaps were identified, which the conducted study intended to address.

Lack of Commuter Interfaces that Combine Seat Availability, Route View, and Pre-boarding Terminal Selection

Most documented AFCS implementations focused on payment and card validation, with limited emphasis on commuter-facing tools that show real-time seat availability and allow

commuters to select their drop-off terminal before boarding. Existing systems often assumed that destination selection occurred at the time of payment or was handled verbally with the driver, rather than through an integrated commuter interface that also displays route information and incoming vehicles. This left a gap in how a commuter-oriented web interface can reduce boarding bottlenecks and fare disputes in public transport operations.

Limited Integration of Tap-based AFCS with Physical Seat Allocation Mechanisms

The reviewed literature on AFCS and smart card technologies typically considers logical capacity control but seldom integrates actual seat locking or seat-level monitoring with the fare collection process. There is a scarcity of prototypes that link a valid tap-based transaction to a physical seat allocation mechanism using servo-controlled seat locks and per-seat infrared sensors. As a result, there is limited guidance on how AFCS can enforce one seat per valid commuter and reflect real-time seat status in both commuter and driver views for vehicles.

Sparse Treatment of In-trip Terminal Extension in Vehicles

Many fare systems in the literature were designed around fixed origin-destination pairs or station-based boarding and alighting, and do not explicitly support in-trip changes of destination in open or semi-open services. There was little work that models an extended terminal mechanism in which a commuter who has already selected a drop-off terminal can request a farther terminal while on board, with automatic recalculation of the additional fare and consistent updating of seat and transaction records. This gap was particularly relevant to vehicles, where passengers may decide mid-trip to extend their travel beyond the original terminal.

Insufficient Attention to Driver Side Real-time Decision Support in Jeepney Operations

Many AFCS designs concentrate on back-office reporting and commuter payment experience, with comparatively less attention to the information needs of drivers in small public utility vehicles. Few studies described driver dashboards that indicate, in real time, which terminals still have boarding or alighting passengers, how many seats remain

available, and when the driver can safely proceed without stopping. This gap affected how AFCS can help reduce driver distraction by removing manual fare computation and supporting more predictable stopping behavior.

Limited Local Prototypes that Use MIFARE DESFire EV2 with Edge Computing for Intermittent Connectivity

While MIFARE DESFire technology was well described in datasheets and foreign case studies, few local prototypes implement DESFire EV2-based AFCS with explicit handling of intermittent connectivity through edge computing. The literature provides limited examples where a Raspberry Pi class controller continues validating taps, computing fares, and logging data locally during connection loss, then automatically synchronizes transactions to web-based dashboards once network connectivity is restored. This gap was important in the Philippine setting, where dead spots are common along routes.

Synthesis

The reviewed literature and policies showed that Philippine transport reform, particularly under the PUVMP and AFCS initiatives, is moving toward vehicles that are cashless, data-driven, and continuously monitored through GPS and related technologies [11], [21]. These references established that automated fare collection is not only technically viable but also a national direction for public transport, which provided a clear justification for developing the AFCS prototype. At the same time, local data analytics efforts and open data initiatives indicated that transport operations are increasingly expected to generate actionable information for planners, operators, and cooperatives [24], [25].

Literature on NFC and MIFARE DESFire clarified why DESFire EV2 was appropriate as the fare medium in this study. DESFire cards supported secure messaging, mutual authentication, and AES-based encryption, which are suitable for transport applications that require both fast validation and strong protection of stored value [27], [28], [29]. NFC standards and the ecosystem shaped by the NFC Forum further demonstrated that short-

range card reader communication is mature enough to support reliable tap-based transactions in public transport [30], [31], [32]. These works provided the cryptographic and communication foundation for using MIFARE DESFire EV2 cards and an ACR1252U reader in the developed AFCS prototype.

The literature on data analytics and computing platforms showed that low-cost devices such as the Raspberry Pi could serve as embedded controllers that interface with sensors, NFC readers, and communication modules while logging data to a database and feeding web-based dashboards [35], [36], [39]. Descriptive and diagnostic analytics were commonly used to summarize ridership, revenue trends, and route performance in smart mobility contexts [33], [34]. These ideas supported the decision to use a Raspberry Pi 4, a database-backed architecture, and web dashboards in the developed AFCS, so that seat availability, boarding and alighting per terminal, and basic revenue and expense indicators can be monitored in near real time.

Empirical studies on cashless fare collection systems and RFID or NFC-based prototypes confirmed that tap-based payment, balance inquiry, and simple reporting can be implemented with affordable hardware and open-source software [43], [44], [45], [46], [47], [48]. Other works on data analytics dashboards in transit operations demonstrated how visualizing driver behavior, route performance, and service quality could improve oversight and decision making in public transport [49], [50], [51]. Research on smart card security explained the move away from vulnerable card families toward more secure options such as MIFARE DESFire, emphasizing the need for robust encryption and sound key management in transport payment systems [51], [52], [53], [54]. Studies that applied ISO/IEC 25010 and standardized usability instruments such as SUS to transport and payment applications showed that it was feasible to evaluate such systems with respect to functional suitability, performance, reliability, security, and usability using structured questionnaires and quantitative metrics [18], [40], [41], [42], [55], [56].

Furthermore, the reviewed studies in [57], [58], [59], [60], [61], [62], [63], [64], [65],

[66], [67], [68], and [69] collectively support the design, development, testing, and evaluation of the five modules of the developed automated fare collection system. For the commuter interface, the literature shows that commuter-facing transport systems must provide accurate and timely balance, route, seat availability, and terminal information to support pre-boarding decisions and reduce passenger uncertainty [57], [58], [59], [60]. These studies justify the evaluation of the module in terms of card balance display accuracy and incoming vehicle information accuracy.

For the tap-based payment process, the reviewed studies emphasize that reliable card reading, correct fare computation, and efficient automated processing are essential functions of smart fare collection systems [57], [61], [62], [64], [65], [66]. These references support the selected evaluation of the module in terms of tap read success rate, terminal-based fare computation accuracy, and automatic seat allocation success rate, since correct validation and post-validation processing are central to automated fare collection.

For the seat monitoring and allocation mechanism, the literature supports the evaluation of actual seat-state detection, consistency of reflected seat availability across interfaces, and physical actuation accuracy, since modern transport monitoring systems rely on accurate occupancy sensing and dependable response mechanisms [65], [67]. These studies justify testing both the sensing and actuation components of the developed module.

For the extended terminal feature, the reviewed references show that modern fare systems must correctly process trip updates, route-dependent charging, and additional fare adjustments when travel conditions change [57], [68]. These studies support the evaluation of the module in terms of extension success rate and deduction correctness, since the developed feature required the system to correctly reflect a farther terminal selection and apply the corresponding additional fare.

For the driver and admin dashboards, the reviewed reference on Firestore real-time synchronization supports the need for consistent and near real-time reflection of operational and financial data across connected interfaces [69]. This justifies the evaluation of the

dashboards in terms of data consistency, particularly for revenue, expenses, and profit.

For the overall evaluation of the system using ISO/IEC 25010:2023, the references show that the standard provides an appropriate framework for assessing systems in terms of functional suitability, performance efficiency, reliability, security, usability, and data quality [70], [71]. Related studies also demonstrate that these quality characteristics can be evaluated through structured questionnaires and mean-based interpretation of respondent ratings, making them suitable for the expert evaluation used in the present study [72], [73].

In addition, the reviewed references support the inclusion of usability and data quality as important dimensions of system evaluation. Recent studies show that usability may be measured through user interaction capability and related response-based instruments, while data quality remains essential in ensuring that stored and reported system records are credible and useful for monitoring and analytics [72], [75]. The use of a standardized reporting format for usability evaluations also strengthens the consistency and interpretability of the assessment process [74].

Patent documents complemented these academic sources by illustrating concrete system architectures that connected card readers, central servers, and visual interfaces. Patents on MIFARE memory access methods and transportation card readers offered examples of how card data could be efficiently verified and managed at the terminal level [76], [77]. Designs for secondary validation units and interactive dashboards demonstrated networked arrangements where field devices send transaction and sensor data to central computing units that generated real-time visualizations for decision support [78], [79]. These patented ideas informed the overall structure of the developed AFCS, where an on-board controller, centralized database, and web-based dashboards worked together to support automated fare collection and operational analytics.

Despite the substantial body of literature on automated fare collection and transport analytics, several needs in the context of public transport operations remained insufficiently addressed. In response to these gaps, the developed AFCS prototype was designed to

integrate the key functions reflected in the research objectives, namely a commuter interface; a tap-based payment process; a seat monitoring and allocation; an extended terminal feature; and driver and admin dashboards for operational monitoring and financial analytics.

CHAPTER III

RESEARCH METHODOLOGY

Project/Research Design

This study used a **developmental research design**, specifically the **Design and Development Research (DDR)** [80] approach, which guided the design, development, and evaluation of the automated fare collection system (AFCS) prototype. Developmental research was appropriate because the primary output was a working technological solution that was iteratively refined based on identified requirements, prototype testing, and quality evaluation, rather than a purely observational or correlational model.

Within this framework, the DDR approach structured the study into systematic phases that include: (1) analysis of user needs and system requirements in the context of vehicles, (2) design of the hardware and software architecture, (3) development and integration of the prototype components, and (4) evaluation of the resulting system against predefined objectives and quality criteria. The main artifacts in this study were the commuter web interface, the e-bike-mounted AFCS device that uses MIFARE DESFire EV2 smart cards and an NFC reader, the servo and infrared-based seat allocation mechanism, the extended terminal feature, and the web-based driver and administrator dashboards.

The research design aligned with the specific objectives of the study. Objective 1 covered the design and implementation of the commuter interface that displays balance, incoming vehicle information, seat availability, and terminal selection. Objective 2 focused on the tap-based payment process, in which the commuter's previously selected terminal in the interface is used to compute the fare, deduct the amount from the card, and trigger seat allocation. Objective 3 concerned the development of the seat monitoring and allocation mechanism that uses servo-controlled seat locks and infrared sensors to reflect real-time seat status. Objective 4 focused on the driver and administrator dashboards that provide operational status, boarding and alighting per terminal, and revenue, expense, and profit

analytics using data captured from the prototype. Finally, Objective 5 involved the evaluation of the integrated system using selected ISO/IEC 25010:2023 product quality characteristics, namely functional suitability, performance efficiency, reliability, security, usability, and data quality.

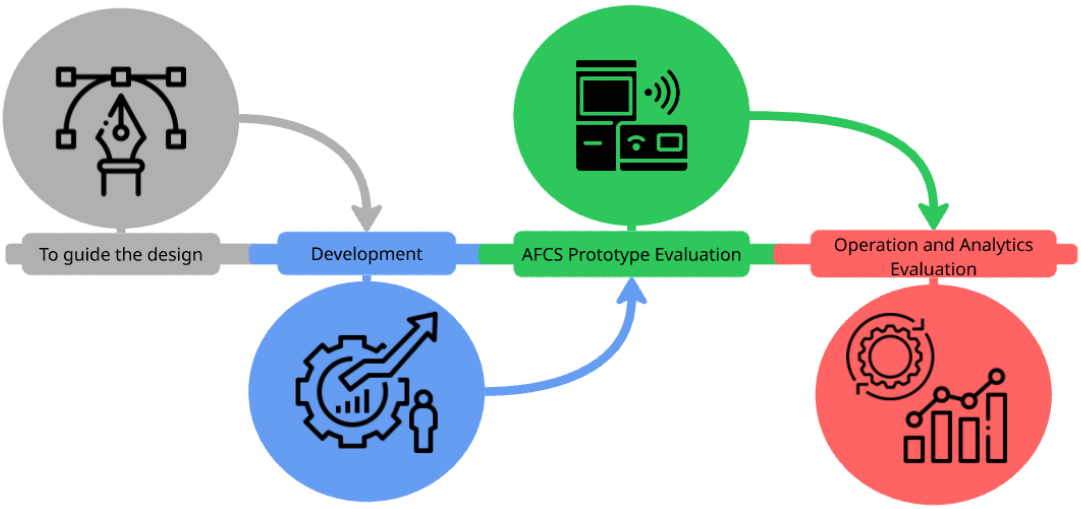


Fig. 2. Developmental Research Design Phases

Through the phases of developmental research design illustrated in Figure 2, the study aimed to produce a validated prototype of the automated fare collection system (AFCS) that demonstrated not only the technical feasibility of integrating automated fare collection, seat allocation, terminal management, and profit analytics for vehicles but was also evaluated against recognized quality standards.

System Architecture

This study adopted an Internet of Things (IoT) architecture with the following layers: device layer, edge layers, network layer, cloud layer, business layer, and application layer.

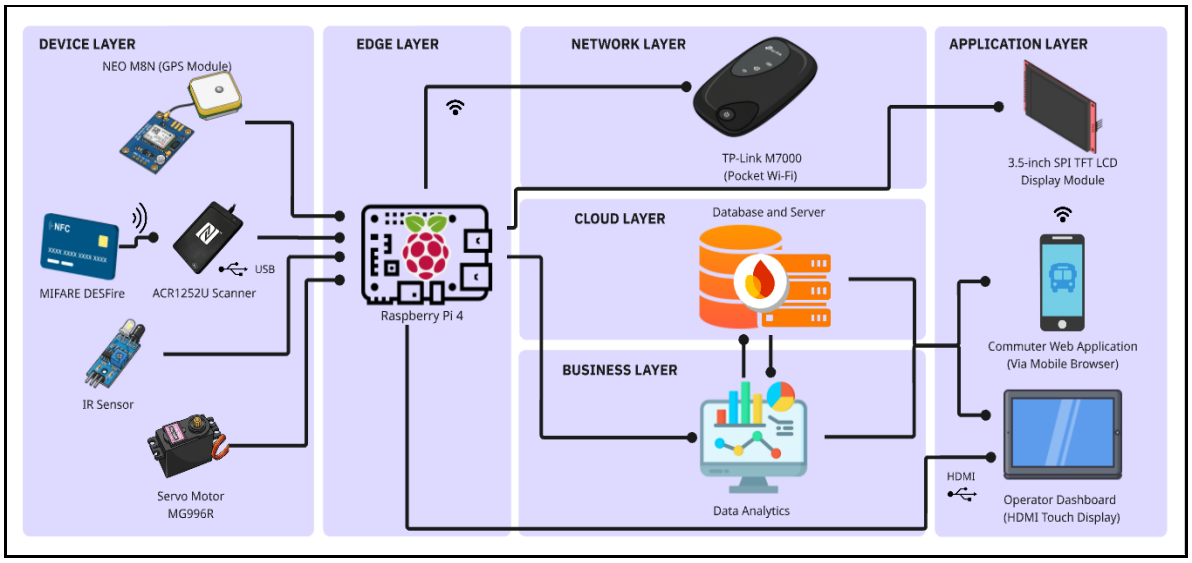


Fig. 3. IoT Architecture of the Developed AFCS

Figure 3 presented the IoT-based system architecture of the developed AFCS with seat allocation, terminal selection, and profit analytics dashboards.

The device layer contained the field devices, which include the NEO M8N GPS module, the MIFARE DESFire smart card and ACR1252U NFC scanner, the infrared seat occupancy sensor, and the MG996R servo motor that actuates the seat locking mechanism.

The edge layer was represented by the Raspberry Pi 4, which performs local processing of tap-based events, seat allocation, and trip logging. When Internet connectivity is unavailable, the system displays a system unavailable status on the TFT screen, denies card validation, and resumes normal operation once connectivity is restored.

The network layer provided wireless connectivity through the TP-Link M7000 pocket

Wi-Fi, which links the edge device to the remote server infrastructure.

The cloud layer was implemented using Firebase Firestore as the centralized real-time database and Firebase Cloud Functions for server-side logic and automation. Firestore stores the registered commuter card records and status (valid or invalid), commuter profiles, route, terminal, and fare table definitions, tap-based transaction logs with computed fares, seat status and allocation history, expense categories and expense entries, and device and operational logs. Cloud Functions enforce consistent backend rules such as card validation against the registration database, secure transaction handling and data integrity checks, and automatic aggregation of revenue and expense data for the real-time analytics dashboards.

The business layer performed data analytics over the stored records in order to compute revenue, expenses, and profit indicators for cooperative-level assessment.

Finally, the application layer delivered these functions to end users through web-based interfaces, specifically a commuter web application accessed via a mobile browser, a 3.5-inch SPI TFT LCD module for on-board information, and an HDMI touch display that serves as the operator dashboard.

Block Diagram

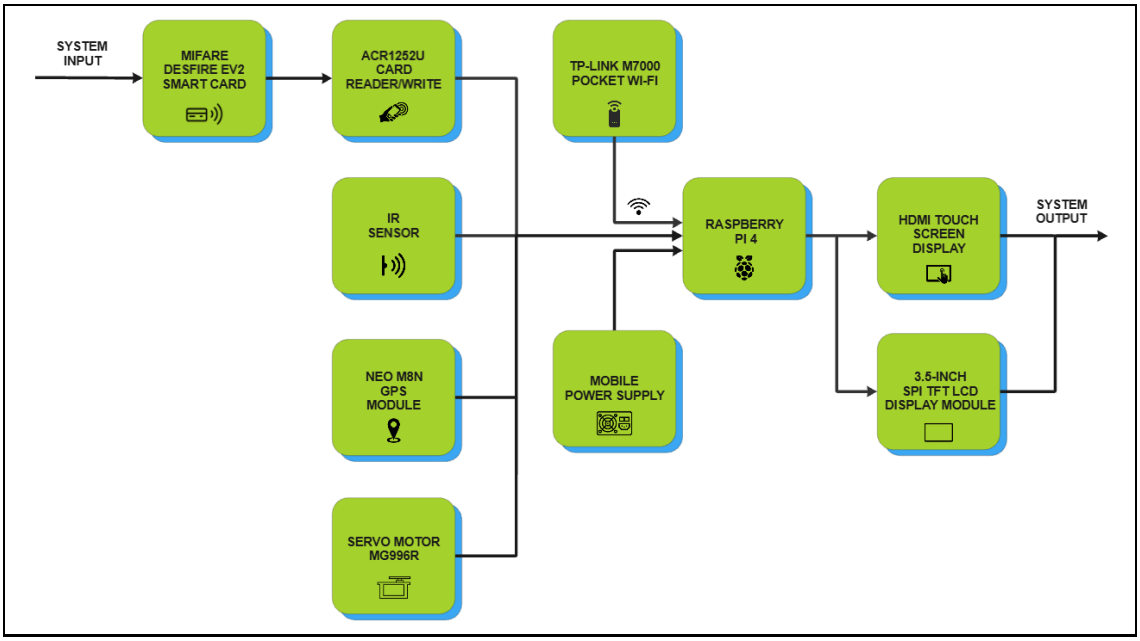


Fig. 4. Block Diagram of the Developed AFCS

Figure 4 presented the hardware block diagram of the developed AFCS prototype. The system input originates from the MIFARE DESFire EV2 smart card, which is tapped on the ACR1252U NFC card reader and writer. The ACR1252U forwards the card identifier and transaction data to the Raspberry Pi 4, which serves as the main controller.

The Raspberry Pi 4 also receives signals from the infrared (IR) sensor that monitors seat occupancy and from the NEO M8N GPS module that provides location and movement data for the simulated incoming vehicle.

In addition, the Raspberry Pi 4 drives the MG996R servo motor, which acts as the seat lock mechanism that opens or closes a seat based on the validity of the tap and the current seat allocation state.

Network connectivity is provided through the TP-Link M7000 pocket Wi-Fi, which

connects the Raspberry Pi 4 to the backend server for transaction logging and dashboard synchronization. System outputs are displayed through two visual interfaces: an HDMI touch screen display for the driver, and a 3.5-inch SPI TFT LCD module or mobile-web view for the commuter interface. Together, these components implement the end-to-end flow of developed AFCS, from card tap and sensor input, through embedded processing and network communication, to visual feedback for commuters and drivers.

Project Development

This section described the technical design and development of the developed AFCS prototype. The project adopted the Das V-Modell as its overarching development model, in which each specification and design activity on the left side of the V was paired with a corresponding test and validation activity on the right side, as shown in Figure 5.

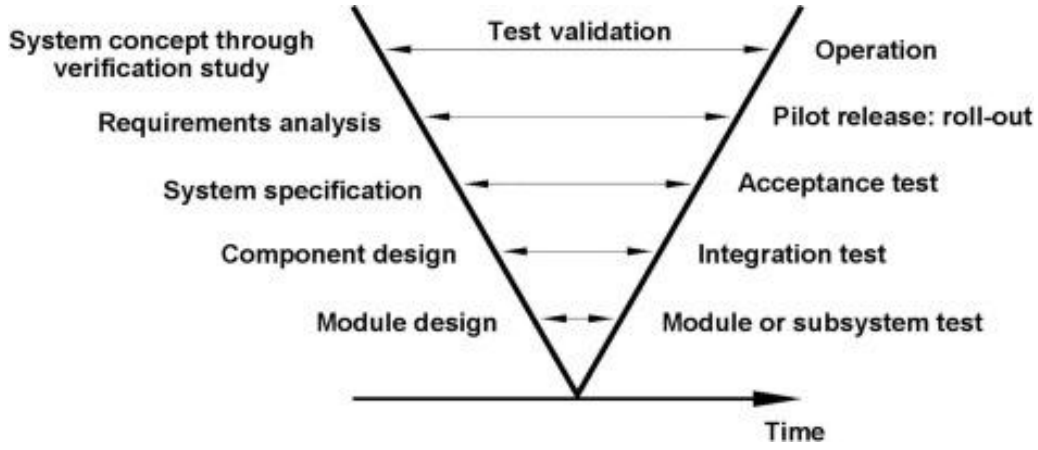


Fig. 5. Das V-Modell [81]

In this study, the V-Modell was instantiated through six concrete project phases, namely Planning, Hardware Design, Software Design, Development, Testing, and Evaluation, as illustrated in Figure 6. These phases provided a structured and iterative

process for specifying requirements, designing hardware and software, implementing the prototype, and systematically verifying and evaluating its performance and quality.

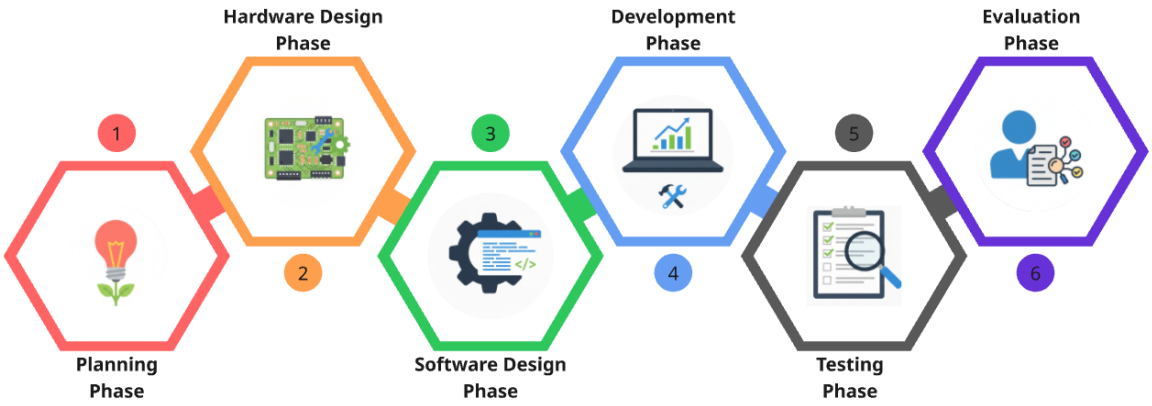


Fig. 6. Project Development Phases

Planning Phase

1. The researchers first reviewed the finalized research objective, scope, and limitations to determine the required technicalities for the commuter interface, tap-based payment process, seat monitoring and allocation mechanism, extended terminal feature, driver and admin dashboards, and ISO/IEC 25010-based evaluation.
2. The researchers identified the major hardware and software components required for the development of the system, which include the Raspberry Pi 4 as the processing and control unit, the ACR1252U as the card reader, the TFT display as the visual interface, Firebase as the database platform, Hostinger as the hosting platform, the power station with AC and DC output as the power source, the MG996R servo motor as the actuator, and the infrared sensor as the detection sensor
3. Using a Gantt chart (see Appendix H, p. 190), the researchers scheduled the subsequent phases, allocating time for hardware design, software design,

development, testing, and evaluation. Milestones were defined for hardware readiness, minimum viable software, integrated prototype, and evaluation readiness.

Hardware Design Phase

1. A list of hardware components was prepared that includes all key components and other materials needed in developing the prototype for the developed AFCS (see Appendix E, p. 161). The corresponding unit costs and total project cost were presented separately (see Appendix J, p. 193).
2. The researchers measured the e-bike and the two-seat configuration, then drafted a scaled prototype layout in AutoCAD that showed the planned positions of the seat monitoring allocation mechanism, tap-based payment process, project box and power station, driver's dashboard, and external GPS antenna (see Figures 7–11).

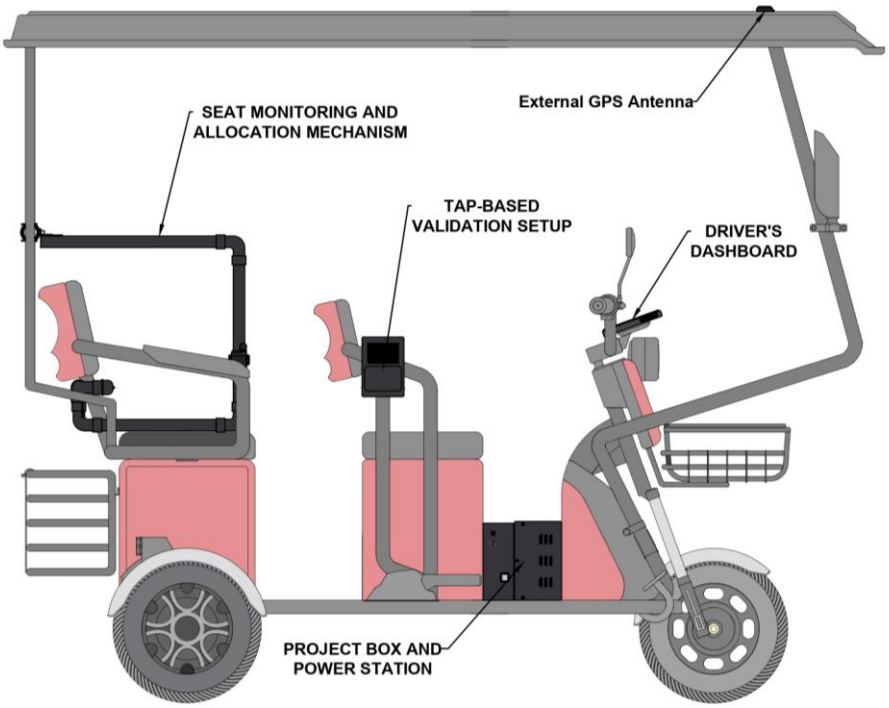


Fig. 7. Overall Layout of the Developed AFCS Prototype

Figure 7 illustrated the e-bike with attached modules, the seat monitoring and allocation mechanism, the tap-based payment process, and the driver's dashboard. The figure also showed the project box and power station and the external GPS antenna.

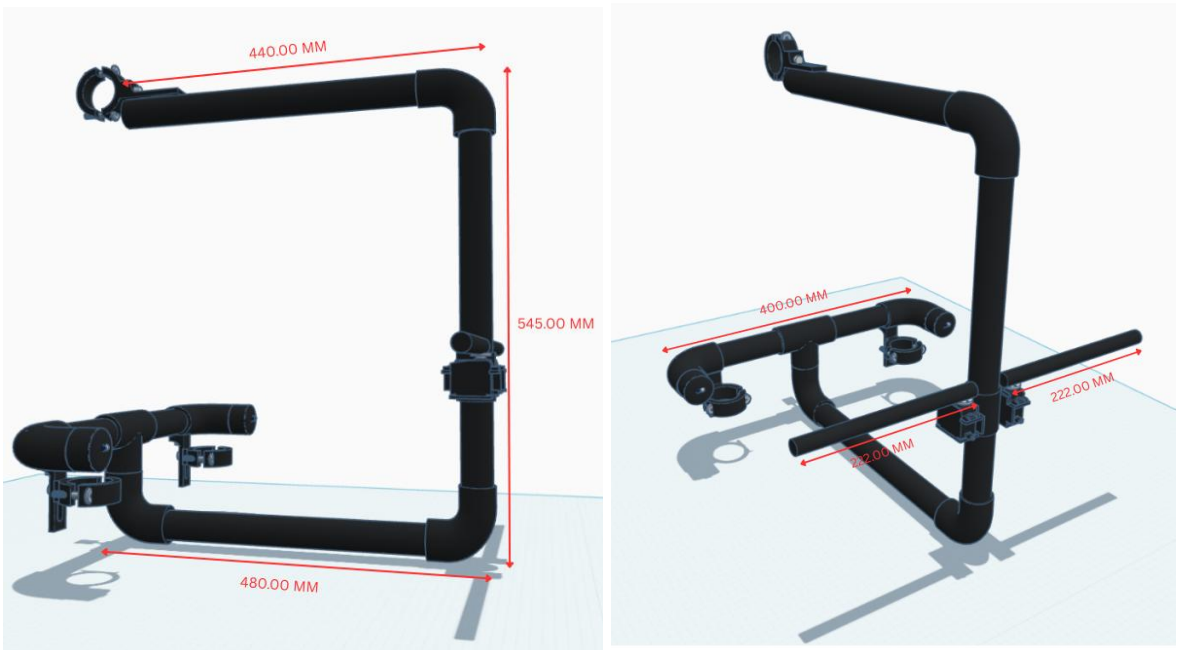


Fig. 8. Seat Monitoring and Allocation Mechanism Layout

Figure 8 detailed the measurement of the designed and developed seat monitoring and allocation mechanism, which integrates two infrared sensors and two servos.

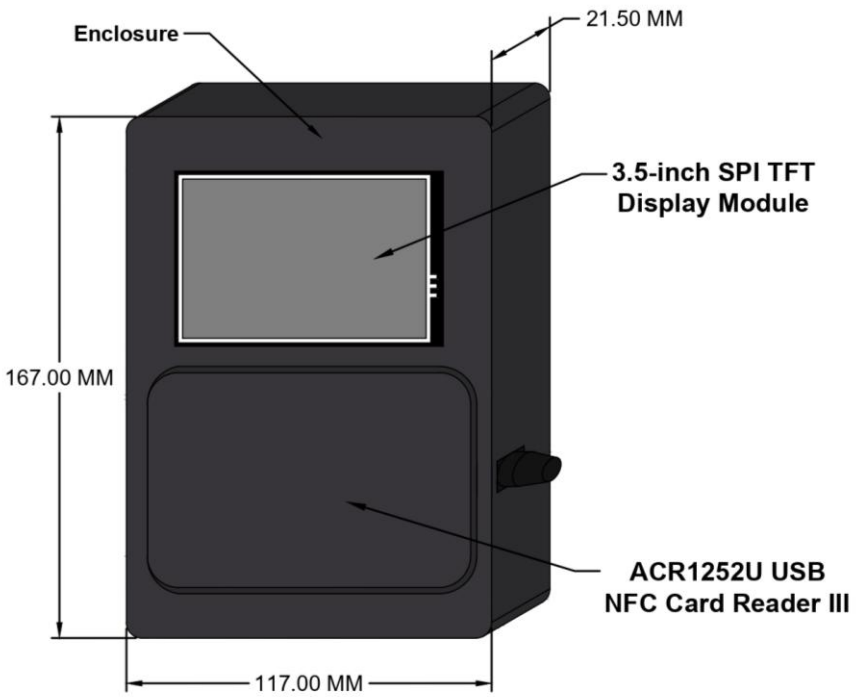


Fig. 9. Tap-Based Payment Process Layout

Figure 9 structured the enclosure of the two hardware components, the display module and the card reader, which are the major interfacing components for the tap-based payment process.

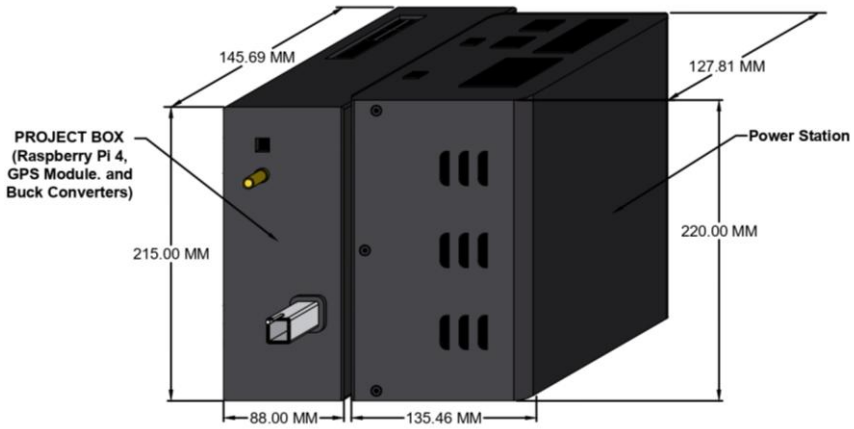


Fig. 10. Project Box and Power Station Layout

Figure 10 showed the housing of the main controller of the developed system and its power supply. The main controller includes the Raspberry Pi 4, GPS Module, and the buck converters. To supply these components, a dedicated power station is needed which outputs both AC and DC.

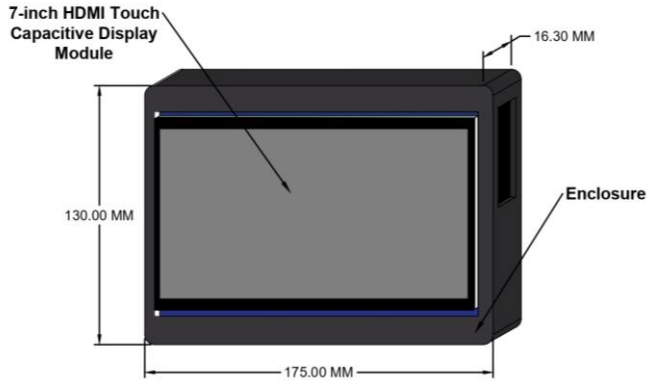


Fig. 11. Driver's Dashboard Layout

Figure 11 showed the casing of the driver's dashboard which encloses the HDMI touch capacitive display module.

3. A detailed circuit diagram was created using Miro to show all electrical connections between the Raspberry Pi GPIO pins and the IR sensors, servo motors, GPS module, TFT display, and supporting power rails (see Figure 12). USB connections for the ACR1252U and HDMI connections for the 7-inch display were also included.

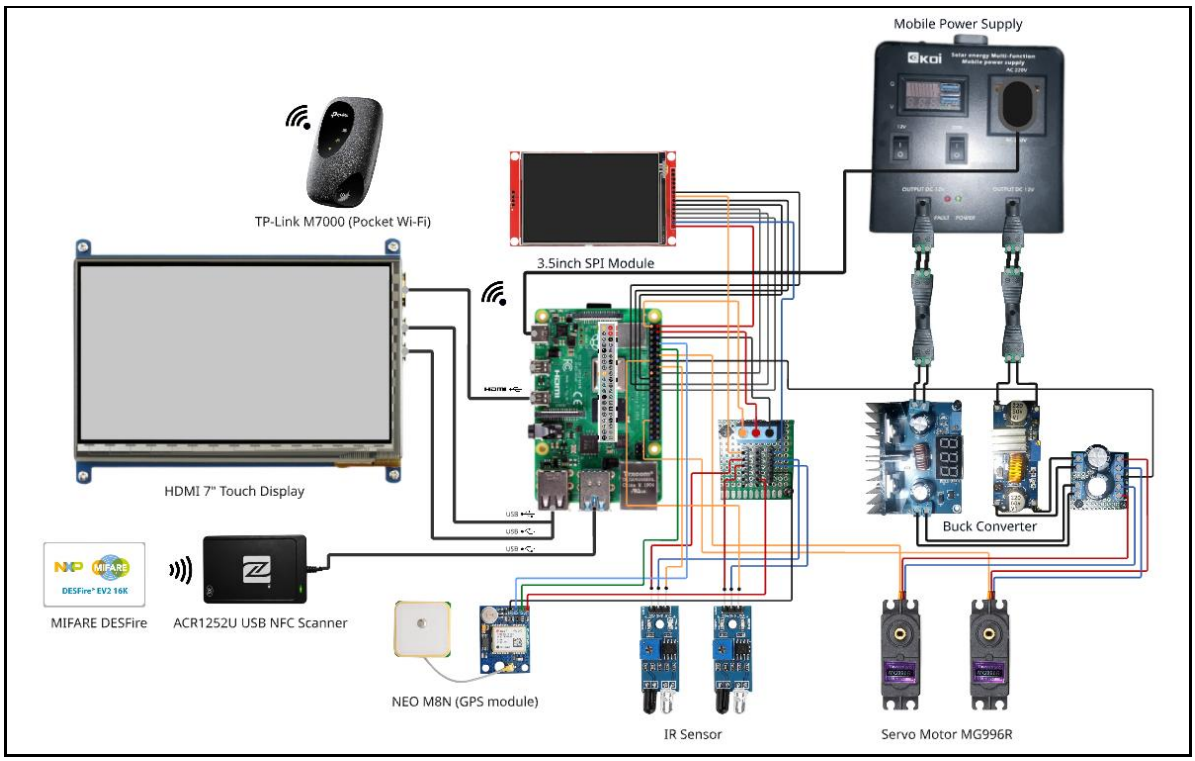


Fig. 12. Circuit Diagram of the Developed AFCS

4. The researchers prepared a pin mapping table that assigned specific GPIO pins for each IR sensor, servo control signal, and GPS UART lines. Power rails at 5 V and 3.3 V will be allocated carefully to avoid overloading any single source. Furthermore, Table 1 sets instructions for the overall wiring and connections for better circuitry of the developed AFCS.

Table 1

Pin Definitions

PIN LABEL IN SCHEMATIC	RASPBERRY PI PIN (BCM)	PHYSICAL PIN NO.	DIRECTION / TYPE	CONNECTED DEVICE / FUNCTION
GPS_RX	GPIO14	Pin 8	Output	UART transmit signal from Raspberry Pi to GPS module
GPS_TX	GPIO15	Pin 10	Input	UART receive signal from GPS module to Raspberry Pi
IR1_OUT	GPIO23	Pin 16	Input	Digital input from IR sensor for Seat 1 occupancy detection
IR2_OUT	GPIO27	Pin 13	Input	Digital input from IR sensor for Seat 2 occupancy detection
SERVO1_SIG	GPIO18	Pin 12	Output (PWM)	PWM control signal for Seat 1 locking servo motor
SERVO2_SIG	GPIO13	Pin 33	Output (PWM)	PWM control signal for Seat 2 locking servo motor
5V	5V	Pin 2	Power	5 V supply for TFT and GPS
3V3	3V3	Pin 1	Power	3.3 V logic supply for IR and Backlight
GND	GND	Various	Ground	Common ground reference for Raspberry Pi and all external devices

5. Power distribution for the Raspberry Pi, displays, NFC reader, servos, and sensors was designed to ensure stable operation. The researchers used a dedicated power station for the overall system.
6. Before full assembly, individual components were tested on a temporary setup. For instance, each IR sensor was powered and observed to verify detection behavior, each servo was driven using a simple PWM script, and the GPS module were tested to confirm valid National Marine Electronics Association (NMEA) outputs.
7. Any issues observed during bench testing, such as signal noise or undervoltage under load, was used to revise the circuit diagram and layout. Only when the design passes bench tests will the researchers be allowed to proceed to full assembly.

Software Design Phase

1. The researchers decomposed the system into modules: commuter web interface, driver dashboard, administrator dashboard, Firebase/Firestore database, seat monitoring logic, extended terminal logic, and analytics computations. In order to implement these modules, certain technologies and libraries were consolidated in software specifications (see Appendix F, p. 163).
2. Operational flowcharts were created using draw.io to organize the overall system operations from start to end. These comprised the login flowchart, commuter flowchart, driver flowchart, and admin flowchart (see Figures 14–18 in the Operation/Testing Procedure section).
3. A data flow diagram (DFD) was prepared in Miro to describe how data moves between commuters, drivers, administrators, the Raspberry Pi, backend server, and database (see Figure 13).

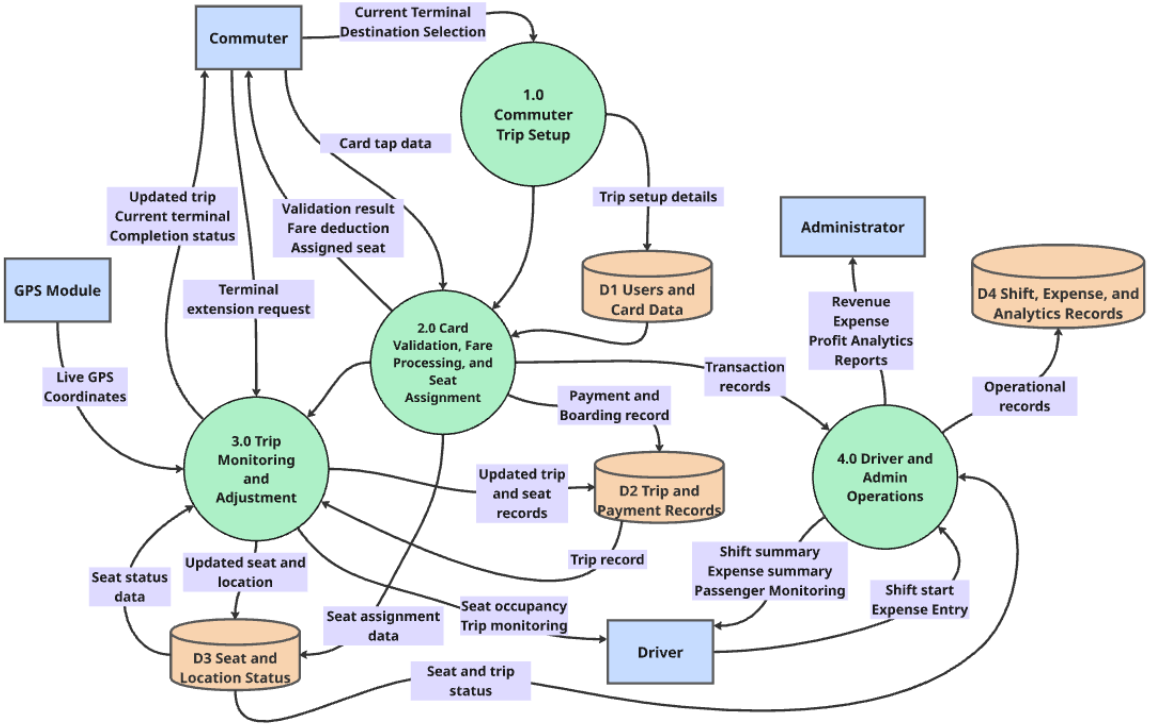


Fig. 13. Data Flow Diagram of the Developed AFCS

Figure 13 showed the data flow diagram of the developed AFCS. It illustrated how data move among external entities, processing modules, and data stores from account registration and balance management to trip handling, fare computation, and dashboard reporting, while enforcing security at each stage.

4. The researchers identified the required cloud backend services and data flows of the developed system using Firebase. These included user authentication, card validation, tap-based processing, seat status updates, extended terminal requests, trip logging, revenue and expense recording, and data retrieval for the commuter, driver, and administrator interfaces. The corresponding Firestore collections, Realtime Database paths, and Cloud Function triggers were also planned to support

real-time and automated system operations.

5. Wireframes were drafted to define the layout and flow of the commuter web application using a mobile-first approach. The planned screens included login and verification, card balance display, available incoming vehicle, seat availability, origin and destination terminal selection, route extension, and tap feedback. The interface was designed to be responsive and user-friendly for mobile devices (see Appendix L, p. 201). In addition, the developed commuter card design was prepared as part of the automated fare payment feature of the system (see Appendix L, p. 255).
6. The driver and administrator dashboards were planned to present operational and financial information through organized digital interfaces. The driver dashboard was designed to display waiting passengers per terminal, onboarded passengers, seat occupancy, active route information, shift controls, and daily revenue and expenses, while the administrator dashboard was planned to show fleet seat monitoring, terminal activity, revenue, expenses, profit, and report generation features. The arrangement of summary cards, tables, and charts was sketched before implementation.

Development Phase

The development phase of the study was subdivided into the modules that represented the first four objectives of the research. These are the commuter interface, tap-based payment process, seat monitoring and allocation mechanism, and driver and admin dashboards, which were developed using defined procedures (see Appendix H, p. 183).

Testing Phase

The testing phase was conducted to verify whether each module performs its intended functions correctly, consistently, and in coordination with the other parts of the system. Each module was tested and refined iteratively based on its specific features, expected outputs, and integration with the overall AFCS. In order to achieve the modules, the researchers conducted the following.

1. Performed repeated testing to verify login and verification flow, card balance display, seat availability display, terminal selection, fare display, top-up process, and real-time dashboard updates.
2. Tested valid and invalid tap scenarios. Performed repeated tests using registered and unregistered cards to verify that the system correctly detects the card, validates the commuter, and displays the proper tap result. Enhanced the Python firmware through repeated testing and debugging until the tap-based payment process performed consistently and satisfied the objective.
3. Tested and refined the mechanism iteratively. Performed repeated hardware and software testing to verify correct seat assignment, gate movement, sensor detection, event logging, and Firebase synchronization, then refine the mechanism until the objective was achieved.
4. Tested and refined both dashboards iteratively. Performed repeated testing to verify authentication, real-time monitoring, route updating, expense recording, analytics display, and report generation, then refined both dashboards until the objective was achieved.

Evaluation Phase

The evaluation phase was carried out after the development and testing of the commuter interface, tap-based payment process, seat monitoring and allocation mechanism, and driver and admin dashboards. In this phase, each developed feature was assessed using structured test results supported by appropriate formulas and interpreted using the corresponding evaluation criteria. Furthermore, the overall developed system was evaluated through Objective 5 using the ISO/IEC 25010 software quality model, focusing on functional suitability, performance efficiency, reliability, security, usability, and data quality. Through this phase, the researchers determined whether each objective has been achieved and whether the system as a whole satisfied the required quality standards.

Operation/Testing Procedure

Operational Flowcharts

The operational flowcharts of the developed AFCS and served as the basis for the testing procedures discussed in this chapter. Each flowchart illustrated the step-by-step sequence of actions, decisions, and data exchanges involved in the major user roles and system components. The actual process flow of the developed system, starts from the login process for commuter, driver, and administrator flowcharts show how users are authenticated, how trips and fare deductions are processed, how the process of extended terminal feature works, and how operational and financial information is presented in real time. Collectively, these diagrams clarify the expected behavior of the system and highlight the key decision points and processes that must be verified during testing.

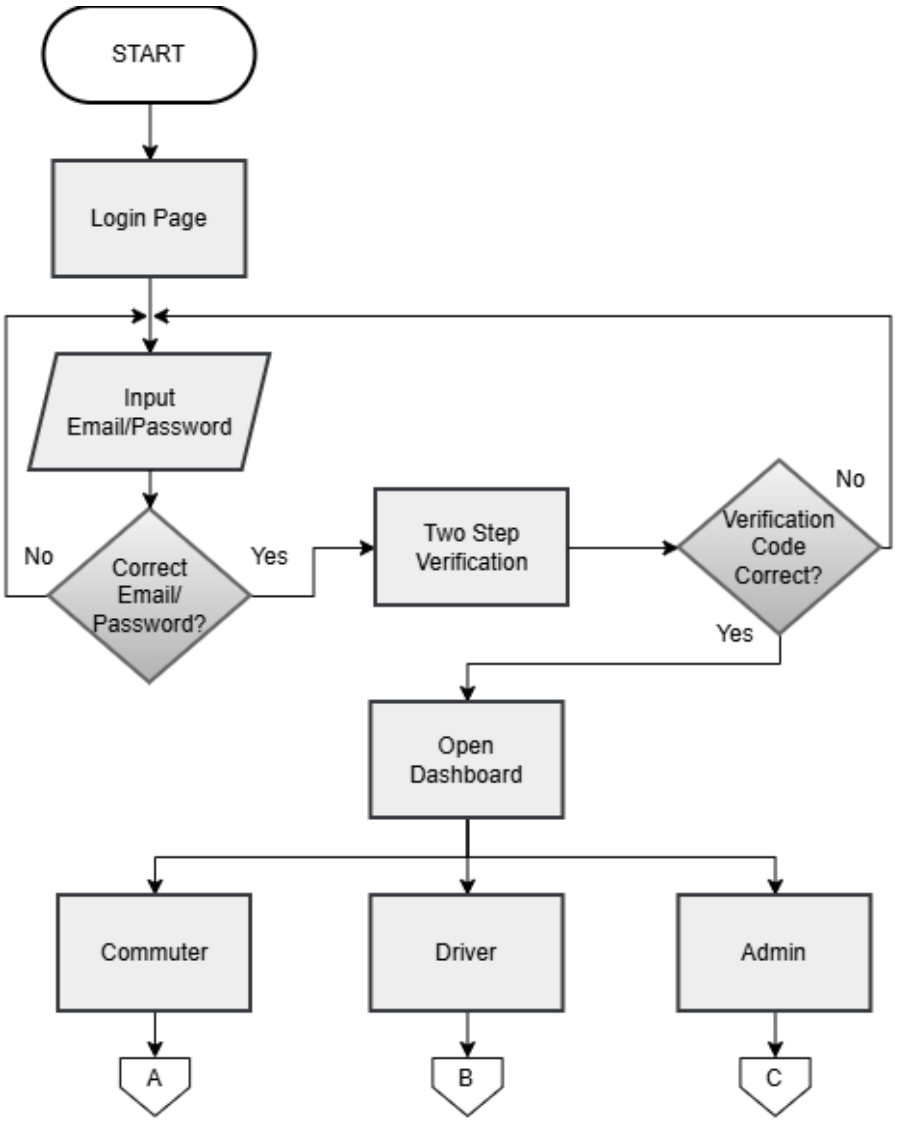


Fig. 14. Process Flow of the Login Page

As shown in Figure 14, the flow would begin when the user opens the login page and enters the registered email address and password. The system first checks whether the entered login credentials are correct. If the email address or password is incorrect, the

process returns to the login input step and requires the user to enter the credentials again. If the credentials are valid, the system proceeds to the two-step verification stage.

At the two-step verification stage, the user enters the verification code sent by the system. The system then checks whether the entered verification code is correct. If the verification code is invalid, the process loops back and requires the user to re-enter the correct code. Once the verification code is successfully validated, the system opens the dashboard.

After successful login and verification, the user is directed to the appropriate dashboard based on the account type. As shown in the flowchart, the process may continue to the commuter, driver, or admin flow, which are represented by the connected flowchart points A, B, and C, respectively. This flow ensures that only users with valid credentials and correct verification codes are granted access to the system.

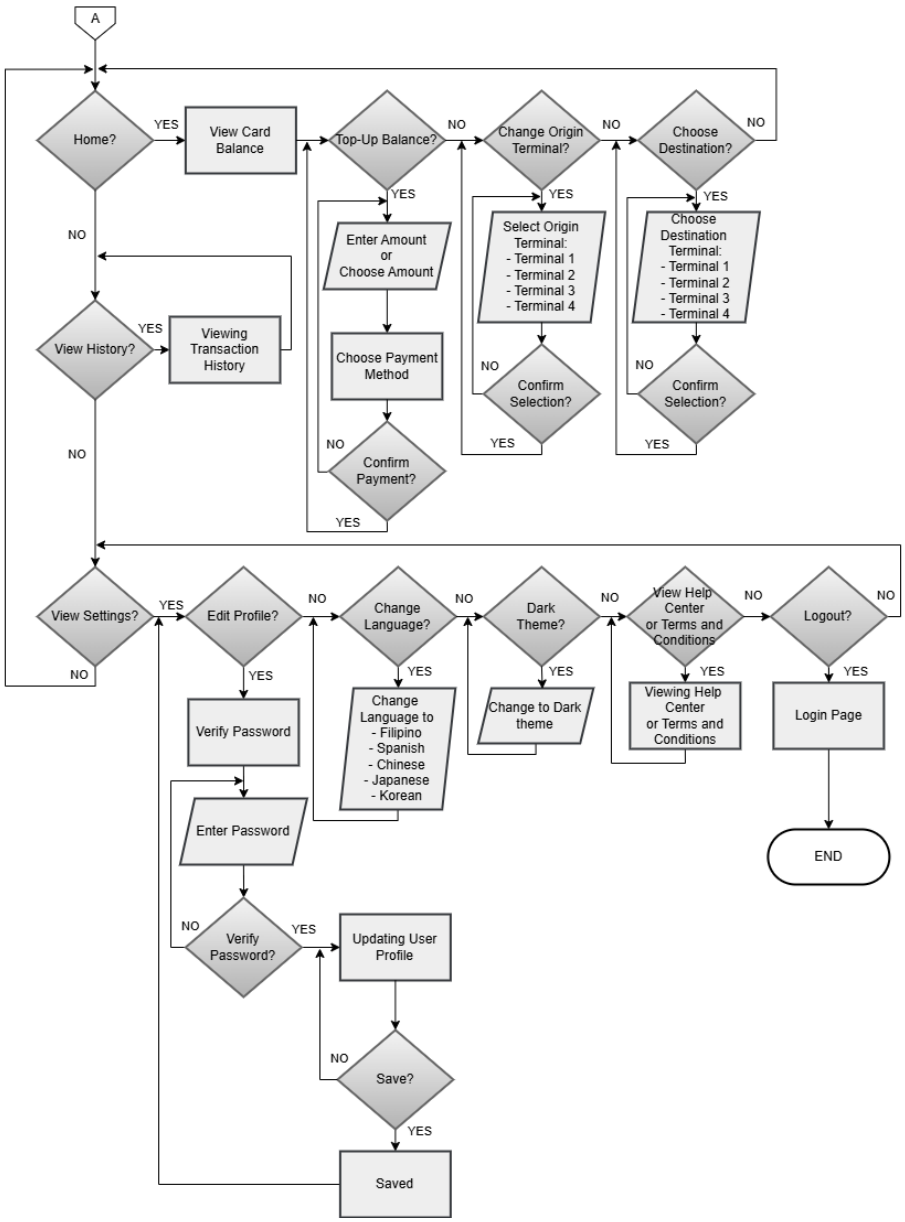


Fig. 15. Process Flow of the Commuter Interface

In the commuter interface, after the user has successfully logged in through connector **A** from the login flowchart, the flow began by checking whether the commuter selects the

Home page. If **yes**, the system displays the card balance and allows the commuter to perform home-page actions such as topping up the balance, changing the origin terminal, and choosing a destination terminal. In the top-up process, the commuter enters or selects an amount, chooses a payment method, and confirms the payment before the system returns to the dashboard. For route-related actions, the commuter may select an origin terminal or choose a destination terminal from the available terminals, after which the system asks for confirmation before saving the selection and returning to the main dashboard flow.

If the commuter chooses to view the **History** page, the system displays the transaction history and then returns to the dashboard flow. If the commuter chooses the **Settings** page, the commuter may edit the profile, change the language, switch the theme, view the help center or terms and conditions, or log out of the system.

Within the **Settings** page, the flowchart showed that profile editing requires password verification before any profile changes can be saved. The commuter may also change the interface language to any of the available options and switch the theme to dark mode. In addition, the commuters may open the help center or terms and conditions for guidance and reference. If the commuter chooses to log out, the system redirects the user to the login page and ends the session. Overall, the flowchart presented the navigation flow among the three main commuter pages, namely **Home, History, and Settings**, together with the functions available within each page.

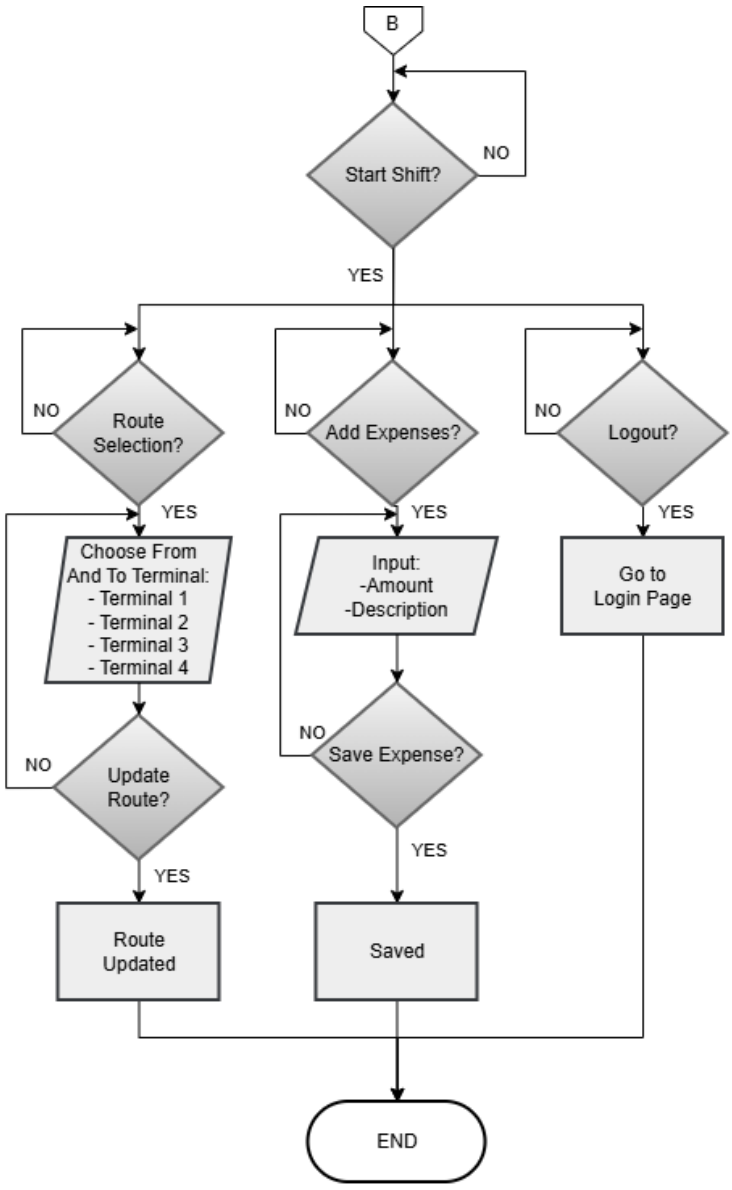


Fig. 16. Process Flow of the Driver Dashboard

In the driver dashboard, after the user has successfully logged in through connector **B** from the login flowchart, the flow began by checking whether the driver selects **Start Shift**.

If the driver does not start the shift, the flow remains at the same decision point. Once the driver starts the shift, the process proceeds to the main driver functions, namely **route selection**, **adding expenses**, and **logout**.

If the driver chooses **Route Selection**, the system allows the driver to choose the origin and destination terminals from the available terminal options. The system then checks whether the driver confirms the route update. If confirmed, the route is updated; otherwise, the flow returns to the route selection process. If the driver chooses **Add Expenses**, the system requires the driver to input the amount and description of the expense. The system then checks whether the expense entry will be saved. If confirmed, the expense record is saved; otherwise, the flow returns to the expense entry process.

If the driver chooses **Logout**, the system redirects the user to the login page, after which the process ends. Overall, the revised flowchart showed that the driver must first start the shift before accessing the main dashboard functions, and it clearly presented the operational flow for route updating, expense recording, and logout.

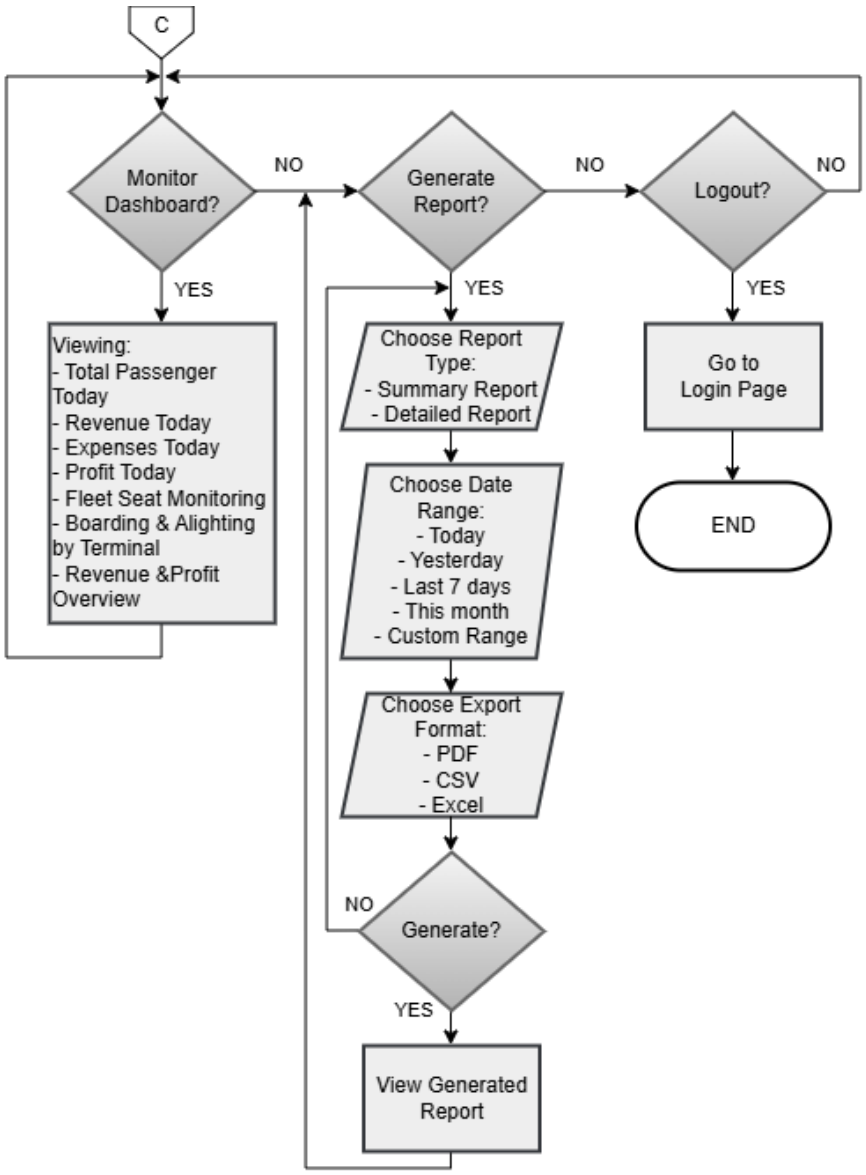


Fig. 17. Process Flow of the Admin Dashboard

In the admin dashboard, after the user has successfully logged in through connector **C** from the login flowchart, the flow began when the administrator chooses to monitor the

dashboard. In this process, the system displays the main dashboard information, which includes the total passengers for the day, revenue for the day, expenses for the day, profit for the day, fleet seat monitoring, boarding and alighting activity by terminal, and the revenue and profit overview. After viewing these details, the process returns to the main admin dashboard flow.

When the administrator chooses to generate a report, the system first requires the selection of the report type, which may be either a summary report or a detailed report. The administrator then selects the desired date range, such as today, yesterday, the last seven days, this month, or a custom range. After that, the administrator selects the export format, which may be PDF, CSV, or Excel. The system then checks whether the administrator confirms the report generation. If confirmed, the generated report is displayed for viewing.

When the administrator chooses to log out, the system redirects the user to the login page and ends the session. Overall, the flowchart showed that the admin dashboard supports two main functions: monitoring operational and financial data, and generating reports based on the selected report type, date range, and file format.

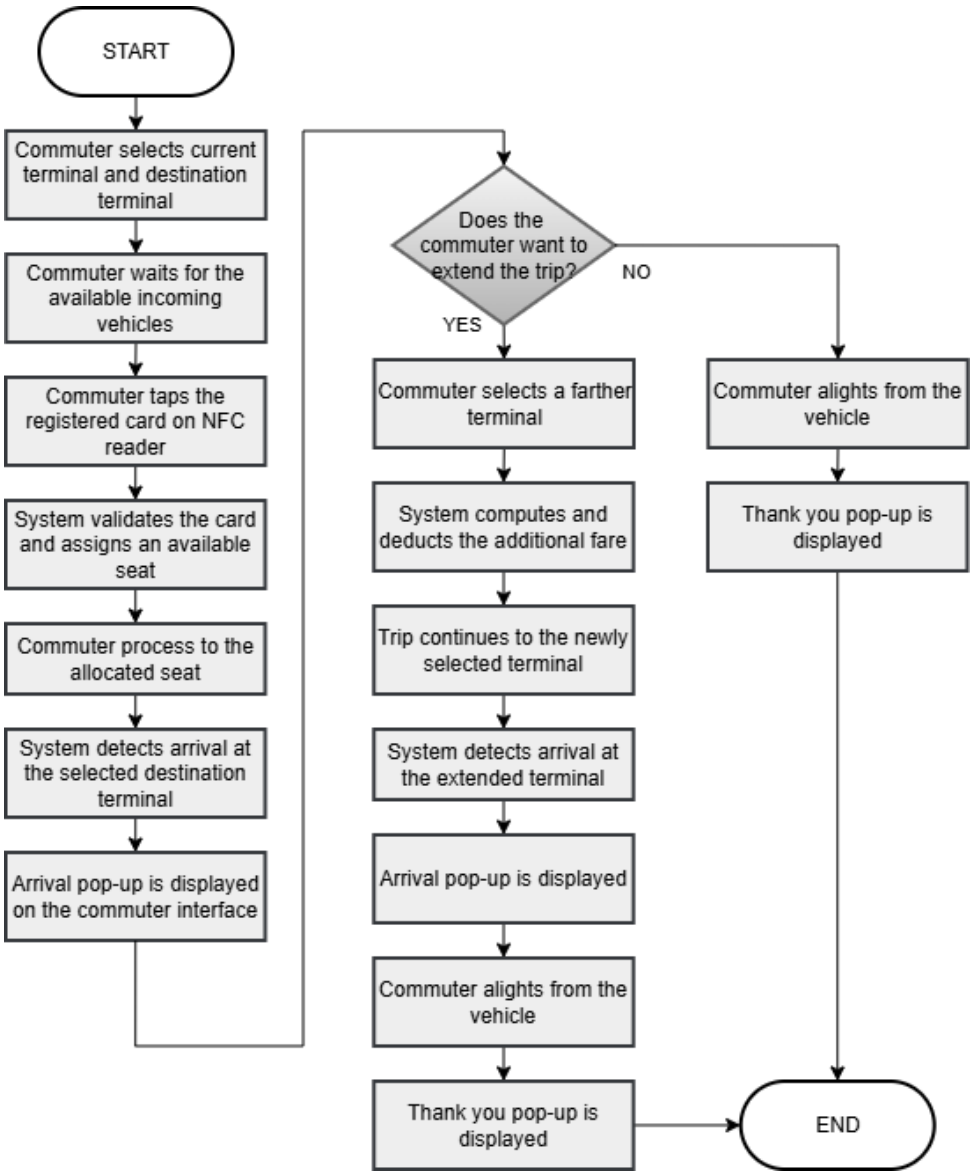


Fig. 18. Process Flow of the Commuter Extended Terminal Feature

For the commuter extended terminal feature, it began when the commuter selects the current terminal and destination terminal, waits for the available incoming vehicle, and taps the

registered card on the NFC reader. The system then validates the card and assigns an available seat, allowing the commuter to proceed to the allocated seat. Once the vehicle reaches the selected destination terminal, the system displays an arrival pop-up asking whether the commuter wants to extend the trip. If the commuter chooses to continue to a farther terminal, the system allows the selection of an extended terminal, computes and deducts the additional fare, and continues the trip until the newly selected terminal is reached. Upon arrival, the system displays another arrival pop-up, after which the commuter alights from the vehicle and a thank you pop-up is shown. If the commuter does not extend the trip, the commuter simply alights at the original destination terminal and the process ends with the display of the thank you pop-up.

Procedures of Testing

The testing procedures involved the conduct of structured tests for each major feature of the system to verify whether the intended functions operated accurately and consistently. A default set of **10 registered cards** was used, consisting of **six active cardholders** and **four inactive cardholders**. In addition, some RFID cards would be used as unregistered cards.

Furthermore, the cash values displayed in the interface were **simulated values only** and were used exclusively for testing. The number of test cases per module would be a **minimum of 100**. The results were documented in Google Sheets (see Appendix G, p. 165), with each worksheet assigned to a corresponding system feature. The tests covered the commuter interface, tap-based payment process, seat monitoring and allocation mechanism, and driver and admin dashboards were as follow:

1. **Commuter Interface**

1.1. Card Balance Display Accuracy. The card balance in the commuter interface was evaluated using the following accuracy percentage formula:

$$Card\ Balance\ Display\ Accuracy\ (\%) = \frac{N_{balance,correct}}{N_{balance,total}} \times 100 \qquad \text{(equation 1)}$$

Where:

$N_{balance,total}$ = total number of balance display checks

$N_{balance,correct}$ = number of correct checks

1.2. Incoming Vehicle Information Display Accuracy. The details of the incoming vehicle in the commuter interface were evaluated using the accuracy percentage formula:

$$Incoming\ Vehicle\ Information\ Accuracy\ (\%) = \frac{N_{incoming\ vehicle,correct}}{N_{incoming\ vehicle,total}} \times 100 \quad \text{(equation 2)}$$

Where:

$N_{incoming\ vehicle,total}$ = the total number of incoming vehicle information checks

$N_{incoming\ vehicle,correct}$ = the number of correct checks

2. Tap-based payment process

2.1. Tap Success Rate. To measure whether the system successfully detected and read the commuter card during each tap attempt.

$$Tap\ Success\ Rate\ (\%) = \frac{N_{tap,success}}{N_{tap,total}} \times 100 \quad \text{(equation 3)}$$

Where:

$N_{tap,total}$ = total number of tap attempts

$N_{tap,success}$ = number of tap attempts correctly processed

2.2. Terminal-Based Fare Computation Accuracy. To measure whether the system correctly computed the fare based on the selected origin and destination terminals.

$$Fare\ Accuracy\ Rate\ (\%) = \frac{N_{fare,correct}}{N_{fare,valid}} \times 100 \quad \text{(equation 4)}$$

Where:

$N_{fare,valid}$ = total number of valid taps

$N_{fare,correct}$ = number of correct transactions

2.3. Automatic Seat Allocation Success Rate. To measure whether the system successfully assigned an available seat after a valid tap.

$$Seat\ Allocation\ Success\ Rate\ (\%) = \frac{N_{alloc,correct}}{N_{alloc,total}} \times 100 \qquad \text{(equation 5)}$$

Where:

$N_{alloc,total}$ = total number of successful seat allocations

$N_{alloc,correct}$ = number of correct cases

2.4. Invalid Reason Classification Accuracy Rate. To measure whether the system accurately displayed invalid reasons with corresponding conditions.

$$Invalid\ Reason\ Accuracy\ Rate\ (\%) = \frac{N_{correct}}{N_{total,tests}} \times 100 \qquad \text{(equation 6)}$$

Where:

$N_{total,test}$ = total number of tests

$N_{correct}$ = number of correct cases

2.5. System Unavailable Display Accuracy Rate. To measure whether the system accurately displayed the system unavailable display.

$$System\ Unavailable\ Accuracy\ Rate\ (\%) = \frac{N_{correct}}{N_{total,tests}} \times 100 \qquad \text{(equation 7)}$$

Where:

$N_{total,test}$ = total number of tests

$N_{correct}$ = number of correct cases

3. Seat Monitoring and Allocation Mechanism

3.1. Seat Occupancy Detection Accuracy. To measure whether the seat monitoring mechanism correctly detected the actual occupancy status of the assigned

$$Seat\ Detection\ Accuracy\ (\%) = \frac{N_{detect,correct}}{N_{detect,total}} \times 100 \qquad \text{(equation 8)}$$

Where:

$N_{total,test}$ = total number of tests

$N_{correct}$ = number of correct cases

3.2. Driver Seat Status View Consistency Rate. To measure whether the seat status shown in the driver dashboard consistently matched the actual seat condition and recorded seat events.

$$Driver\ View\ Consistency\ Rate\ (\%) = \frac{N_{driver,consistent}}{N_{driver,total}} \times 100 \qquad \text{(equation 9)}$$

Where:

$N_{driver,total}$ = total number of driver dashboard checks

$N_{driver,consistent}$ = number of checks where driver view is equal to ground-truth physical status.

3.3. Commuter Seat Status View Consistency Rate. To measure whether the seat status shown in the commuter interface consistently matched the actual seat condition and assigned seat status.

$$Commuter\ View\ Consistency\ Rate\ (\%) = \frac{N_{comm,consistent}}{N_{comm,total}} \times 100 \qquad \text{(equation 10)}$$

Where:

$N_{comm,total}$ = total number of commuter interface checks

$N_{comm,consistent}$ = number of checks where commuter view is equal to ground-truth physical status.

3.4. Seat Lock/Unlock Actuation Accuracy. To measure whether the seat mechanism correctly performed the intended lock/unlock actions based on seat allocation events.

$$Seat\ Lock/Unlock\ Accuracy\ (\%) = \frac{N_{lock,correct}}{N_{lock,total}} \times 100 \qquad \text{(equation 11)}$$

Where:

$N_{lock,total}$ = total number of events

$N_{lock,correct}$ = number of events where the seat actuation state is correct.

4. Driver and Admin Dashboards

4.1. Dashboard Data Consistency Rate. To measure whether the revenue, expenses, and profit values displayed in the driver dashboard were consistent with those displayed in the admin dashboard.

$$Dashboards\ Consistency\ Rate\ (\%) = (\frac{Matching\ Dashboard\ Values}{Total\ Value\ Checks}) \times 100 \quad \text{(equation 12)}$$

4.2. Driver Alert Pop-up Accuracy Rate. To measure whether the alert pop-up is triggered accurately on the driver dashboard.

$$Pop-up\ Accuracy\ Rate\ (\%) = \frac{N_{correct}}{N_{total,tests}} \times 100 \quad \text{(equation 13)}$$

Where:

$N_{total,tests}$ = total number of trial runs

$N_{correct}$ = number of events where the alert pop-up is accurately displayed.

4.3. Indicator State Correctness Rate. To measure whether the In Destination, Violator, and Extended values displayed in the driver dashboard were correct.

$$Indicator\ State\ Correctness\ Rate\ (\%) = \frac{N_{correct}}{N_{total,tests}} \times 100 \quad \text{(equation 14)}$$

Where:

$N_{total,tests}$ = total number of trial runs

$N_{correct}$ = number of events where the indicator state is accurately displayed.

4.4. Driver Monitoring Non-Repeat Compliance Rate. To measure whether if either the alert pop-up or any of the indicator state was repeated or not.

$$Driver\ Monitoring\ Compliance\ Rate\ (\%) = \frac{N_{correct}}{N_{total,tests}} \times 100 \quad \text{(equation 15)}$$

Where:

$N_{total,tests}$ = total number of trial runs

$N_{correct}$ = number of events where the indicator state and alert pop-up is not repeated.

Evaluation Procedure

The developed AFCS was evaluated to determine whether it achieved the stated research objectives. The evaluation was conducted through technical evaluation and respondent-based evaluation.

The technical evaluation was based on the results of the structured testing procedures. These tests assessed the major functions of the system, including card balance display accuracy, terminal selection, card validation, terminal-based fare computation, automatic seat allocation, seat monitoring, driver dashboard monitoring, and administrator dashboard analytics. The results were used to determine whether the developed system performed its intended functions within the scope of the prototype.

The respondent-based evaluation was conducted using questionnaires based on the ISO/IEC 25010:2023 software product quality model. 12 expert respondents answered an online evaluation instrument composed of 30 items covering functional suitability, performance efficiency, reliability, security, usability, and data quality. For the user evaluation, 100 respondents evaluated the commuter interface and tap-based payment process under Objectives 1 and 2, while another 100 respondents evaluated the seat monitoring mechanism, driver dashboard, and administrator dashboard under Objectives 3 and 4. The user evaluation focused on usability, particularly interaction capability.

After collecting and encoding the responses, the researchers computed the mean score and standard deviation for each questionnaire item and for each grouped quality characteristic.

Mean Score

The mean score was computed to determine the average rating of the respondents for each item and quality characteristic using the formula:

$$\bar{x} = \frac{\sum x_i}{n} \qquad \text{(equation 16)}$$

Where:

$\bar{x}$ = mean score

Σx_i = sum of all ratings

n = total number of respondents.

Standard Deviation

The standard deviation was computed to determine the consistency or variation of the respondents' ratings using the formula:

$$SD = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}} \qquad \text{(equation 17)}$$

Where:

SD = standard deviation

x_i = each individual rating

$\bar{x}$ = mean score

n-1 = degrees of freedom.

Evaluation Criteria

The evaluation criteria defined how the results from the **technical evaluation** and **respondent-based evaluation** were interpreted to determine whether the developed AFCS achieved the research objectives. Since the study used both structured technical testing and ISO/IEC 25010:2023-based questionnaires, two sets of criteria were applied. The technical evaluation measured the actual performance of the system features through accuracy rates,

success rates, and consistency results, while the respondent-based evaluation measured the users' and experts' level of agreement regarding the system's usability and overall quality.

For the technical evaluation, the results from the structured testing procedures were interpreted using the scale shown in Table 2.

Table 2

Evaluation Criteria for Technical Evaluation

PERCENTAGE RANGE	QUALITATIVE RATING	INTERPRETATION
≥ 99%	Excellent	The system feature performed highly effectively and consistently.
95%–98.99%	Very Good	The system feature performed effectively with minor inconsistencies.
90%–94.99%	Good	The system feature performed at an acceptable level.
80%–89.99%	Fair	The system feature performed below the desired level and requires improvement.
< 80%	Poor	The system feature did not meet the expected level of performance.

Source: [30], [57]

A result of Good or higher was considered acceptable for the technical evaluation. The percentage score for each technical test was computed using the formula indicated in the testing procedures. The technical evaluation covered the system functions under Objectives 1 to 4, including card balance display accuracy, terminal selection accuracy, tap-based card validation, fare computation accuracy, automatic seat allocation, seat monitoring accuracy, driver dashboard monitoring, administrator dashboard analytics, and related system output consistency.

For the respondent-based evaluation, the expert and user responses were interpreted using a 5-point Likert scale as indicated in Tables 3 and 4. The expert evaluation covered functional suitability, performance efficiency, reliability, security, usability, and data quality, while the user evaluation focused on usability, particularly interaction capability.

Table 3

Evaluation Criteria for Respondent-Based Evaluation

RANGE OF MEAN	VERBAL INTEPRETATION	EVALUATION INTERPRETATION
4.21 - 5.00	Strongly Agree	The respondents strongly agreed that the evaluated objective or quality characteristic was achieved.
3.41 - 4.20	Agree	The respondents agreed that the evaluated objective or quality characteristic was achieved.
2.61 - 3.40	Neutral	The respondents were neutral regarding whether the evaluated objective or quality characteristic was achieved.
1.81 - 2.60	Disagree	The respondents disagreed that the evaluated objective or quality characteristic was achieved.
1.00 - 1.80	Strongly Disagree	The respondents strongly disagreed that the evaluated objective or quality characteristic was achieved.

Source: [18]

A respondent-based evaluation result interpreted as Agree or higher was considered acceptable. For the user evaluation of Objectives 1 and 2, Questions 3 to 5 were used to evaluate Objective 1, which focused on the commuter interface for displaying card balance details, seat availability, and terminal selection before boarding. Questions 6 to 9 were used to evaluate Objective 2, which focused on the tap-based payment process, including card tapping, card validation feedback, automatic seat allocation, and fare deduction.

For the user evaluation of Objectives 3 and 4, Questions 2 and 4 were used to evaluate Objective 3, which focused on the seat monitoring and allocation mechanism, particularly in monitoring passengers waiting to board and displaying seat status indicators. Questions 5 to 7 were used to evaluate Objective 4, which focused on the driver and administrator dashboards for operational monitoring and financial analytics, including revenue monitoring, expense recording, and profit computation.

For Objective 5, the expert evaluation was interpreted based on the ISO/IEC 25010:2023 quality characteristics, namely functional suitability, performance efficiency, reliability, security, usability, and data quality. Each quality characteristic was evaluated using five questionnaire items, and the mean score was used to determine the experts' level of agreement regarding the overall quality of the developed system.

The standard deviation was reported after the mean score for each questionnaire item and for each grouped objective or quality characteristic. A lower standard deviation indicated that the respondents' ratings were more consistent, while a higher standard deviation indicated greater variation in their responses. The standard deviation was used as a supporting measure and not as a separate pass-or-fail criterion.

Through these evaluation criteria, the researchers were able to interpret both the measured technical performance and the perceived quality of the developed AFCS. This provided a consistent basis for determining whether the system achieved its intended functions and met the expected quality characteristics stated in the research objectives.

Instruments and Techniques Used

This section presented the instruments and techniques used in the study for the development, testing, and evaluation of the automated fare collection system (AFCS).

Research Instruments

1. **Structured tests.** These were used for Objectives 1 to 4 to record the test name, procedure, expected result, observed result, number of trials, and remarks for each structured test. These sheets served as the primary record for measurable system performance. The accomplished test records were encoded, organized, tabulated, and summarized using Google Sheets (see Appendix G, p. 165 for representative raw logs; full records are archived digitally).
2. **ISO/IEC 25010:2023 expert evaluation kit.** The expert evaluation was done through online using Gmail, FB Messenger, Google Drive, and Google Forms. It served as the main expert-based assessment tool for the overall developed AFCS (see Appendix B, p. 136).
3. **User evaluation questionnaires (ISO/IEC 25010:2023 Interaction Capability).** Separate questionnaires were used for end-users to evaluate usability-related aspects of the developed prototype based on their actual interaction with the system. These

questionnaires focused on the user side of system quality and interaction capability (see Appendix B, p. 136).

4. **Ethical safeguards.** Before participation, evaluators were briefed on the study's objectives and procedures through a formal letter of request (see Appendix A, p. 133) and with an informed consent form (see Appendix C, p. 142). Participation was entirely voluntary, with the option to withdraw at any time. To protect confidentiality, identifying information was either coded or hidden from the dataset, and findings are reported only in aggregate form. The study also adhered to the institution's ethical guidelines and received approval from the Research Ethics Review Committee (RERC).
5. **Gannt Chart.** The Gantt chart was used as a project management instrument to present the schedule, sequence, and duration of the major activities involved in the study. It helped monitor the progress of the research from planning up to testing and evaluation (see Appendix H, p. 190).

Sampling Methods Used

1. **Non-probability sampling.** The study used non-probability sampling since the respondents were selected based on availability, relevance, and their ability to use or evaluate the prototype.
2. **Purposive sampling.** This served as the main sampling method because the respondents had to meet specific criteria relevant to the system being evaluated, such as professional background for experts.
3. **Convenience sampling.** This was applied in the user evaluation by gathering data from available and willing people who are willing to listen to the live prototype demonstration, prototype trying, and answering the provided questionnaire.
4. **Quota sampling.** This was used by setting a target number of respondents for each group to ensure representation in the evaluation, specifically 12 expert evaluators and 100 end-users (see Appendix D, p. 146 for computation of quota for end-user evaluation), using the Cochran's formula [82], [83].

$$n_o = \frac{Z^2 p(1-p)}{e^2} \qquad \text{(equation 18)}$$

Where:

Z= Z-score for confidence level

p= population proportion (as a decimal)

e= margin of error (as a decimal)

n_0= required sample size (before any finite population correction)

Statistical Tools

1. **Five-point Likert scale.** This was used in the expert and user evaluation questionnaires to measure the respondents' level of agreement with each evaluation statement.

Table 4

5-point Likert Scale

SCALE	RANGE OF MEAN	VERBAL INTERPRETATION	DESCRIPTION
5	4.21 – 5.00	Strongly Agree	The respondent strongly agreed that the evaluated system feature or quality characteristic was achieved.
4	3.41 – 4.20	Agree	The respondent agreed that the evaluated system feature or quality characteristic was achieved.
3	2.61 – 3.40	Neutral	The respondent was neutral regarding whether the evaluated system feature or quality characteristic was achieved.
2	1.81 – 2.60	Disagree	The respondent disagreed that the evaluated system feature or quality characteristic was achieved.
1	1.00 – 1.80	Strongly Disagree	The respondent strongly disagreed that the evaluated system feature or quality characteristic was achieved.

Source: [18]

2. **Descriptive and Reliability Statistics.** The study employed basic descriptive measures to summarize responses and trials and indicate variability

Software Tools

1. **Simulation Web.** This refers to a separate web page or browser tab used by the developer to simulate the actual process flow of the system. It was used to manually trigger and observe certain transactions and system events for testing, validation, and debugging purposes (see Appendix L, p. 258).
2. **AutoCAD, Tinkercad, Canva, draw.io, Miro, Mendeley, Google Workspace, GPT5.2, and Grammarly.** These tools were used for diagrams, designs and layouts, and documentation and collaboration.
3. ***systemd-journald timestamps***. Secure shell commands to the firmware to record logs locally and persistently, while ***journalctl*** outputs the live timestamps of every system event.

Hardware Tools

1. **Multimeter.** Used to measure voltage, continuity, and electrical connections during circuit checking and troubleshooting.
2. **Wiring and assembly tools (wire stripper, soldering kit, tweezer).** Utilized collectively for preparing wires, securing electrical connections, and handling small components during precise hardware assembly.
3. **Finishing and fabrication tools (grinder, cutter, black spray paint, dedicated toolbox).** Employed for shaping and cutting materials, applying protective coating, and organizing components within a portable enclosure to ensure durability and system integrity.
4. **Three-wheel e-bike.** Served as the actual testing vehicle of the prototype, enabling evaluation of the automated fare collection system and seat monitoring mechanism under intended operating conditions.

CHAPTER IV

RESULTS AND DISCUSSION

This chapter detailed the technical description of the designed and developed automated fare collection system (AFCS) with seat allocation, terminal selection, and profit analytics dashboards, the structure of the project and the organization behind it, the limitations and capabilities of the project, and the results and discussion of evaluating each project module and research objectives.

Project Technical Description

This section presented the technical description of the developed automated fare collection system (AFCS). The system was developed to improve fare collection, passenger seat management, terminal-based trip handling, and operational monitoring by combining both hardware and software components into a unified platform. Specifically, the prototype integrates near-field communication (NFC) smart card technology, a Raspberry Pi-based embedded controller, infrared sensors, servo-driven seat control, global positioning system (GPS) location tracking, cloud database services, and web-based dashboards to support automated and synchronized transport functions.

Through this integration, the system enables commuters to access a mobile-friendly interface to view card balance, check seat availability, and select terminals before boarding. During boarding, the embedded device performs card validation, fare computation, fare deduction, and automatic seat allocation. The system also supports real-time monitoring of seat occupancy and trip progress, while operational data are transmitted to the driver and administrator dashboards for monitoring revenue, expenses, and profit. In this way, the developed prototype provides a more structured and efficient alternative to conventional fare collection and passenger monitoring practices.



Fig. 19. Mounted AFCS Prototype on an E-Bike

Figure 19 showed the mounted AFCS prototype on the e-bike, indicating the actual placement of the major operational components of the system. These components include the **seat monitoring and allocation mechanism** (A), **tap-based payment process** (B), **central controller box** (C), **power station** (D), and **driver's dashboard** (E). This figure serves as an overall view of the developed prototype prior to the discussion of each module in the succeeding subsections.

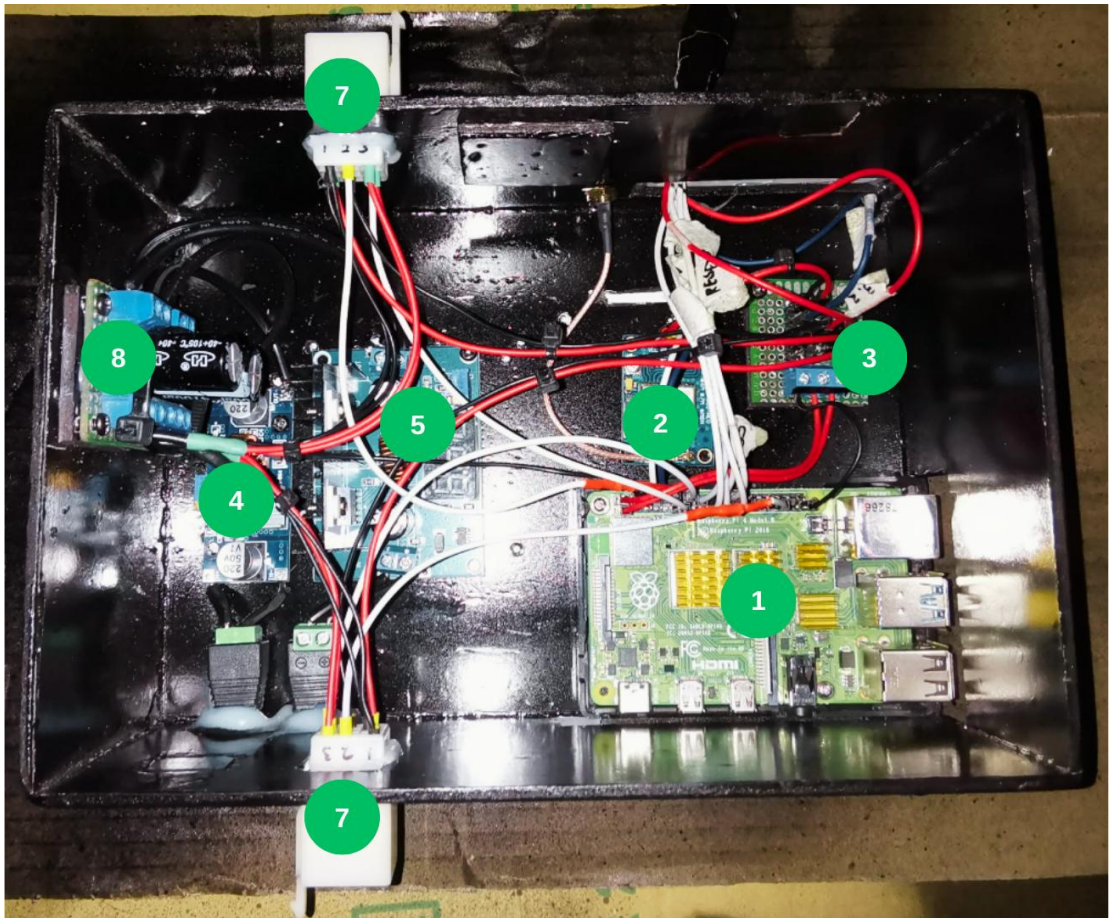


Fig. 20. Central Controller of the AFCS prototype

Figure 20 showed the central controller and the installed configuration of the embedded components that support the operation of the developed AFCS prototype. The labeled components include the **Raspberry Pi 4 Model B, 8GB** (1), **NEO-M8N GPS module** (2), **customized power rail for 5 V, 3.3 V, and GND from the Raspberry Pi 4** (3), **DC-DC step-down buck converters** (4) and (5), **wire connector for Servo 2 and IR Sensor 2** (6), **wire connector for Servo 1 and IR Sensor 1** (7), and the **customized capacitor rail for the buck converters connected to the servo motors** (8). These components were installed inside the enclosure to organize the embedded wiring, power distribution, signal

connections, and controller integration of the prototype.

The Raspberry Pi 4 served as the main controller of the system and handled the core processing functions of the prototype. The GPS module supported location-based functions, while the customized power and capacitor rails helped provide organized and stable electrical distribution within the enclosure. The two buck converters were used to step down the power input to the level required by the servo motors, and the wire connectors provided structured output connections for the paired servo motors and IR sensors.

The **software integration** enabled the hardware components, cloud services, and user interfaces to operate as one coordinated system. At the embedded level, the Raspberry Pi OS and Python-based firmware handled card reading, validation, fare computation, seat allocation, sensor monitoring, servo control, display output, GPS reading, and database communication. These functions were integrated with Firebase services, particularly Firestore for centralized data storage and synchronization, and Authentication for user access control. Through this setup, data from the NFC reader, seat sensors, and GPS module were processed by the main controller and transmitted to the cloud database, allowing the commuter interface and the driver and administrator dashboards to receive updated operational information in near real time. **Source codes** for the software integration comprised the codes for card validation, fare computation, seat monitoring logic, terminal extension logic, and dashboard data synchronization (see Appendix I, p. 191).

Along with the central controller of the accomplished AFCS, the five corresponding modules are: **commuter interface, tap-based payment process, seat monitoring and allocation mechanism, extended terminal feature, and driver and admin dashboards**. Each module was developed to perform a specific function within the overall system and to operate in coordination with the other modules.

Commuter Interface

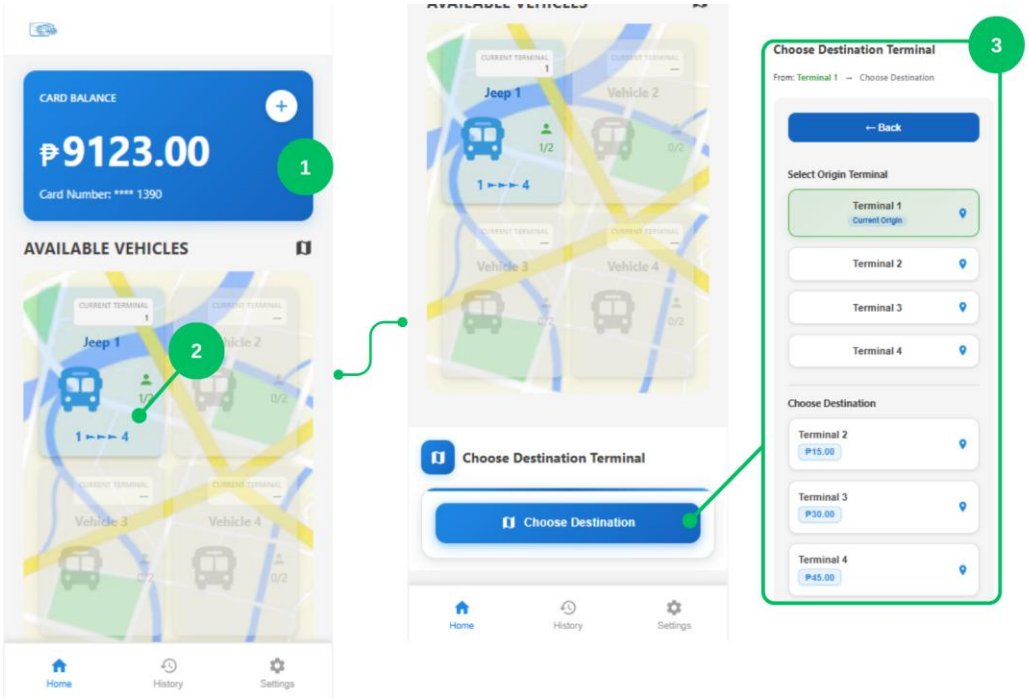


Fig. 21. Commuter Interface

Figure 21 showed the developed commuter interface, which presented the key information needed by commuters before boarding the incoming vehicle. The interface displays the **card balance details** (1), **seat availability of the available vehicles** (2), and **destination terminal selection** (3). These features were arranged to guide the commuter through the required pre-boarding steps in a clear and logical sequence.

This module was developed using React, Vite, Node.js, npm, Firebase Authentication, Cloud Firestore, Hostinger, and Apache/.htaccess. React was used to build an interactive and responsive commuter interface, while Vite was used to support faster development and efficient building of the web application. Node.js and npm were used to

manage the development environment and project dependencies. Firebase Authentication was used to support secure commuter access, while Cloud Firestore was used as the centralized database for storing and synchronizing card details, seat availability, terminal information, and other system records in near real time. Hostinger, together with Apache and .htaccess, was used to deploy the web application and support proper page routing in the hosted interface. These technologies were used so that the commuter interface could be accessible, responsive, and connected to live system data.

The card balance is placed at the upper portion of the interface so that the commuter can immediately check whether the available balance is sufficient for the intended trip. The seat availability section is shown in the middle of the interface to provide real-time information on the currently deployed vehicles, including their available seats, route direction, and current terminal. The destination terminal selection is provided as the next step so that the commuter can choose the intended drop-off point before boarding. This interface structure was developed to help commuters review their balance, check vehicle availability, and select a destination in an organized manner before proceeding to the tap-based validation process.

Tap-Based Payment Process



Fig. 22. Tap-Based Payment Process

Figure 22 showed the developed tap-based payment process, which performs the key transaction functions after the commuter completes the pre-boarding steps in the commuter interface. The module supports **commuter card validation** (1), **terminal-based fare computation** (2), and **automatic seat allocation after successful validation** (3). These functions were integrated into one module so that the boarding transaction can be completed in a single, guided process.

This module used the ACR1252U NFC smart card reader and writer, MIFARE DESFire EV2 smart cards, 3.5-inch SPI TFT LCD display module, Raspberry Pi 4, Raspberry Pi OS, and Python-based firmware. The ACR1252U was used because it can read the commuter smart cards during tap-based transactions, while the MIFARE DESFire EV2 cards were used as the secure contactless medium for commuter identification and validation. The TFT display was used to provide immediate visual feedback to the commuter, such as valid

or invalid card status. The Raspberry Pi 4 served as the main controller of the module, while Raspberry Pi OS and Python were used to support card reading, fare computation, display output, and communication with the cloud database. These technologies were used to allow the validation process, fare deduction, and seat allocation logic to operate in one coordinated embedded setup.

After the commuter selects the current terminal and destination in the commuter interface, the commuter taps the smart card on the reader for validation. Once the card is confirmed as valid, the system computes the fare based on the selected destination terminal and then proceeds with automatic seat allocation according to the available seat. If the validation is unsuccessful, the display immediately shows the invalid status, preventing the boarding process from continuing. This module was designed to combine payment validation, fare processing, and seat assignment into a coordinated transaction flow that supports faster and more organized boarding.

The automatic seat allocation shown in annotation **(3)** serves as the transition point between the tap-based payment process and the next module of the system. After a successful tap, the system does not only confirm payment and deduct the correct fare, but also assigns the commuter to an available seat in the prototype. The actual implementation of this assigned seating process, including seat-status detection and controlled seat access, is discussed further in the succeeding figure under the seat monitoring and allocation mechanism**.**

Seat Monitoring Allocation Mechanism

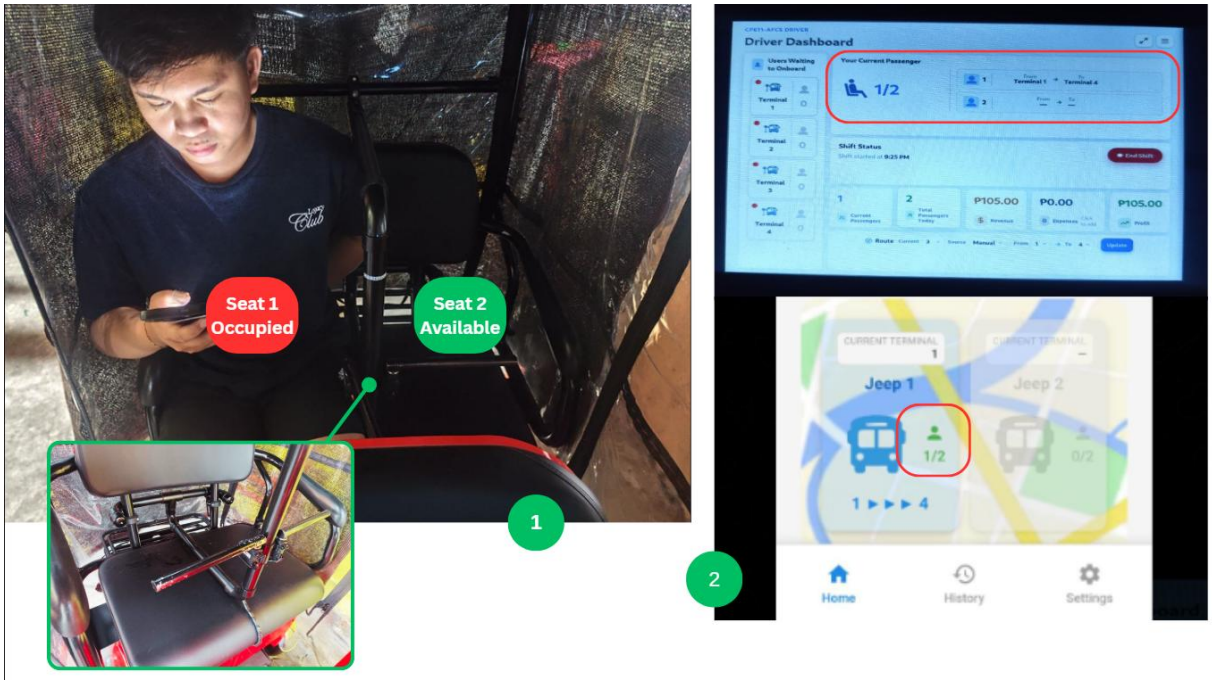


Fig. 23. Seat Monitoring and Allocation Mechanism

Figure 23 showed the developed seat monitoring and allocation mechanism mounted on the e-bike prototype. The figure presented the **physical seat monitoring and allocation mechanism** installed at the passenger seating area (1) and the **seat status indicators reflected in the driver dashboard and commuter interface** (2). These parts were developed to connect the actual seat condition inside the prototype with the seat availability information shown to both the commuter and the driver.

This module used the **Raspberry Pi 4**, **Raspberry Pi OS**, **Python-based firmware**, **infrared sensors**, **servo motors**, and **Cloud Firestore**. The infrared sensors were used to detect whether a seat is occupied or available, while the servo motors were used to control the opening and closing of the assigned seat gates. The Raspberry Pi 4 served as the main

controller that processed the seat logic, and Python was used because it supports sensor reading, servo control, and real-time hardware coordination. Cloud Firestore was used to store and synchronize seat status updates so that the same seat information could be reflected in the commuter interface and driver dashboard in near real time. The physical frame of the mechanism was constructed using PVC pipes because they are lightweight, easy to assemble, and suitable for prototype mounting on the e-bike structure.

The seat monitoring and allocation mechanism operates after the tap-based validation process is completed successfully. Once a commuter is validated, the system assigns the next available seat and opens the corresponding seat gate. When the commuter occupies the assigned seat, the paired infrared sensor detects the presence of the passenger and updates the seat status to occupied. This updated status is then reflected in the interfaces as seat indicators such as **0/2, 1/2, or 2/2**. When the commuter leaves the seat, the system detects that the seat has become clear, updates the status back to available, and closes the corresponding seat gate. This logic operation is illustrated in Figure 24.

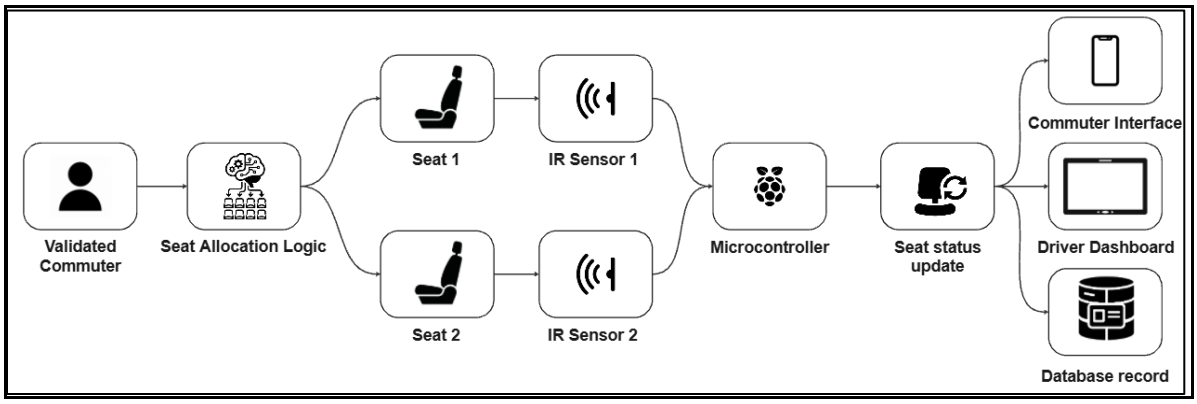


Fig. 24. Logic Operation of Seat Monitoring and Allocation Mechanism

Extended Terminal Feature

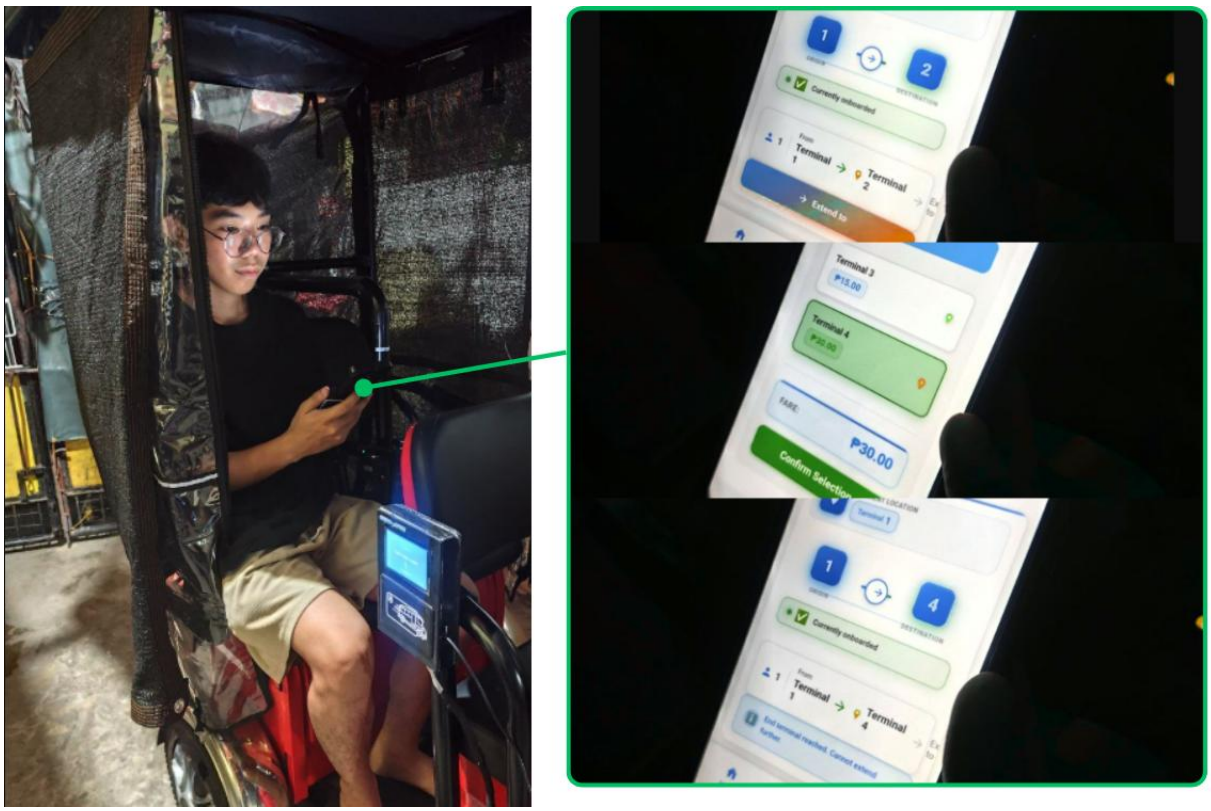


Fig. 25. Extended Terminal Feature

Figure 25 showed the developed extended terminal feature within the commuter interface, which allows a commuter to change to a farther destination terminal after boarding, with the corresponding fare adjustment applied by the system. The figure presented the actual use of the feature inside the prototype and provides an overview of how terminal extension is accessed through the commuter interface while the commuter is already onboard. This module was developed to address cases where a commuter needs to continue to a farther terminal beyond the originally selected destination.

This module used the same core web and cloud technologies applied in the commuter interface, particularly React, Vite, Node.js, npm, Cloud Firestore, and Firestore Cloud Functions. React was used to build the interactive interface for terminal extension, while Vite supported efficient development and updating of the web application. Node.js and npm were used to manage the development environment and required packages. Cloud Firestore was used to store the current trip details, updated destination, and fare-related records, while Firestore Cloud Functions were used to process the extension request and apply the corresponding fare adjustment in a centralized and controlled manner. These technologies were used so that destination changes could be reflected properly in the system while keeping trip and fare records synchronized.

The extended terminal feature was developed as an added commuter function that becomes available after boarding. Through this feature, the commuter can request a farther terminal within the same route direction, and the system updates the trip record and additional fare accordingly.

Driver and Admin Dashboards

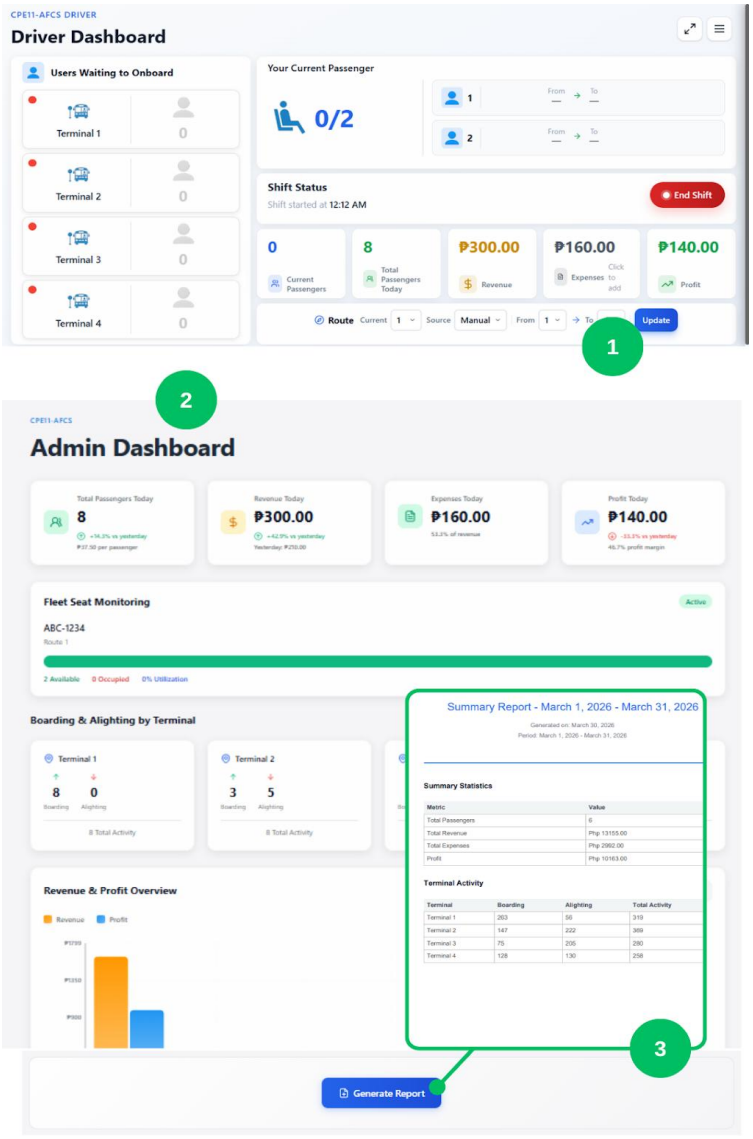


Fig. 26. Driver and Admin Dashboards

Figure 26 showed the developed **driver and administrator dashboards**, which were

designed to support **operational monitoring** and **financial analytics** of the developed AFCS. The figure presented the **driver dashboard** (1), the **admin dashboard** (2), and the **generated report for financial analytics** (3). These parts were developed so that operational data from the prototype can be monitored through role-specific dashboards, while summarized financial records can also be generated for reporting purposes.

This module was developed using React, Vite, Node.js, npm, Firebase Authentication, Cloud Firestore, Firestore Cloud Functions, Hostinger, and Apache/.htaccess. React was used to build responsive dashboard interfaces that can display changing operational and financial data in an organized layout. Vite was used to support efficient development and fast updating of the web applications, while Node.js and npm were used to manage the project environment and required dependencies. Firebase Authentication was used to provide controlled access for driver and administrator users, while Cloud Firestore served as the centralized source of passenger, trip, seat, revenue, and expense records. Firestore Cloud Functions were used to process backend events and maintain updated system records for dashboard display. Hostinger, together with Apache and .htaccess, was used to deploy the dashboards as accessible web applications with proper routing support. These technologies were used so that the dashboards could present synchronized data, support user access control, and provide a practical browser-based monitoring platform.

The driver dashboard was developed to support operational monitoring during active use of the prototype, such as viewing current passengers, seat status, route-related information, and trip activity. The admin dashboard was developed to provide a broader monitoring view of system records, including passenger activity, fleet seat monitoring, and summarized operational results. In addition, the report generation function was developed to present financial analytics records in a more organized output, including revenue, expenses, and profit. This module was therefore developed to provide both real-time operational visibility and summarized financial reporting as part of the overall AFCS.

Project Structure/Organization

The development of the automated fare collection system (AFCS) with seat allocation, terminal selection, and profit analytics dashboards was accomplished through the coordinated efforts of the research team under the guidance of the thesis adviser. To ensure an organized workflow during the study, the group adopted a project structure that assigned specific responsibilities to each member for documentation, testing, programming, calibration, hardware design layout, software design layout, and component canvassing. This distribution of functions helped maintain proper coordination in the planning, design, development, integration, testing, and evaluation of the prototype.

The organizational structure of the project team developing the AFCS is shown in Figure 27. The figure presented the thesis adviser, the researcher leader, and the researchers, along with their respective functional responsibilities.

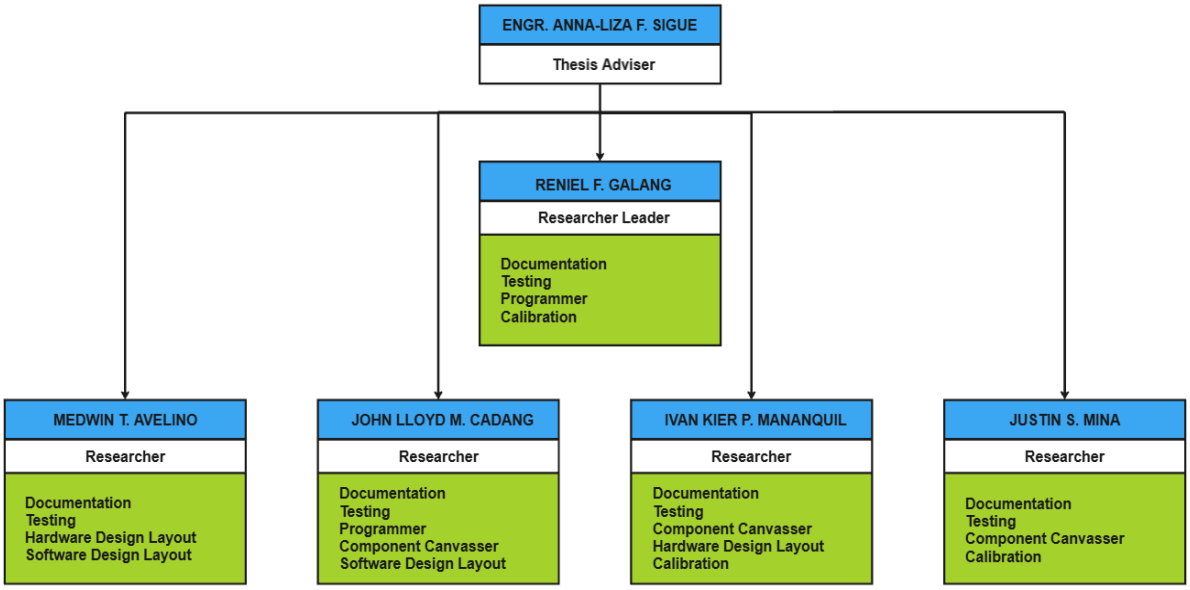


Fig. 27. Organizational Structure of the Development of the AFCS

The roles and responsibilities indicated in Figure 27 are defined as follows:

1. **Adviser**. The adviser provided academic and technical guidance throughout the study. This role included reviewing the project's direction and ensuring that the design, implementation, documentation, and evaluation remained aligned with the research objectives.
2. **Researcher Leader**. Supervised the overall progress of the project and coordinated the tasks of the team. This role involved consolidating outputs, monitoring completion of assigned responsibilities, and ensuring that the different parts of the study were properly integrated.
3. **Researcher**. The researchers served as the primary contributors to the study. They participated in the design, development, testing, documentation, and refinement of the prototype according to their assigned responsibilities.
4. **Documentation**. This role focused on preparing, organizing, and revising the written components of the study. It included drafting manuscript sections, recording project progress, and compiling technical descriptions and findings.
5. **Testing**. This role involved conducting functional tests and validating the prototype's performance. It included executing test procedures, recording observations, identifying issues, and verifying whether the developed system met the intended objectives.
6. **Programmer**. Responsible for implementing, refining, and maintaining the software components of the prototype. This included the development of the embedded logic, interface behavior, database-related processes, and other system functions required for AFCS operation. Responsible for implementing, refining, and maintaining the software components of the prototype. The role also covered GPS calibration support, preparation of SSH operational documentation, and the continuous debugging, troubleshooting, and improvement of the source codes to ensure reliable system performance.
7. **Calibration**. This role involved adjusting and verifying the prototype's settings to ensure

proper, consistent operation. It included checking hardware responses, sensor behavior, and related system outputs during testing and actual use.

8. **Hardware Design Layout**. This role focused on the physical arrangement, structural design, and hardware fabrication of the prototype components. It included planning device placement, wiring organization, mechanical assembly, and the physical integration of the prototype onto the e-bike platform to ensure the developed system was both functional and properly installed.
9. **Software Design Layout**. This role involved planning and arranging the software interfaces, logic flow, implementation, deployment, and system simulation testing. It included organizing the layouts of user interfaces and dashboards, implementing required software functions, deploying the developed interfaces, and conducting simulation-based tests to verify clarity, usability, and functional consistency.
10. **Component Canvasser**. This role involved identifying, sourcing, and comparing the components required for the project. It included checking prices, availability, and suitability of materials needed for the development of the prototype.

Project Limitations and Capabilities

This section presented the operational capabilities and limitations of the developed automated fare collection system (AFCS). The capabilities refer to the functions and services the prototype can perform based on the modules it implements. In contrast, the limitations refer to the conditions and constraints that restrict its operation, scope, and practical deployment.

Operational Capabilities of the Prototype

The developed AFCS was designed to perform integrated transport automation functions through its embedded controller, sensing mechanisms, cloud database, and synchronized user interfaces. Its capabilities are reflected in the implementation of the commuter interface, tap-based payment process, seat monitoring and allocation mechanism,

extended terminal feature, and driver and admin dashboards. As an integrated prototype, the system is capable of the following:

1. It displays commuter card balance, seat availability, and terminal selection options before boarding through the commuter interface.
2. It performs tap-based card reading and validation using the implemented NFC-based setup.
3. It handles invalid taps through invalid reason displays, including system unavailable.
4. It computes terminal-based fares according to the commuter's selected route or destination terminal.
5. It automatically allocates an available seat after successful card validation.
6. It controls seat access through servo-operated seat gates linked to the assigned seat.
7. It detects seat occupancy and vacancy through infrared sensor-based monitoring.
8. It updates seat availability information across connected interfaces through cloud-based data synchronization.
9. It allows commuters to extend their trip to a farther terminal with a corresponding fare adjustment.
10. It provides the driver with real-time operational monitoring information, including the number of commuters waiting to board, the current passengers and their routes, occupied seats, revenue, expenses, and profit.
11. It provides pop-up messages to commuter interface upon arrival and alert pop-ups and indicators for driver dashboard for operational monitoring.
12. It provides the administrator with monitoring and analytics functions through dashboard-based summaries and visualizations.
13. It stores and synchronizes operational records, such as tap events, seat events, and trip-related data, using the centralized database.
14. It supports GPS-based trip and terminal monitoring as part of the prototype's route-handling functions.

Project Limitations

Despite the implemented functions of the developed prototype, the system remained subject to several limitations related to its hardware setup, software logic, prototype scale, and operating environment. These limitations do not negate the system's functionality, but rather define the present boundaries of its implementation. As an integrated prototype, the system has the following limitations:

1. It operates on a limited-seating setup and does not reflect the full seating capacity of an actual public transport vehicle.
2. It depends on proper environmental conditions, correct placement, and minimal obstruction. Sensors' performance are affected by extreme sunlight and dark clothing.
3. It relies on proper servo alignment, stable voltage supply, and reliable wiring connections for consistent seat gate operation.
4. It depends on GPS signal availability, which may vary due to weak reception or environmental obstructions.
5. It may experience delayed reflected updates in the dashboards and the commuter interface when internet connectivity or cloud communication becomes unstable.
6. It limits the extended terminal feature to further terminal selection within the system's defined route logic. It depends on the intended sequence of use, such as terminal selection before boarding, successful card tapping, and occupancy of the assigned seat, for proper operation.
7. It limits the developed interfaces to the implemented user roles of commuter, driver, and administrator.
8. It uses simulated monetary values for top-up and transactions and does not yet process actual monetary amounts.
9. It follows the prototype's configured route, terminal, and seat allocation rules, which may required further modification for more complex transport scenarios.

Project Evaluation

This section presented the evaluation results of the developed automated fare collection system (AFCS) in relation to the five research objectives of the study. Objectives 1 to 4 were evaluated through technical evaluation and respondent-based evaluation to determine whether the implemented system modules performed their intended functions and were usable from the perspective of the target users. Objective 5 was evaluated using the ISO/IEC 25010:2023 software product quality model, focusing on functional suitability, performance efficiency, reliability, security, usability, and data quality. For transparency and reproducibility, the complete timestamped raw test records and detailed data gathered from user and expert respondents are archived (see Appendix G, p. 165).

The results were presented through tables, computed scores, standard deviations, interpretations, and brief discussions to show both the measured performance and perceived quality of the developed prototype. The **technical evaluation** provided evidence of the system's actual functional performance, while the **respondent-based evaluation** presented the assessment of end users and experts to the developed prototype and system. These results collectively determine the extent to which the developed AFCS achieved the intended objectives of the study

Technical Evaluation

This subsection provided the results of the system's actual functional performance through structured testing procedures and conditions. It included the table results for the commuter interface, tap-based payment process, seat monitoring and allocation mechanism, and driver and admin dashboards.

Table 5

Technical Evaluation Results for Commuter Interface

TEST INDICATOR	TEST CONDITION	SCORE	RATING	INTERPRETATION
Card Balance Display Accuracy	Balance Loading	100%	Excellent	The commuter interface accurately displayed the updated balance after balance loading.
	Fare Deduction	100%	Excellent	The commuter interface accurately displayed the deducted fare and updated balance after riding.
Incoming Vehicle Information Accuracy	Route	100%	Excellent	The commuter interface accurately displayed the route of the incoming vehicle.
	Seats Available	100%	Excellent	The commuter interface accurately displayed the available seat count of the incoming vehicle.
	Current Terminal	100%	Excellent	The commuter interface accurately displayed the current terminal of the incoming vehicle.

Table 5 presented the technical evaluation results for the commuter interface in terms of card balance display accuracy and incoming vehicle information accuracy. The results show that all five test conditions, namely balance loading, fare deduction, route, seats available, and current terminal, obtained an accuracy rate of 100% and were rated Excellent. This indicates that the commuter interface accurately displayed the required pre-boarding information needed by commuters before using the tap-based payment process, supporting the literature stating that commuter-facing transport systems should provide accurate and timely pre-boarding information to reduce passenger uncertainty and guide commuter decisions [57], [58], [59], [60].

Table 6

Technical Evaluation Results for Tap-Based Payment Process

TEST INDICATOR	MEASUREMENT BASIS	SCORE	RATING	INTERPRETATION
Tap Read Success Rate	Successful card reads over total tap attempts	98.08%	Very Good	The tap-based payment process read commuter cards effectively, with only minor unsuccessful tap attempts.
Terminal-Based Fare Computation Accuracy	Correct fare computations over total fare computation trials	100%	Excellent	The system accurately computed the correct fare based on the selected terminal interval.
Automatic Seat Allocation Success Rate	Correct seat allocations over total successful validation trials	99.05%	Excellent	The system allocated seats highly effectively after successful card validation.
Invalid Reason Classification Accuracy	Correct invalid reason classifications over total invalid tap cases	100%	Excellent	The system accurately displayed the correct invalid reason for rejected tap transactions.
System Unavailable Display Accuracy	Correct unavailable screen displays over total connectivity interruption trials	93%	Good	The system displayed the unavailable status at an acceptable level during connectivity interruption trials.

Table 6 presented the technical evaluation results for the tap-based payment process. The results show that terminal-based fare computation accuracy and invalid reason classification accuracy both obtained 100%, while automatic seat allocation success rate obtained 99.05%, all rated Excellent. Tap read success rate obtained 98.08% and was rated Very Good, while system unavailable display accuracy obtained 93% and was rated Good. These results indicate that the tap-based payment process generally performed effectively in validating cards, computing fares, allocating seats, identifying invalid tap reasons, and displaying the system unavailable status during connectivity interruptions, which aligned with the literature emphasizing reliable card reading, correct fare computation, and efficient automated processing as essential functions of smart fare collection systems [57], [61], [62], [64], [65], [66].

Table 7

Technical Evaluation Results for Seat Monitoring and Allocation Mechanism

TEST INDICATOR	MEASUREMENT BASIS	SCORE	RATING	INTERPRETATION
Seat Occupancy Detection Accuracy	Correct seat occupancy detections over total seat detection trials	93.40%	Good	The system detected seat occupancy at an acceptable level.
Driver Seat Status View Consistency	Correct driver dashboard seat status updates over total status view trials	93.40%	Good	The driver dashboard displayed seat status updates at an acceptable level.
Commuter Seat Status View Consistency	Correct commuter interface seat status updates over total status view trials	93.40%	Good	The commuter interface displayed seat status updates at an acceptable level.
Seat Lock and Unlock Actuation Accuracy	Correct seat lock and unlock actions over total actuation trials	99.06%	Excellent	The seat lock and unlock mechanism performed highly effectively and consistently.

Table 7 presented the technical evaluation results for the seat monitoring and allocation mechanism. The results show that seat occupancy detection accuracy, driver seat status view consistency, and commuter seat status view consistency each obtained a score of 93.40%, which was rated Good. Meanwhile, the seat lock and unlock actuation accuracy obtained a score of 99.06%, which was rated Excellent. These results indicate that the system generally provided acceptable seat detection and seat status display performance, while the seat lock and unlock mechanism performed highly effectively and consistently, supporting the reviewed studies that highlight the importance of accurate occupancy sensing and dependable response mechanisms in transport monitoring systems [65], [67].

Table 8

Technical Evaluation Results for Driver and Admin Dashboards

TEST INDICATOR	MEASUREMENT BASIS	SCORE	RATING	INTERPRETATION
Dashboard Data Consistency Rate	Correct dashboard data displays over total consistency trials	100%	Excellent	The dashboards displayed consistent system data across the tested records.
Driver Alert Pop-Up Accuracy	Correct alert pop-up displays over total alert trials	100%	Excellent	The driver dashboard accurately displayed alert pop-ups during the tested conditions.
Indicator State Correctness	Correct indicator states over total indicator trials	100%	Excellent	The dashboards indicators accurately reflected the correct system states.
Driver Monitoring Non-Repeat Compliance	Correct non-repeated monitoring updates over total monitoring trials	100%	Excellent	The driver monitoring function complied with the non-repeat requirement during the tested trials.

Table 8 presented the technical evaluation results for the driver and administrator dashboards. The results show that dashboard data consistency rate, driver alert pop-up accuracy, indicator state correctness, and driver monitoring non-repeat compliance all obtained a score of 100.00%, which was rated Excellent. These results indicate that the dashboard functions consistently displayed system data, alerts, indicators, and monitoring updates during the conducted trials, which followed the reviewed reference on Firestore real-time synchronization, supporting the need for consistent and near real-time reflection of operational and financial data across connected interfaces [69].

Respondent-Based Evaluation

This subsection presented the respondent-based evaluation of the developed prototype and system based on the assessments of end users and experts. The user evaluation focused on usability in terms of interaction capability (UIC), which refers to how easily and clearly users interacted with the system features during prototype use. It involved 100 end-user respondents for Objectives 1 and 2, which covered the commuter interface and tap-based payment process, and another 100 end-user respondents for Objectives 3 and 4, which covered the seat monitoring and allocation mechanism, driver dashboard, and administrator dashboard. For Objective 5, the evaluation was based on the responses of 12 expert respondents using the ISO/IEC 25010:2023 software product quality model. The results are presented using the computed mean, standard deviation (SD), verbal interpretation, and objective-based discussion.

Table 9

User Evaluation Results for Commuter Interface

STATEMENT	MEAN	SD	EVALUATION INTERPRETATION
Viewing important information such as balance, incoming vehicle, route, available seats, and current terminal was clear and easy to understand.	4.59	0.68	The respondents strongly agreed that the commuter interface clearly displayed important pre-boarding information.
Topping up my balance using the interface was straightforward and easy to complete.	4.57	0.76	The respondents strongly agreed that the commuter interface made balance loading easy to perform.
Selecting my destination terminal and confirming my current terminal was easy and not confusing.	4.51	0.77	The respondents strongly agreed that the commuter interface allowed easy terminal selection and current terminal confirmation.
Overall Objective 1: Commuter Interface	4.56	0.74	The respondents strongly agreed that Objective 1 was achieved based on usability in terms of interaction capability.

Table 9 presented the user evaluation results for Objective 1, which focused on the commuter interface. The results show that the three items obtained mean scores of 4.59, 4.57, and 4.51, all falling under the Strongly Agree range based on the respondent-based evaluation criteria. The overall mean score of 4.56 with a standard deviation of 0.74 indicates that the

respondents strongly agreed that the commuter interface was clear and usable in displaying important pre-boarding information, supporting balance loading, and allowing terminal selection before boarding. These results were crucial to avoid confusion and dissatisfaction among users [59], [60].

Table 10

User Evaluation Results for Tap-Based Payment Process

STATEMENT	MEAN	SD	EVALUATION INTERPRETATION
After tapping, the system clearly informed the user whether the card was valid or rejected and why.	4.51	0.76	The respondents strongly agreed that the system clearly displayed card validation results and rejection reasons after tapping.
When validated, the seat allocation result was clear and easy to notice.	4.57	0.79	The respondents strongly agreed that the system clearly presented the seat allocation result after successful validation.
The fare deduction, including the deducted amount and updated balance, was clear and understandable.	4.65	0.70	The respondents strongly agreed that the system clearly displayed the fare deduction details and updated balance.
Overall Objective 2: Tap-Based Payment Process	4.58	0.75	The respondents strongly agreed that Objective 2 was achieved based on usability in terms of interaction capability.

Table 10 presented the user evaluation results for Objective 2, which focused on the tap-based payment process. The results show that the three items obtained mean scores of 4.51, 4.57, and 4.65, all falling under the Strongly Agree range based on the respondent-based evaluation criteria. The overall mean score of 4.58 with a standard deviation of 0.75 indicates that the respondents strongly agreed that the tap-based payment process was clear and usable in displaying card validation feedback, presenting seat allocation results, and showing fare deduction details with the updated balance. These were consistent with the studies of [65] and [66], where the seat allocation's relevance depended on correct status recognition after successful validation of payment, during the trial runs coordinated with the actual users.

Table 11

User Evaluation Results for Seat Monitoring and Allocation Mechanism

STATEMENT	MEAN	SD	EVALUATION INTERPRETATION
Monitoring upcoming passengers who are waiting to be boarded was clear and easy to track.	4.63	0.60	The respondents strongly agreed that the system clearly supported the monitoring of upcoming passengers waiting to board.
Seat status monitoring, including who is seated and seat availability, was easy to understand at a glance.	4.65	0.61	The respondents strongly agreed that the system clearly presented seat status information and seat availability.
Overall Objective 3: Seat Monitoring and Allocation Mechanism	4.64	0.61	The respondents strongly agreed that Objective 3 was achieved based on usability in terms of interaction capability.

Table 11 presented the user evaluation results for Objective 3, which focused on the seat monitoring and allocation mechanism. The results show that the two items obtained mean scores of 4.63 and 4.65, both falling under the Strongly Agree range based on the respondent-based evaluation criteria. The overall mean score of 4.64 with a standard deviation of 0.61 indicates that the respondents strongly agreed that the system was clear and usable in monitoring upcoming passengers, displaying seat status, and showing seat availability. These results were assessed by the users if the system monitored the actual seat accurately under operational conditions, similar to related studies [65], [67].

Table 12

User Evaluation Results for Driver and Admin Dashboards

STATEMENT	MEAN	SD	EVALUATION INTERPRETATION
Monitoring revenue, total passengers, and trip summaries was easy to read and understand.	4.58	0.62	The respondents strongly agreed that the dashboards clearly presented revenue, total passengers, and trip summaries.
Recording expenses was easy to do, and the system confirmed that the expense was saved correctly.	4.35	0.61	The respondents strongly agreed that the dashboards supported easy expense recording and provided clear confirmation after saving.
The profit computation, including revenue minus expenses, was clearly presented and easy to verify.	4.67	0.59	The respondents strongly agreed that the dashboards clearly presented profit computation and made it easy to verify.
Overall Objective 4: Driver and Admin Dashboards	4.53	0.61	The respondents strongly agreed that Objective 4 was achieved based on usability in terms of interaction capability.

Table 12 presented the user evaluation results for Objective 4, which focused on the driver and administrator dashboards for operational monitoring and financial analytics. The results show that the three items obtained mean scores of 4.58, 4.35, and 4.67, all falling under the Strongly Agree range based on the respondent-based evaluation criteria. The overall mean score of 4.53 with a standard deviation of 0.61 indicates that the respondents strongly agreed that the dashboards were clear and usable in monitoring revenue, total passengers, trip summaries, expense records, and profit computation. These results were consistent in the studies of [59] and [60], where the usability and satisfaction of the system were crucial based on the answers of the users during the evaluation.

Table 13

Expert Evaluation Results for Functional Suitability

STATEMENT	MEAN	SD	EVALUATION INTERPRETATION
The system correctly validates smart card taps and accepts only authorized or registered cards.	4.83	0.39	The expert respondents strongly agreed that the system correctly validated smart card taps and accepted only authorized or registered cards.
The system correctly computes and applies fares based on the selected destination terminal and the configured fare matrix or rules.	4.83	0.39	The expert respondents strongly agreed that the system correctly computed and applied fares based on the selected destination terminal and fare rules.
The system correctly performs seat allocation and seat status updates based on sensor inputs and allocation rules.	4.67	0.65	The expert respondents strongly agreed that the system correctly performed seat allocation and seat status updates based on sensor inputs and allocation rules.
The system provides the required functions for commuter, driver, and admin roles without functional errors.	4.67	0.49	The expert respondents strongly agreed that the system provided the required functions for commuter, driver, and administrator roles.
The system accurately records key operational outputs, including transactions, trip logs, seat events, and dashboard indicators needed for monitoring and reporting.	4.67	0.49	The expert respondents strongly agreed that the system accurately recorded key operational outputs needed for monitoring and reporting.
Overall Functional Suitability	4.73	0.25	The expert respondents strongly agreed that the developed system achieved functional suitability.

Table 13 presented the expert evaluation results for Functional Suitability. The results showed that all five items obtained mean scores ranging from 4.67 to 4.83, all falling under the

Strongly Agree range based on the respondent-based evaluation criteria. The overall mean score of 4.73 with a standard deviation of 0.25 indicated that the expert respondents strongly agreed that the system was functionally suitable in terms of smart card validation, fare computation, seat allocation, role-based functions, and recording of operational outputs. In a recent study about jeepney fare payment application, the expert ratings in terms of Functionality garnered a mean of 4.33, interpreted as Excellent [70].

Table 14

Expert Evaluation Results for Performance Efficiency

STATEMENT	MEAN	SD	EVALUATION INTERPRETATION
The system confirms a card tap promptly and shows results with minimal delay.	4.50	0.52	The expert respondents strongly agreed that the system confirmed card taps promptly and displayed results with minimal delay.
Seat status changes and related interface updates appear quickly after a real seat event occurs.	4.67	0.65	The expert respondents strongly agreed that seat status changes and interface updates appeared quickly after seat events occurred.
The dashboards load and refresh operational data and analytics within acceptable response times under normal use.	4.83	0.39	The expert respondents strongly agreed that the dashboards loaded and refreshed operational data and analytics within acceptable response times.
The embedded device and backend services run with stable performance without excessive CPU or memory usage.	4.42	0.67	The expert respondents strongly agreed that the embedded device and backend services operated with stable performance.
Data synchronization and updates remain responsive when network connectivity is available.	4.50	0.67	The expert respondents strongly agreed that data synchronization and updates remained responsive when network connectivity was available.
Overall Performance Efficiency	4.58	0.32	The expert respondents strongly agreed that the developed system achieved performance efficiency.

Table 14 presented the expert evaluation results for Performance Efficiency. The results show that all five items obtained mean scores ranging from 4.42 to 4.83, all falling under the Strongly Agree range based on the respondent-based evaluation criteria. The overall mean score of 4.58 with a standard deviation of 0.32 indicated that the expert respondents strongly agreed that the system performed efficiently in terms of card tap confirmation, seat status updates, dashboard response time, device and backend stability, and data synchronization. In

a recent study about jeepney fare payment application, the expert ratings in terms of Performance Efficiency garnered a mean of 4.00, interpreted as Very Good [70].

Table 15

Expert Evaluation Results for Reliability

STATEMENT	MEAN	SD	EVALUATION INTERPRETATION
The system performs consistently during continuous operation without frequent crashes or failures.	4.67	0.49	The expert respondents strongly agreed that the system performed consistently during continuous operation without frequent crashes or failures.
Transaction and monitoring data remain accurate and intact even during network interruptions.	4.17	0.83	The expert respondents agreed that transaction and monitoring data remained accurate and intact during network interruptions.
The system can recover and resume normal operation after minor faults or interruptions.	4.42	0.79	The expert respondents strongly agreed that the system could recover and resume normal operation after minor faults or interruptions.
Logs and dashboard outputs remain dependable for operational monitoring and decision-making.	4.50	0.52	The expert respondents strongly agreed that the logs and dashboard outputs remained dependable for operational monitoring and decision-making.
The system maintains consistent synchronization behavior with minimal downtime or duplicated records.	4.25	0.87	The expert respondents strongly agreed that the system maintained consistent synchronization behavior with minimal downtime or duplicated records.
Overall Reliability	4.40	0.58	The expert respondents strongly agreed that the developed system achieved reliability.

Table 15 presented the expert evaluation results for Reliability. The results show that four items obtained mean scores ranging from 4.25 to 4.67, all falling under the Strongly Agree range based on the respondent-based evaluation criteria, while one item obtained a mean score of 4.17, which falls under the Agree range. The overall mean score of 4.40 with a standard deviation of 0.58 indicated that the expert respondents strongly agreed that the system was reliable in terms of continuous operation, recovery after interruptions, dependable logs and dashboard outputs, and synchronization behavior. Although the item on data accuracy during network interruptions obtained the lowest mean, it remained within the acceptable Agree range. In a recent study about jeepney fare payment application, the expert ratings in terms of Reliability garnered a mean of 4.25, interpreted as Excellent [70].

Table 16

Expert Evaluation Results for Security

STATEMENT	MEAN	SD	EVALUATION INTERPRETATION
The system's smart card security mechanisms, such as secure authentication and encryption, protect fare transactions.	4.42	0.67	The expert respondents strongly agreed that the smart card security mechanisms protected fare transactions.
User data and stored values, such as balances and transaction records, are protected from unauthorized access or alteration.	4.50	0.80	The expert respondents strongly agreed that user data and stored values were protected from unauthorized access or alteration.
The system enforces role-based access control for commuter, driver, and administrator accounts to restrict sensitive functions and data.	4.67	0.65	The expert respondents strongly agreed that the system enforced role-based access control for restricting sensitive functions and data.
Communication between devices, backend services, and databases uses secure methods to protect transmitted data.	4.50	0.80	The expert respondents strongly agreed that secure communication methods were used to protect transmitted data.
Audit logs and access controls support accountability and traceability of key activities.	4.50	0.67	The expert respondents strongly agreed that audit logs and access controls supported accountability and traceability of key activities.
Overall Security	4.52	0.59	The expert respondents strongly agreed that the developed system achieved security.

Table 16 presented the expert evaluation results for Security. The results show that all five items obtained mean scores ranging from 4.42 to 4.67, all falling under the Strongly Agree range based on the respondent-based evaluation criteria. The overall mean score of 4.52 with a standard deviation of 0.59 indicated that the expert respondents strongly agreed that the system achieved security in terms of smart card transaction protection, user data protection, role-based access control, secure communication, and audit-based accountability. According to the report of World Bank, smartcard ticketing could deliver savings through reduced fraud, shorter dwell times at stops, and less cash handling [73].

Table 17

Expert Evaluation Results for Usability

STATEMENT	MEAN	SD	EVALUATION INTERPRETATION
The system interfaces are intuitive and easy to navigate for their intended users.	4.58	0.51	The expert respondents strongly agreed that the system interfaces were intuitive and easy to navigate for their intended users.
The system provides clear feedback and messages during important actions, such as tap validation, seat allocation, and error cases.	4.67	0.65	The expert respondents strongly agreed that the system provided clear feedback and messages during important actions.
The dashboard layout and visualizations make operational status and analytics easy to understand.	4.42	0.67	The expert respondents strongly agreed that the dashboard layout and visualizations made operational status and analytics easy to understand.
New users can operate the system with minimal training or assistance.	4.58	0.67	The expert respondents strongly agreed that new users could operate the system with minimal training or assistance.
The interface design supports efficient workflow and helps minimize user input errors.	4.42	0.67	The expert respondents strongly agreed that the interface design supported efficient workflow and helped minimize user input errors.
Overall Usability	4.53	0.52	The expert respondents strongly agreed that the developed system achieved usability.

Table 17 presented the expert evaluation results for Usability. The results show that all five items obtained mean scores ranging from 4.42 to 4.67, all falling under the Strongly Agree range based on the respondent-based evaluation criteria. The overall mean score of 4.53 with a standard deviation of 0.52 indicated that the expert respondents strongly agreed that the system achieved usability in terms of interface navigation, feedback messages, dashboard layout, ease of operation, workflow efficiency, and error minimization. In a recent study about jeepney fare payment application, the expert ratings in terms of Usability garnered a mean of 4.22, interpreted as Excellent [70].

Table 18

Expert Evaluation Results for Data Quality

STATEMENT	MEAN	SD	EVALUATION INTERPRETATION
The data stored by the system, including users, cards, transactions, trips, seats, and expenses, are accurate and consistent across modules.	4.75	0.45	The expert respondents strongly agreed that the data stored by the system were accurate and consistent across modules.
The system validates inputs and prevents invalid or inconsistent records, such as impossible seat counts, duplicate events, or incorrect values.	4.58	0.51	The expert respondents strongly agreed that the system validated inputs and prevented invalid or inconsistent records.
The collected data are complete enough to support reporting, auditing, and analytics requirements.	4.42	0.67	The expert respondents strongly agreed that the collected data were complete enough to support reporting, auditing, and analytics requirements.
Recorded timestamps and event sequences reflect real operations and allow traceability of transactions and seat events.	4.67	0.65	The expert respondents strongly agreed that the recorded timestamps and event sequences supported traceability of transactions and seat events.
The data displayed on dashboards match the data stored in the database and update in a timely manner.	4.67	0.49	The expert respondents strongly agreed that the dashboard data matched the stored database records and updated in a timely manner.
Overall Data Quality	4.62	0.48	The expert respondents strongly agreed that the developed system achieved data quality.

Table 18 presented the expert evaluation results for Data Quality. The results show that all five items obtained mean scores ranging from 4.42 to 4.75, all falling under the Strongly Agree range based on the respondent-based evaluation criteria. The overall mean score of 4.62 with a standard deviation of 0.48 indicated that the expert respondents strongly agreed that the system achieved data quality in terms of data accuracy, consistency, input validation, completeness, traceability, dashboard-data matching, and timely updates. Evidence supporting the present study's Data Quality discussion was provided by a smart-card transaction study that processed 100,000 transactions and reported that 98,632 were error-free, while identifying an example error pattern as consecutive swiping activity occurring more than three times in the same bus trip; this supported the need for transaction validation and cleaning so that stored logs remain credible for dashboards, reports, and analytics [74].

CHAPTER V

SUMMARY OF FINDINGS, CONCLUSION, AND RECOMMENDATION

This study focused on the design, development, and evaluation of an automated fare collection system (AFCS) with seat allocation, terminal selection, and profit analytics dashboards. The developed system aimed to support public transport operations by providing a commuter interface for viewing card balance, seat availability, and terminal selection; a tap-based payment process for card validation, fare computation, and seat allocation; a seat monitoring and allocation mechanism for tracking seat status; and driver and administrator dashboards for operational monitoring and financial analytics. An e-bike prototype was used to model the system operation at a small scale.

The study used a developmental research design through the Design and Development Research (DDR) approach. This approach guided the researchers in identifying system requirements, designing the prototype, developing the hardware and software components, integrating the system modules, and evaluating the completed prototype. The developed AFCS was evaluated through technical testing and respondent-based evaluation to determine whether the system achieved the stated research objectives.

Objectives 1 to 4 were evaluated through technical testing and user evaluation. The technical evaluation measured the actual performance of the system in terms of accuracy rates, success rates, and consistency of major system functions, with 100 trial runs per testing type. The user evaluation focused on usability in terms of interaction capability, with 100 end-user respondents evaluating Objectives 1 and 2, and another 100 end-user respondents evaluating Objectives 3 and 4. Objective 5 was evaluated by 12 expert respondents using the ISO/IEC 25010:2023 software product quality model, specifically in terms of functional suitability, performance efficiency, reliability, security, usability, and data quality.

Summary of Significant Findings

Based on the five research objectives of the study, the findings included the technical evaluation results gathered from the structured testing and the respondent-based evaluation results gathered from end-user and expert respondents.

1. The findings of the study showed that the commuter interface accurately displayed the updated balance after 100 trials of balance loading with a score of 100% (Excellent), the commuter interface accurately displayed the deducted fare and updated balance after 100 trials of riding with a score of 100% (Excellent), the commuter interface accurately displayed the route after 100 checks of the route of the incoming vehicle with a score of 100% (Excellent), the commuter interface accurately displayed the available seat count after 100 checks of the available seat of the incoming vehicle with a score of 100% (Excellent), and the commuter interface accurately displayed the current terminal after 100 checks of the current terminal of the incoming vehicle with a score of 100% (Excellent). The results from technical evaluation indicated that the researchers had successfully designed and developed a commuter interface that displays card balance details, seat availability, and terminal selection before boarding the incoming vehicle.

   The respondent-based evaluation further supported these findings, as the 100 end-users strongly agreed that the commuter interface was clear, accessible, and effective in facilitating balance loading, terminal selection, and pre-boarding information access. The obtained weighted means ranging from 4.51 to 4.59, with relatively low standard deviations, indicated a high level of user acceptance and consistency in respondents' perceptions regarding the usability and interaction capability of the developed interface. These findings suggested that the prototype provided a user-centered and operationally functional commuter experience aligned with the intended objectives of the study.

Despite the favorable results, the findings should be interpreted within the scope and limitations of the developed prototype. The system was implemented using a limited-seating configuration, predefined route and terminal logic, simulated monetary transactions, and role-specific interfaces limited to commuters, drivers, and administrators. In addition, the effectiveness of several system functions depended on stable internet connectivity, reliable cloud synchronization, proper hardware alignment, and adequate environmental conditions for sensor and GPS performance. Consequently, while the prototype effectively demonstrated the feasibility and operational capability of an integrated smart transportation management system, further refinement, scalability testing, and real-world deployment evaluation are necessary to support broader implementation in actual public transportation environments.

2. The findings of the study revealed that after 100 trial-runs, the tap-based payment process read commuter cards effectively, with only minor unsuccessful tap attempts with a score of 98.08% (Very Good), the system accurately computed the correct fare based on the selected terminal interval with a score of 100% (Excellent), the system allocated seats highly effectively after successful card validation with a score of 99.05% (Excellent), the system accurately displayed the correct invalid reason for rejected tap transactions with a score of 100% (Excellent), and the system displayed the unavailable status at an acceptable level during connectivity interruption trials with a score of 93% (Good). These results indicated that the tap-based payment process generally performed effectively in validating cards, computing fares, allocating seats, identifying invalid tap reasons, and displaying the system unavailable status during connectivity interruptions.

   The respondent-based evaluation further supported these findings, as the 100 end-users strongly agreed that the system clearly displayed card validation results, rejection reasons, seat allocation information, fare deductions, and updated commuter

balances. The obtained weighted means ranging from 4.51 to 4.65, accompanied by relatively low standard deviations, indicated a high degree of user acceptance, usability, and consistency in respondent perceptions regarding the interaction capability of the tap-based payment process. These results suggested that the developed prototype provided an understandable, responsive, and user-friendly transaction experience for commuters during the boarding process.

Despite the positive outcomes, the findings should be interpreted within the operational scope and limitations of the prototype. The system operated under a predefined route structure, limited seating configuration, and simulated monetary transaction environment. In addition, the effectiveness of the tap-based payment and synchronization functions depended on stable internet connectivity, reliable cloud communication, proper hardware alignment, and uninterrupted NFC sensor operation. Connectivity interruptions occasionally affected the reflection of system status updates, while the implemented transaction logic remained limited to the configured operational conditions of the prototype. Therefore, although the developed system successfully demonstrated the feasibility and functional capability of automated commuter validation and fare management, further enhancements and large-scale testing are necessary to support real-world public transportation deployment.

3. The findings of the study demonstrated that after 100 trial-runs, the seat monitoring and allocation mechanism detected seat occupancy at an acceptable level with a score of 93.40% (Good), the driver dashboard displayed seat status updates at an acceptable level with a score of 93.40% (Good), the commuter interface displayed seat status updates at an acceptable level with a score of 93.40% (Good), and the seat lock and unlock mechanism performed highly effectively and consistently with a score of 99.06% (Excellent). These results indicated that the system generally provided acceptable seat detection and seat status display performance, while the seat lock and unlock mechanism performed highly effectively and consistently.

The respondent-based evaluation further supported these findings, as the 100 end-users strongly agreed that the system effectively supported the monitoring of passengers waiting to board and clearly presented seat status and seat availability information. The obtained weighted means ranging from 4.63 to 4.65, with relatively low standard deviations, reflected a high level of user acceptance and consistency in respondents' perceptions regarding the usability and interaction capability of the implemented seat monitoring and allocation mechanisms. These findings suggest that the prototype provided a responsive and user-centered approach to automated seat management within the public transportation setting.

Despite the favorable results, the findings should be interpreted within the scope and limitations of the developed prototype. The seat monitoring mechanism relied on proper infrared sensor positioning, stable environmental conditions, minimal obstructions, and accurate hardware calibration for consistent seat detection performance. Sensor accuracy may be affected by environmental factors such as extreme lighting conditions and dark-colored objects or clothing. In addition, the prototype operated using a limited-seating configuration and predefined seat allocation logic, which may not fully represent the complexity of actual large-scale public transportation systems. Consequently, although the developed prototype effectively demonstrated the feasibility and operational capability of automated seat monitoring and allocation, further refinement and scalability testing are recommended for broader real-world implementation.

4. The findings of the study revealed that the driver and admin dashboards displayed consistent system data across the 100 tested records with a score of 100% (Excellent), the driver dashboard accurately displayed 100 alert pop-ups during the tested conditions with a score of 100% (Excellent), the dashboard indicators accurately reflected the correct system states after 100 trials with a score of 100% (Excellent), and the driver monitoring function complied with the non-repeat requirement during the 100

tested trials with a score of 100% (Excellent). The results from technical evaluation indicated that the researchers had successfully designed and developed driver and admin dashboard for operational monitoring and financial analytics.

The respondent-based evaluation further supported these findings, as the 100 end-users strongly agreed that the dashboards clearly presented revenue information, passenger counts, trip summaries, expense recording functions, and profit computation details. The obtained weighted means ranging from 4.35 to 4.67, with relatively low standard deviations, indicated a high level of user acceptance and consistency in respondents' perceptions regarding the usability, clarity, and interaction capability of the driver and administrator dashboards. In particular, the respondents strongly agreed that the dashboards simplified the verification of operational and financial information, suggesting that the system effectively enhanced monitoring and decision-support functions for transport operations.

Despite the favorable outcomes, the findings should be interpreted within the scope and limitations of the developed prototype. The implemented dashboards relied on stable cloud synchronization, internet connectivity, and centralized database communication to maintain real-time data consistency across interfaces. Delays in reflected updates and operational notifications may occur during connectivity interruptions or unstable network conditions. In addition, the financial analytics functions were based on simulated monetary transactions and predefined operational data generated within the prototype environment. The developed dashboards were likewise limited to the implemented user roles of driver and administrator and operated under the configured route, seating, and transaction logic of the study. Therefore, while the prototype successfully demonstrated the feasibility and effectiveness of integrated operational monitoring and financial analytics in a smart transportation environment, further enhancement and scalability testing are recommended for practical large-scale implementation.

5. The findings of the study revealed that the developed AFCS achieved a high level of software quality based on the evaluation conducted by the 12 expert respondents using the ISO/IEC 25010:2023 quality standards. In terms of Functional Suitability, the system obtained an overall mean score of 4.73, interpreted as Strongly Agree, indicating that the prototype effectively performed its intended functions related to smart card validation, fare computation, seat allocation, role-based operations, and operational data recording. The expert respondents confirmed that the system consistently executed the required commuter, driver, and administrator functions with high accuracy and completeness, demonstrating the operational capability of the developed prototype within the configured transportation environment.

   In terms of Performance Efficiency, the system attained an overall mean score of 4.58, interpreted as Strongly Agree, indicating that the prototype responded efficiently in processing commuter transactions, updating seat status information, refreshing dashboard analytics, and synchronizing operational data. The findings demonstrated that the embedded hardware devices, backend services, and cloud-based communication mechanisms were capable of supporting responsive and timely system operations under normal connectivity conditions. These results suggest that the prototype successfully integrated real-time monitoring and transaction processing functions with acceptable operational responsiveness.

   The evaluation results for Reliability yielded an overall mean score of 4.40, interpreted as Strongly Agree, indicating that the system consistently maintained operational stability, synchronization behavior, and recovery capability during continuous use and minor interruptions. Although the indicator related to maintaining data accuracy during network interruptions obtained the lowest mean score among the reliability criteria, it remained within the acceptable Agree range. This finding suggests that while the prototype generally demonstrated dependable operation and data integrity, network-dependent functions may still require further enhancement to

strengthen resilience during unstable connectivity conditions.

In terms of Security, the system obtained an overall mean score of 4.52, interpreted as Strongly Agree, indicating that the prototype effectively implemented smart card transaction protection, role-based access control, secure communication, audit logging, and accountability mechanisms. These findings confirmed that the developed system incorporated appropriate security measures for protecting operational data, restricting unauthorized access, and supporting traceability of critical system activities within the scope of the prototype implementation.

The system likewise demonstrated a high level of Usability, obtaining an overall mean score of 4.53, interpreted as Strongly Agree. The expert respondents agreed that the interfaces were intuitive, easy to navigate, and capable of providing clear operational feedback and understandable dashboard visualizations. Furthermore, the findings indicated that the prototype supported efficient workflow execution and minimized user input errors, suggesting that the developed interfaces were user-centered and suitable for the intended commuter, driver, and administrator interactions.

Lastly, the system achieved a high evaluation in terms of Data Quality, with an overall mean score of 4.62, interpreted as Strongly Agree. The findings confirmed that the developed prototype maintained accurate, consistent, and traceable operational records across system modules while supporting timely dashboard synchronization and reliable database updates. The implemented validation mechanisms effectively prevented invalid records and ensured that the collected operational data were sufficient for monitoring, reporting, auditing, and analytics purposes.

Overall, the evaluation findings demonstrated that the developed AFCS successfully satisfied the major software quality characteristics defined under ISO/IEC 25010:2023. However, the findings should still be interpreted within the operational scope and limitations of the prototype, including its limited-seating configuration, predefined route and terminal logic, simulated monetary transactions, dependence on

stable internet connectivity and GPS availability, and controlled testing environment. Consequently, while the system demonstrated strong functional and software quality performance, further refinement, scalability testing, and real-world deployment evaluation are recommended to support broader implementation in actual public transportation systems.

Conclusions

The study successfully designed, developed, and evaluated an automated fare collection system (AFCS) with seat allocation, terminal selection, and profit analytics dashboards, achieving its five research objectives at the prototype level. Based on the findings and their implications, the following conclusions are drawn per research objective.

1. The system effectively supported accurate commuter information display, automated balance updating, route and terminal presentation, and seat availability monitoring through the integration of NFC-based transactions and cloud-connected synchronization mechanisms. The respondent-based evaluation further confirmed that the commuter interface was understandable, accessible, and operationally effective for intended users. However, the system remained limited to predefined routes, limited seating configurations, simulated monetary transactions, and role-specific operations. In addition, the operational effectiveness of the commuter interface depended heavily on stable internet connectivity, reliable cloud synchronization, and proper GPS and hardware performance.
2. The system effectively supported reliable NFC card reading, accurate fare computation, automated seat allocation, invalid tap detection, and synchronized commuter balance updating during boarding operations. The respondent-based evaluation further confirmed that the tap-based transaction process was understandable, responsive, and user-friendly for commuters. However, the system remained affected by connectivity

interruptions and synchronization delays during unstable network conditions. In addition, the implemented transaction logic and payment environment remained limited to simulated monetary transactions and predefined operational conditions within the prototype setup.

3. The implemented infrared sensor-based monitoring system and servo-operated seat control mechanism effectively supported synchronized seat occupancy detection, seat status reflection, and controlled seat access operations. The respondent-based evaluation further confirmed that the implemented seat management mechanisms were usable, responsive, and operationally effective for commuters and system operators. However, the seat monitoring mechanism remained susceptible to environmental conditions, sensor positioning, hardware calibration issues, and object reflectivity variations, which affected the consistency of seat occupancy detection and seat status synchronization.
4. The developed automated fare collection system successfully achieved the intended objectives related to operational monitoring and financial analytics through the implemented driver and administrator dashboards. The dashboards effectively supported synchronized operational records, accurate monitoring indicators, event notifications, and financial tracking functions within the configured prototype environment. The respondent-based evaluation further confirmed that the dashboards were understandable, usable, and effective for operational verification and monitoring activities. However, the implemented dashboards remained dependent on stable internet connectivity, centralized cloud synchronization, and simulated operational data. In addition, some dashboard interaction and visualization components may still require refinement to improve usability clarity and advanced monitoring capability.
5. The developed automated fare collection system successfully achieved a high level of software quality based on the ISO/IEC 25010:2023 evaluation conducted by the expert respondents. The system effectively supported commuter validation, fare computation,

seat allocation, operational synchronization, secure role-based access, dashboard monitoring, and operational data recording functions. The findings further confirmed that the developed interfaces were understandable, responsive, and capable of supporting reliable operational workflows within the configured transportation environment. However, the system remained dependent on network stability and cloud-based synchronization mechanisms, which affected resilience during connectivity interruptions. In addition, the implemented security and recovery mechanisms may still require further enhancement to strengthen protection, synchronization reliability, and operational recovery capability under more complex real-world transportation conditions.

Recommendations

Based on the findings and conclusions of the study, the researchers propose the following recommendations for future researchers and developers, categorized according to the five research objectives of the study.

1. To address the limitations related to predefined operational conditions, simulated monetary transactions, and dependence on stable connectivity, transport operators should establish clearer onboarding procedures, commuter guidance materials, and operational support mechanisms before system deployment. Future researchers should develop and evaluate scalable route configurations, real-world payment integration, offline-capable transaction handling, and more robust synchronization mechanisms suitable for actual transportation environments.
2. To strengthen the reliability of the tap-based payment and commuter validation processes, transport operators should improve hardware durability, enclosure compactness, and device placement to reduce accidental damage and maintain stable NFC operation during commuter interaction. Operators should also implement stable network infrastructure and operational recovery procedures to minimize

synchronization interruptions during connectivity loss. Future researchers should develop offline-capable validation logic, local transaction buffering, automatic synchronization recovery mechanisms, and real-world electronic payment integration to reduce dependence on continuous internet connectivity.

3. To improve the consistency and reliability of the seat monitoring and seat allocation mechanisms, transport operators should adopt more durable industrial-grade or automotive-grade sensing technologies and regularly perform sensor calibration, maintenance, and positioning verification. Operators should also consider implementing multiple sensing mechanisms per seat to reduce incorrect occupancy detection caused by environmental conditions and object reflectivity variations. Commuters should avoid obstructing or damaging seat monitoring components during boarding and seating operations. Future researchers should evaluate advanced sensor-fusion techniques, redundant sensing configurations, filtering algorithms, and large-scale seat monitoring implementations under actual transportation conditions.
4. To strengthen the operational monitoring and financial analytics capability of the developed dashboards, transport operators should maintain reliable cloud synchronization infrastructure, establish operational data validation procedures, and continuously monitor system connectivity to support consistent real-time dashboard reflection. Operators should also expand the use of collected transportation data for operational planning and decision-support activities. Future researchers should develop advanced analytics

   engines capable of supporting predictive passenger demand analysis, forecasting, route optimization, occupancy estimation, and intelligent transportation decision-support mechanisms for future smart public transportation systems.
5. Future researchers should prioritize strengthening reliability and security by developing and validating robust network-resilience and data-protection mechanisms: design and test offline data protection and local transaction queueing with automatic retry and

duplicate-suppression logic, implement comprehensive synchronization logs and recovery-state tracking to shorten downtime and clarify reconciliation steps, and evaluate timeout, backoff, and failover strategies under realistic connectivity scenarios; concurrently, researchers should harden security through stronger authentication and role-based access control, encrypted storage of sensitive records, secure device-to-backend communication, regular penetration and vulnerability testing, and expanded audit logging of access attempts, transaction changes, synchronization events, and administrative actions to ensure traceability and accountability.

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APPENDICES

Appendix A

Letter Request

Republic of the Philippines
University of Cabuyao
(Pamantasan ng Cabuyao)
College of Engineering
Katapatan Mutual Homes, Brgy. Banay-banay, City of Cabuyao, Laguna 4025

PNC:AS-LE-435.42
February 18, 2026

IT Professional / Computer Engineer

Good day!

We are fourth-year Bachelor of Science in Computer Engineering students from Pamantasan ng Cabuyao (University of Cabuyao). We are currently developing our design project entitled "**Automated Fare Collection System with Seat Allocation, Terminal Selection, and Profit Analytics Dashboard**."

In line with our research objectives, we would like to respectfully invite you to serve **as one of our Expert Evaluators**. Your evaluation will help us assess the overall quality of our prototype using ISO/IEC 25010:2023 in terms of functional suitability, performance efficiency, reliability, security, usability, and data quality.

For the evaluation procedure, you will (1) watch a recorded video demonstration of the prototype, (2) access the provided website links of the system, and (3) answer the ISO/IEC 25010:2023 evaluation questionnaire form. The evaluation may be completed at your most convenient time on **any day from March 9 to March 14, 2026**.

If you are willing to participate, we will send the video, instructions, website links, and the questionnaire form according to your preferred schedule. Rest assured that all responses will be kept confidential and will be used solely for academic research purposes.

Thank you very much for your time and consideration.

Respectfully yours,

AVELINO, MEDWIN T.
Researcher

CADANG, JOHN LLOYD M.
Researcher

GALANG, RENIEL F.
Researcher

MANANQUIL, IVAN KIER P.
Researcher

MINA, JUSTIN S.
Researcher

Noted by:

ENGR. ANNA-LIZA F. SIGUE
Research Adviser

ENGR. MARY GRACE P. BEAÑO
Research Instructor

Conforme:

I hereby signify my voluntary agreement to serve as an Expert Evaluator for the above-mentioned study and to conduct the evaluation within the agreed period.

IT Professional / Computer Engineer

rfg/02182026

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Republic of the Philippines
University of Cabuyao
(Pamantasan ng Cabuyao)
College of Engineering
Katapatan Mutual Homes, Brgy. Banay-banay, City of Cabuyao, Laguna 4025

PNC:AS-LE-27
January 8, 2026

DR. ROBERTO F. RAÑOLA JR.
OIC- University President
University of Cabuyao (Pamantasan ng Cabuyao)

THRU: **DR. GEORGE V. LAMBOT**
Vice President for Academics and Student Services

DR. JESSICA V. REY
Director, Research and Development Department
Vice President for Research Planning and Extension

Dear Dr. Rañola:

Dangal greetings!

Respectfully submitted for your approval is the research study entitled "[Research Title]" to be conducted by different programs; BS Computer Engineering, BS Electronics Engineering and BS Industrial Engineering from the College of Engineering. Please see attached letter.

In connection with this study, we respectfully request your approval and signature to allow the evaluation of the research paper, administration of the survey questionnaire, and gathering of necessary data from the identified respondents. All information to be collected will be used strictly for academic and research purposes and will be treated with utmost confidentiality.

We sincerely hope for your kind consideration and support of our academic endeavor. Should you require further details or clarification, we would be glad to coordinate at your convenience.

Thank you for your continued support of academic research.

Respectfully yours,
ENGR. ERICSON A. MANDAYO
Research Coordinator, College of Engineering

Noted by:

ENGR. MARICAR R. CABANSAG
Department Chair, BSECE

ENGR. ALDRIN J. SORIANO
Department Chair, BSCpE

ENGR. JOHN WENDELL E. ARROYO
Department Chair, BSIE

DR. RIZAL M. MOSQUERA
Dean, College of Engineering

DR. GEORGE V. LAMBOT
Vice President for Academics and Student Services

DR. JESSICA V. REY
Vice President for Planning Research Extension

Approved by:
DR. ROBERTO F. RAÑOLA JR.
OIC- University President

eam/01082026

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Appendix B

Validated Research Instrument

ISO 25010 QUESTIONNAIRE FOR EVALUATING THE DESIGN PROJECT OF CPE11:
"Automated Fare Collection System with Seat Allocation, Terminal Selection, and Profit Analytics Dashboards"

Name: _____ Date: _____ Time: _____

Profession: _____ Years in Industry: _____

Good day! Your task as an admin evaluator is to evaluate the overall system using the provided ISO 25010 questionnaire in terms of Functional suitability, Performance Efficiency, Reliability, Security, Usability, and Data Quality.

Please rate the level by putting a check mark (/) on the column that corresponds to your assessment. Use the scale below. Numeric Value Equivalent Rating:

5 - Strongly Agree 4 - Agree 3 - Neutral 2 - Disagree 1 - Strongly Disagree

A. Functionality (Assesses completeness, correctness, and appropriateness of the system's functions)	5	4	3	2	1
1. The system correctly validates smart card taps and accepts only authorized or registered cards.					
2. The system correctly computes and applies fares based on the selected destination terminal and the configured fare matrix or rules.					
3. The system correctly performs seat allocation and seat status updates based on sensor inputs and allocation rules.					
4. The system provides the required functions for commuter, driver, and admin roles without functional errors.					
5. The system accurately records key operational outputs (transactions, trip logs, seat events, and dashboard indicators) needed for monitoring and reporting.					

B. Performance Efficiency (Assesses time behavior, responsiveness, and resource utilization)	5	4	3	2	1
1. The system confirms a card tap promptly and shows results with minimal delay.					
2. Seat status changes and related interface updates appear quickly after a real seat event occurs.					
3. The dashboards load and refresh operational data and analytics within acceptable response times under normal use.					
4. The embedded device and backend services run with stable performance without excessive CPU or memory usage.					
5. Data synchronization and updates remain responsive when network connectivity is available.					

ISO 25010 QUESTIONNAIRE FOR EVALUATING THE DESIGN PROJECT OF CPE11:
"Automated Fare Collection System with Seat Allocation, Terminal Selection, and Profit Analytics Dashboards"

C. Reliability (Assesses stability, fault tolerance, recoverability, and availability)	5	4	3	2	1
1. The system performs consistently during continuous operation without frequent crashes or failures.					
2. Transaction and monitoring data remain accurate and intact even during network interruptions.					
3. The system can recover and resume normal operation after minor faults or interruptions.					
4. Logs and dashboard outputs remain dependable for operational monitoring and decision-making.					
5. The system maintains consistent synchronization behavior with minimal downtime or duplicated records.					

D. Security (Assesses protection of data, access control, and transaction safety)	5	4	3	2	1
1. The system's smart card security mechanisms (for example, secure authentication and encryption) protect fare transactions.					
2. User data and stored values (for example, balances and transaction records) are protected from unauthorized access or alteration.					
3. The system enforces role-based access control (commuter, driver, admin) to restrict sensitive functions and data.					
4. Communication between devices, backend services, and databases uses secure methods to protect transmitted data.					
5. Audit logs and access controls support accountability and traceability of key activities.					

E. Usability (Assesses ease of use, learnability, clarity of feedback, and interface design)	5	4	3	2	1
1. The system interfaces are intuitive and easy to navigate for their intended users.					
2. The system provides clear feedback and messages during important actions (for example, tap validation, seat allocation, and error cases).					
3. The dashboard layout and visualizations make operational status and analytics easy to understand.					
4. New users can operate the system with minimal training or assistance.					
5. The interface design supports efficient workflow and helps					

ISO 25010 QUESTIONNAIRE FOR EVALUATING THE DESIGN PROJECT OF CPE11:
"Automated Fare Collection System with Seat Allocation, Terminal Selection, and Profit Analytics Dashboards"

minimize user input errors.					

F. **Data Quality** (Assesses accuracy, completeness, consistency, timeliness, validity, and traceability of data)	5	4	3	2	1
1. The data stored by the system (users, cards, transactions, trips, seats, expenses) are accurate and consistent across modules.					
2. The system validates inputs and prevents invalid or inconsistent records (for example, impossible seat counts, duplicate events, or incorrect values).					
3. The collected data are complete enough to support reporting, auditing, and analytics requirements.					
4. Recorded time stamps and event sequences reflect real operations and allow traceability of transactions and seat events.					
5. The data displayed on dashboards match the data stored in the database and update in a timely manner.					

G. **Open Ended Questions**
1. What improvements can you suggest to enhance the system's functional suitability and performance efficiency during real operations?
2. What potential issues or risks do you foresee regarding the system's reliability, security, or data quality?
3. What interface or workflow improvements would you recommend to make the system easier and clearer for users?

☐ I certify that my responses above are based on my honest evaluation and observations. I understand that the data will be treated confidentially and used solely for academic research purposes.

Signature over Printed Name: _____

Date: _____

Commuter Usability Evaluation (ISO/IEC 25010:2023 Interaction Capability)

Project Title: "Automated Fare Collection System with Seat Allocation, Terminal Selection, and Profit Analytics Dashboards"

Instructions: Mark one answer per item based on your experience after trying the prototype.

Scale:
1 – Strongly Disagree (Lubos na Hindi Sumasang-ayon)
2 – Disagree (Hindi Sumasang-ayon)
3 – Neutral (Neutral)
4 – Agree (Sumasang-ayon)
5 – Strongly Agree (Lubos na Sumasang-ayon)

Name (optional): _____ **Age:** _____ **Sex:** _____ **City/Area:** _____

Prototype tried: ☐ Yes

No	Statement (English / Tagalog)	1	2	3	4	5
1	I immediately understood the purpose of the commuter interface and what I should do first. / Agad kong naintindihan kung para saan ang commuter interface at kung ano ang unang dapat gawin.	☐	☐	☐	☐	☐
2	Logging in and accessing my profile (including editing profile details) was easy to do. / Madali lang mag-login at i-access ang profile ko (kasama ang pag-edit ng profile).	☐	☐	☐	☐	☐
3	Viewing important information (balance, available modern jeep, route, available seats, and current terminal) was clear and easy to understand. / Malinaw at madaling intindihin ang mga importanteng impormasyon (balance, available modern jeep, ruta, available seats, at current terminal).	☐	☐	☐	☐	☐
4	Topping up my balance using the interface was straightforward and easy to complete. / Madali at diretsong gawin ang pag-top-up ng balance gamit ang interface.	☐	☐	☐	☐	☐
5	Selecting my destination terminal and confirming my current terminal was easy and not confusing. / Madali at hindi nakakalito ang pagpili ng destination terminal at pag-confirm ng current terminal.	☐	☐	☐	☐	☐
6	During boarding, tapping the card felt quick and simple. / Sa pag-board, mabilis at madali ang pag-tap ng card.	☐	☐	☐	☐	☐
7	After tapping, the system clearly informed me whether my card was valid or rejected and why. / Pagkatapos mag-tap, malinaw na ipinapakita ng system kung valid o rejected ang card at kung bakit.	☐	☐	☐	☐	☐
8	When validated, the seat allocation result was clear and easy to notice. / Kapag validated, malinaw at madaling mapansin ang resulta ng seat allocation.	☐	☐	☐	☐	☐
9	The fare deduction (including showing the deducted amount and updated balance) was clear and understandable. / Malinaw at madaling intindihin ang fare deduction (kasama ang deducted amount at updated balance).	☐	☐	☐	☐	☐
10	Extending to a farther terminal was easy to perform, and the added fare deduction was clearly explained. / Madali gawin ang pag-extend sa mas malayong terminal, at malinaw ang paliwanag sa dagdag na fare at deduction.	☐	☐	☐	☐	☐

☐ I certify that my responses reflect my honest evaluation based on my experience with the prototype.

Signature (optional): _____ **Date:** _____

Driver Usability Evaluation (ISO/IEC 25010:2023 Interaction Capability)

Project Title: "Automated Fare Collection System with Seat Allocation, Terminal Selection, and Profit Analytics Dashboards"

Instructions: Mark one answer per item based on your experience after trying the prototype.

Scale:
1 – Strongly Disagree (Lubos na Hindi Sumasang-ayon)
2 – Disagree (Hindi Sumasang-ayon)
3 – Neutral (Neutral)
4 – Agree (Sumasang-ayon)
5 – Strongly Agree (Lubos na Sumasang-ayon)

Name (optional): _____ **Role:** ☐ Driver ☐ Conductor ☐ Other: _____

Years of experience: _____ **Prototype tried:** ☐ Yes

No	Statement (English / Tagalog)	1	2	3	4	5
1	Starting a shift and selecting a route was easy and clear. / Madali at malinaw ang pag-start ng shift at pagpili ng route	☐	☐	☐	☐	☐
2	Monitoring upcoming passengers who are waiting to be boarded was clear and easy to track. / Malinaw at madaling i-track ang mga paparating na pasaherong naghihintay sumakay.	☐	☐	☐	☐	☐
3	The system clearly shows boarding activity (successful tap, rejected tap, and status updates). / Malinaw na ipinapakita ng system ang boarding activity (successful tap, rejected tap, at mga status update).	☐	☐	☐	☐	☐
4	Seat status monitoring (who is seated and seat availability) was easy to understand at a glance. / Madaling makita at maintindihan agad ang seat status (sino ang nakaupo at ilang seats ang available).	☐	☐	☐	☐	☐
5	Monitoring revenue, total passengers, and trip summaries was easy to read and understand. / Madaling basahin at intindihin ang revenue, total passengers, at trip summaries.	☐	☐	☐	☐	☐
6	Recording expenses was easy to do and the system confirmed that the expense was saved correctly. / Madali maglagay ng expense at may malinaw na confirmation na na-save ito nang tama.	☐	☐	☐	☐	☐
7	The profit computation (revenue minus expenses) was clearly presented and easy to verify. / Malinaw ang profit computation (revenue minus expenses) at madaling i-check.	☐	☐	☐	☐	☐
8	The interface helped prevent mistakes (clear prompts, warnings, and confirmations before important actions). / Nakakatulong ang interface para maiwasan ang pagkakamali (malinaw na prompts, warnings, at confirmations bago ang importanteng actions).	☐	☐	☐	☐	☐
9	The driver interface was readable and usable under real operating conditions (time pressure, lighting, and quick checks). / Nababasang maayos at nagagamit ang driver interface kahit sa actual na kondisyon (mabilisang galaw, ilaw, at mabilisang tingin).	☐	☐	☐	☐	☐
10	Ending the shift felt organized, and the system provided clear final summaries or records. / Organisado ang pag-end ng shift at nagbibigay ang system ng malinaw na final summaries o records.	☐	☐	☐	☐	☐

☐ I certify that my responses reflect my honest evaluation based on my experience with the prototype.

Signature (optional): _____ **Date:** _____

Appendix C

Informed Consent Form

Republic of the Philippines
Pamantasan ng Cabuyao
(University of Cabuyao)
Planning, Research, and Extension Division
Research and Development Department
Katapatan Mutual Homes, Brgy. Banay-banay, City of Cabuyao, Laguna 4025

INFORMED CONSENT FORM

Research Project Title	AUTOMATED FARE COLLECTION SYSTEM WITH SEAT ALLOCATION, TERMINAL SELECTION, PROFIT ANALYTICS DASHBOARDS			
Proponents	**Name**	**Email Address**	**Contact Number**	**Department**
Research Lead	Galang, Reniel F.	galangreniel68@gmail.com	0992 515 2549	COE
Member/s	Avelino, Medwin T.	avelinomedwin24@gmail.com	0995 908 9222	COE
	Cadang, John Lloyd M.	cadangjohnlloyd92@gmail.com	0960 321 8164	COE
	Mananquil, Ivan Kier P.	mananquilivankier23@gmail.com	0994 518 0228	COE
	Mina, Justin S.	minajustin613@gmail.com	0997 962 7267	COE
Thesis Adviser (for student research)	Engr. Anna-liza F. Sigue	afsigue@pnc.edu.ph	---	CESD / COE

You are being invited to participate in a research study. Before you decide whether to participate, it is important that you understand why the research is being done and what your participation would involve. Please read the following information carefully and take time to ask any questions that you may have. You are free to choose whether or not to participate in this study. If you do not want to participate, you do not have to give a reason, and your decision will not affect any relationship you may have with the researchers or the institution.

Purpose of the Study	To collect expert technical assessments of an automated payment and intelligent reservation with reporting for modern jeepney commuting and operations. Responses to an ISO/IEC 25010–based questionnaire will be used to evaluate the system across functional suitability, performance efficiency, reliability, security, usability, and data quality and to inform further prototype refinements.
Risks, Benefits, or Discomforts of the Study	The following are the potential risks, benefits, or discomforts of participating in this study: Risks. There may be risks associated with participating in this study. We will take all necessary precautions to minimize these risks. ✔ Physical risks include physical discomfort, pain, injury, illness or disease brought about by the methods and procedures of the research. A physical risk may result from the involvement of physical stimuli such as noise, electric shock, heat, cold, electric magnetic or gravitational fields, etc. Engaging a subject in a social situation which could involve violence may also create a physical risk.

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Research and Development Department
Katapatan Mutual Homes, Brgy. Banay-banay, City of Cabuyao, Laguna 4025

	☐ Psychological risks include the production of negative affective states such as anxiety, depression, guilt, shock and loss of self-esteem and altered behavior. Sensory deprivation, sleep deprivation, use of hypnosis, deception or mental stresses are examples of psychological risks. ☐ Social/Economic risks include alterations in relationships with others that are to the disadvantage of the subject, including embarrassment, loss of respect of others, labeling a subject in a way that will have negative consequences, or in some way diminishing those opportunities and powers a person has by virtue of relationships with others. Economic risks include payment by subjects for procedures not otherwise required, loss of wages or other income and any other financial costs, such as damage to a subject's employability, as a consequence of participation in the research. ☐ Loss of confidentiality ☐ Legal risks exist when the research methods are such that the subject or others will be liable for a violation of the law, either by revealing that the subject or others have or will engage in conduct for which the subject or others may be criminally or civilly liable, or by requiring activities for which the subject or others may be criminally or civilly liable. Benefits. You may benefit from this research. However, we cannot guarantee that you will receive any benefits from participating in this study. ✔ Contribute to the advancement of scientific knowledge ✔ Develop participants' new skills or learn more about themselves. ☐ Receive access to resources, such as support groups or educational materials, that may be beneficial to their health or well-being. ☐ Receive access to interventions or treatments that could improve their health outcomes. ☐ Receive compensation. Discomforts There may be discomfort associated with participating in this study. We will take all necessary precautions to minimize these discomforts. ✔ Time Commitment: Participating in this study may require a significant amount of time and effort on your part, including attending appointments and completing questionnaires or other assessments. ☐ Emotional Distress: Some participants may find that discussing sensitive topics or answering personal questions can cause emotional distress or discomfort. ☐ Physical Discomfort: Depending on the nature of the study, there may be physical discomfort associated with participating, such as

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	experiencing side effects from medication or undergoing a medical procedure. ☐ Confidentiality Concerns: While we will take all reasonable steps to protect your confidentiality, there is always a risk that your information could be inadvertently disclosed. ☐ Unforeseen Risks: Despite our efforts to minimize risks and discomforts, there may be unforeseen risks or discomforts associated with participating in this study that we cannot anticipate
Duration of Participation:	All participants in this study are invited to complete a hands-on testing session of the fare payment prototype, followed immediately by an ISO/IEC 25010-based questionnaire. The entire activity is expected to take approximately **50 to 70 minutes**.
Confidentiality:	All questionnaire responses and test data will be treated as confidential and stored on encrypted university servers with access restricted to the research team. Publications and reports will present anonymized or aggregated results only. A separate, encrypted linkage file that maps participant names to anonymized IDs will be retained for 12 months after the first paper is published to allow verification and follow-up, after which it will be securely deleted. De-identified data may be retained and used for future research unless you opt out on this form. We may be required to disclose information if there is a legal obligation or if there is a clear and imminent risk of serious harm to you or others.
Voluntary Nature of Participation:	Participation is entirely voluntary. You may decline to participate or withdraw at any time without penalty and without affecting your employment or standing. If you withdraw before completing the activity you will receive the prorated reimbursement described in the information sheet. If you request deletion of identifiable data at the time of withdrawal, we will delete identifying information where technically feasible; anonymized responses already included in aggregated analyses may not be retrievable for removal.
Contact Information:	If you have any questions or concerns about the study, you can contact the researchers at **galangreniel68@gmail.com**. If you have any concerns about your rights as a participant, you can contact the Research Ethics Review Committee (RERC) at rdd@pnc.edu.ph
Consent:	By signing below, you indicate that you have read and understood the information provided above and that you voluntarily agree to participate in this study.

Appendix D

Calculations and Statistical Information

Sampling for evaluation of the developed AFCS using ISO/IEC 25010:2023

$$n_o = \frac{Z^2 p(1-p)}{e^2} = \frac{1.96^2 0.50(1-0.50)}{0.10^2} = 96.04$$

Thus, the sample size was rounded to 100 respondents. This target was selected to obtain a stable estimate of mean usability ratings while remaining feasible within the face-to-face demonstration, trials, and pen-and-paper usability evaluations of the prototype. In the study, 100 end-users evaluated Objectives 1 and 2, and another 100 end-users for Objectives 3 and 4. Meanwhile, the 12 experts who evaluated the overall system consisted of seven computer engineers and five information technology professionals, who have relevant experiences in the field of study.

Technical Evaluation

Commuter Interface Testing

$$Card\ Balance\ Display\ Accuracy\ (\%) = \frac{N_{balance,correct}}{N_{balance,total}} \times 100$$

$$Card\ Balance\ Display\ Accuracy\ (\%) = \frac{200}{200} \times 100 = 100 = 100\%$$

$$Incoming\ Jeepney\ Information\ Accuracy\ (\%) = \frac{N_{modern\ jeepney,correct}}{N_{modern\ jeepney,total}} \times 100$$

$$Incoming\ Jeepney\ Information\ Accuracy\ (\%) = \frac{300}{300} \times 100 = 100 = 100\%$$

Tap-Based Payment Process

$$Tap\ Success\ Rate\ (\%) = \frac{N_{tap,success}}{N_{tap,total}} \times 100$$

$$Tap\ Success\ Rate\ (\%) = \frac{102}{104} \times 100 = 98.08 = 98.08\%$$

$$Fare\ Accuracy\ Rate\ (\%) = \frac{N_{fare,correct}}{N_{fare,valid}} \times 100$$

$$Fare\ Accuracy\ Rate\ (\%) = \frac{105}{105} \times 100 = 100 = 100\%$$

$$Seat\ Allocation\ Success\ Rate\ (\%) = \frac{N_{alloc,correct}}{N_{alloc,valid}} \times 100$$

$$Seat\ Allocation\ Success\ Rate\ (\%) = \frac{104}{105} \times 100 = \ 99.05 = \ \ 99.05\%$$

$$Invalid\ Reason\ Accuracy\ Rate\ (\%) = \frac{N_{correct}}{N_{total,tests}} \times 100$$

$$Invalid\ Reason\ Accuracy\ Rate\ (\%) = \frac{140}{140} \times 100 = \ 100 \ = \ 100\%$$

$$System\ Unavailable\ Accuracy\ Rate\ (\%) = \frac{N_{correct}}{N_{total,tests}} \times 100$$

$$System\ Unavailable\ Accuracy\ Rate\ (\%) = \frac{93}{100} \times 100 = \ 93 \ = \ 93\%$$

Seat Monitoring and Allocation Mechanism Testing

$$Seat\ Detection\ Accuracy\ (\%) = \frac{N_{detect,correct}}{N_{detect,total}} \times 100$$

$$Seat\ Detection\ Accuracy\ (\%) = \frac{99}{106} \times 100 = \ 93.40 = \ 93.40\%$$

$$Driver\ View\ Consistency\ Rate\ (\%) = \frac{N_{driver,consistent}}{N_{driver,total}} \times 100$$

$$Driver\ View\ Consistency\ Rate\ (\%) = \frac{99}{106} \times 100 = \ 93.40 = \ 93.40\%$$

$$Commuter\ View\ Consistency\ Rate\ (\%) = \frac{N_{comm,consistent}}{N_{comm,total}} \times 100$$

$$Commuter\ View\ Consistency\ Rate\ (\%) = \frac{99}{106} \times 100 = \ 93.40 = \ 93.40\%$$

$$Seat\ Lock/Unlock\ Accuracy\ (\%) = \frac{N_{lock,correct}}{N_{lock,total}} \times 100$$

$$Seat\ Lock/Unlock\ Accuracy\ (\%) = \frac{105}{106} \times 100 = \ 99.06 = \ \ 99.06\%$$

Driver and Admin Dashboards Testing

$$Dashboards\ Consistency\ Rate\ (\%) = (\frac{Matching\ Dashboard\ Values}{Total\ Value\ Checks}) \times 100$$

$$Dashboards\ Consistency\ Rate\ (\%) = \frac{102}{102} \times 100 = \ 100 \ = \ 100\%$$

$$Pop-up\ Accuracy\ Rate\ (\%) \ = \frac{N_{correct}}{N_{total,tests}} \times 100$$

$$Pop-up\ Accuracy\ Rate\ (\%) \ = \frac{100}{100} \times 100 = \ 100 \ = \ 100\%$$

$$Indicator\ State\ Correctness\ Rate\ (\%) = \frac{N_{correct}}{N_{total,tests}} \times 100$$

$$Indicator\ State\ Correctness\ Rate\ (\%) \ = \frac{100}{100} \times 100 = \ 100 \ = \ 100\%$$

$$Driver\ Monitoring\ Compliance\ Rate\ (\%) \ = \frac{N_{correct}}{N_{total,tests}} \times 100$$

$$Driver\ Monitoring\ Compliance\ Rate\ (\%) \ = \frac{100}{100} \times 100 = \ 100 \ = \ 100\%$$

Respondent-Based Evaluation

User Evaluation Results for Commuter Interface

Statement 1

Statement: Viewing important information such as balance, incoming vehicle, route, available seats, and current terminal was clear and easy to understand.

Mean of Statement 1

$$\bar{x} = \frac{459}{100} = 4.59$$

Standard Deviation of Statement 1

$$SD = \sqrt{\frac{\sum(x_i - 4.59)^2}{100 - 1}}$$
$$SD = \sqrt{\frac{46.19}{99}} = 0.6830561152$$
$$SD = 0.68$$

Statement 2

Statement: Topping up my balance using the interface was straightforward and easy to complete.

Mean of Statement 2

$$\bar{x} = \frac{457}{100} = 4.57$$

Standard Deviation of Statement 2

$$SD = \sqrt{\frac{\sum(x_i - 4.57)^2}{100 - 1}}$$
$$SD = \sqrt{\frac{56.51}{99}} = 0.7555184186$$
$$SD = 0.76$$

Statement 3

Statement: Selecting my destination terminal and confirming my current terminal was easy and not confusing.

Mean of Statement 3

$$\bar{x} = \frac{451}{100} = 4.51$$

Standard Deviation of Statement 3

$$SD = \sqrt{\frac{\sum(x_i - 4.51)^2}{100 - 1}}$$
$$SD = \sqrt{\frac{58.97}{99}} = 0.7719187690$$
$$SD = 0.77$$

User Evaluation Results for Tap-Based Payment Process

Statement 1

Statement: After tapping, the system clearly informed me whether my card was valid or rejected and why.

Mean of Statement 1

$$\bar{x} = \frac{451}{100} = 4.51$$

Standard Deviation of Statement 1

$$SD = \sqrt{\frac{\sum(x_i - 4.51)^2}{100 - 1}}$$
$$SD = \sqrt{\frac{56.97}{99}} = 0.7587203475$$
$$SD = 0.76$$

Statement 2

Statement: When validated, the seat allocation result was clear and easy to notice.

Mean of Statement 2

$$\bar{x} = \frac{457}{100} = 4.57$$

Standard Deviation of Statement 2

$$SD = \sqrt{\frac{\sum(x_i - 4.57)^2}{100 - 1}}$$
$$SD = \sqrt{\frac{62.51}{99}} = 0.7946157193$$
$$SD = 0.79$$

Statement 3

Statement: The fare deduction, including showing the deducted amount and updated balance, was clear and understandable.

Mean of Statement 3

$$\bar{x} = \frac{465}{100} = 4.65$$

Standard Deviation of Statement 3

$$SD = \sqrt{\frac{\sum(x_i - 4.65)^2}{100 - 1}}$$
$$SD = \sqrt{\frac{48.75}{99}} = 0.7017294653$$
$$SD = 0.70$$

User Evaluation Results for Seat Monitoring and Allocation Mechanism

Statement 1

Statement: Monitoring upcoming passengers who are waiting to be boarded was clear and easy to track.

Mean of Statement 1

$$\bar{x} = \frac{463}{100} = 4.63$$

Standard Deviation of Statement 1

$$SD = \sqrt{\frac{\sum(x_i - 4.63)^2}{100 - 1}}$$
$$SD = \sqrt{\frac{35.31}{99}} = 0.5972157622$$
$$SD = 0.60$$

Statement 2

Statement: Seat status monitoring, including who is seated and seat availability, was easy to understand at a glance.

Mean of Statement 2

$$\bar{x} = \frac{465}{100} = 4.65$$

Standard Deviation of Statement 2

$$SD = \sqrt{\frac{\sum(x_i - 4.65)^2}{100 - 1}}$$
$$SD = \sqrt{\frac{36.75}{99}} = 0.6092717958$$
$$SD = 0.61$$

User Evaluation Results for Driver and Administrator Dashboards

Statement 1

Statement: Monitoring revenue, total passengers, and trip summaries was easy to read and understand.

Mean of Statement 1

$$\bar{x} = \frac{458}{100} = 4.58$$

Standard Deviation of Statement 1

$$SD = \sqrt{\frac{\sum(x_i - 4.58)^2}{100 - 1}}$$
$$SD = \sqrt{\frac{38.36}{99}} = 0.6224746963$$
$$SD = 0.62$$

Statement 2

Statement: Recording expenses was easy to do and the system confirmed that the expense was saved correctly.

Mean of Statement 2

$$\bar{x} = \frac{435}{100} = 4.35$$

Standard Deviation of Statement 2

$$SD = \sqrt{\frac{\sum(x_i - 4.35)^2}{100 - 1}}$$
$$SD = \sqrt{\frac{36.75}{99}} = 0.6092717958$$
$$SD = 0.61$$

Statement 3

Statement: The profit computation, revenue minus expenses, was clearly presented and easy to verify.

Mean of Statement 3

$$\bar{x} = \frac{467}{100} = 4.67$$

Standard Deviation of Statement 3

$$SD = \sqrt{\frac{\sum(x_i - 4.67)^2}{100 - 1}}$$
$$SD = \sqrt{\frac{34.11}{99}} = 0.5869799439$$
$$SD = 0.59$$

The following captures present the raw data gathered from the respondents.

			OBJECTIVE 1				OBJECTIVE 2				
USABILITY INTERACTION CAPABILITY (UIC)	UIC1	UIC2	UIC3	UIC4	UIC5	UIC6	UICC7	UIC8	UIC9	UIC10	UIC MEAN
User 1	4	3	4	5	5	5	4	4	5	5	4.4
User 2	5	4	4	5	5	5	4	4	4	4	4.4
User 3	5	5	5	5	5	5	5	4	4	4	4.7
User 4	5	4	5	4	4	4	4	4	5	5	4.4
User 5	5	5	5	5	5	4	4	5	5	5	4.8
User 6	4	4	5	5	5	5	4	5	5	5	4.7
User 7	5	5	5	5	4	4	4	5	5	5	4.7
User 8	5	5	5	4	4	5	5	5	5	5	4.8
User 9	5	5	5	5	4	4	4	5	5	5	4.7
User 10	5	4	4	4	4	4	5	5	5	5	4.5
User 11	5	4	5	5	4	4	4	4	4	5	4.4
User 12	5	5	4	5	4	4	4	5	5	5	4.6
User 13	5	4	4	4	4	5	5	5	5	5	4.6
User 14	4	4	4	4	4	5	4	4	5	5	4.3
User 15	4	4	5	4	4	4	4	4	5	5	4.3
User 16	4	4	4	4	4	5	5	4	4	4	4.2
User 17	4	4	4	4	5	5	4	4	4	4	4.2
User 18	5	5	5	5	4	4	4	4	4	4	4.4
User 19	5	4	4	5	5	4	4	5	5	5	4.6
User 20	5	5	5	5	5	5	5	5	5	5	5
User 21	5	4	4	5	4	4	4	5	5	5	4.5
User 22	5	4	4	4	4	4	4	4	5	4	4.2
User 23	2	3	3	3	3	3	3	3	3	3	2.9
User 24	5	5	5	5	5	5	5	5	5	5	5
User 25	5	5	5	5	5	5	5	5	5	5	5
User 26	5	5	4	5	5	5	5	5	5	4	4.8
User 27	5	5	5	5	5	5	5	5	5	5	5
User 28	4	5	5	5	5	5	5	4	5	5	4.8
User 29	5	5	5	5	5	5	5	5	5	5	5
User 30	3	4	3	3	4	3	3	2	2	3	3
User 31	4	4	3	3	3	3	3	2	3	3	3.1
User 32	5	5	5	5	5	5	5	5	5	5	5
User 33	4	5	5	5	5	5	5	5	5	5	4.9
User 34	5	5	5	5	5	5	5	5	5	5	5
User 35	5	5	5	5	5	5	5	5	5	5	5
User 36	5	5	5	5	5	5	5	5	5	5	5
User 37	5	5	5	5	5	5	5	5	5	5	5
User 38	5	5	5	5	5	5	5	5	5	5	5
User 39	5	4	5	5	5	5	5	5	5	5	4.9
User 40	5	5	5	5	5	5	5	5	5	5	5
User 41	5	5	5	5	5	5	5	5	5	5	5
User 42	5	5	5	5	5	5	5	5	5	5	5
User 43	5	5	5	5	4	5	4	5	4	5	4.7
User 44	5	5	5	5	4	5	5	5	4	5	4.8
User 45	5	5	5	5	5	5	5	5	5	5	5
User 46	5	5	5	5	5	5	5	5	5	5	5
User 47	5	5	5	5	5	5	5	5	5	5	5
User 48	5	5	5	5	5	5	5	5	5	5	5
User 49	5	4	5	5	4	5	4	5	4	5	4.6
User 50	5	5	5	5	5	5	5	5	5	5	5
User 51	5	4	4	5	5	4	5	5	5	5	4.7
User 52	3	3	2	2	2	1	1	1	2	1	1.8
User 53	4	4	3	2	4	2	4	4	4	5	3.6
User 54	4	4	4	4	4	4	4	4	4	4	4
User 55	3	3	4	3	2	3	2	4	3	4	3.1
User 56	5	5	5	5	5	5	5	5	5	5	5
User 57	5	5	5	5	5	5	5	5	5	5	5
User 58	5	5	5	5	5	5	5	5	5	5	5
User 59	1	2	2	3	1	3	3	2	3	2	2.2
User 60	5	5	5	5	5	5	5	5	5	5	5
User 61	4	4	4	4	4	4	4	4	4	4	4
User 62	4	4	4	4	4	4	4	4	4	4	4
User 63	4	4	4	4	4	4	4	4	4	4	4
User 64	4	2	4	3	4	5	5	4	5	5	4.1
User 65	5	5	5	5	5	5	5	5	5	5	5
User 66	5	5	5	5	5	5	5	5	5	5	5
User 67	5	5	5	5	5	5	5	5	5	5	5
User 68	5	5	5	5	4	5	4	4	5	4	4.6
User 69	5	4	5	5	4	5	5	5	5	5	4.8
User 70	5	5	4	5	5	5	4	5	5	5	4.8
User 71	3	3	4	3	5	5	4	4	5	4	4
User 72	2	3	3	2	2	1	2	2	2	3	2.2
User 73	5	5	5	5	5	5	5	5	5	5	5
User 74	5	5	5	5	5	5	5	5	5	5	5
User 75	5	5	5	5	5	5	4	5	5	5	4.9
User 76	5	4	5	4	5	5	4	5	4	5	4.6
User 77	5	5	5	5	5	5	5	5	5	5	5
User 78	5	5	5	4	5	4	5	5	5	4	4.7

User 79	5	4	5	5	4	5	5	5	4	5	4.7
User 80	4	4	5	5	4	5	5	5	5	5	4.7
User 81	5	5	4	4	5	5	5	4	5	5	4.7
User 82	5	5	5	5	4	5	5	4	5	4	4.7
User 83	5	5	5	4	4	5	5	4	5	5	4.7
User 84	5	5	5	5	5	5	5	5	5	5	5
User 85	5	5	5	5	5	5	5	5	5	5	5
User 86	5	5	5	5	5	5	5	5	5	5	5
User 87	4	4	4	4	5	5	4	5	4	4	4.3
User 88	5	5	5	5	5	5	5	5	5	5	5
User 89	4	4	5	5	5	5	5	4	5	5	4.7
User 90	5	5	5	5	5	5	5	5	5	5	5
User 91	5	5	5	5	5	5	5	5	5	5	5
User 92	5	5	5	5	5	5	5	5	5	5	5
User 93	4	4	5	5	5	5	5	5	5	5	4.8
User 94	4	5	4	4	5	5	4	5	5	5	4.6
User 95	5	5	5	5	5	5	5	5	5	5	5
User 96	5	5	5	5	5	5	5	5	5	5	5
User 97	5	5	5	5	5	5	5	5	5	5	5
User 98	5	5	5	5	5	5	5	5	5	5	5
User 99	5	4	5	5	4	4	4	5	5	4	4.5
User 100	5	5	4	4	4	4	5	5	4	4	4.4
MEAN per Question	**4.62**	**4.51**	**4.59**	**4.57**	**4.51**	**4.59**	**4.51**	**4.57**	**4.65**	**4.66**	**4.578**
Rounded Up	**4.62**	**4.51**	**4.59**	**4.57**	**4.51**	**4.59**	**4.51**	**4.57**	**4.65**	**4.66**	**4.58**
Standard Deviation	**0.7491409221**	**0.7034546786**	**0.6830561152**	**0.7555184186**	**0.771918769**	**0.8052228501**	**0.7587203475**	**0.7946157193**	**0.7017294653**	**0.6994947672**	**0.6392198528**

		OBJECTIVE 3			OBJECTIVE 4						
USABILITY INTERACTION CAPABILITY (UIC)	**UIC1**	**UIC2**	**UIC3**	**UIC4**	**UIC5**	**UIC6**	**UICC7**	**UIC8**	**UIC9**	**UIC10**	**UIC MEAN**
User 1	1	1	1	2	1	2	1	2	3	2	1.6
User 2	4	4	4	5	5	4	5	4	4	5	4.4
User 3	3	4	5	5	5	4	5	5	5	5	4.6
User 4	5	5	5	5	5	5	5	5	5	5	5
User 5	5	5	4	4	5	5	5	5	5	5	4.8
User 6	4	5	4	4	5	5	5	5	5	5	4.7
User 7	5	5	5	5	5	5	5	5	5	4	4.9
User 8	5	5	5	5	4	4	4	5	5	5	4.7
User 9	5	5	5	5	5	5	5	5	5	5	5
User 10	5	5	5	4	5	5	5	5	5	5	4.9
User 11	4	5	5	4	4	4	4	5	5	5	4.5
User 12	5	5	4	5	5	4	5	5	5	5	4.8
User 13	5	5	4	4	5	5	5	5	4	4	4.6
User 14	4	5	4	5	5	5	5	4	5	5	4.7
User 15	5	5	5	5	5	5	5	5	5	5	5
User 16	5	5	5	5	4	3	4	5	5	5	4.6
User 17	4	4	3	5	5	5	5	5	4	5	4.5
User 18	5	5	5	5	4	4	5	5	5	5	4.8
User 19	5	5	5	5	4	4	5	5	5	5	4.8
User 20	5	5	5	5	5	5	5	5	5	5	5
User 21	5	5	4	5	4	5	5	5	4	5	4.7
User 22	5	5	5	5	4	4	5	5	5	5	4.8
User 23	4	4	4	4	4	4	4	4	4	4	4
User 24	5	4	5	5	4	4	5	5	4	5	4.6
User 25	4	4	5	5	4	5	5	5	4	5	4.6
User 26	5	4	5	5	5	4	4	4	5	5	4.6
User 27	5	5	5	5	5	5	5	5	5	5	5
User 28	4	5	4	5	4	4	5	5	4	5	4.5
User 29	5	4	5	5	4	4	5	5	5	5	4.7
User 30	5	5	4	5	5	5	5	5	5	5	4.9
User 31	5	4	3	4	4	4	5	4	3	3	3.9
User 32	5	4	3	5	5	4	4	4	4	4	4.2
User 33	4	4	3	5	5	3	5	5	3	4	4.1
User 34	4	5	4	5	5	4	5	5	4	5	4.6

User 35	5	5	3	4	5	5	5	5	3	5	4.5
User 36	5	5	3	5	5	4	5	5	4	4	4.5
User 37	4	4	3	5	4	4	5	4	5	5	4.3
User 38	4	4	4	5	5	4	5	5	4	5	4.5
User 39	4	5	5	5	5	5	4	4	4	4	4.5
User 40	5	5	3	5	5	4	5	5	4	4	4.5
User 41	4	4	4	4	5	5	4	5	5	5	4.5
User 42	4	4	3	5	4	4	5	5	3	5	4.2
User 43	5	5	3	5	5	3	5	4	4	4	4.3
User 44	5	4	3	4	5	4	5	5	5	5	4.5
User 45	4	5	4	5	4	5	4	4	5	5	4.5
User 46	5	5	4	4	4	4	5	4	5	5	4.5
User 47	4	4	3	5	5	4	5	5	4	4	4.3
User 48	4	5	4	5	5	4	5	5	5	4	4.6
User 49	4	4	3	5	4	5	5	5	4	5	4.4
User 50	5	5	3	5	5	5	4	5	3	4	4.4
User 51	5	5	4	4	5	4	5	5	5	5	4.7
User 52	5	5	4	5	5	4	5	4	4	5	4.6
User 53	4	4	4	4	4	4	4	4	4	4	4
User 54	5	5	5	5	4	4	5	4	5	5	4.7
User 55	5	5	4	5	5	5	5	5	5	5	4.9
User 56	5	5	4	5	5	5	5	5	5	5	4.9
User 57	4	4	4	4	4	5	4	4	4	4	4.1
User 58	5	5	5	5	5	4	4	5	5	5	4.8
User 59	5	5	5	5	5	4	4	5	5	4	4.7
User 60	5	5	5	5	4	5	5	5	4	5	4.8
User 61	4	4	4	5	4	4	4	4	4	4	4.1
User 62	5	4	4	5	4	4	4	4	5	5	4.4
User 63	4	4	3	4	4	3	4	4	4	4	3.8
User 64	5	5	4	5	5	4	5	4	5	5	4.7
User 65	4	4	4	4	4	4	4	4	4	4	4
User 66	5	5	4	5	5	4	5	5	5	5	4.8
User 67	4	5	4	4	5	4	4	4	5	4	4.3
User 68	5	5	5	5	5	4	5	5	5	5	4.9
User 69	4	4	4	5	4	4	4	4	4	4	4.1
User 70	5	4	5	5	5	4	5	5	5	5	4.8
User 71	5	5	5	5	5	5	5	5	5	5	5
User 72	5	5	4	5	5	5	5	5	5	5	4.9
User 73	4	5	5	5	4	5	5	5	5	5	4.8
User 74	5	5	5	5	5	5	5	5	5	5	5
User 75	5	5	5	5	5	4	5	5	5	5	4.9
User 76	5	4	5	5	5	4	5	5	5	5	4.8
User 77	4	5	5	5	5	4	5	5	5	5	4.8
User 78	5	5	5	5	5	5	5	5	5	5	5
User 79	5	5	4	5	5	4	5	5	5	5	4.8
User 80	5	4	4	5	5	4	5	5	5	5	4.7
User 81	5	5	5	5	5	5	5	5	5	5	5
User 82	4	5	4	3	3	5	5	4	5	5	4.3
User 83	4	4	4	4	4	5	4	3	4	4	4
User 84	3	5	5	3	4	5	4	3	5	5	4.2
User 85	4	5	5	4	5	4	4	3	5	5	4.4
User 86	5	5	5	3	5	4	5	4	5	5	4.6
User 87	4	5	5	4	4	5	5	3	4	5	4.4
User 88	4	5	4	5	5	5	5	3	5	5	4.6
User 89	4	5	5	4	5	4	4	3	5	4	4.3
User 90	5	4	4	4	4	4	4	4	5	5	4.3
User 91	4	5	5	4	4	4	5	5	5	5	4.6
User 92	5	4	5	5	5	5	5	3	5	5	4.7
User 93	4	5	5	3	5	4	4	4	4	5	4.3
User 94	3	5	4	4	4	4	4	3	5	4	4
User 95	4	5	5	5	5	5	5	3	5	5	4.7
User 96	5	4	4	5	5	5	4	3	4	5	4.4
User 97	4	5	4	5	5	5	4	4	4	4	4.4
User 98	4	5	4	5	4	4	5	5	4	5	4.5
User 99	5	4	5	5	4	4	5	5	5	5	4.7
User 100	5	5	4	5	5	5	5	5	5	5	4.9
MEAN per Question	4.51	4.63	4.25	4.65	4.58	4.35	4.67	4.5	4.57	4.71	4.542
Rounded Up	4.51	4.63	4.25	4.65	4.58	4.35	4.67	4.5	4.57	4.71	4.54
STANDARD DEVIATIO	0.6589707309	0.5972157622	0.7833494518	0.6092717958	0.6224746963	0.6092717958	0.5869799439	0.7177405626	0.6072790782	0.537389876	0.4161342052

Functional Suitability (FS)

RESPONSES	FS1	FS2	FS3	FS4	FS5	EXPERT MEAN
Expert 1	5	5	5	5	4	4.8
Expert 2	5	5	3	5	5	4.6
Expert 3	4	5	5	5	5	4.8
Expert 4	4	5	4	4	5	4.4
Expert 5	5	4	5	4	4	4.4
Expert 6	5	5	5	4	4	4.6
Expert 7	5	5	5	5	5	5
Expert 8	5	4	5	5	5	4.8
Expert 9	5	5	5	5	5	5
Expert 10	5	5	4	4	4	4.4
Expert 11	5	5	5	5	5	5
Expert 12	5	5	5	5	5	5
Mean per question	4.833333333	4.833333333	4.666666667	4.666666667	4.666666667	4.733333333
Standard deviation	0.3892494721	0.3892494721	0.6513389473	0.4923659639	0.4923659639	0.246182982

Performance Efficiency (PE)

RESPONSES	PE1	PE2	PE3	PE4	PE5	EXPERT MEAN
Expert 1	4	4	4	4	4	4
Expert 2	4	3	5	5	4	4.2
Expert 3	5	5	5	4	3	4.4
Expert 4	4	5	5	5	4	4.6
Expert 5	4	5	5	4	4	4.4
Expert 6	5	4	5	4	5	4.6
Expert 7	5	5	5	5	5	5
Expert 8	5	5	5	3	5	4.6
Expert 9	5	5	5	5	5	5
Expert 10	4	5	5	5	5	4.8
Expert 11	5	5	5	5	5	5
Expert 12	4	5	4	4	5	4.4
Mean per question	4.5	4.666666667	4.833333333	4.416666667	4.5	4.583333333
Standard deviation	0.5222329679	0.6513389473	0.3892494721	0.6685579234	0.6741998625	0.3242707436

Reliability (RE)

RESPONSES	RE1	RE2	RE3	RE4	RE5	EXPERT MEAN
Expert 1	4	3	3	4	3	3.4
Expert 2	5	4	4	4	4	4.2
Expert 3	5	3	3	5	3	3.8
Expert 4	5	5	4	5	5	4.8
Expert 5	4	4	5	4	4	4.2
Expert 6	4	3	4	4	3	3.6
Expert 7	5	5	5	5	5	5
Expert 8	5	5	5	5	5	5
Expert 9	5	5	5	5	5	5
Expert 10	5	4	5	4	4	4.4
Expert 11	5	5	5	5	5	5
Expert 12	4	4	5	4	5	4.4
Mean per question	4.666666667	4.166666667	4.416666667	4.5	4.25	4.4
Standard deviation	0.4923659639	0.8348471099	0.7929614611	0.5222329679	0.8660254038	0.5783990444

Security (SE)

RESPONSES	SE1	SE2	SE3	SE4	SE5	EXPERT MEAN
Expert 1	4	3	4	3	3	3.4
Expert 2	4	4	5	5	4	4.4
Expert 3	3	3	3	3	5	3.4
Expert 4	4	5	5	5	5	4.8
Expert 5	4	5	4	5	4	4.4
Expert 6	4	4	5	4	4	4.2
Expert 7	5	5	5	5	5	5
Expert 8	5	5	5	5	5	5
Expert 9	5	5	5	5	5	5
Expert 10	5	5	5	5	4	4.8
Expert 11	5	5	5	5	5	5
Expert 12	5	5	5	4	5	4.8
Mean per question	4.416666667	4.5	4.666666667	4.5	4.5	4.516666667
Standard deviation	0.6685579234	0.7977240352	0.6513389473	0.7977240352	0.6741998625	0.5874959703

Usability (US)

RESPONSES	US1	US2	US3	US4	US5	EXPERT MEAN
Expert 1	4	3	3	3	3	3.2
Expert 2	5	5	5	4	4	4.6
Expert 3	5	4	5	5	5	4.8
Expert 4	5	5	4	5	5	4.8
Expert 5	4	4	4	5	4	4.2
Expert 6	5	5	5	5	4	4.8
Expert 7	5	5	5	5	5	5
Expert 8	4	5	4	5	5	4.6
Expert 9	5	5	5	5	5	5
Expert 10	4	5	4	4	4	4.2
Expert 11	5	5	5	5	5	5
Expert 12	4	5	4	4	4	4.2
Mean per question	4.583333333	4.666666667	4.416666667	4.583333333	4.416666667	4.533333333
Standard deviation	0.5149286505	0.6513389473	0.6685579234	0.6685579234	0.6685579234	0.5210711578

Data Quality (DQ)

RESPONSES	DQ1	DQ2	DQ3	DQ4	DQ5	EXPERT MEAN
Expert 1	4	4	4	3	4	3.8
Expert 2	4	4	3	4	4	3.8
Expert 3	5	5	5	5	5	5
Expert 4	5	5	4	5	4	4.6
Expert 5	4	4	4	4	4	4
Expert 6	5	5	4	5	5	4.8
Expert 7	5	5	5	5	5	5
Expert 8	5	5	5	5	5	5
Expert 9	5	5	5	5	5	5
Expert 10	5	4	5	5	5	4.8
Expert 11	5	5	5	5	5	5
Expert 12	5	4	4	5	5	4.6
Mean per question	4.75	4.583333333	4.416666667	4.666666667	4.666666667	4.616666667
Standard deviation	0.4522670169	0.5149286505	0.6685579234	0.6513389473	0.4923659639	0.4783177592

Appendix E

Hardware Specifications

COMPONENT	SPECIFICATION	PURPOSE
Portable Power Station	Portable rechargeable power source with AC outlet and two DC barrel outputs	Serves as the dedicated main power source of the prototype. It supplies AC power for the Raspberry Pi 4 through its adapter and provides DC outputs for the two buck converters that power the servo motors.
Raspberry Pi 4	Model B, 8GB RAM	Serves as the main processing unit installed in the e-bike. It runs the automated fare collection logic, processes card transactions, interfaces with the NFC reader, GPS, sensors, servo motors, and displays, and transmits real-time data to Firebase/Firestore.
Smart Card	MIFARE DESFire EV2, 4KB	Acts as the secure smart card used by commuters for tap-based fare validation. It stores the card identifier used for registered user verification and secure transaction handling.
NFC Smart Card Reader and Writer	ACR1252U, USB interface	Reads and authenticates the MIFARE DESFire EV2 smart cards, enabling commuter card tapping for validation and fare transaction processing.
Driver Display	7-inch HDMI Touch Capacitive Display Module	Serves as the driver-facing interactive display for accessing and viewing the Driver's Dashboard of the system.
Passenger Display	3.5-inch SPI TFT LCD Display Module	Serves as the passenger-facing display that provides real-time visual feedback during card validation and fare transaction events.
GPS Module	NEO-M8N GPS Module	Provides the real-time location of the incoming vehicle for terminal detection, route monitoring, and location-based system functions.
GPS External Antenna	SMA Male	Improves GPS signal reception for more stable and accurate location tracking.
Mobile Wi-Fi Device	TP-Link M7000 4G LTE Mobile Wi-Fi	Provides internet connectivity for the Raspberry Pi 4, enabling real-time transmission of transaction, seat, and GPS data to the cloud database.
SIM Card	SMART 5G + 4G/LTE Ready SIM Card	Provides mobile data connectivity for the TP-Link M7000, allowing the system to communicate with the cloud platform in real time.

Appendix F

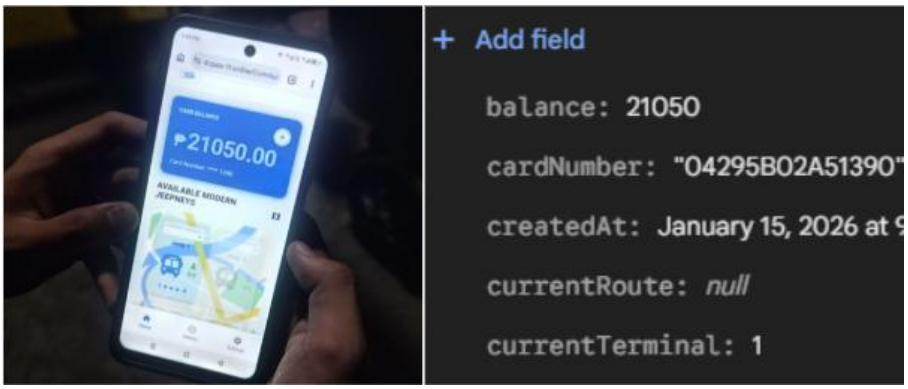
Software Specifications

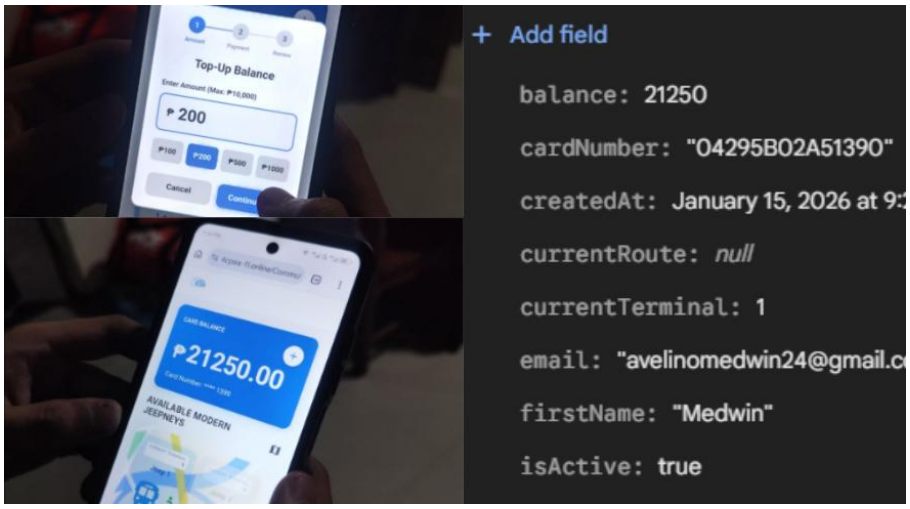
LAYER	TECHNOLOGY / LIBRARIES	PURPOSE
Development Environment Layer	Cursor AI	AI-assisted code editor used in developing and refining the system source code
	Visual Studio Code	Source code editor used for coding, editing, and managing the project files
	Windows PowerShell	Command-line interface used for running commands, project setup, file access, and remote device management
Edge Device Layer	Raspberry Pi OS	Operating system for the Raspberry Pi 4 (RP4)
	Python	Main programming language for the firmware
Cloud Backend Layer	Firebase Authentication	User authentication for commuter, driver, and admin interfaces
	Cloud Firestore	For centralized database of the overall system
	Firestore Cloud Functions	Processes tap events, seat events, fare deduction, and GPS for current terminal
Application Layer	Node.js	Runtime environment for web development tools and backend services
	Npm	Package manager for installing project dependencies
	Vite	Build tool and development server for the web applications
	React	Frontend library for the commuter, driver, and admin interfaces
Deployment Layer	Hostinger	Hosting provider for the deployed web applications
	Apache / .htaccess	Web hosting configuration for routing and deployment support

Appendix G

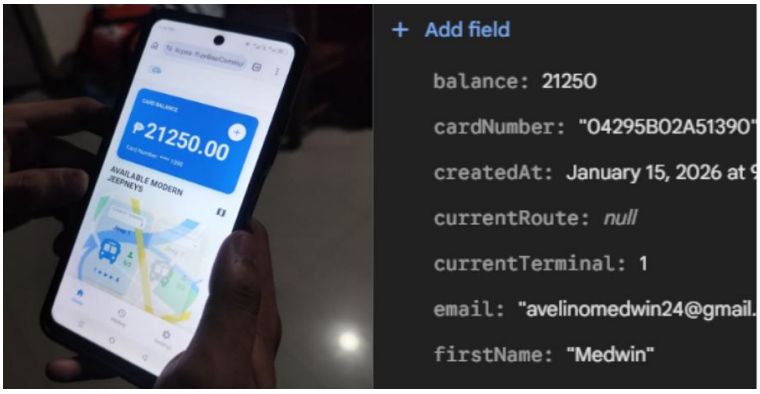
Actual Testing and Results

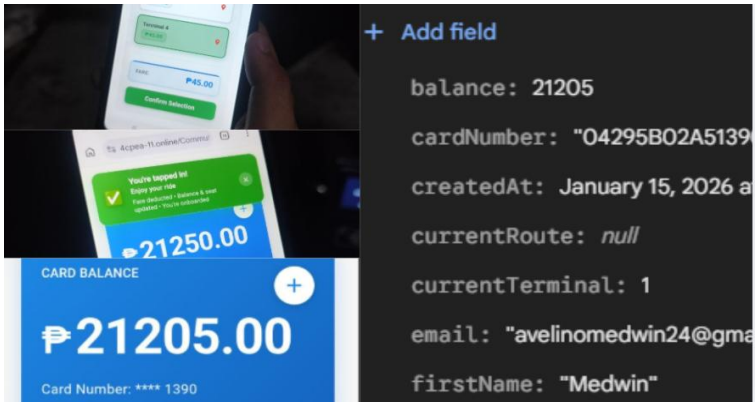
Objective 1: Commuter Interface





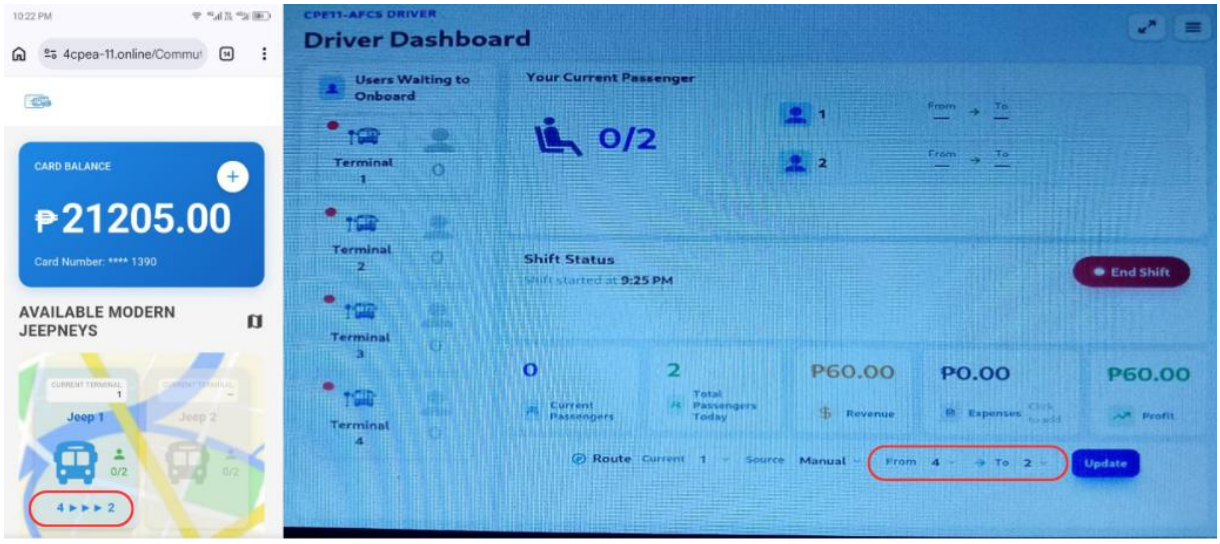
O1-T01 Card Balance Display Accuracy Test (Balance Loading)								
TESTING				BALANCE		DECISION		
Timestamp	Testing Type	Testing Number	Card Name	Commuter Interface	Database	Instance	Tester Name(s)	Remarks
3/7/2026 15:03:55	Balance Loa…	1	niel	₱11,953.00	₱11,953.00	Correct	Reniel	
3/7/2026 15:05:24	Balance Loa…	2	niel	₱11,955.00	₱11,955.00	Correct	Reniel	
3/7/2026 15:06:39	Balance Loa…	3	niel	₱11,958.00	₱11,958.00	Correct	Reniel	
3/7/2026 15:07:22	Balance Loa…	4	niel	₱11,962.00	₱11,962.00	Correct	Reniel	
3/7/2026 15:07:59	Balance Loa…	5	niel	₱11,967.00	₱11,967.00	Correct	Reniel	
3/7/2026 15:08:34	Balance Loa…	6	niel	₱11,973.00	₱11,973.00	Correct	Reniel	
3/7/2026 15:09:05	Balance Loa…	7	niel	₱11,980.00	₱11,980.00	Correct	Reniel	
3/7/2026 15:09:43	Balance Loa…	8	niel	₱11,988.00	₱11,988.00	Correct	Reniel	
3/7/2026 15:10:26	Balance Loa…	9	niel	₱11,997.00	₱11,997.00	Correct	Reniel	
3/7/2026 15:10:58	Balance Loa…	10	niel	₱12,007.00	₱12,007.00	Correct	Reniel	
3/7/2026 15:11:28	Balance Loa…	11	niel	₱12,107.00	₱12,107.00	Correct	Reniel	
3/7/2026 15:12:30	Balance Loa…	12	niel	₱12,307.00	₱12,307.00	Correct	Reniel	
3/7/2026 15:13:09	Balance Loa…	13	niel	₱12,807.00	₱12,807.00	Correct	Reniel	





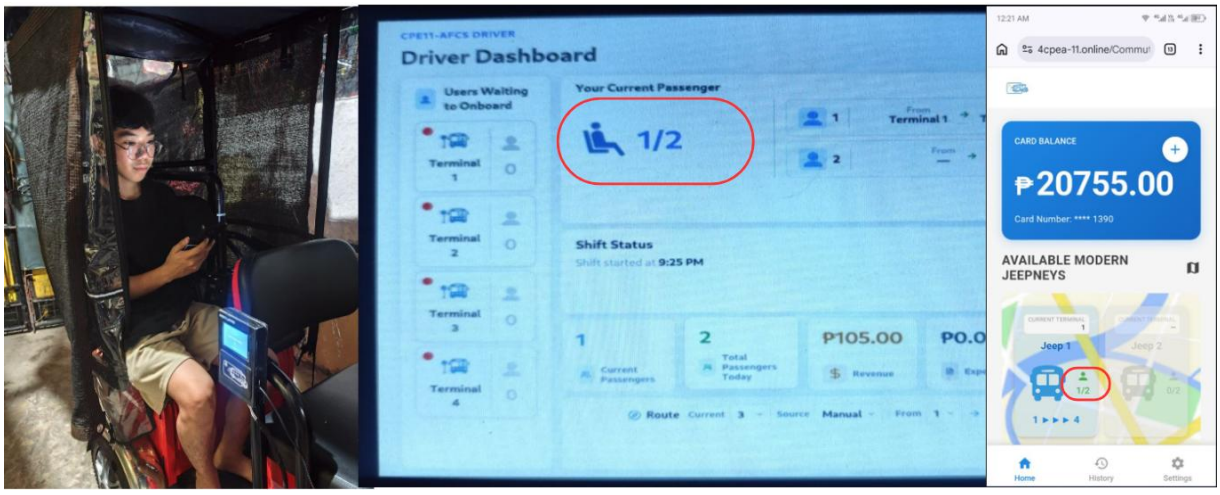
O1-T01 Card Balance Display Accuracy Test (Fare Deduction)

TESTING				BALANCE		DECISION		
Timestamp	Testing Type	Testing Number	Card Name	Commuter Interface	Database	Instance	Tester Name(s)	Remarks
3/12/2026 19:33:38	Fare Deduction	1	med	₱21,335.00	₱21,335.00	Correct	Medwin	CT: T1, CS: S1, R: T1-T2
3/12/2026 19:35:26	Fare Deduction	2	med	₱21,305.00	₱21,305.00	Correct	Medwin	CT: T1, CS: S1, R: T1-T3
3/12/2026 19:36:30	Fare Deduction	3	med	₱21,260.00	₱21,260.00	Correct	Medwin	CT: T1, CS: S1, R: T1-T4
3/12/2026 19:40:05	Fare Deduction	4	med	₱21,245.00	₱21,245.00	Correct	Medwin	CT: T1, CS: S1, R: T1-T2
3/12/2026 19:40:35	Fare Deduction	5	med	₱21,215.00	₱21,215.00	Correct	Medwin	CT: T1, CS: S1, R: T1-T3
3/12/2026 19:42:02	Fare Deduction	6	med	₱21,200.00	₱21,200.00	Correct	Medwin	CT: T1, CS: S1, R: T1-T2
3/12/2026 19:42:41	Fare Deduction	7	med	₱21,185.00	₱21,185.00	Correct	Medwin	CT: T1, CS: S1, R: T1-T2
3/12/2026 19:44:31	Fare Deduction	8	med	₱21,170.00	₱21,170.00	Correct	Medwin	CT: T1, CS: S1, R: T1-T2
3/12/2026 19:45:01	Fare Deduction	9	med	₱21,140.00	₱21,140.00	Correct	Medwin	CT: T1, CS: S1, R: T1-T3
3/12/2026 19:45:40	Fare Deduction	10	med	₱21,095.00	₱21,095.00	Correct	Medwin	CT: T1, CS: S1, R: T1-T4
3/12/2026 19:49:36	Fare Deduction	11	med	₱21,080.00	₱21,080.00	Correct	Medwin	CT: T1, CS: S2, R: T1-T2
3/12/2026 19:50:24	Fare Deduction	12	med	₱21,065.00	₱21,065.00	Correct	Medwin	CT: T1, CS: S2, R: T1-T2
3/12/2026 19:51:16	Fare Deduction	13	med	₱21,035.00	₱21,035.00	Correct	Medwin	CT: T1, CS: S2, R: T1-T3

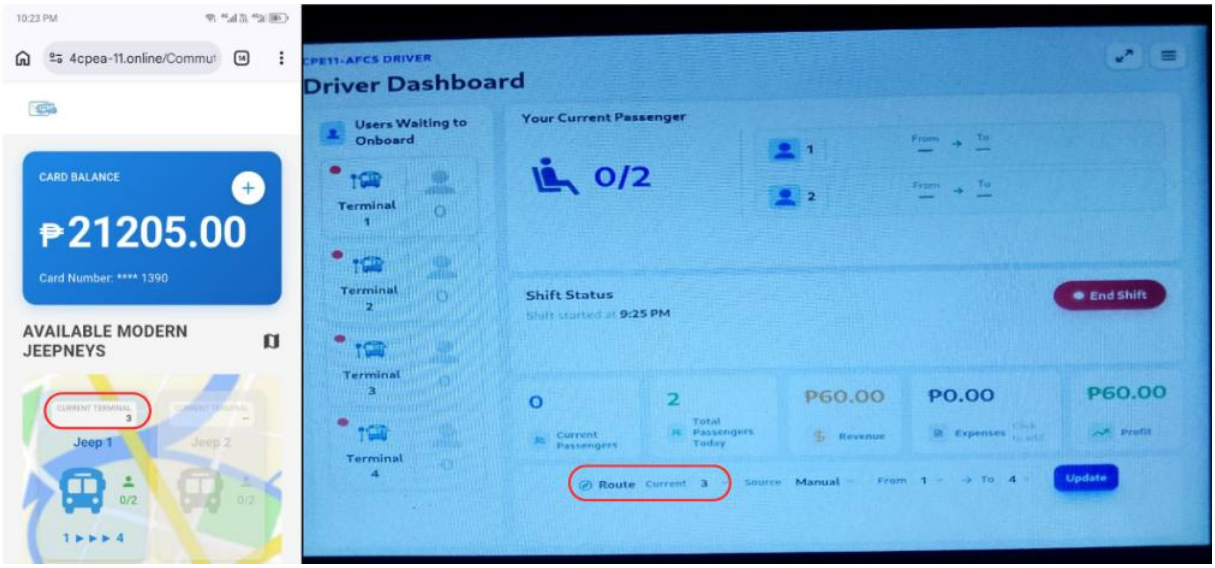


Route Display Accuracy Test

TESTING				DISPLAY		DECISION		
Timestamp	Testing Type	Testing Number	Card Name	Commuter Interface	Actual	Instance	Tester Name(s)	Remarks
3/12/2026 22:24:55	Route	1	med	T1 -> T4	T1 -> T4	Correct	Medwin	
3/12/2026 22:25:57	Route	2	med	T1 -> T3	T1 -> T3	Correct	Medwin	
3/12/2026 22:29:56	Route	3	med	T1 -> T2	T1 -> T2	Correct	Medwin	
3/12/2026 22:31:22	Route	4	med	T2 -> T3	T2 -> T3	Correct	Medwin	
3/12/2026 22:31:58	Route	5	med	T2 -> T4	T2 -> T4	Correct	Medwin	
3/12/2026 22:32:23	Route	6	med	T3 -> T4	T3 -> T4	Correct	Medwin	



Seat Display Accuracy Test								
TESTING				DISPLAY		DECISION		
Timestamp	Testing Type	Testing Number	Card Name	Commuter Interface	Actual	Instance	Tester Name(s)	Remarks
3/12/2026 22:50:54	Seat	1	med	0/2	0/2	Correct	med	
3/12/2026 22:55:01	Seat	2	med	1/2	1/2	Correct	med	
3/12/2026 22:57:40	Seat	3	med	2/2	2/2	Correct	med	
3/12/2026 22:58:41	Seat	4	med	1/2	1/2	Correct	med	
3/12/2026 22:58:48	Seat	5	med	0/2	0/2	Correct	med	
3/12/2026 23:04:28	Seat	6	med	1/2	1/2	Correct	med	



Terminal Display Accuracy Test								
TESTING				DISPLAY		DECISION		
Timestamp	Testing Type	Testing Number	Card Name	Commuter Interface	Actual	Instance	Tester Name(s)	Remarks
3/12/2026 23:32:20	Terminal	1	med	Current Terminal 1	Current Terminal 1	Correct	med	
3/12/2026 23:32:20	Terminal	2	med	Current Terminal 2	Current Terminal 2	Correct	med	
3/12/2026 23:32:20	Terminal	3	med	Current Terminal 3	Current Terminal 3	Correct	med	
3/12/2026 23:32:20	Terminal	4	med	Current Terminal 4	Current Terminal 4	Correct	med	
3/12/2026 23:32:20	Terminal	5	med	Current Terminal 1	Current Terminal 1	Correct	med	
3/12/2026 23:32:20	Terminal	6	med	Current Terminal 4	Current Terminal 4	Correct	med	

Objective 2: Tap-based payment process



O2-T01 Tap Read Success Rate Test							
TESTING			RESULTS		DECISION		
Timestamp	Testing Number	Card Name	expected	actual	Instance	Tester Name(s)	Remarks
3/5/2026	1	jc	Valid	Valid	Correct	Niel, JL	
	2	cj29	Valid	Valid	Correct	Niel, JL	
	3	kier	Valid	Valid	Correct	Kier, Med, Niel	
	4	niel	Valid	Valid	Correct	Kier, Med, Niel	
3/6/2026 20:05:24	5	cj29	Valid	Valid	Correct	Kier, Med, Niel	
3/6/2026 20:05:24	6	niel	Valid	Valid	Correct	Kier, Med, Niel	
3/6/2026 20:05:24	7	jc	Valid	Invalid	Incorrect	Kier, Med, Niel	seat full
3/6/2026 20:06:30	8	jc	Valid	Valid	Correct	Kier, Med, Niel	
3/6/2026 20:06:52	9	justin	Valid	Valid	Correct	Kier, Med, Niel	
3/6/2026 20:07:46	10	med	Valid	Valid	Correct	Kier, Med, Niel	
3/6/2026 20:07:53	11	kier	Valid	Valid	Correct	Kier, Med, Niel	
3/6/2026 20:08:38	12	niel	Valid	Valid	Correct	Kier, Med, Niel	
3/6/2026 20:09:15	13	cj29	Valid	Invalid	Incorrect	Kier, Med, Niel	commuter1 still boarding
3/6/2026 20:10:05	14	jc	Valid	Valid	Correct	Kier, Med, Niel	
3/6/2026 20:10:29	15	justin	Valid	Valid	Correct	Kier, Med, Niel	
3/6/2026 20:10:42	16	med	Valid	Valid	Correct	Kier, Med, Niel	
3/6/2026 20:11:17	17	kier	Valid	Valid	Correct	Kier, Med, Niel	
3/6/2026 20:12:40	18	unknown card	Invalid	Invalid	Correct	Kier, Med, Niel	
3/6/2026 20:12:46	19	unknown card	Invalid	Invalid	Correct	Kier, Med, Niel	

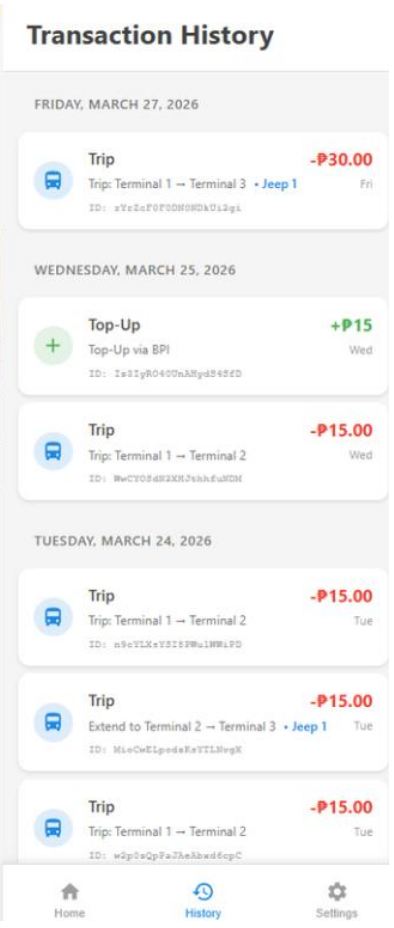


rYrZcFOFODNONDkUi2gi

+ Start collection

+ Add field

amount: -30
balanceAfter: 14499
createdAt: March 27, 2026 at 4:22:30 PM UTC+8
description: "Trip: Terminal 1 → Terminal 3"
jeepneyId: "jeep1"
route
from: 1
to: 3
timestamp: March 27, 2026 at 4:22:30 PM UTC+8
type: "trip"
userId: "ulsKhAzJECgPVcwJnDePCP1E7Ok1"



O2-T02 Terminal-Based Fare Computation Accuracy Test / O2-T03 Automatic Seat Allocation Success Rate Test

TESTING			FARE		DECISION			SEAT ALLOCATION			DECISION		
Timestamp	Testing Number	Card Name	Expected	Actual	Instance	Tester Name	Remarks	Expected		Actual	Instance	Tester Name	Remarks
3/5/2026	1	jc	₱15.00	₱15.00	Correct	jc		seat1		Seat 1	Correct	jc	
	2	jc	₱45.00	₱45.00	Correct	jc		seat1		seat1	Correct	jc	
	3	cj29	₱15.00	₱15.00	Correct	cj29		seat1		seat1	Correct	cj29	
	4	kier	₱30.00	₱30.00	Correct	kier		seat1		seat1	Correct	kier	
	5	niel	₱15.00	₱15.00	Correct	niel		seat2		seat2	Correct	niel	
3/13/2026 4:18:24	6	kier	₱15.00	₱15.00	Correct	kier		seat1		seat1	Correct	kier	
3/13/2026 6:22:47	7	kier	₱15.00	₱15.00	Correct	kier		seat1		seat1	Correct	kier	
3/13/2026 6:22:47	8	niel	₱15.00	₱15.00	Correct	niel		seat2		seat2	Correct	niel	
3/13/2026 6:22:47	9	kier	₱30.00	₱30.00	Correct	kier		seat1		seat1	Correct	kier	
3/13/2026 9:42:32	10	kier	₱15.00	₱15.00	Correct	kier		seat1		seat1	Correct	kier	
3/13/2026 9:44:41	11	tin	₱30.00	₱30.00	Correct	tin		seat2		seat2	Correct	tin	
3/13/2026 9:47:33	12	tin	₱15.00	₱15.00	Correct	tin		seat1		seat1	Correct	tin	
3/13/2026 9:48:25	13	kier	₱30.00	₱30.00	Correct	kier		seat1		seat1	Correct	kier	
3/13/2026 9:56:51	14	kier	₱45.00	₱45.00	Correct	kier		seat1		seat1	Correct	kier	
3/13/2026 9:56:51	15	tin	₱15.00	₱15.00	Correct	tin		seat2		seat2	Correct	tin	
3/13/2026 9:56:51	16	kier	₱15.00	₱15.00	Correct	kier		seat1		seat1	Correct	kier	
3/13/2026 9:56:51	17	tin	₱45.00	₱45.00	Correct	tin		seat2		seat2	Correct	tin	
3/13/2026 10:04:45	18	niel	₱15.00	₱15.00	Correct	niel		seat1		none	Incorrect	niel	
3/13/2026 10:22:02	19	kier	₱15.00	₱15.00	Correct	kier		seat1		seat1	Correct	kier	
3/13/2026 10:22:02	20	niel	₱30.00	₱30.00	Correct	niel		seat2		seat2	Correct	niel	
3/13/2026 10:22:02	21	niel	₱15.00	₱15.00	Correct	niel		seat1		seat1	Correct	niel	
3/13/2026 10:22:02	22	kier	₱15.00	₱15.00	Correct	kier		seat2		seat2	Correct	kier	
3/13/2026 10:27:16	23	niel	₱15.00	₱15.00	Correct	niel		seat1		seat1	Correct	niel	
3/13/2026 10:27:16	24	kier	₱30.00	₱30.00	Correct	kier		seat2		seat2	Correct	kier	
3/13/2026 10:27:21	25	niel	₱15.00	₱15.00	Correct	niel		seat1		seat1	Correct	niel	





O2-T04 Invalid Reason Classification Accuracy Test

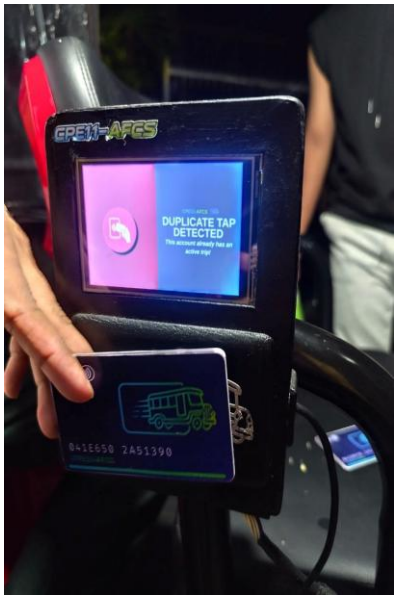
TESTING			RESULTS			DECISION		
Timestamp	Testing Number	Testing Type	Card Name	expected	actual	Instance	Tester	Remarks
4/14/2026 15:13:47	1	destination_not_selected	niel	destination_not_selected	destination_not_selected	Correct	Med/JL/Tin	
4/14/2026 15:13:56	2	destination_not_selected	niel	destination_not_selected	destination_not_selected	Correct	Med	
4/14/2026 15:14:06	3	destination_not_selected	niel	destination_not_selected	destination_not_selected	Correct	Med	
4/14/2026 15:14:12	4	destination_not_selected	niel	destination_not_selected	destination_not_selected	Correct	Med	
4/14/2026 15:15:06	5	destination_not_selected	tin	destination_not_selected	destination_not_selected	Correct	Med	
4/14/2026 15:15:27	6	destination_not_selected	tin	destination_not_selected	destination_not_selected	Correct	Med	
4/14/2026 15:15:31	7	destination_not_selected	tin	destination_not_selected	destination_not_selected	Correct	Med	
4/14/2026 15:15:36	8	destination_not_selected	tin	destination_not_selected	destination_not_selected	Correct	Med	
4/14/2026 15:16:10	9	destination_not_selected	kier	destination_not_selected	destination_not_selected	Correct	Med	
4/14/2026 15:16:16	10	destination_not_selected	kier	destination_not_selected	destination_not_selected	Correct	Med	
4/14/2026 15:17:12	11	destination_not_selected	kier	destination_not_selected	destination_not_selected	Correct	Med	
4/14/2026 15:17:23	12	destination_not_selected	kier	destination_not_selected	destination_not_selected	Correct	Med	
4/14/2026 15:17:54	13	destination_not_selected	med	destination_not_selected	destination_not_selected	Correct	Med	
4/14/2026 15:18:19	14	destination_not_selected	med	destination_not_selected	destination_not_selected	Correct	Med	
4/14/2026 15:18:24	15	destination_not_selected	med	destination_not_selected	destination_not_selected	Correct	Med	
4/14/2026 15:18:29	16	destination_not_selected	med	destination_not_selected	destination_not_selected	Correct	Med	
4/14/2026 15:21:15	17	destination_not_selected	eljey	destination_not_selected	destination_not_selected	Correct	Med	
4/14/2026 15:21:22	18	destination_not_selected	eljey	destination_not_selected	destination_not_selected	Correct	Med	
4/14/2026 15:21:36	19	destination_not_selected	jeyel	destination_not_selected	destination_not_selected	Correct	Med	
4/14/2026 15:21:40	20	destination_not_selected	jeyel	destination_not_selected	destination_not_selected	Correct	Med	
4/14/2026 15:52:33	21	inactive_account	none	inactive_account	inactive_account	Correct	Med	
4/14/2026 15:52:39	22	inactive_account	none	inactive_account	inactive_account	Correct	Med	
4/14/2026 15:52:45	23	inactive_account	none	inactive_account	inactive_account	Correct	Med	
4/14/2026 15:52:49	24	inactive_account	none	inactive_account	inactive_account	Correct	Med	
4/14/2026 15:53:11	25	inactive_account	none	inactive_account	inactive_account	Correct	Med	
4/14/2026 15:53:18	26	inactive_account	none	inactive_account	inactive_account	Correct	Med	
4/14/2026 15:53:23	27	inactive_account	none	inactive_account	inactive_account	Correct	Med	
4/14/2026 15:53:34	28	inactive_account	none	inactive_account	inactive_account	Correct	Med	
4/14/2026 15:53:37	29	inactive_account	none	inactive_account	inactive_account	Correct	Med	
4/14/2026 15:53:48	30	inactive_account	none	inactive_account	inactive_account	Correct	Med	

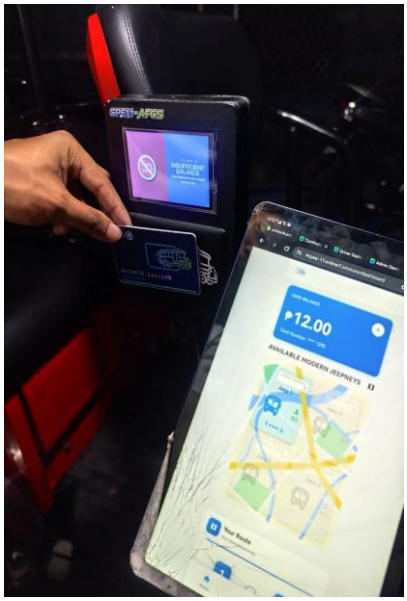




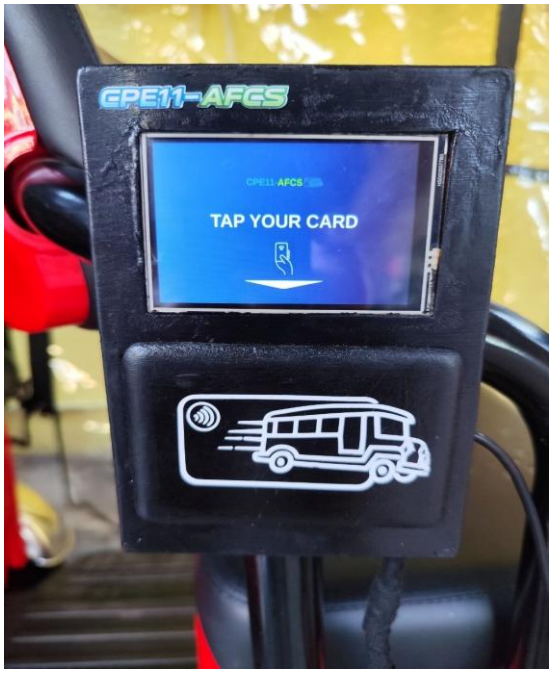
4/14/2026 15:59:18	50	unregistered_account	unknown	unregistered_account	unregistered_account	Correct	Med	
4/14/2026 15:59:28	51	unregistered_account	unknown	unregistered_account	unregistered_account	Correct	Med	
4/14/2026 15:59:33	52	unregistered_account	unknown	unregistered_account	unregistered_account	Correct	Med	
4/14/2026 15:59:38	53	unregistered_account	unknown	unregistered_account	unregistered_account	Correct	Med	
4/14/2026 15:59:42	54	unregistered_account	unknown	unregistered_account	unregistered_account	Correct	Med	
4/14/2026 15:59:49	55	unregistered_account	unknown	unregistered_account	unregistered_account	Correct	Med	
4/14/2026 16:00:00	56	unregistered_account	unknown	unregistered_account	unregistered_account	Correct	Med	
4/14/2026 16:00:06	57	unregistered_account	unknown	unregistered_account	unregistered_account	Correct	Med	
4/14/2026 16:00:12	58	unregistered_account	unknown	unregistered_account	unregistered_account	Correct	Med	
4/14/2026 16:00:17	59	unregistered_account	unknown	unregistered_account	unregistered_account	Correct	Med	
4/14/2026 16:00:21	60	unregistered_account	unknown	unregistered_account	unregistered_account	Correct	Med	
4/14/2026 16:08:54	61	no_seats_available	niel	no_seats_available	no_seats_available	Correct	Med	
4/14/2026 16:09:02	62	no_seats_available	niel	no_seats_available	no_seats_available	Correct	Med	
4/14/2026 16:09:12	63	no_seats_available	niel	no_seats_available	no_seats_available	Correct	Med	
4/14/2026 16:09:16	64	no_seats_available	niel	no_seats_available	no_seats_available	Correct	Med	
4/14/2026 16:13:42	65	no_seats_available	tin	no_seats_available	no_seats_available	Correct	Med	
4/14/2026 16:13:45	66	no_seats_available	tin	no_seats_available	no_seats_available	Correct	Med	
4/14/2026 16:13:53	67	no_seats_available	tin	no_seats_available	no_seats_available	Correct	Med	
4/14/2026 16:13:59	68	no_seats_available	tin	no_seats_available	no_seats_available	Correct	Med	
4/14/2026 16:15:21	69	no_seats_available	kier	no_seats_available	no_seats_available	Correct	Med	
4/14/2026 16:15:27	70	no_seats_available	kier	no_seats_available	no_seats_available	Correct	Med	
4/14/2026 16:15:34	71	no_seats_available	kier	no_seats_available	no_seats_available	Correct	Med	







4/14/2026 18:05:26	95	passenger_still_boarding	med	passenger_still_boarding	passenger_still_boarding	Correct	Med	
4/14/2026 18:05:36	96	passenger_still_boarding	med	passenger_still_boarding	passenger_still_boarding	Correct	Med	
4/14/2026 18:06:05	97	passenger_still_boarding	niel	passenger_still_boarding	passenger_still_boarding	Correct	Med	
4/14/2026 18:06:48	98	passenger_still_boarding	niel	passenger_still_boarding	passenger_still_boarding	Correct	Med	
4/14/2026 18:07:30	99	passenger_still_boarding	tin	passenger_still_boarding	passenger_still_boarding	Correct	Med	
4/14/2026 18:08:01	100	passenger_still_boarding	tin	passenger_still_boarding	passenger_still_boarding	Correct	Med	
4/14/2026 18:10:53	101	insufficient_balance	niel	insufficient_balance	insufficient_balance	Correct	Med	
4/14/2026 18:10:58	102	insufficient_balance	niel	insufficient_balance	insufficient_balance	Correct	Med	
4/14/2026 18:11:03	103	insufficient_balance	niel	insufficient_balance	insufficient_balance	Correct	Med	
4/14/2026 18:11:10	104	insufficient_balance	niel	insufficient_balance	insufficient_balance	Correct	Med	
4/14/2026 18:12:41	105	insufficient_balance	tin	insufficient_balance	insufficient_balance	Correct	Med	
4/14/2026 18:12:47	106	insufficient_balance	tin	insufficient_balance	insufficient_balance	Correct	Med	
4/14/2026 18:12:52	107	insufficient_balance	tin	insufficient_balance	insufficient_balance	Correct	Med	
4/14/2026 18:12:57	108	insufficient_balance	tin	insufficient_balance	insufficient_balance	Correct	Med	
4/14/2026 18:13:42	109	insufficient_balance	kier	insufficient_balance	insufficient_balance	Correct	Med	
4/14/2026 18:13:47	110	insufficient_balance	kier	insufficient_balance	insufficient_balance	Correct	Med	
4/14/2026 18:13:55	111	insufficient_balance	kier	insufficient_balance	insufficient_balance	Correct	Med	
4/14/2026 18:14:00	112	insufficient_balance	kier	insufficient_balance	insufficient_balance	Correct	Med	
4/14/2026 18:15:04	113	insufficient_balance	med	insufficient_balance	insufficient_balance	Correct	Med	
4/14/2026 18:15:45	114	insufficient_balance	med	insufficient_balance	insufficient_balance	Correct	Med	
4/14/2026 18:16:02	115	insufficient_balance	med	insufficient_balance	insufficient_balance	Correct	Med	
4/14/2026 18:16:07	116	insufficient_balance	med	insufficient_balance	insufficient_balance	Correct	Med	
4/14/2026 18:17:34	117	insufficient_balance	eljey	insufficient_balance	insufficient_balance	Correct	Med	
4/14/2026 18:17:40	118	insufficient_balance	eljey	insufficient_balance	insufficient_balance	Correct	Med	
4/14/2026 18:18:21	119	insufficient_balance	jeyel	insufficient_balance	insufficient_balance	Correct	Med	
4/14/2026 18:18:28	120	insufficient_balance	jeyel	insufficient_balance	insufficient_balance	Correct	Med	
4/14/2026 18:21:28	121	duplicate_tap_detected	niel	duplicate_tap_detected	duplicate_tap_detected	Correct	Med	
4/14/2026 18:22:00	122	duplicate_tap_detected	niel	duplicate_tap_detected	duplicate_tap_detected	Correct	Med	
4/14/2026 18:22:03	123	duplicate_tap_detected	niel	duplicate_tap_detected	duplicate_tap_detected	Correct	Med	
4/14/2026 18:22:08	124	duplicate_tap_detected	niel	duplicate_tap_detected	duplicate_tap_detected	Correct	Med	
4/14/2026 18:23:09	125	duplicate_tap_detected	tin	duplicate_tap_detected	duplicate_tap_detected	Correct	Med	
4/14/2026 18:23:12	126	duplicate_tap_detected	tin	duplicate_tap_detected	duplicate_tap_detected	Correct	Med	
4/14/2026 18:23:17	127	duplicate_tap_detected	tin	duplicate_tap_detected	duplicate_tap_detected	Correct	Med	
4/14/2026 18:23:21	128	duplicate_tap_detected	tin	duplicate_tap_detected	duplicate_tap_detected	Correct	Med	

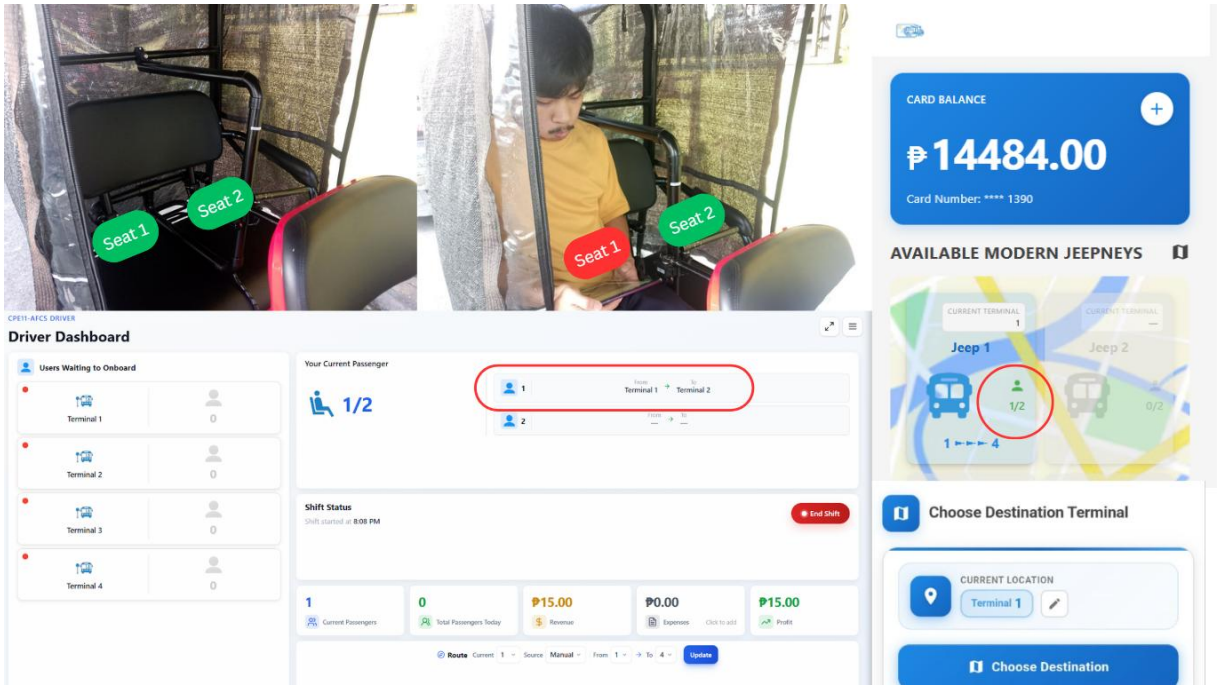




O2-T05 System Unavailable Display Accuray Test / O2-T06 Recovery Time Test

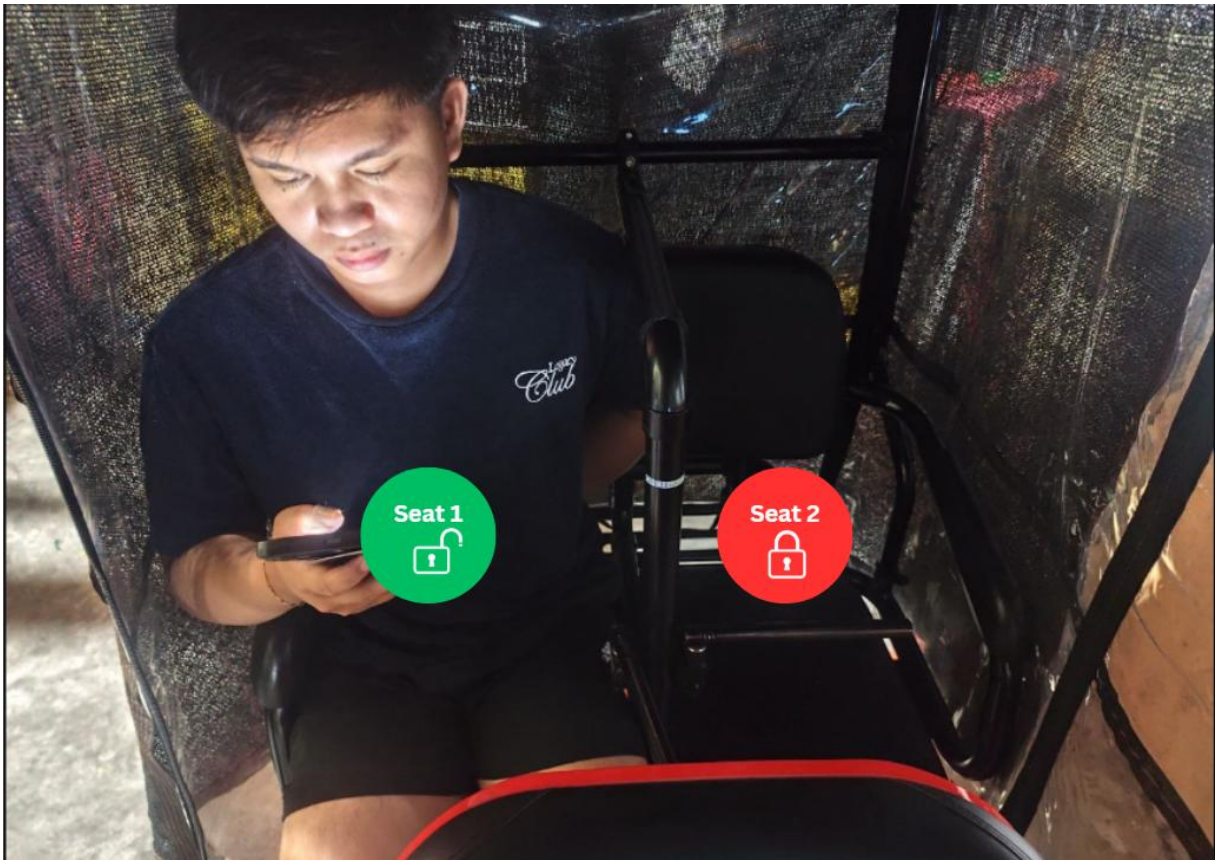
TESTING		RESULTS				RECOVERY TIME			
Timestamp	Testing Number	Expected Display	Actual Display	Instance	Notes / Remarks	System Unavailable Time	System Recovery Time	Time elapsed	Seconds
4/14/2026 20:32:27	1	System Unavailable Display	System Unavailable Display	Correct		20:04:15	20:04:49	0:00:34	34
4/14/2026 20:40:19	2	System Unavailable Display	System Unavailable Display	Correct		20:24:29	20:24:38	0:00:09	9
4/14/2026 20:43:36	3	System Unavailable Display	System Unavailable Display	Correct		20:28:26	20:28:30	0:00:04	4
4/14/2026 20:45:04	4	System Unavailable Display	System Unavailable Display	Correct		20:32:21	20:34:47	0:02:26	146
4/14/2026 21:02:18	5	System Unavailable Display	System Unavailable Display	Correct		20:34:59	20:37:31	0:02:32	152
4/14/2026 21:02:18	6	System Unavailable Display	System Unavailable Display	Correct		20:40:12	20:43:21	0:03:09	189
4/14/2026 21:02:18	7	System Unavailable Display	System Unavailable Display	Correct		20:43:30	20:43:40	0:00:10	10
4/14/2026 21:02:18	8	System Unavailable Display	System Unavailable Display	Correct		20:49:19	20:51:03	0:01:44	104
4/14/2026 21:02:18	9	System Unavailable Display	System Unavailable Display	Correct		20:51:36	20:52:52	0:01:16	76
4/14/2026 20:57:23	10	System Unavailable Display	System Unavailable Display	Correct		20:53:11	20:53:36	0:00:25	25
4/14/2026 20:58:42	11	System Unavailable Display	System Unavailable Display	Correct		20:54:24	20:56:02	0:01:38	98
4/14/2026 21:02:18	12	System Unavailable Display	System Unavailable Display	Correct		20:56:17	20:57:20	0:01:03	63
4/14/2026 21:02:18	13	System Unavailable Display	System Unavailable Display	Correct		20:57:30	20:58:35	0:01:05	65
4/14/2026 21:02:18	14	System Unavailable Display	System Unavailable Display	Correct		20:58:42	20:59:00	0:00:18	18
4/14/2026 21:02:18	15	System Unavailable Display	System Unavailable Display	Correct		20:59:06	21:00:30	0:01:24	84
4/14/2026 21:02:18	16	System Unavailable Display	System Unavailable Display	Correct		21:00:49	21:00:58	0:00:09	9
4/14/2026 21:03:07	17	System Unavailable Display	System Unavailable Display	Correct		21:01:07	21:02:55	0:01:48	108
4/14/2026 21:03:56	18	System Unavailable Display	System Unavailable Display	Correct		21:03:03	21:03:31	0:00:28	28
4/14/2026 21:06:12	19	System Unavailable Display	System Unavailable Display	Correct		21:03:52	21:05:54	0:02:02	122
4/14/2026 21:10:45	20	System Unavailable Display	System Unavailable Display	Correct		21:06:07	21:06:19	0:00:12	12
4/14/2026 21:12:24	21	System Unavailable Display	Default	Incorrect	Delay	21:06:32	21:08:00	0:01:28	88
4/14/2026 21:13:51	22	System Unavailable Display	System Unavailable Display	Correct		21:10:42	21:10:48	0:00:06	6
4/14/2026 21:17:36	23	System Unavailable Display	Default	Incorrect	Delay	21:13:32	21:16:51	0:03:19	199

Objective 3: Seat Monitoring and Allocation Mechanism



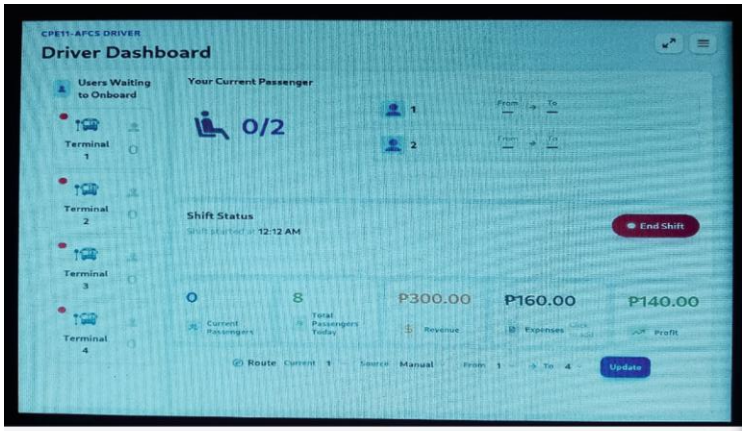
O3-T01 Seat Occupancy Detection Accuracy Test;
O3-T02 Driver Seat Status View Consistency Test;
O3-T03 Commuter Seat Status View Consistency Test

TESTING			RESULTS				DECISION		
Timestamp	Testing Number	Card Name	Expected	RP4 Logs	Database	Driver's Dashboard	Instance	Tester Name(s)	Remarks
2/20/2026	1	jc	seat1 occupied	seat1 available	seat1 available	seat1 available	Incorrect	jc	Black shirt si commuter
	2	cj29	seat1 occupied	seat1 occupied	seat1 occupied	seat1 occupied	Correct	cj29	
	3	kier	seat1 occupied	seat1 occupied	seat1 occupied	seat1 occupied	Correct	kier	
	4	niel	seat2 occupied	seat2 occupied	seat2 occupied	seat2 occupied	Correct	niel	
	5	kier	seat1 available	seat1 available	seat1 available	seat1 occupied	Incorrect	kier	driver full screen not refreshing
	6	niel	seat1 available	seat1 available	seat1 available	seat1 occupied	Incorrect	niel	driver full screen not refreshing
3/13/2026 6:22:47	7	kier	seat1 occupied	seat1 occupied	seat1 occupied	seat1 occupied	Correct	kier	
3/13/2026 6:22:47	8	kier	seat1 occupied	seat1 occupied	seat1 occupied	seat1 occupied	Correct	kier	
3/13/2026 6:22:47	9	niel	seat 2 occupied	seat 2 occupied	seat 2 occupied	seat 2 occupied	Correct	niel	
3/13/2026 6:22:47	10	kier	pending timeout	pending timeout	pending timeout	pending timeout	Correct	kier	
3/13/2026 9:43:02	11	kier	seat1 occupied	seat1 occupied	seat1 occupied	seat1 occupied	Correct	kier	
3/13/2026 9:44:58	12	tin	seat2 occupied	seat2 occupied	seat2 occupied	seat2 occupied	Correct	tin	
3/13/2026 9:45:50	13	kier	seat1 available	seat1 available	seat1 available	seat1 available	Correct	kier	



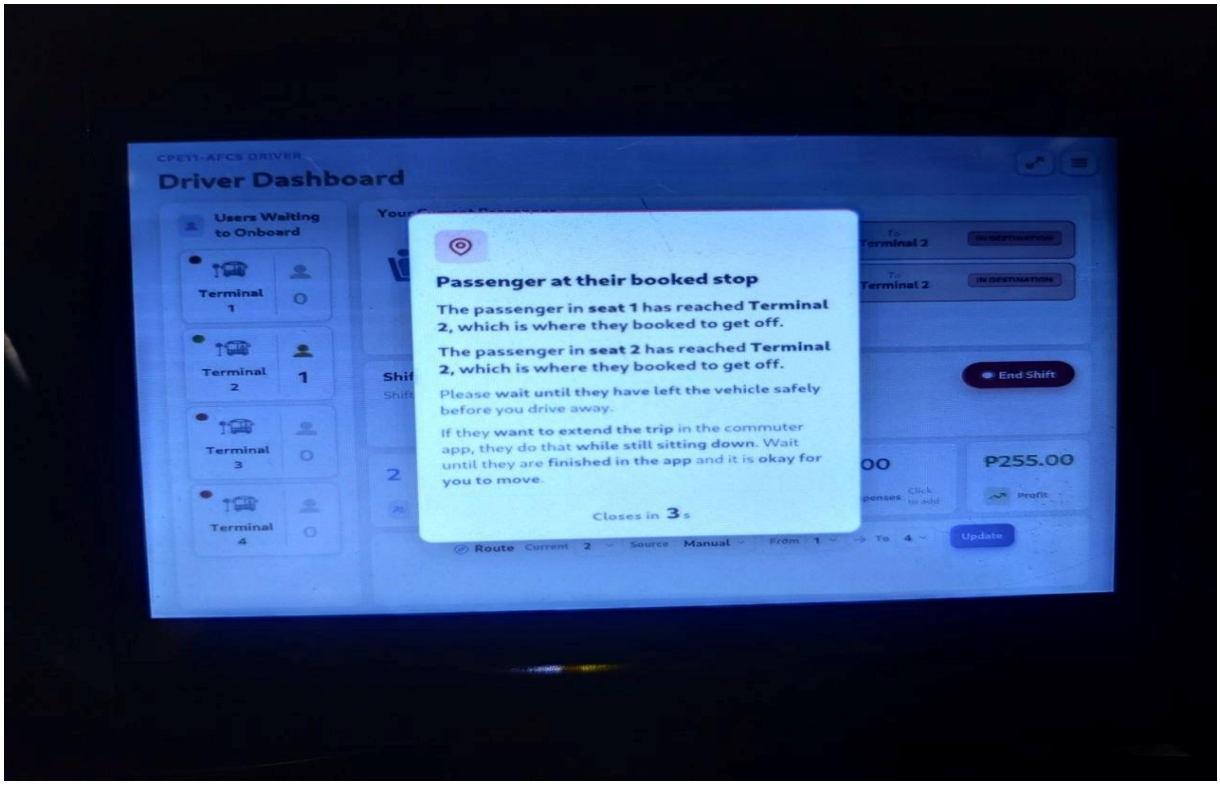
O3-T04 Seat Lock and Unlock Actuation Accuracy Test								
TESTING			RESULTS			DECISION		
Timestamp	Testing Number	Card Name		Expected	Actual	Instance	Tester Name(s)	Remarks
3/5/2026	1	jc		seat1	seat1	Correct	jc	
	2	cj29		seat1	seat1	Correct	cj29	
	3	kier		seat1	seat1	Correct	kier	
	4	niel		seat2	seat2	Correct	niel	
	5	kier		seat1	seat1	Correct	kier	
	6	niel		seat1	seat1	Correct	niel	
3/6/2026 19:33:59	8	kier		seat2	seat2	Correct	kier	
12/30/1899 0:00:00	9	kier		seat1	seat1	Correct	kier	
12/30/1899 0:00:00	10	niel		seat1	seat1	Correct	niel	
12/30/1899 0:00:00	11	kier		seat2	seat2	Correct	kier	
3/13/2026 9:44:26	12	kier		seat1	seat1	Correct	kier	
3/13/2026 9:45:31	13	tin		seat1	seat1	Correct	tin	
3/13/2026 9:47:20	14	tin		seat1	seat1	Correct	tin	
3/13/2026 9:49:01	15	kier		seat2	seat2	Correct	kier	
3/13/2026 9:56:51	16	kier		seat1	seat1	Correct	kier	
3/13/2026 9:56:51	17	tin		seat2	seat2	Correct	tin	

Objective 4: Driver and Admin Dashboards



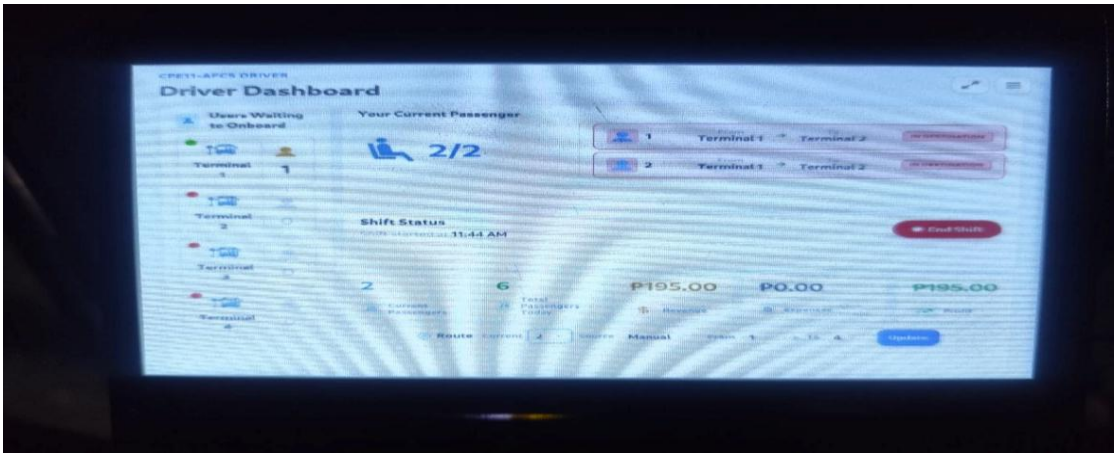
O5-T01 Dashboard Data Consistency Rate Test

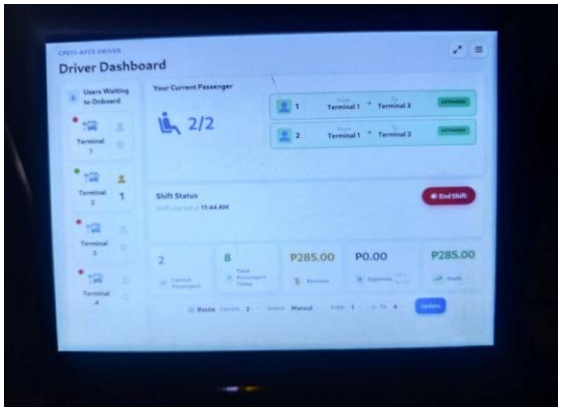
TESTING		DRIVER			ADMIN			DECISION		
Timestamp	Testing Number	Revenue	Expenses	Profit	Revenue	Expenses	Profit	Instance	Tester Name(s)	Remarks
3/12/2026	1	₱180.00	₱0.00	₱180.00	₱180.00	₱0.00	₱180.00	Correct	kier	
	2	₱225.00	₱0.00	₱225.00	₱225.00	₱0.00	₱225.00	Correct	niel	
3/13/2026 4:14:58	3	₱1,695.00	₱0.00	₱1,695.00	₱1,695.00	₱0.00	₱1,695.00	Correct	tin	
3/14/2026 0:29:34	4	₱1,275.00	₱1,500.00	-₱225.00	₱1,275.00	₱1,500.00	-₱225.00	Correct	tin	
3/15/2026 8:31:40	5	₱2,160.00	₱0.00	₱2,160.00	₱2,160.00	₱0.00	₱2,160.00	Correct	tin	
3/15/2026 10:11:31	6	₱2,175.00	₱35.00	₱2,140.00	₱2,175.00	₱35.00	₱2,140.00	Correct	tin	
3/15/2026 10:36:13	7	₱2,250.00	₱55.00	₱2,195.00	₱2,250.00	₱55.00	₱2,195.00	Correct	tin	
3/15/2026 10:36:13	8	₱2,295.00	₱105.00	₱2,190.00	₱2,295.00	₱105.00	₱2,190.00	Correct	niel	
3/15/2026 10:36:13	9	₱2,310.00	₱117.00	₱2,193.00	₱2,310.00	₱117.00	₱2,193.00	Correct	niel	
3/15/2026 10:36:13	10	₱2,355.00	₱117.00	₱2,238.00	₱2,355.00	₱117.00	₱2,238.00	Correct	niel	
3/15/2026 10:36:13	11	₱2,415.00	₱127.00	₱2,288.00	₱2,415.00	₱127.00	₱2,288.00	Correct	niel	
3/15/2026 10:36:13	12	₱2,475.00	₱147.00	₱2,328.00	₱2,475.00	₱147.00	₱2,328.00	Correct	niel	
3/15/2026 10:37:15	13	₱2,490.00	₱147.00	₱2,343.00	₱2,490.00	₱147.00	₱2,343.00	Correct	niel	
3/15/2026 10:37:30	14	₱2,505.00	₱160.00	₱2,345.00	₱2,505.00	₱160.00	₱2,345.00	Correct	niel	
3/15/2026 10:39:05	15	₱2,595.00	₱160.00	₱2,435.00	₱2,595.00	₱160.00	₱2,435.00	Correct	niel	
3/15/2026 10:45:13	16	₱2,640.00	₱200.00	₱2,440.00	₱2,640.00	₱200.00	₱2,440.00	Correct	niel	

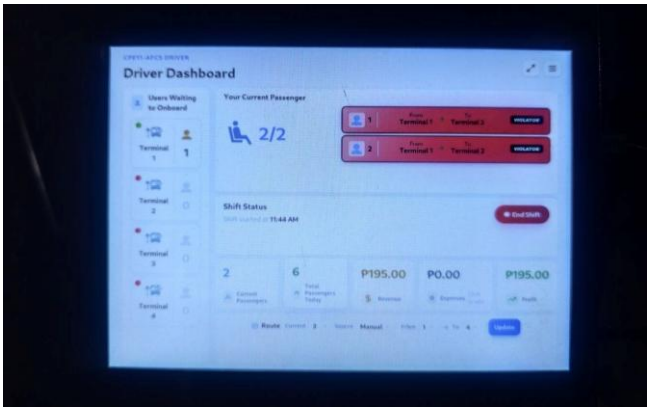


O5-T02 Driver Alert Pop-up Accuracy Test

TESTING		POP-UP CORRECT			DECISION	
Timestamp	Testing Number	Seat	Expected	Actual	Instance	Remarks
4/14/2026 16:08:06	1	Seat 1	Trigge…	Trigge…	Correct	
4/14/2026 16:09:02	2	Seat 2	Trigge…	Trigge…	Correct	
4/14/2026 16:18:39	3	Seat 1	Trigge…	Trigge…	Correct	
4/14/2026 16:18:40	4	Seat 2	Trigge…	Trigge…	Correct	
4/14/2026 16:20:24	5	Seat 1	Trigge…	Trigge…	Correct	
4/14/2026 16:20:24	6	Seat 2	Trigge…	Trigge…	Correct	
4/14/2026 17:39:00	7	Seat 1	Trigge…	Trigge…	Correct	
4/14/2026 17:39:00	8	Seat 2	Trigge…	Trigge…	Correct	
4/14/2026 17:56:42	9	Seat 1	Trigge…	Trigge…	Correct	
4/14/2026 17:56:43	10	Seat 2	Trigge…	Trigge…	Correct	
4/14/2026 18:40:36	11	Seat 1	Trigge…	Trigge…	Correct	
4/14/2026 18:41:33	12	Seat 2	Trigge…	Trigge…	Correct	
4/14/2026 18:47:45	13	Seat 1	Trigge…	Trigge…	Correct	
4/14/2026 18:48:17	14	Seat 2	Trigge…	Trigge…	Correct	
4/14/2026 19:13:02	15	Seat 1	Trigge…	Trigge…	Correct	
4/14/2026 19:13:02	16	Seat 2	Trigge…	Trigge…	Correct	

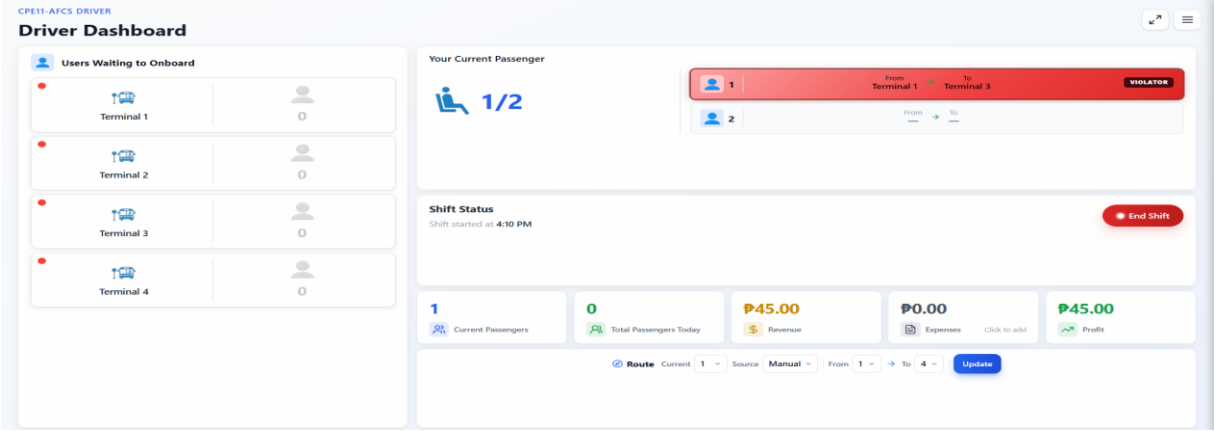


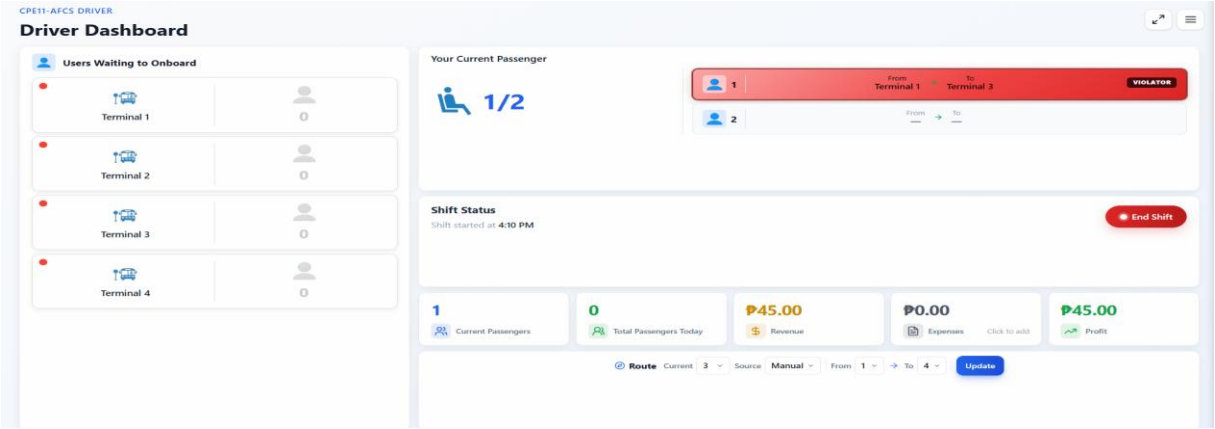




O5-T03 Indicator State Correctness Test

TESTING		INDICATOR STATE			DECISION	
Timestamp	Testing Number	Seat	Expected	Actual	Instance	Remarks
4/14/2026 16:14:02	1	Seat 1	In Destination	In Destination	Correct	
4/14/2026 16:14:03	2	Seat 2	In Destination	In Destination	Correct	
4/14/2026 16:14:56	3	Seat 1	Violator	Violator	Correct	
4/14/2026 16:14:56	4	Seat 2	Violator	Violator	Correct	
4/14/2026 16:17:11	5	Seat 1	In Destination	In Destination	Correct	
4/14/2026 16:17:11	6	Seat 2	In Destination	In Destination	Correct	
4/14/2026 16:17:50	7	Seat 1	Extended	Extended	Correct	
4/14/2026 16:17:52	8	Seat 2	Extended	Extended	Correct	
4/14/2026 16:20:48	9	Seat 1	In Destination	In Destination	Correct	
4/14/2026 16:20:49	10	Seat 2	In Destination	In Destination	Correct	
4/14/2026 16:21:28	11	Seat 1	Violator	Violator	Correct	
4/14/2026 16:21:30	12	Seat 2	Violator	Violator	Correct	
4/14/2026 17:38:52	13	Seat 1	In Destination	In Destination	Correct	
4/14/2026 17:38:54	14	Seat 2	In Destination	In Destination	Correct	
4/14/2026 17:40:22	15	Seat 1	Extended	Extended	Correct	
4/14/2026 17:40:23	16	Seat 2	Extended	Extended	Correct	
4/14/2026 17:41:00	17	Seat 1	Violator	Violator	Correct	
4/14/2026 17:41:01	18	Seat 2	Violator	Violator	Correct	





O5-T04 Driver Monitoring Non-Repeat Compliance Test

TESTING		NON-REPEAT COMPLIANCE		DECISION	
Timestamp	Testing Number	Seat	Repeated?	Instance	Remarks
4/14/2026 16:13:37	1	Seat 1	No	Correct	
4/14/2026 16:13:37	2	Seat 2	No	Correct	
4/14/2026 16:18:44	3	Seat 1	No	Correct	
4/14/2026 16:19:03	4	Seat 2	No	Correct	
4/14/2026 16:21:41	5	Seat 1	No	Correct	
4/14/2026 16:21:41	6	Seat 2	No	Correct	
4/14/2026 17:40:48	7	Seat 1	No	Correct	
4/14/2026 17:40:49	8	Seat 2	No	Correct	
4/14/2026 17:58:40	9	Seat 1	No	Correct	

Structured Functional Testing Records

Document Details

File Type: Google Sheets (online spreadsheet)

Contents: Complete testing logs for the system evaluation

Access: Link-based access (view only, unless otherwise granted)

Spreadsheet Link: Use the QR code or the boxed link below.

https://docs.google.com/spreadsheets/d/15x8ByZlV--VPqIrcLwZTYOdT-DqcO0s_Df7N0DWUncc/edit?usp=sharing

QR Code

Scan to open the testing logs.

Quick Guide

- Raw trial logs and computed results for structured functional testing (Objectives 1 to 5)
- Supporting computations and summaries used for table generation and interpretation
- Timestamped logs

Note: The spreadsheet serves as the complete evidence record for the system tests and contains all supporting computations used in the study.

Appendix H

Project Structure and Assessment

The developed automated fare collection system (AFCS) with seat allocation, terminal selection, and profit analytics dashboards were subdivided into four modules. These were the commuter interface, tap-based payment process, seat monitoring and allocation mechanism, and driver and admin dashboards.

1. The commuter interface was developed as a mobile-first web application using React and Vite, integrated with Firebase services for authentication, database management, and real-time updates, using the following steps.
    1.1. Create a new Vite React project for the commuter web application and prepare the default project structure, including the src, public, and configuration files.
    1.2. Install the required dependencies for routing, Firebase services, icon components, and React development support.
    1.3. Configure Firebase Authentication, Cloud Firestore, Realtime Database, and Cloud Functions in a centralized configuration file for the commuter application.
    1.4. Define the required Firestore collections and helper modules for users, jeepneys, transactions, and verification codes to support commuter-related operations.
    1.5. Implement email and password login, two-factor authentication, and trusted-device checking to secure commuter access.
    1.6. Develop the main commuter pages, including login, verification, dashboard, top-up, origin selection, and destination selection pages, using a mobile-first layout.
    1.7. Create a fare calculation module that computes travel cost based on the selected origin and destination terminals before route confirmation.
    1.8. Connect the interface to Firebase listeners to display real-time seat availability, commuter status updates, and tap confirmation events.

1.9. Configure protected routes and navigation flow for login, verification, dashboard, and terminal-selection pages.

1.10. Add language support, light and dark theme settings, and navigation components to improve usability and accessibility.

1.11. Configure the project for production build and prepare the output files and routing support needed for deployment to the hosting platform.

2. The tap-based payment process was developed by integrating the Raspberry Pi 4, NFC card reader, TFT display, and Firebase-based backend services to validate commuter cards, process tap events, compute fares, and return tap results, using the following steps.

2.1. Prepare and assemble the enclosure based on the planned CAD design to securely hold the Raspberry Pi 4, TFT display, NFC card reader, and related wiring.

2.2. Fix the Raspberry Pi 4, TFT display, NFC card reader, and power connections in their assigned positions inside the casing for stable operation and accessibility.

2.3. Connect the TFT display pins to the assigned SPI and GPIO pins of the Raspberry Pi 4 based on the planned pin configuration.

2.4. Attach the NFC card reader to the Raspberry Pi 4 through the USB interface and verify that the device is recognized by the system.

2.5. Prepare the Raspberry Pi environment by installing the required Python libraries for Firebase, serial communication, NFC smart card reading, TFT display control, image handling, and GPIO control.

2.6. Develop the initial Python program that initializes the TFT display, detects the NFC card reader, reads card UIDs on tap, and keeps the device ready for continuous card scanning.

2.7. Create the required display images such as default.png, valid.png, and invalid.png, and organize them for use as visual feedback on the TFT screen.

2.8. Update the Python program so that the TFT display shows the default screen during idle state, the valid screen for accepted taps, and the invalid screen for rejected taps.

2.9. Connect the Raspberry Pi firmware to the same Firebase project used in Module 1 so that card tap data, device records, and tap results can be stored and synchronized with the web applications.

2.10. Configure the tap process to check whether the tapped card UID is registered, linked to a commuter account, and eligible for boarding based on the commuter's current route selection and status.

2.11. Plan the tap process to trigger fare computation based on the selected terminals, update the commuter's trip status, and assign an available seat after successful validation.

2.12. Configure the system to record tap events in Firebase and return the corresponding tap result to the TFT display and connected dashboards.

3. The seat monitoring and allocation mechanism was developed by fabricating the seat structure, installing the seat-control hardware, integrating the Raspberry Pi control logic, and connecting the mechanism to Firebase for real-time seat-status updates, using the following steps.

3.2. Build the seat monitoring and allocation structure using the planned PVC pipes and fittings to form the seat dividers and mounting support for the hardware components.

3.3. Install the servo motors and IR sensors in their assigned positions on the fabricated frame based on the planned seat layout.

3.4. Wire the servo motors and IR sensors to the assigned GPIO pins of the Raspberry Pi 4 and secure the power and signal connections.

3.5. Mount the completed seat monitoring and allocation mechanism inside the 3-wheeled e-bike and secure it for actual operation.

3.6. Adjust the servo opening and closing positions and test the IR sensor detection response to ensure proper gate movement and seat occupancy detection.

3.7. Configure the Raspberry Pi firmware so that Servo 1 and IR Sensor 1 correspond to Seat 1, while Servo 2 and IR Sensor 2 correspond to Seat 2.

3.8. Extend the tap-based validation firmware so that a successful tap will assign an available seat and open the corresponding seat gate.

3.9. Configure the firmware so that, after a seat gate opens, the assigned IR sensor will wait for passenger detection and update the seat status accordingly.

3.10. Define the seat states and transitions, such as reserved, occupied, and available, so that the system can track boarding and alighting accurately.

3.11. Configure the Raspberry Pi to log each seat-related event, including seat assignment, gate opening, seat occupancy detection, and seat release.

3.12. Integrate the firmware with Firebase so that seat events and seat status updates will be transmitted to the database for real-time display in the commuter and driver interfaces.

4. The driver and administrator dashboards were developed as separate web applications to support operational monitoring, financial analytics, and report generation, using the following steps.

4.1. Set up two Vite React projects to separate driver functions from administrator functions and to simplify access control and deployment.

4.2. Add the libraries needed for routing, Firebase integration, icon components, styling, and data visualization.

4.3. Set up Tailwind CSS and related build files to support the layout and styling of both dashboard interfaces.

4.4. Connect both applications to the same Firebase project used by the commuter interface so they can access authentication, database, and real-time data services.
4.5. Create database-access modules that will handle real-time passenger monitoring, jeepney data, shift records, expenses, terminal activity, revenue, profit, and report data retrieval.
4.6. Implement login, two-factor authentication, and trusted-device checking for both the driver and administrator accounts.
4.7. Create the driver dashboard structure to display terminals, waiting passengers, onboarded passengers, seat count, route controls, shift controls, and daily statistics.
4.8. Add real-time logic that will count waiting commuters per terminal and display the current terminal activity to the driver.
4.9. Display onboarded passengers, assigned routes, current seat occupancy, and indicators for passengers with extended trips.
4.10. Add controls for starting and ending the shift and record the corresponding timestamps in Firebase.
4.11. Display daily passenger count, revenue, expenses, and profit, and connect the values to real-time database updates.
4.12. Add terminal selection controls that will allow the driver to update the active route of the vehicle.
4.13. Add a card that will allow the driver to record operational expenses and save them to Firebase.
4.14. Create the admin dashboard structure to display key performance indicators.
4.15. Display total passengers, revenue, expenses, and profit, together with comparison values based on previous records.
4.16. Add real-time displays for vehicle seat utilization, boarding activity, alighting activity, and total terminal activity.

4.17. Add chart components and period filters to display financial trends in daily, weekly, monthly, and yearly views.

4.18. Add report controls that will allow the administrator to generate summary or detailed reports in PDF or CSV format.

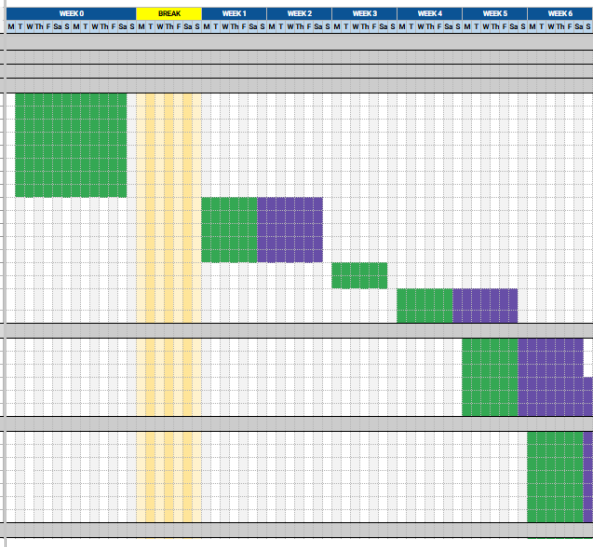
4.19. Set up the routes for login, verification, and dashboard pages, and restrict dashboard access to authenticated users only.

4.20. Generate the production build of the driver and admin applications and prepare the output files for hosting and deployment.

The project development phase of the study was successfully accomplished utilizing the Gantt Chart organized by the researchers, as shown below.

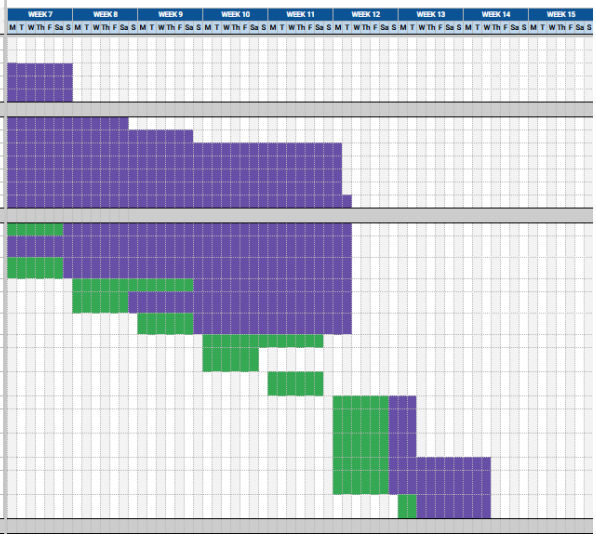
GANTT CHART for PROJECT DEVELOPMENT

WBS NUMBER	TASK TITLE	TASK OWNER	START DATE	DUE DATE	EXTENTED DUE DATE	DURATION (DAYS)	PCT OF TASK COMPLETE	STATUS
1	PLANNING PHASE							
2	HARDWARE DESIGN PHASE							
3	SOFTWARE DESIGN PHASE							
4	DEVELOPMENT PHASE						100.00%	Completed
4.1	Plan project, set up repository, and define standards		12/15/25	12/27/25		12	100.00%	Completed
4.2	Procure components and label inventory		12/15/25	12/27/25		12	100.00%	Completed
4.3	Install RPI OS and configure Wi-Fi and peripherals		12/15/25	12/27/25		12	100.00%	Completed
4.4	Configure NFC reader and implement tap validation baseline		12/15/25	12/27/25		12	100.00%	Completed
4.5	Wire IR sensors and servos for seat allocation		12/15/25	12/27/25		12	100.00%	Completed
4.6	Calibrate seat allocation and implement control logic		12/15/25	12/27/25		12	100.00%	Completed
4.7	Set up GPS module and log location data		12/15/25	12/27/25		12	100.00%	Completed
4.8	Initially improve the interfaces for commuter, driver, and admin		12/15/25	12/27/25		12	100.00%	Completed
4.9	Integrate TFT display and test UI output		1/5/26	1/10/26	1/17/26	5	100.00%	Completed
4.10	Assemble power system and add protection		1/5/26	1/10/26	1/17/26	5	100.00%	Completed
4.11	Design database schema and set up centralized database		1/5/26	1/10/26	1/17/26	5	100.00%	Completed
4.12	Develop backend APIs (auth, validation, trips, transactions)		1/5/26	1/10/26	1/17/26	5	100.00%	Completed
4.13	Build front-end dashboards (commuter, driver, admin)		1/5/26	1/10/26	1/17/26	5	100.00%	Completed
4.14	Draft real-time analytics and dashboard charts		1/19/26	1/24/26		5	100.00%	Completed
4.15	Start end-to-end tap-in/out to analytics pipeline		1/19/26	1/24/26		5	100.00%	Completed
4.16	Fabricate the hardware components into a working prototype		1/26/26	1/31/26	2/7/26	5	100.00%	Completed
4.17	Integrate all working codes into one system		1/26/26	1/31/26	2/7/26	5	100.00%	Completed
5	TESTING PHASE						100.00%	Completed
5.1	Define test cases and scenarios		2/2/26	2/7/26	2/14/26	5	100.00%	Completed
5.2	Conduct unit tests on hardware modules		2/2/26	2/7/26	2/14/26	5	100.00%	Completed
5.3	Conduct unit tests on software modules		2/2/26	2/7/26	2/14/26	5	100.00%	Completed
5.4	Perform integration testing of complete flows		2/2/26	2/7/26	2/22/26	5	100.00%	Completed
5.5	Test extended terminal and edge processing scenarios		2/2/26	2/7/26	2/22/26	5	100.00%	Completed
5.6	Record and analyze test results		2/2/26	2/7/26	2/22/26	5	100.00%	Completed
6	EVALUATION PHASE						100.00%	Completed
6.1	Prepare evaluation instruments and criteria		2/9/26	2/14/26	2/28/26	5	100.00%	Completed
6.2	Recruit evaluators using purposive sampling		2/9/26	2/14/26	3/7/26	5	100.00%	Completed
6.3	Print evaluation questionnaires		2/9/26	2/14/26	3/23/26	5	100.00%	Completed
6.4	Administer evaluation questionnaires		2/9/26	2/14/26	3/23/26	5	100.00%	Completed
6.5	Collect and consolidate system metrics		2/9/26	2/14/26	3/23/26	5	100.00%	Completed
6.6	Analyze evaluation data		2/9/26	2/14/26	3/23/26	5	100.00%	Completed
6.7	Identify improvements and limitations		2/9/26	2/14/26	3/24/26	5	100.00%	Completed
7	FINAL PAPER WRITING PHASE						100.00%	Completed



GANTT CHART for PROJECT DEVELOPMENT

WBS NUMBER	TASK TITLE	TASK OWNER	START DATE	DUE DATE	EXTENTED DUE DATE	DURATION (DAYS)	PCT OF TASK COMPLETE	STATUS
5.2	Conduct unit tests on hardware modules		2/2/26	2/7/26	2/14/26	5	100.00%	Completed
5.3	Conduct unit tests on software modules		2/2/26	2/7/26	2/14/26	5	100.00%	Completed
5.4	Perform integration testing of complete flows		2/2/26	2/7/26	2/22/26	5	100.00%	Completed
5.5	Test extended terminal and edge processing scenarios		2/2/26	2/7/26	2/22/26	5	100.00%	Completed
5.6	Record and analyze test results		2/2/26	2/7/26	2/22/26	5	100.00%	Completed
6	EVALUATION PHASE						100.00%	Completed
6.1	Prepare evaluation instruments and criteria		2/9/26	2/14/26	2/28/26	5	100.00%	Completed
6.2	Recruit evaluators using purposive sampling		2/9/26	2/14/26	3/7/26	5	100.00%	Completed
6.3	Print evaluation questionnaires		2/9/26	2/14/26	3/23/26	5	100.00%	Completed
6.4	Administer evaluation questionnaires		2/9/26	2/14/26	3/23/26	5	100.00%	Completed
6.5	Collect and consolidate system metrics		2/9/26	2/14/26	3/23/26	5	100.00%	Completed
6.6	Analyze evaluation data		2/9/26	2/14/26	3/23/26	5	100.00%	Completed
6.7	Identify improvements and limitations		2/9/26	2/14/26	3/24/26	5	100.00%	Completed
7	FINAL PAPER WRITING PHASE						100.00%	Completed
7.1	Chapter IV – Results and Discussion		2/9/26	2/21/26	3/24/26	12	100.00%	Completed
7.1.1	Write Chapter IV discussing testing results, simulation data, and dashboard visuals.	All Members	2/9/26	2/14/26	3/24/26	5	100.00%	Completed
7.1.2	Interpret outcomes and link findings to research objectives.	All Members	2/16/26	2/21/26	3/24/26	5	100.00%	Completed
7.2	Chapter V – Summary, Conclusions, and Recommendations		2/23/26	3/7/26	3/24/26	14	100.00%	Completed
7.2.1	Write Chapter V summarizing key findings, conclusions, and recommendations.	All Members	2/23/26	2/28/26	3/24/26	5	100.00%	Completed
7.2.2	Highlight future development and scalability of the system.	All Members	3/2/26	3/7/26	3/24/26	5	100.00%	Completed
5.3	Formatting, Consolidation, and Proofreading		3/9/26	3/21/26		12	100.00%	Completed
5.3.1	Combine all thesis chapters into one formatted document following IEEE format.	All Members	3/9/26	3/14/26		5	100.00%	Completed
5.3.2	Proofread, check citations, ensure clarity and grammatical accuracy.	All Members	3/16/26	3/21/26		5	100.00%	Completed
5.4	Final Adviser Consultation and Revisions		3/23/26	3/28/26	3/31/26	5	100.00%	Completed
5.4.1	Present full manuscript (Ch. 1–5) and prototype summary to adviser.	All Members	3/23/26	3/28/26	3/31/26	5	100.00%	Completed
5.4.2	Apply revisions based on adviser's and peer reviewers' comments	All Members	3/23/26	3/28/26	3/31/26	5	100.00%	Completed
5.5	Final Defense Presentation		3/23/26	3/28/26	4/8/26	5	100.00%	Completed
5.5.1	Prepare defense PowerPoint covering Chapters 1–5 and project demonstration.	All Members	3/23/26	3/28/26	4/8/26	5	100.00%	Completed
5.5.2	Present prototype to panel; address Q&A and gather evaluation results.	All Members	3/30/26	3/31/26	4/8/26	0	100.00%	Completed
	TOTAL PROGRESS		12/15/25	2/14/26		59	100.00%	Completed



Appendix I

Source Code

CPE11-AFCS Source Code Repository

Repository Details

Platform: GitHub

Branch: main

Access: Public repository (online)

Repository Link: Use the QR code or the boxed link below.

https://github.com/elle-pnc/4CPEA-11/tree/main

QR Code

Scan to open the repository.

Repository Contents (Quick Guide)

- Commuter interface (front-end source files)
- Driver dashboard (front-end source files)
- Administrator dashboard (front-end source files)
- Raspberry Pi firmware (Python scripts for validation, seat monitoring, GPS, and Firestore integration)
- Backend or cloud logic (if included in the repository)

Note: The complete implementation, commit history, and updates are maintained in the repository for version control and traceability.

Appendix J

Project Costing

Hardware Components

COMPONENT	QUANTITY	UNIT COST	TOTAL COST	REMARKS
Raspberry Pi 4 (Model B), 8 GB	1pc	₱9,095	₱9,095	Main embedded controller
MIFARE DESFire EV2 (4 KB) smart cards	10pcs	₱174	₱1,740	Secure contactless fare media
ACR1252U USB NFC Smart Card Reader and Writer	1pc	₱4,560	₱4,560	NFC reader for tap-based validation
7-inch HDMI Touch Capacitive Display Module	1pc	₱2,099	₱2,099	Driver dashboard display
3.5-inch SPI TFT LCD Display Module	1pc	₱665	₱665	On-board commuter information display
NEO-M8N GPS Module	1pc	₱550	₱550	Location and movement data
GPS External Antenna SMA Male	1pc	₱150	₱150	External GPS reception
Sintra Board	20 pcs	₱20	₱400	Casing of each module
TP-Link M7000 4G LTE Mobile Wi-Fi	1pc	₱1,645	₱1,645	Mobile Internet for Raspberry Pi
Power Station	1pc	₱3,398	₱3,398	Main power source for prototype
DC-DC Step-Down Buck Converter	2pc	₱69	₱138	Regulated 6 V supply for modules
SMART 5G + 4G/LTE Ready SIM Card	1pc	₱49	₱49	Data SIM for TP-Link M7000
IR Proximity Sensor Module	2pcs	₱30	₱60	Seat occupancy detection
Servo TowerPro MG996R	2pcs	₱286	₱572	Seat locking mechanism
		SUBTOTAL	**₱25,121**	

Software Components

SOFTWARE TOOLS	DAYS	TOTAL COST	REMARKS
ChatGPT Plus	120	₱4,400	Used for writing support, documentation, and code review and improvements
Cursor	120	₱14,348	Code editor used for web and backend development
Hostinger	365	₱1,668	Hosting sites for live deployment of the systems
Firebase	~	₱100	Served as the centralized database
	SUBTOTAL	₱20,416	

Overall Cost

Hardware Components Subtotal	₱25,121
Software Components Subtotal	₱20,416
TOTAL	**₱45,537**

Appendix K

Certificates

PNC:PRE-FO-80 rev.1 03212024

Republic of the Philippines
Pamantasan ng Cabuyao
(UNIVERSITY OF CABUYAO)
Planning, Research, and Extension Division
Research and Development Department
Katapatan Mutual Homes, Brgy. Banay-banay, City of Cabuyao, Laguna 4025

CERTIFICATION

This is to certify that **Engr. Anna-liza F. Sigue** a teaching personnel of the College of Engineering served as a ADVISER during the Project Study Oral Defense of the Bachelor of Science in Computer Engineering (BSCpE) students with research title: **Automated Fare Collection System with Seat Allocation, Terminal Selection, and Profit Analytics Dashboards**.

Given this **8th** day of **April**, **2026** at Pamantasan ng Cabuyao, Katapatan Homes Subdivision, Brgy. Banay-banay, Cabuyao City, Laguna.

DR. JESSICA D. REY
Director
Research and Development Department

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Republic of the Philippines
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Planning, Research, and Extension Division
Research and Development Department
Katapatan Mutual Homes, Brgy. Banay-banay, City of Cabuyao, Laguna 4025

CERTIFICATION

This is to certify that **Engr. Maricar R. Cabansag** a teaching personnel of the College of Engineering served as a PANELIST during the Project Study Oral Defense of the Bachelor of Science in Computer Engineering (BSCpE) students with research title: **Automated Fare Collection System with Seat Allocation, Terminal Selection, and Profit Analytics Dashboards**.

Given this **8th** day of **April**, **2026** at Pamantasan ng Cabuyao, Katapatan Homes Subdivision, Brgy. Banay-banay, Cabuyao City, Laguna.

DR. JESSICA D. REY
Director
Research and Development Department

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Research and Development Department
Katapatan Mutual Homes, Brgy. Banay-banay, City of Cabuyao, Laguna 4025

CERTIFICATION

This is to certify that **Engr. Helcy Alon** a teaching personnel of the College of Engineering served as a PANELIST during the Project Study Oral Defense of the Bachelor of Science in Computer Engineering (BSCpE) students with research title: **Automated Fare Collection System with Seat Allocation, Terminal Selection, and Profit Analytics Dashboards**.

Given this **8th** day of **April**, **2026** at Pamantasan ng Cabuyao, Katapatan Homes Subdivision, Brgy. Banay-banay, Cabuyao City, Laguna.

DR. JESSICA D. REY
Director
Research and Development Department

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Research and Development Department
Katapatan Mutual Homes, Brgy. Banay-banay, City of Cabuyao, Laguna 4025

CERTIFICATION

This is to certify that **Engr. Miguel Darren Ronquez** a teaching personnel of the College of Engineering served as a PANELIST during the Project Study Oral Defense of the Bachelor of Science in Computer Engineering (BSCpE) students with research title: **Automated Fare Collection System with Seat Allocation, Terminal Selection, and Profit Analytics Dashboards**.

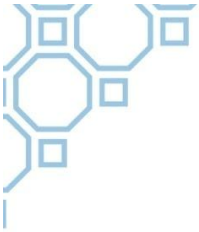
Given this **8th** day of **April**, **2026** at Pamantasan ng Cabuyao, Katapatan Homes Subdivision, Brgy. Banay-banay, Cabuyao City, Laguna.

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Research and Development Department

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Appendix L

User Manual





CPE11-AFCS

Commuter User Manual

CPE11-AFCS | Commuter User Manual

page

Table of Contents

1 Login Page

4 Dashboard Overview

6 Top-Up Balance & Card Balance

9 Available Vehicle

11 Choose your destination & Transaction History

15 Settings

17 Commuter Process Flow
- Extended Terminal Feature

CPE11-AFCS

Commuter
User Manual

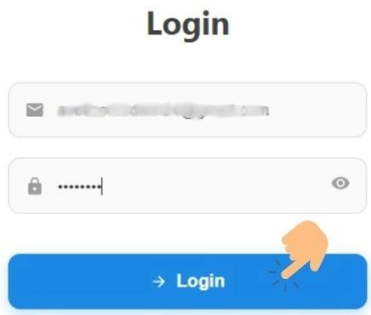
Login Page

CPE11-AFCS
Commuter User Manual
Login Page

Step 1

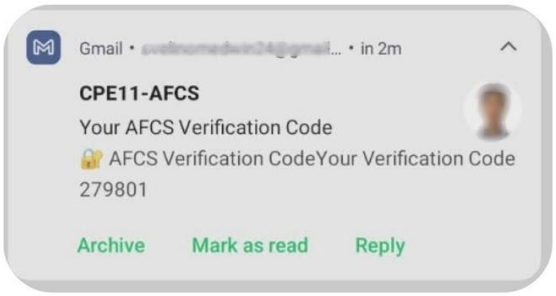
Enter the registered Email and Password, then click Login.





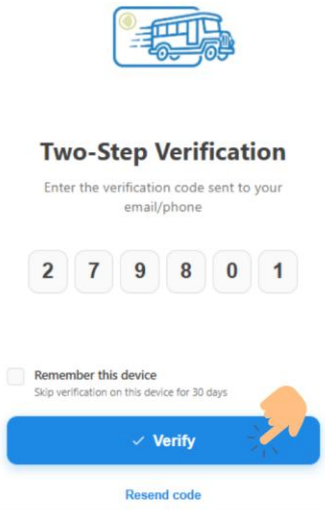
Step 2

Wait for the verification code sent to your Gmail account.



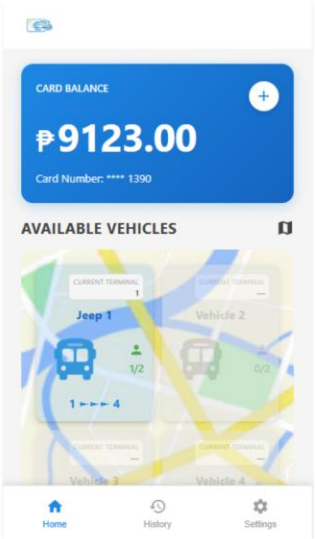
Step 3

Enter the code for verification, then click Verify to Continue. If no code is received, click Resend Code.



Step 4

Once the verification is successful, you will be redirected to the Commuter Dashboard.





CPE11-AFCS

Commuter
User Manual

Dashboard Overview



Dashboard Overview

1. Top-Up Balance
2. Card Balance
3. Available Vehicles
4. Choose your destination
5. Home page
6. Transaction History
7. Settings



CPE11-AFCS

Commuter
User Manual

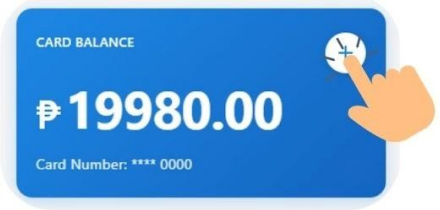
Top-Up Balance and Card Balance

CPE11-AFCS
Commuter User Manual
Top-Up Balance and Card Balance

The Top Up Balance function is used to add more funds to the card, while the Card Balance section shows the commuter's available funds. These features help the user quickly check the remaining balance and reload the card when needed before traveling.

Step 1

If you want to Top-Up, click the "+" button.



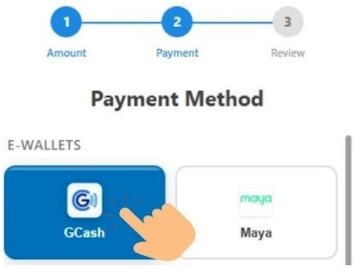
Step 2

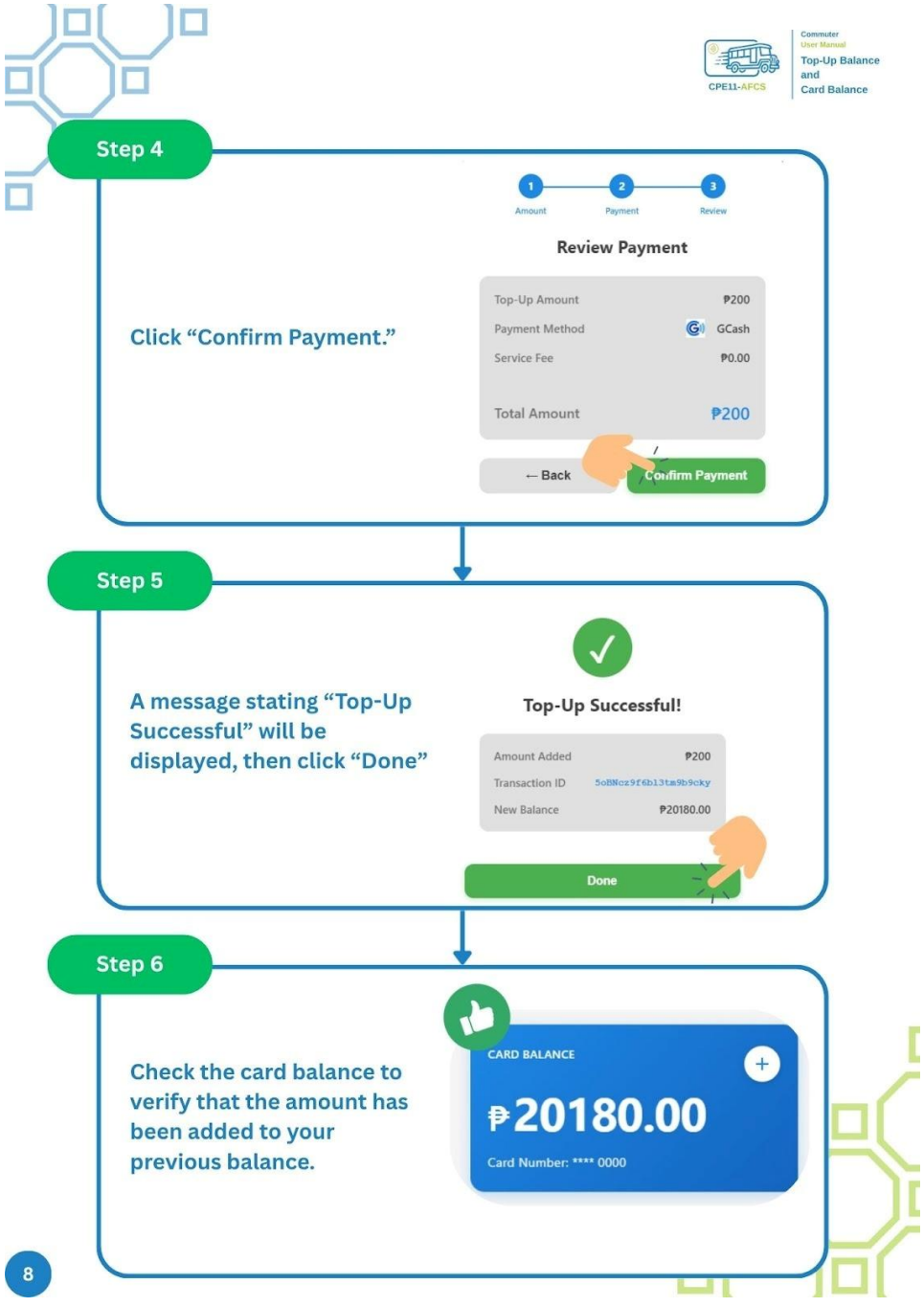
Enter the desired amount (maximum of PHP 10,000), then click "Continue."



Step 3

Select a payment method, either E-wallets, Banks, or Credit/Debit Cards, then click "Continue"




CPE11-AFCS
Commuter User Manual
Top-Up Balance and Card Balance
Step 4
Click "Confirm Payment."
1 Amount
2 Payment
3 Review
Review Payment
Top-Up Amount ₱200
Payment Method GCash
Service Fee ₱0.00
Total Amount ₱200
← Back
Confirm Payment
Step 5
A message stating "Top-Up Successful" will be displayed, then click "Done"
Top-Up Successful!
Amount Added ₱200
Transaction ID 5oBNcz9f6b13tm9b9cky
New Balance ₱20180.00
Done
Step 6
Check the card balance to verify that the amount has been added to your previous balance.
CARD BALANCE
₱20180.00
Card Number: **** 0000
8



CPE11-AFCS

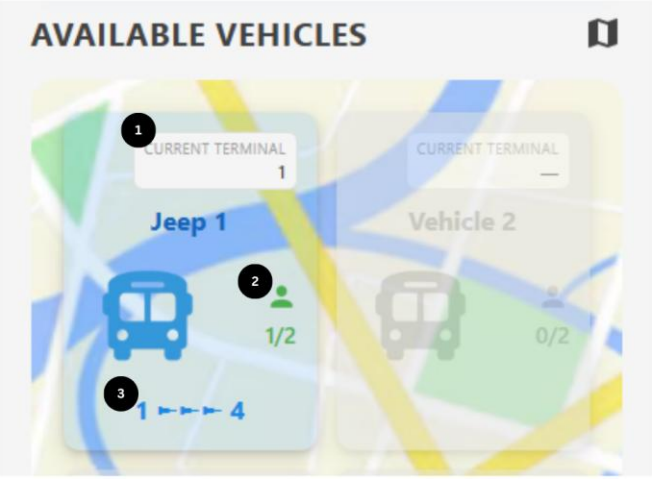
Commuter
User Manual

Available Vehicles

CPE11-AFCS

Commuter User Manual
Available Vehicles

The Available Vehicles section shows the vehicles that are currently available, including their present terminal and seat availability, so the commuter can quickly see which vehicle is ready for boarding.



1. Current Terminal

The Current Terminal indicator shows the terminal where the available vehicles is currently located.

2. Seat Availability

The Seat Availability indicator shows the number of available seats and total seat capacity for that vehicle.

3. Route Indicator

The Route Indicator shows the terminal points covered by the available vehicles, helping the commuter quickly identify its travel route.



CPE11-AFCS

Commuter
User Manual

Choose your destination & Transaction History

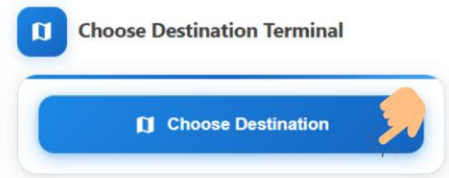
CPE11-AFCS

Commuter
User Manual
Choose your destination & Transaction History

The Choose Your Destination function allows the commuter to select the intended terminal before traveling so the system can prepare the correct trip and fare details.

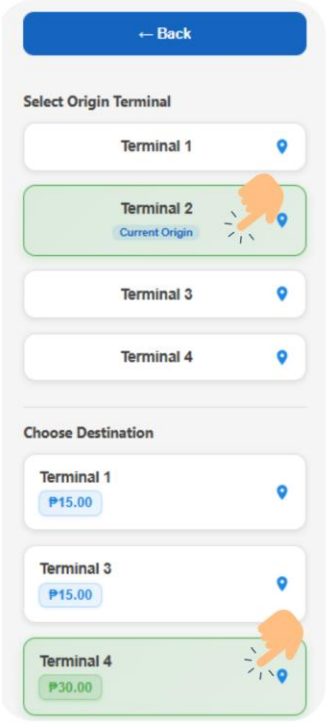
Step 1

Click "Choose Destination."



Step 2

Select your origin terminal, then choose your destination.




CPE11-AFCS
Commuter
User Manual
Choose your destination & Transaction History
Step 3
Click "Confirm Selection."
FARE:
₱15.00
Confirm Selection
Step 4
Wait for the boarding process to begin. If you wish to cancel, click "Cancel." or "Change Route" if you want to change your route.
Your Route
Your selected journey
2
ORIGIN
4
DESTINATION
Waiting for boarding
From
Terminal 2
Terminal 4
Extend to
Change Route
Cancel

CPE11-AFCS

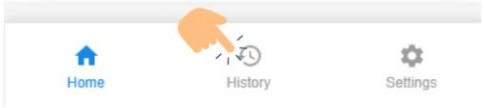
Commuter
User Manual

Choose your destination & Transaction History

The Transaction History section shows the commuter's past trip records, including the route, fare deducted, available vehicles, and time of transaction, making it easy to review previous travel activities.

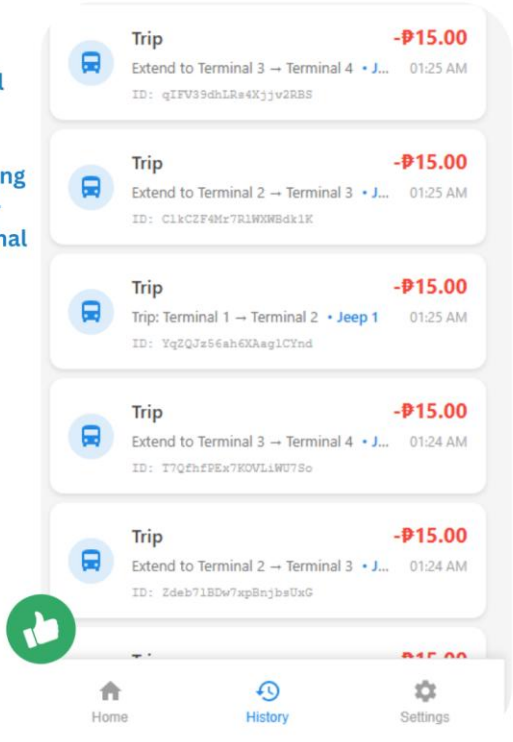
Step 1

Click "History".



Step 2

The system will display the transaction history, including your trips, top-ups, and terminal extensions

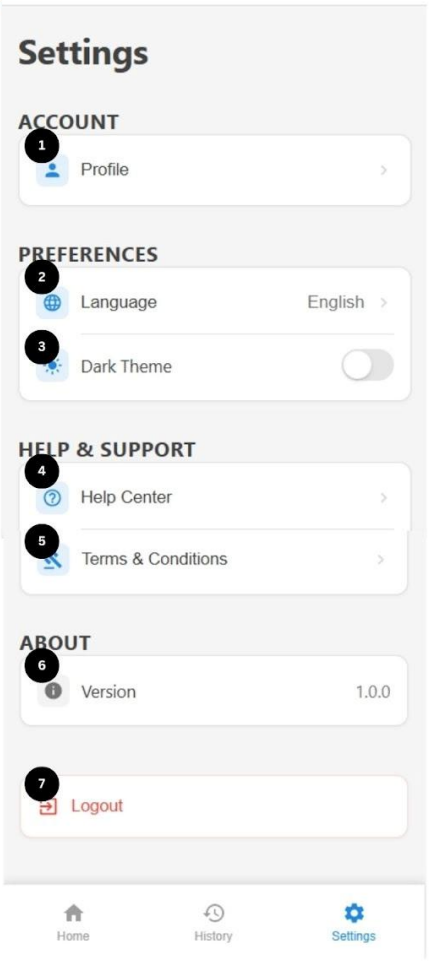




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Commuter
User Manual

Settings



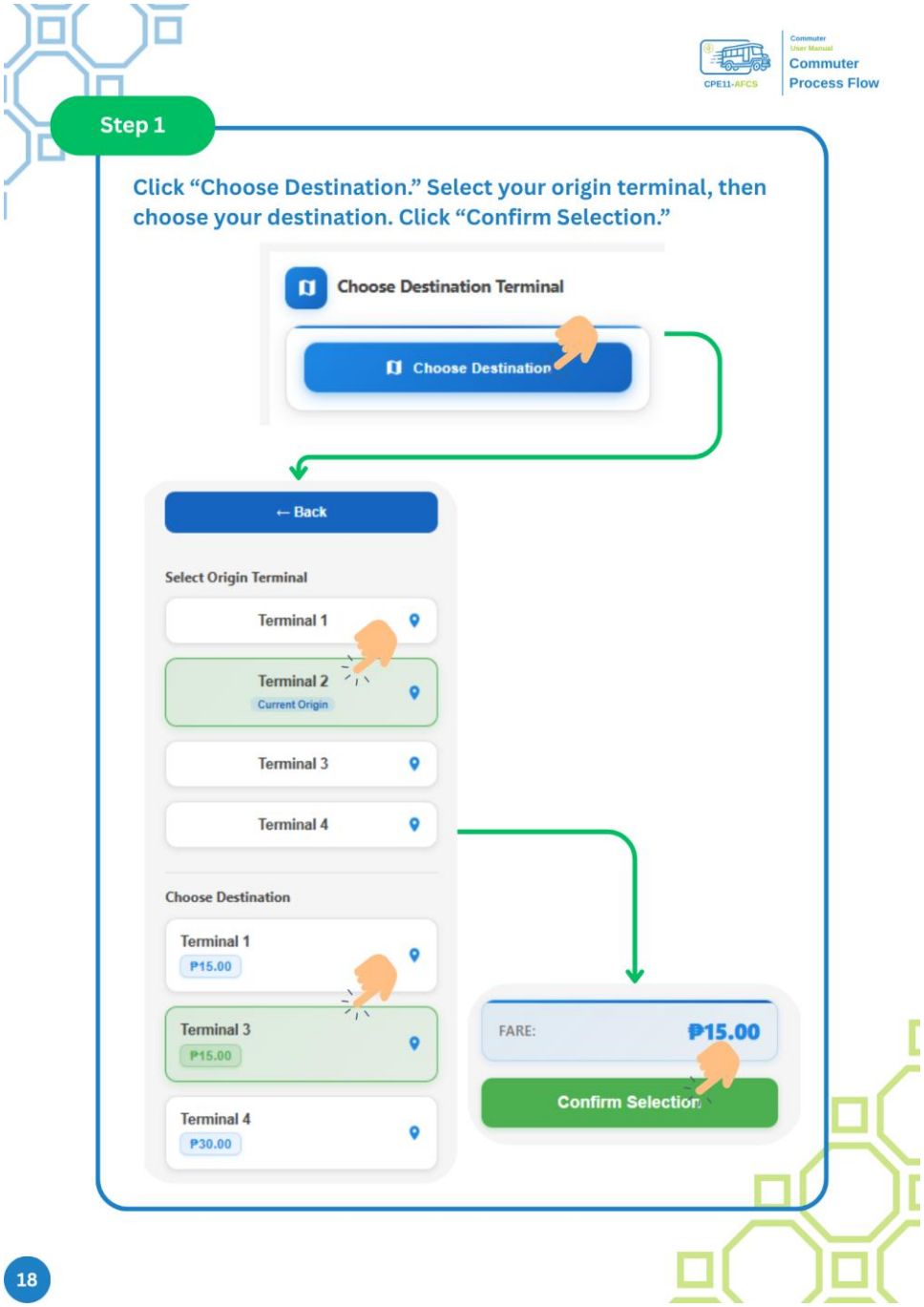
1. Profile - allows the commuter to view or manage account information.
2. Language option - used to select the preferred language of the dashboard
3. Dark Theme - toggle changes the interface appearance
4. Help Center - provides user assistance
5. Terms & Conditions - section shows the rules and policies of the system
6. Version indicator-displays the current application version
7. Logout button - used to securely sign out



CPE11-AFCS

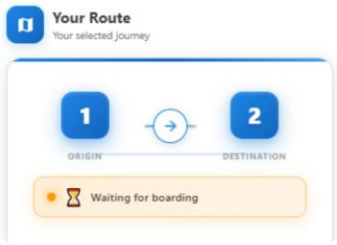
Commuter
User Manual

Commuter Process Flow


CPE11-AFCS
Commuter
Process Flow
Step 1
Click “Choose Destination.” Select your origin terminal, then choose your destination. Click “Confirm Selection.”
Choose Destination Terminal
Choose Destination
← Back
Select Origin Terminal
Terminal 1
Terminal 2
Current Origin
Terminal 3
Terminal 4
Choose Destination
Terminal 1
₱15.00
Terminal 3
₱15.00
Terminal 4
₱30.00
FARE:
₱15.00
Confirm Selection

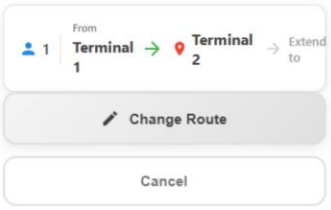
Step 3

Wait for the available vehicles to arrive while the dashboard shows the waiting for boarding status.



Step 4

While waiting, the commuter may choose to change route or cancel the boarding request.



Step 5

Once the available vehicles arrives, tap the card on the scanner for validation.

A



Step 5

B



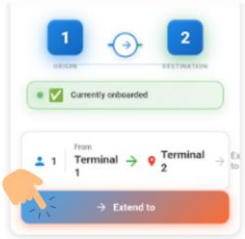
Step 6

After successful validation, proceed to the assigned available seat.



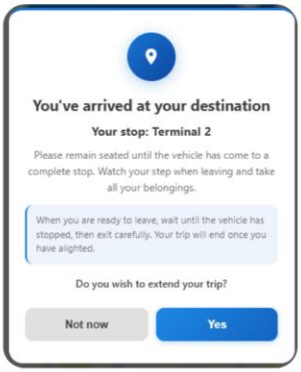
Step 7

During the trip, you may choose to extend your destination to a farther terminal if needed. The full extended terminal process can be found on page 22.



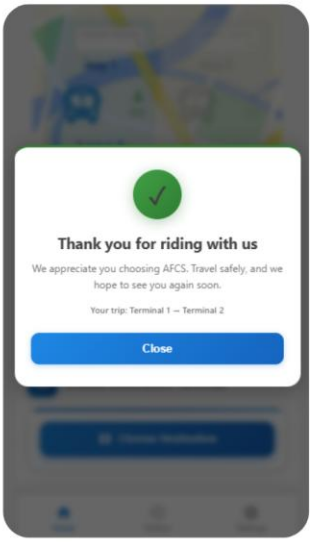
Step 8

Once the commuter reaches the selected terminal, a message stating "You've arrived at your destination" will appear on the commuter interface. If you click "Yes," you may proceed to the extended terminal. The process is described on page 22.



Step 9

After getting off at the terminal, a message stating "Thank you for riding with us" will appear on the commuter interface, indicating a successful trip.

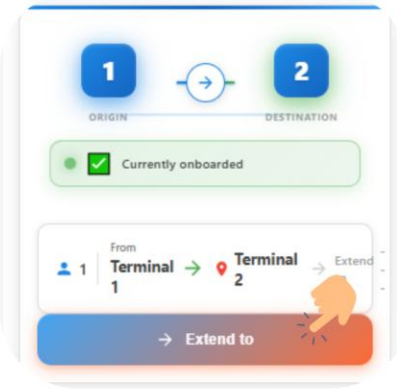


CPE11-AFCS Commuter User Manual
Commuter Process Flow

Extended Terminal Feature

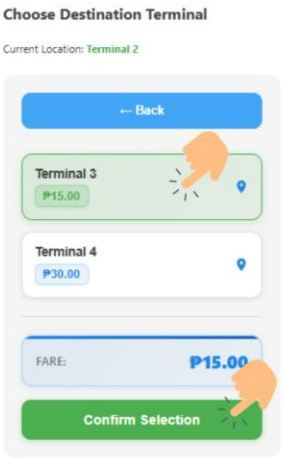
Step 1

If the commuter decides to extend their trip to a farther terminal, click "Extend To."



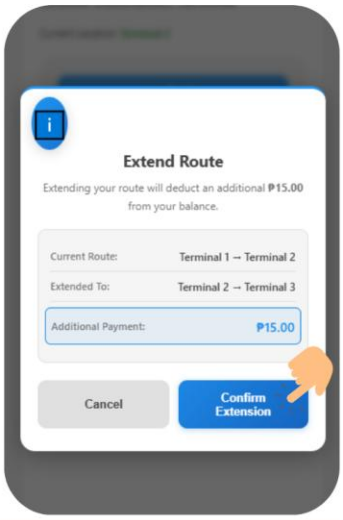
Step 2

Select the new destination terminal where you will exit, Click "Confirm Selection."



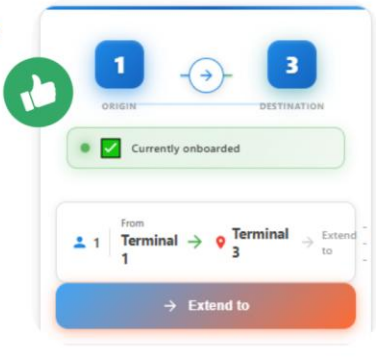
Step 3

Click “Confirm Extension” to proceed. If you change your mind, click “Cancel.”



Step 4

Your terminal has been successfully extended.





CPE11-AFCS

Driver User Manual

page

Table of Contents

1 Login Page

4 Dashboard Overview

9 Driver Process Flow



CPE11-AFCS

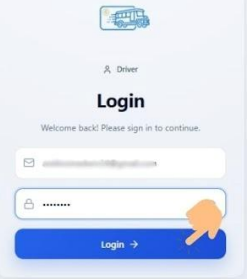
Driver
User Manual

Login Page

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User Manual
Login Page

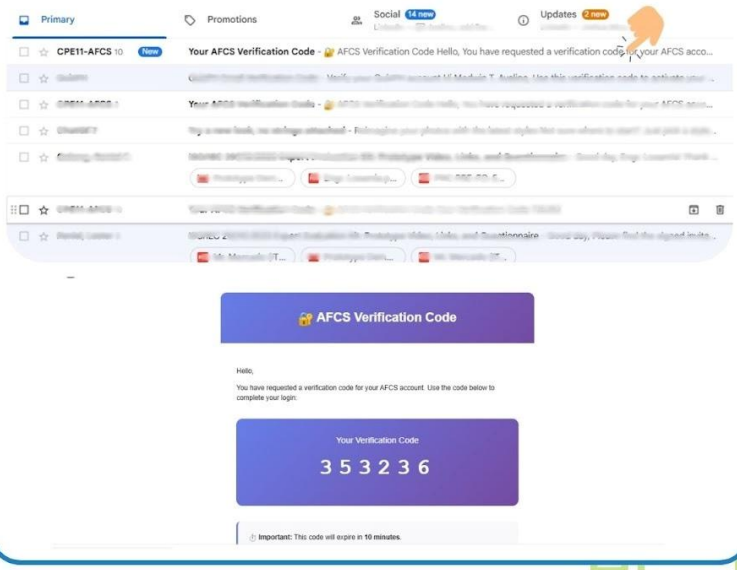
Step 1

Enter the registered Email and Password, then click Login.



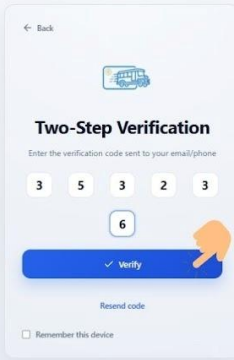
Step 2

Wait for the verification code sent to your Gmail account.



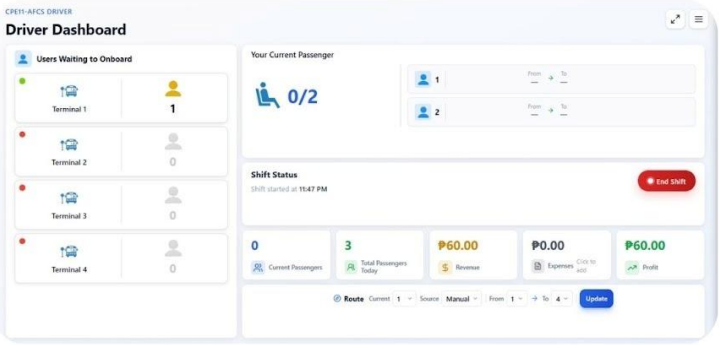
Step 3

Enter the code for verification, then click Verify to Continue. If no code is received, click Resend Code.



Step 4

Once the verification is successful, you will be redirected to the Commuter Dashboard.



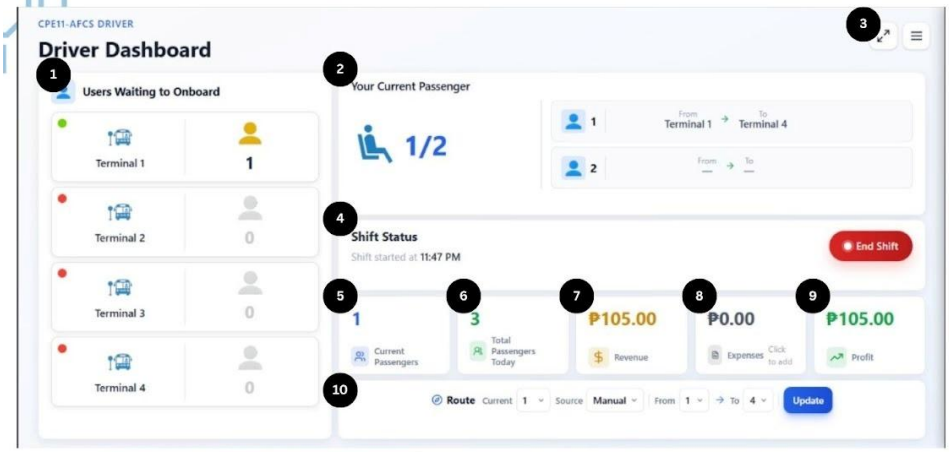


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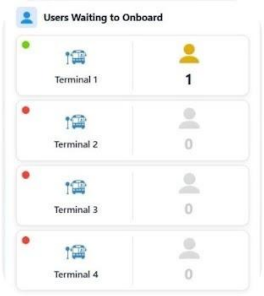
Dashboard Overview

Dashboard Overview



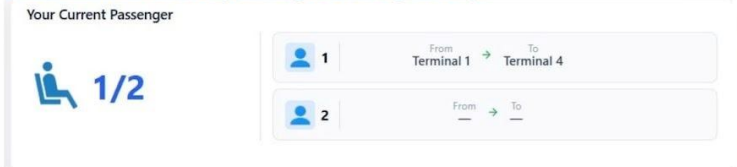
1. Users Waiting to Onboard.

This section shows the terminals where passengers are currently waiting to board. It helps the driver identify where pick-up activity is needed.



2. Your Current Passenger.

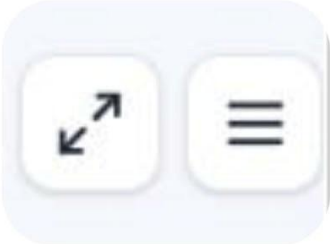
This section shows the passengers who are currently assigned inside the vehicle, including their seat count and trip details. It helps the driver monitor active passengers during the trip.



Dashboard Overview

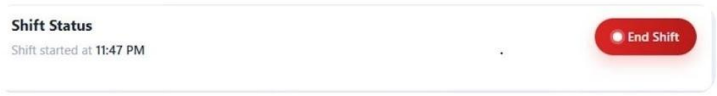
3. Dashboard Controls.

This feature provides dashboard viewing options such as expanding the screen or opening the menu. It helps the driver manage the dashboard display more conveniently.



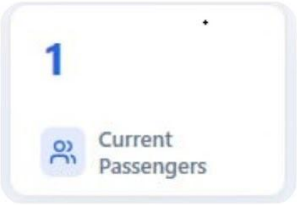
4. Shift Status.

This section shows the terminals where passengers are currently waiting to board. It helps the driver identify where pick-up activity is needed.



5. Current Passengers.

This card displays the number of passengers currently inside the vehicle. It gives the driver a quick view of the present passenger count.



Dashboard Overview

6. Total Passengers Today.

This card shows the total number of passengers served during the day. It helps the driver monitor daily passenger activity.



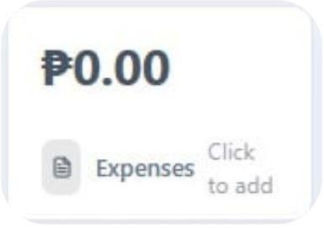
7. Revenue.

This card displays the total fare revenue collected. It helps the driver check the amount earned from passenger trips.



8. Expenses.

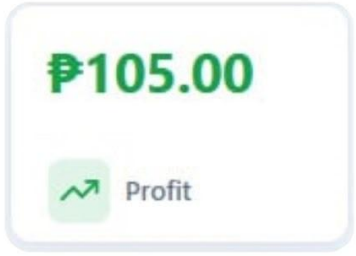
This card shows the recorded operational expenses. It allows the driver to monitor trip-related costs and add expense entries when needed.



Dashboard Overview

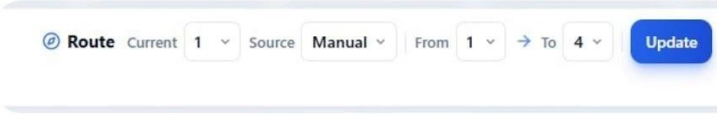
9. Profit.

This card displays the calculated profit based on revenue and expenses. It helps the driver quickly see the current financial result of the operation.



10. Route Controls.

This section allows the driver to set or update the current route information, including the source and destination terminals. It helps ensure that the dashboard reflects the correct travel route.





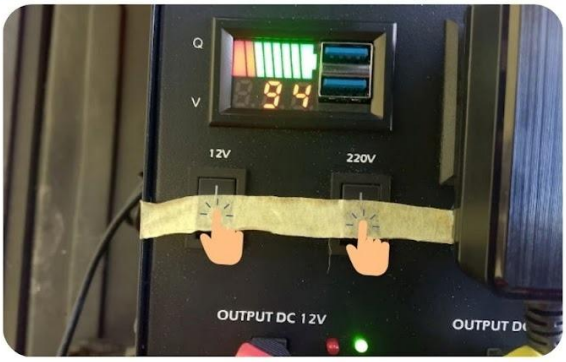
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Driver Process Flow

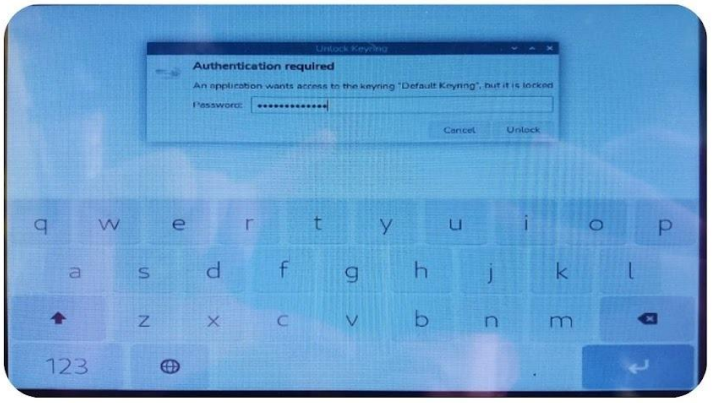
Step 1

Switch on the two buttons of the power supply to start the system.



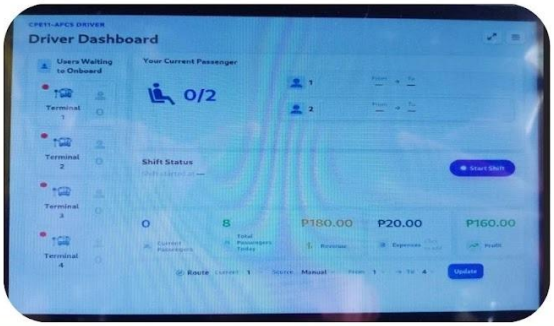
Step 2

Once the system is on, enter the password “whahaha_cpe11” on the driver interface screen.



Step 3

After the password is entered, the system will automatically log in the driver account and proceed to the dashboard.



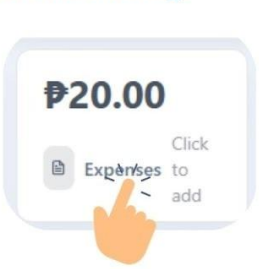
Step 4

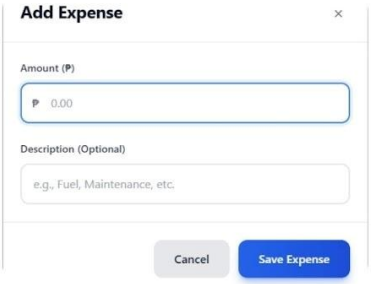
And click "Start shift" to start



Step 5

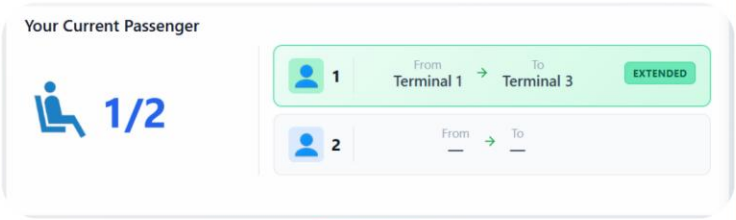
Whenever the driver spends money during the trip, click Expenses and enter the amount so it can be recorded for admin monitoring.



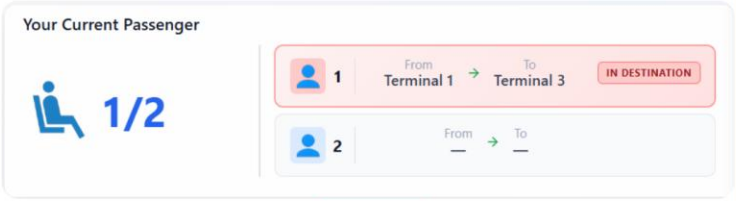


Current Passenger Status Indicator

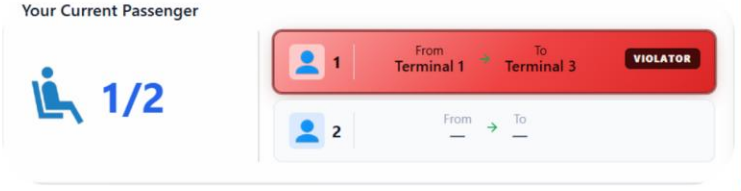
If the commuter successfully extends to another terminal, the driver dashboard will display a corresponding indicator.



If the commuter reaches the selected terminal and gets off as intended, the driver dashboard will display a corresponding indicator.

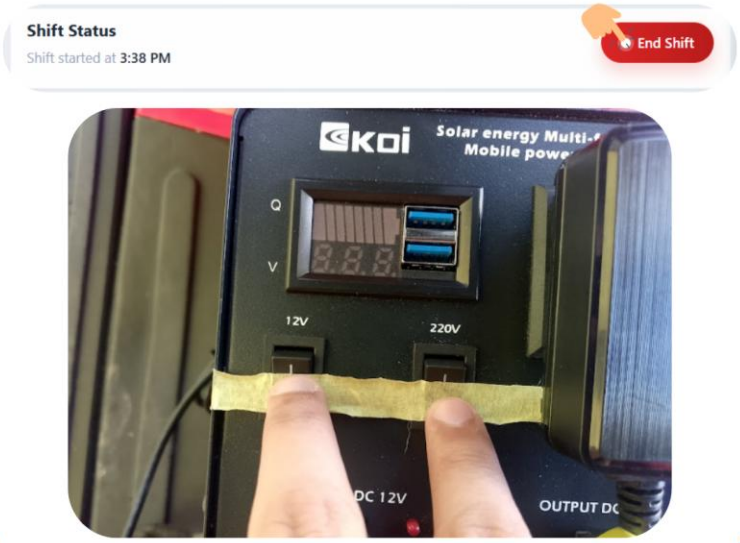


If the commuter does not get off at the selected terminal and also does not get off at the next stop, the commuter will be classified as a “violator.” The driver dashboard will then display the appropriate indicator.



Step 6

After the duty, click "End shift" to end, then "switch off the system



DRIVER ADVISORY

If the system displays 'System Unavailable,' the driver should be aware that this may be caused by wiring errors, component issues, Wi-Fi connection problems, or other technical faults.





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8 Generate Report Process



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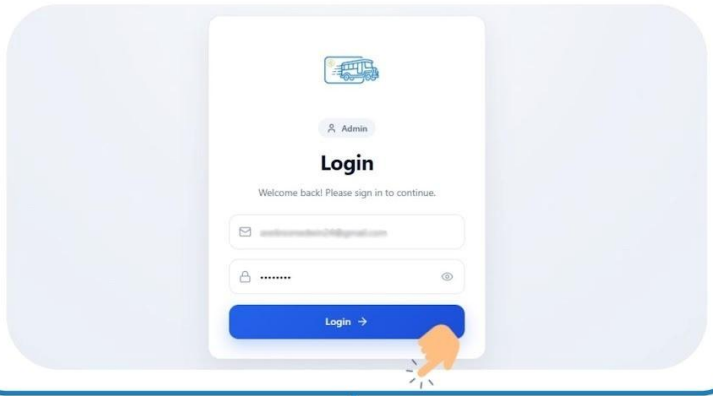
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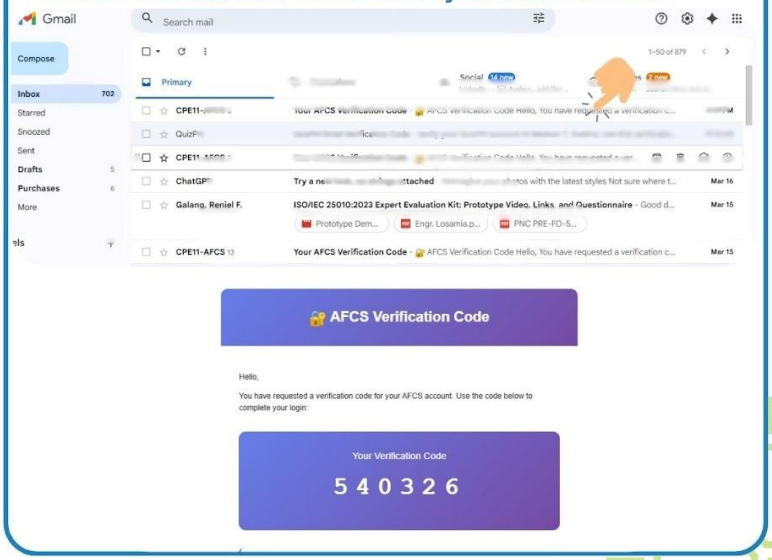
Step 1

Enter the registered Email and Password, then click Login.



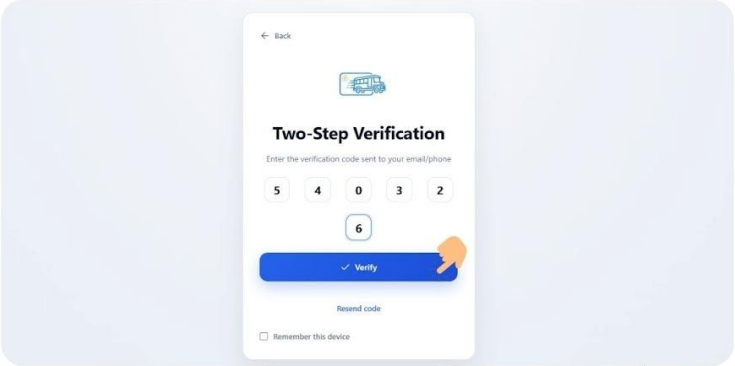
Step 2

Wait for the verification code sent to your Gmail account.



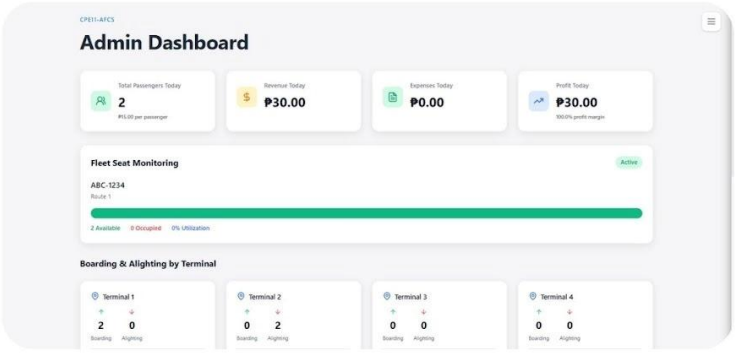
Step 3

Enter the code for verification, then click Verify to Continue. If no code is received, click Resend Code.



Step 4

Once the verification is successful, you will be redirected to the Commuter Dashboard.



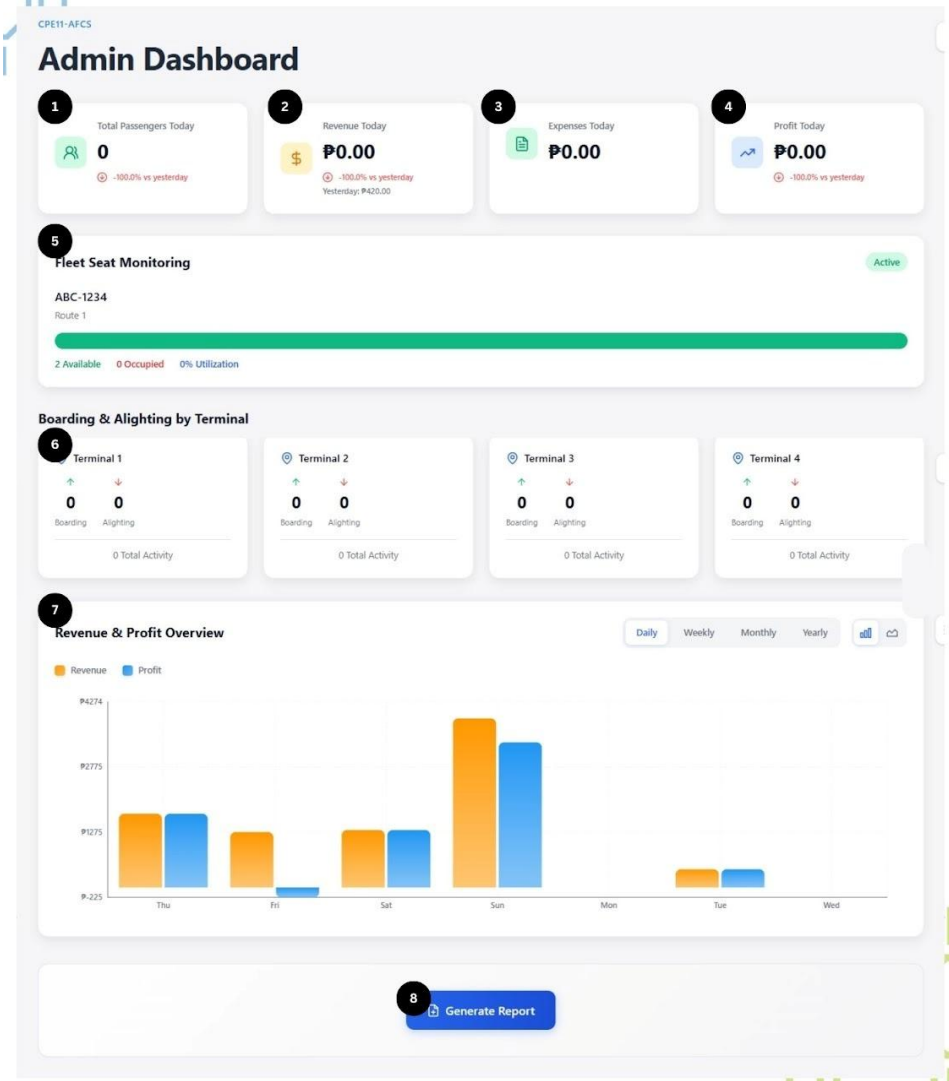


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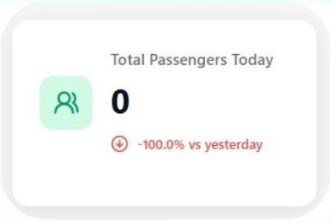
Dashboard Overview

Dashboard Overview



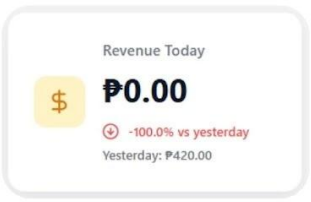
1. Total Passengers Today

This shows the total number of passengers recorded for the current day. It helps the administrator quickly monitor daily passenger activity.



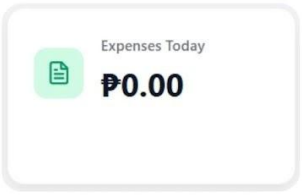
2. Revenue Today

This displays the total revenue collected for the current day. It allows the administrator to track how much fare income has been generated.



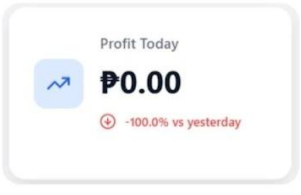
3. Expenses Today

This shows the total expenses entered for the day. It helps the administrator monitor daily operational costs.



4. Profit Today

This displays the computed profit for the day based on revenue and expenses. It gives the administrator a quick view of the system's daily financial performance.



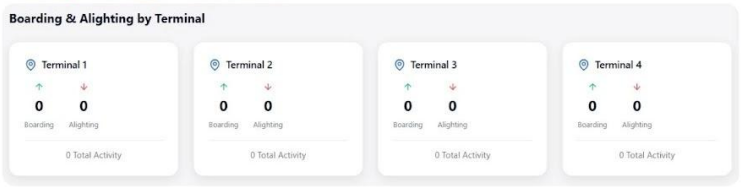
5. Fleet Seat Monitoring

This section shows the current seat status of a vehicle, including available seats, occupied seats, and utilization rate. It helps the administrator monitor seating capacity and usage in real time.



6. Boarding and Alighting by Terminal

This section shows the number of passengers boarding and alighting at each terminal. It helps the administrator track terminal activity and passenger movement.



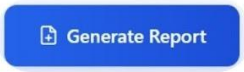
7. Revenue and Profit Overview

This chart presents the revenue and profit data across selected time periods such as daily, weekly, monthly, or yearly. It helps the administrator analyze financial trends and compare performance over time.



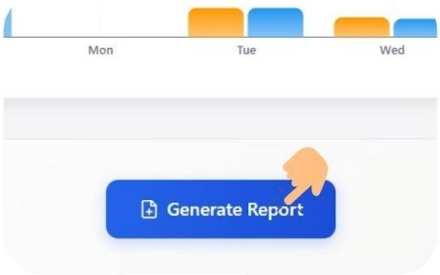
8. Generate Report

This button is used to create a report based on the recorded system data. It helps the administrator produce summarized records for monitoring, documentation, and decision-making.



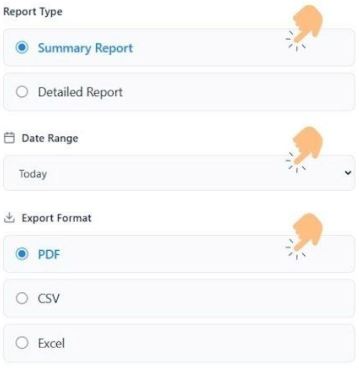
Step 1

Click the Generate Report button on the Admin Dashboard.



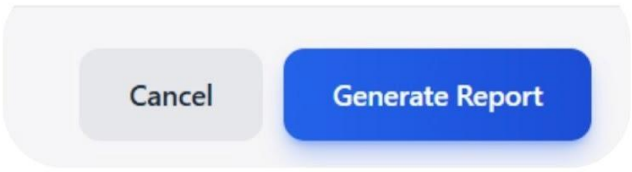
Step 2

In the report window, select the Report Type, Date Range, and Export Format.



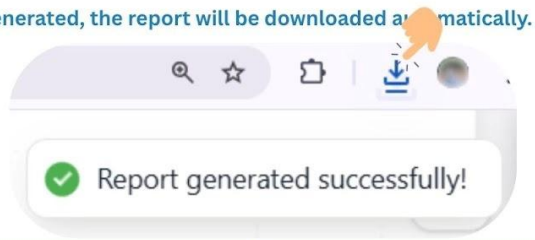
Step 3

Click Generate Report to continue, or click Cancel if you do not want to proceed.



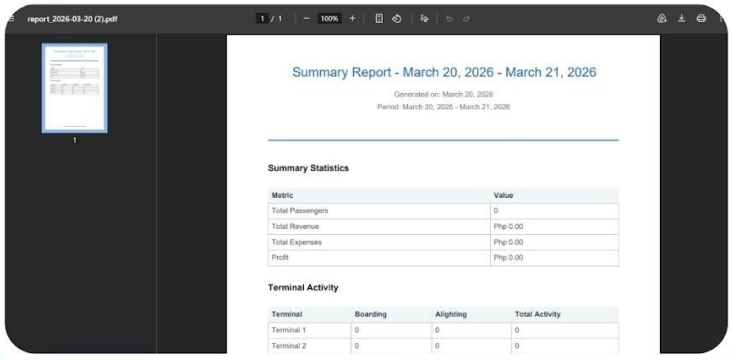
Step 4

Once generated, the report will be downloaded automatically.



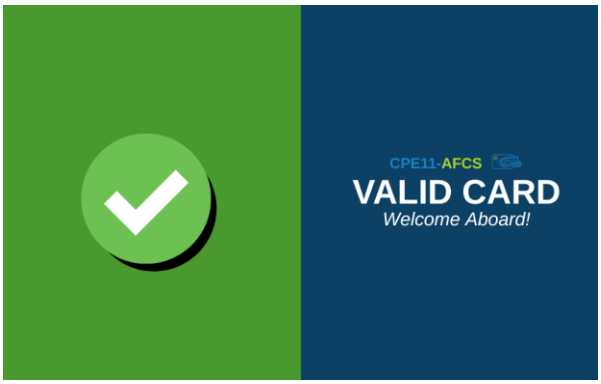
Step 5

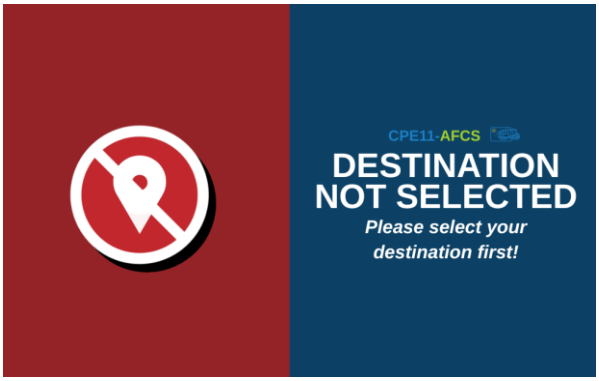
Open the downloaded file to review the generated report.



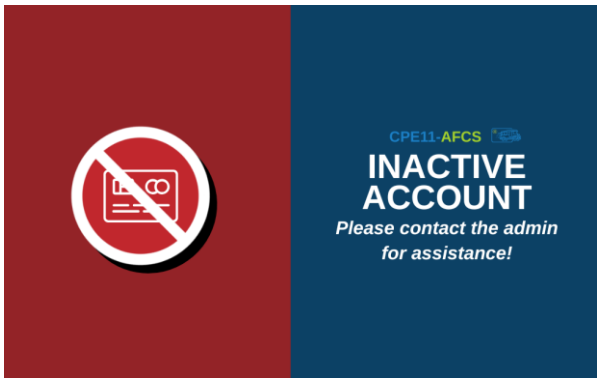
Tap-based Payment Process Interfaces



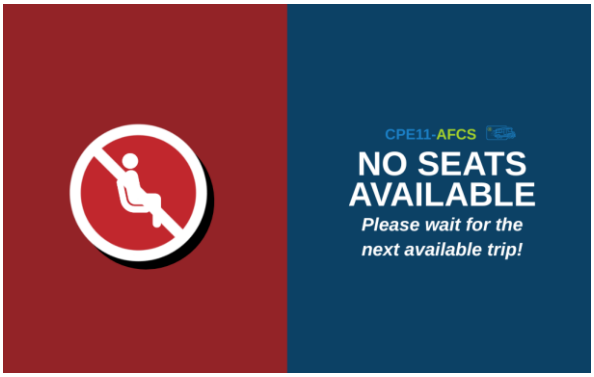






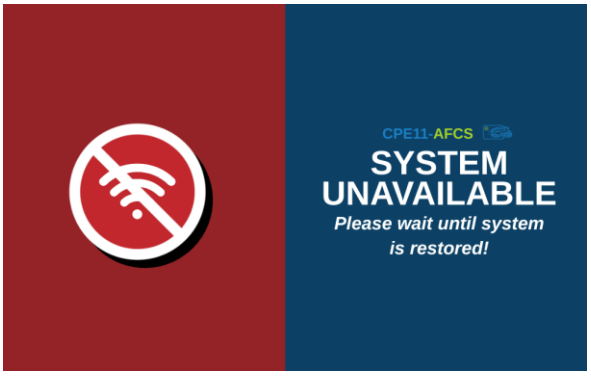


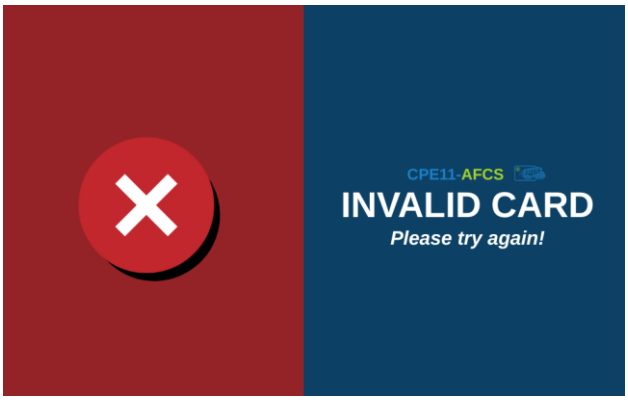



CPE11-AFCS
NO SEATS AVAILABLE
Please wait for the next available trip!


CPE11-AFCS
PASSENGER STILL BOARDING
Please wait for the passenger to be seated!

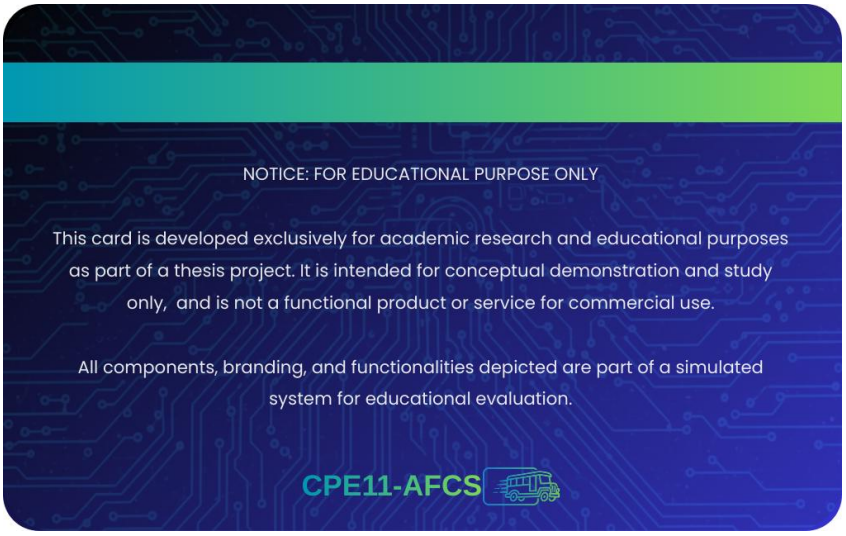

CPE11-AFCS
UNREGISTERED ACCOUNT
Please register your card first!


CPE11-AFCS
SYSTEM UNAVAILABLE
Please wait until system is restored!

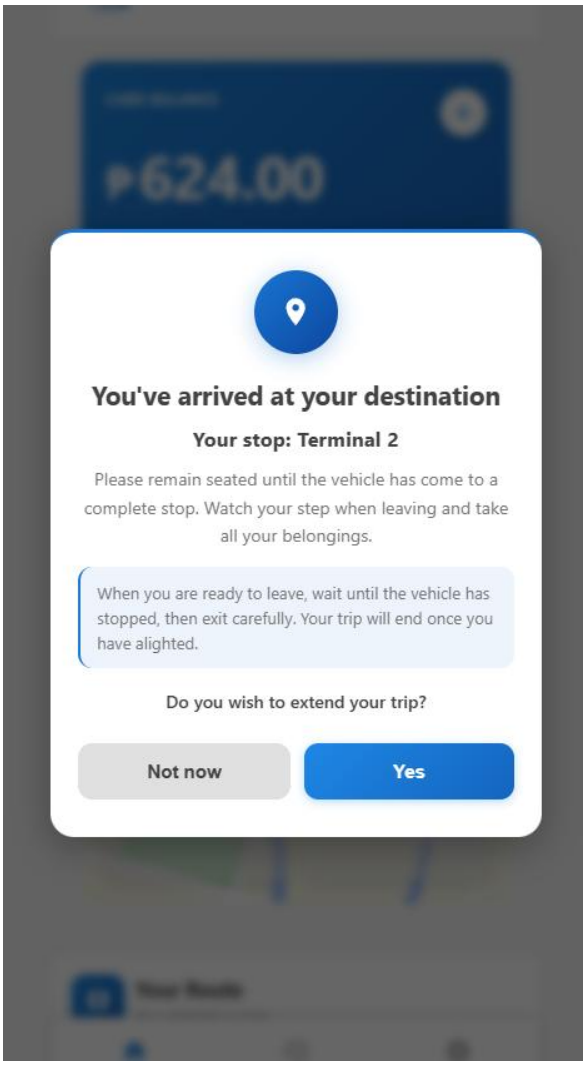

CPE11-AFCS
INVALID CARD
Please try again!

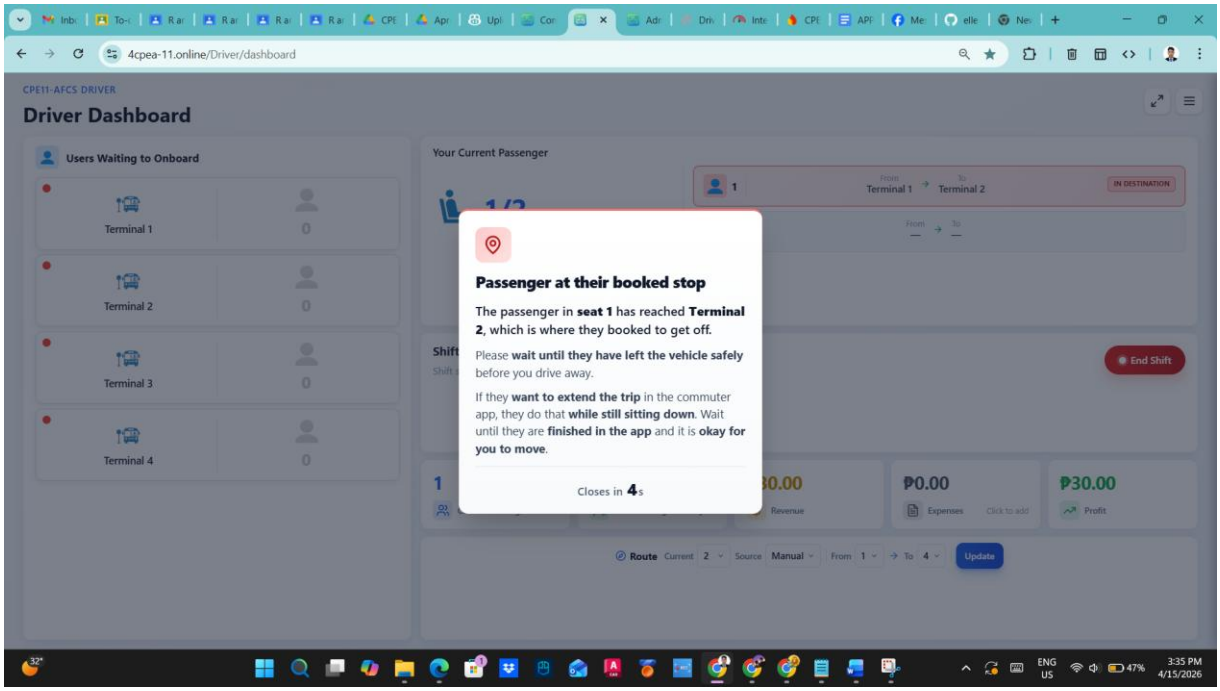
Smart Card Design

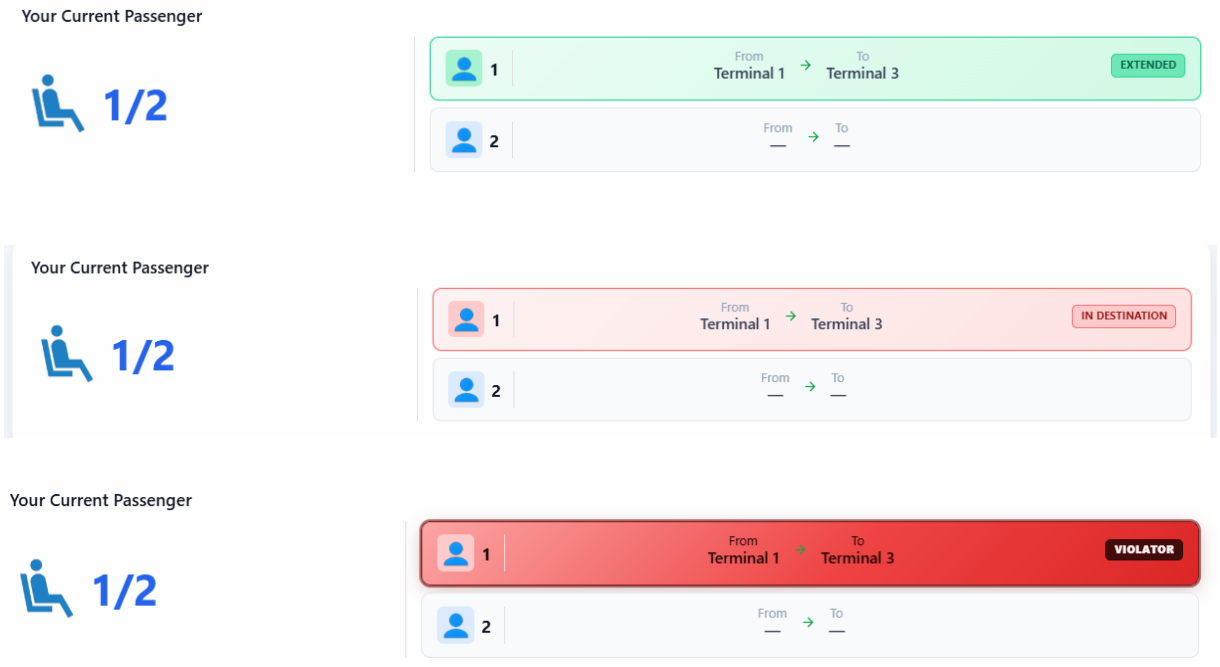




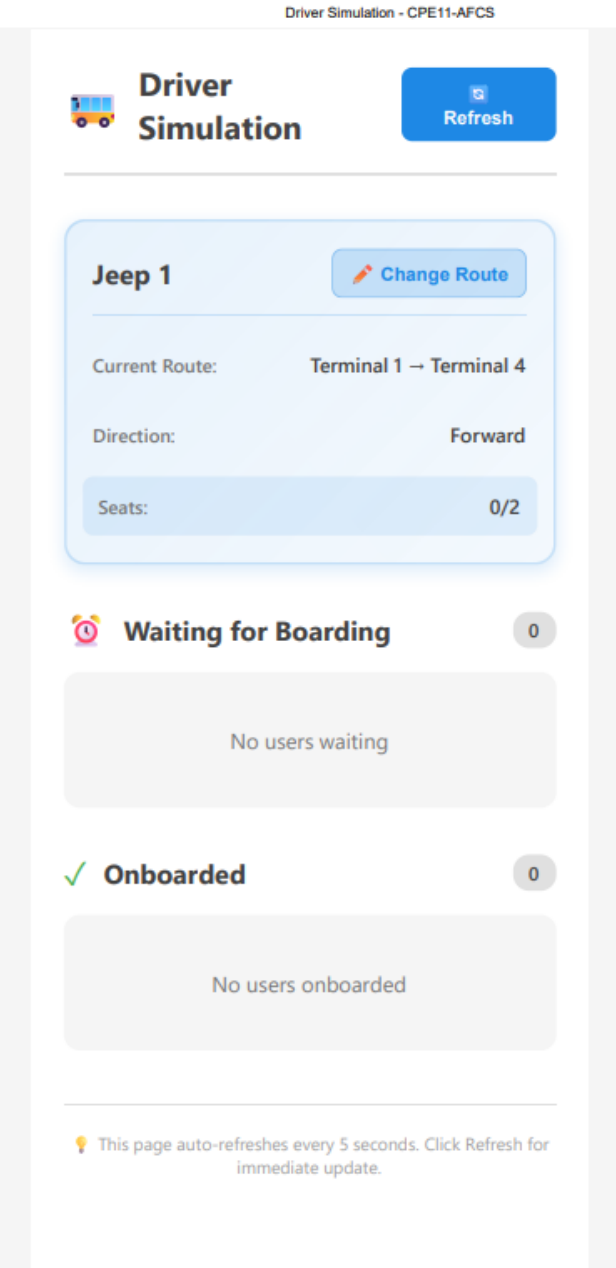
Pop-up Messages and Alerts and Indicators




4cpea-11.online/Driver/dashboard
CPE11-AFCS DRIVER
Driver Dashboard
Users Waiting to Onboard
Terminal 1
Terminal 2
Terminal 3
Terminal 4
Your Current Passenger
Terminal 1
Terminal 2
IN DESTINATION
Passenger at their booked stop
The passenger in seat 1 has reached Terminal 2, which is where they booked to get off.
Please wait until they have left the vehicle safely before you drive away.
If they want to extend the trip in the commuter app, they do that while still sitting down. Wait until they are finished in the app and it is okay for you to move.
Closes in 4s
End Shift
₱0.00
Expenses
₱30.00
Profit
Route
Source
Manual
Update


Your Current Passenger
1/2
1
From
Terminal 1
To
Terminal 3
EXTENDED
2
From
—
To
—
Your Current Passenger
1/2
1
From
Terminal 1
To
Terminal 3
IN DESTINATION
2
From
—
To
—
Your Current Passenger
1/2
1
From
Terminal 1
To
Terminal 3
VIOLATOR
2
From
—
To
—

Driver Simulation Web App



Appendix M

Reports of Plagiarism Software and Language Software

Report: Untitled

Chapter 1

by 4CPE - GROUP 11

General metrics

35,373	5,108	217	20 min 25 sec	39 min 17 sec
characters	words	sentences	reading time	speaking time

Score

90

This text scores better than 90% of all texts checked by Grammarly

Writing Issues

72	16	56
Issues left	Critical	Advanced

Plagiarism

1 %

28 sources

1% of your text matches 28 sources on the web or in archives of academic publications

Report: Chapter 2

Chapter 2

By 4CPE - GROUP 11

General metrics

39,640	5,417	216	21 min 40 sec	41 min 40 sec
characters	words	sentences	reading time	speaking time

Score

87

This text scores better than 87% of all texts checked by Grammarly

Writing Issues

97	10	87
Issues left	Critical	Advanced

Plagiarism

1 %

39 sources

1% of your text matches 39 sources on the web or in archives of academic publications

Report: Chapter 3

Chapter 3

by 4CPE - GROUP 11

General metrics

63,331	8,838	574	35 min 21 sec	1 hr 7 min
characters	words	sentences	reading time	speaking time

Score

82

This text scores better than 82% of all texts checked by Grammarly

Writing Issues

225	51	174
Issues left	Critical	Advanced

Plagiarism

1 %

32 sources

1% of your text matches 32 sources on the web or in archives of academic publications

Report: Chapter 4

Chapter 4

by 4CPE - GROUP 11

General metrics

46,326	6,561	367	26 min 14 sec	50 min 28 sec
characters	words	sentences	reading time	speaking time

Score

88

This text scores better than 88% of all texts checked by Grammarly

Writing Issues

94 Issues left

✓ Critical

94 Advanced

Plagiarism

1 %

46 sources

1% of your text matches 46 sources on the web or in archives of academic publications

Report: Chapter 5

Chapter 5

by 4CPE - GROUP 11

General metrics

8,371	1,156	68	4 min 37 sec	8 min 53 sec
characters	words	sentences	reading time	speaking time

Score

93

This text scores better than 93% of all texts checked by Grammarly

Writing Issues

9	✓	9
Issues left	Critical	Advanced

Plagiarism

✓ This text seems 100% original. Grammarly found no matching text on the Internet or in ProQuest's databases.

Appendix N

Bionote of the Student Researchers

Ivan Kier P. Mananquil (mananquilivankier23@gmail.com) is a Bachelor of Science in Computer Engineering student at the University of Cabuyao (Pamantasan ng Cabuyao), specializing in hardware systems, software analysis, and practical problem-solving. In the group's thesis, "AUTOMATED FARE COLLECTION SYSTEM WITH SEAT ALLOCATION, TERMINAL SELECTION, AND ANALYTICS DASHBOARDS," he serves as one of the primary prototype and hardware integration engineers. His responsibilities include assembling and wiring the fare validation terminal, assisting in the integration of the Raspberry Pi–based controller with the NFC reader-writer and peripheral components, and supporting enclosure layout, materials selection, and physical installation for scale models and simulated setups. He helps ensure that the hardware aspect of the system is safe, reliable, and maintainable in line with the project's ISO/IEC 25010:2023–oriented evaluation.

Throughout his academic journey, Ivan has strengthened his competencies through certifications and industry exposure. He holds a Safety Officer certification from Tamkene Phils., Inc., underscoring his commitment to safe and responsible technical work. As a freelancer at RRC Media, he handled graphics design, and engineering-related tasks, enhancing his documentation and presentation skills. With a solid foundation in Python, Java, and C++, plus expertise in circuit design, simulation, prototyping, data management, project management, and UI/UX design, he serves as a versatile contributor to the thesis team.

Ivan has contributed to several notable academic and prototype projects, serving as Prototype Engineer and Materials Manager for an Automated Teller Machine (ATM) System and a Point-of-Sale Cashier System, and handling Documentation and UI/UX Design for the RSIGCI Web Development project. He also played key roles in ILAW (Internet-Linked Appliances and Wireless Control), a 6-DOF Robotic Arm, and an Automated Greenhouse Blinds and Irrigation System, further sharpening his skills in embedded control and system integration. An active member of the Institute of Computer Engineers of the Philippines (ICpEP) and the Association of Computer Engineering Students (ACoES), he continues to grow his professional network and expertise as he prepares to contribute to hardware–software integrated systems and smart, automated solutions in computer engineering.

John Lloyd M. Cadang (cadangjohnlloyd92@gmail.com) is a Bachelor of Science in Computer Engineering student at the University of Cabuyao (Pamantasan ng Cabuyao), whose academic focus leans strongly toward software development and full-stack solutions. Guided by the values of discipline and responsibility he serves as one of the core software developers in the thesis “AUTOMATED FARE COLLECTION SYSTEM WITH SEAT ALLOCATION, TERMINAL SELECTION, AND ANALYTICS DASHBOARDS.” Within the study, he is primarily responsible for the design and implementation of the commuter-facing web application and the operator/admin dashboard—handling user interface and experience (UI/UX), front-end and back-end logic, database integration, and the synchronization of fare transactions between the Raspberry Pi–based validation device and the cloud-hosted management system. His work emphasizes functional, secure, and user-centered software modules aligned with ISO/IEC 25010:2023 quality characteristics, particularly in terms of functionality, usability, reliability, and security.

John Lloyd specializes in front-end and full-stack development, with strong knowledge of HTML, CSS, JavaScript, and Java, supported by experience in database management, UI/UX, and graphic design—making him the team member most focused on and knowledgeable in software development for the project’s commuter app and operator dashboard. He also holds a Safety Officer 2 certification from Tamkene Phils., Inc. (2025) and has gained professional experience in Digital Product Marketing as a self-employed developer under NiZoAna (2023–present) and as a Computer Engineering Intern in the Technical Seal Department of RSIGCI (2025), where he was exposed to real-world engineering workflows and technical documentation.

He has led and contributed to multiple academic and prototype projects—including a Point-of-Sale Cashier System, the RSIGCI Web Development project, ILAW (Internet-Linked Appliances and Wireless Control), a 6-DOF Robotic Arm, an Automated Greenhouse System, and a Banking Management System—showcasing his ability to integrate software, hardware, and user-centered design into working prototypes. An active member of ICpEP and ACoES at the University of Cabuyao, he continues to build his leadership and technical skills as he prepares for a career in computer engineering, with a strong focus on full-stack development and technology solutions for transportation and enterprise systems.

Medwin T. Avelino (avelinomedwin24@gmail.com) is a Bachelor of Science in Computer Engineering student at the University of Cabuyao (Pamantasan ng Cabuyao) with strong interests in hardware systems, embedded design, and technology-driven solutions for transportation. In the group's thesis, "AUTOMATED FARE COLLECTION SYSTEM WITH SEAT ALLOCATION, TERMINAL SELECTION, AND ANALYTICS DASHBOARDS," he serves as a key hardware, systems prototyping member and is the primary designer of the study's system flowcharts, diagrams, and architecture. His responsibilities include circuit design, simulation, and implementation of the fare validation terminal, integrating components such as the Raspberry Pi–based controller, NFC reader-writer, and peripheral interfaces. He also supports the wiring layout, power management, enclosure planning, and bench testing of the prototype, ensuring that the device-level design aligns with the system's reliability, safety, and maintainability requirements under ISO/IEC 25010:2023.

Medwin specializes in circuit design, simulation, and prototyping, with proficiency in Python, Java, C++, and electronic design tools such as Proteus and KiCad, enabling him to turn system specifications into working, testable hardware modules for the cashless fare system. His background in basic web development, documentation, and graphics/video editing supports the team's technical reports and presentations. He further strengthened his technical preparation and safety awareness by earning a Safety Officer 2 certification from Tamkene Phils., Inc. and completing an on-site internship at REIGN-NAN Sales Industry & General Contractor, Inc. (RSIGCI), where he was exposed to real-world engineering workflows and workplace standards.

Beyond the thesis, Medwin has worked in the Warehouse Department and has served as a freelance graphics designer and video editor for RRC Media, further developing both his technical and creative abilities. He has contributed to several academic and prototype projects—including a Point-of-Sale Cashier System, RSIGCI Website Development, a 6-DOF Robotic Arm, and an Automated Greenhouse Blinds and Irrigation System—often handling documentation, design, and hardware-related tasks across the project lifecycle. An active member of the ICpEP and ACoES, he continues to sharpen his collaboration, leadership, and engineering capabilities as he prepares for a professional career in computer engineering, with interest in embedded systems and intelligent transportation solutions.

Reniel F. Galang (galanreniel68@gmail.com) is a Bachelor of Science in Computer Engineering student at the University of Cabuyao (Pamantasan ng Cabuyao) whose academic interests center on secure embedded systems, data-driven applications, and technology for public transportation. Guided by his values, he has consistently taken on leadership and project management roles in various coursework and research projects, and currently serves as the lead researcher and overall team leader of the study “AUTOMATED FARE COLLECTION SYSTEM WITH SEAT ALLOCATION, TERMINAL SELECTION, AND ANALYTICS DASHBOARDS.” In this thesis, he is responsible for coordinating the research team, leading stakeholder and requirements-gathering activities with transport cooperatives, overseeing system architecture design, and aligning the implementation with the Modified V-Model with Iterative Prototyping used for the project’s hardware–software integration. He also contributes to the design and evaluation of the NFC-enabled fare validation prototype using MIFARE DESFire EV2, ACR1252U NFC reader-writers, Raspberry Pi 4 microcontrollers, and a centralized analytics dashboard assessed under ISO/IEC 25010:2023.

Reniel’s technical background supports the full stack of the thesis system—from device-level development to analytics. He is proficient in Python and C++, with experience in documentation, testing and debugging, circuit design and simulation, and basic data analytics using tools like NumPy, AutoCAD, and Proteus for schematic design and prototype visualization. His certifications in Data Analytics Essentials (Cisco Networking Academy | DICT, 2025) and Lean Six Sigma White Belt (Ask Lex PH Academy, 2025) further equip him to design performance metrics, optimize processes, and contribute to the study’s descriptive and diagnostic analytics for jeepney operations in Laguna.

Beyond the thesis, Reniel has led several academic and prototype projects including a Point-of-Sale Cashier System, the RSIGCI Website, a 6-DOF Robotic Arm, and an Automated Greenhouse System-strengthening his skills in systems integration, UI design, and project coordination. He has also gained professional experience at PSI, RSIGCI, and RRC Media, and remains active in ICpEP and ACoES as he prepares for a career in computer engineering focused on smart mobility and data-driven public transport systems.

Justin S. Mina (minajustin613@gmail.com) is a Bachelor of Science in Computer Engineering student at the University of Cabuyao (Pamantasan ng Cabuyao), whose values of faith and integrity, guide his academic and professional pursuits. In the group's thesis, "AUTOMATED FARE COLLECTION SYSTEM WITH SEAT ALLOCATION, TERMINAL SELECTION, AND ANALYTICS DASHBOARDS," he serves as the documentation member and logistics coordinator. His responsibilities include organizing and maintaining research documents, structuring chapters and related literature using tools like Mendeley, preparing testing and evaluation forms aligned with ISO/IEC 25010:2023 and user-centered design principles, and assisting in the preparation of system diagrams, figures, and tables. As logistics coordinator, he helps manage schedules, materials, and coordination for prototype activities, stakeholder engagements, and data-gathering sessions, ensuring that the team's work is systematically documented and efficiently executed.

Justin has a solid technical foundation in Python, C++, and Java, with strengths in problem-solving, testing and debugging, and circuit design and simulation. He is proficient in engineering and productivity tools such as AutoCAD, Proteus, and Mendeley, as well as Filmora for video editing, allowing him to bridge technical content with clear visual and written communication for reports, presentations, and demonstrations. To strengthen his professional preparation, he obtained Safety Officer 2 certification under Tamkene Phils., Inc. (2025) and completed an on-site internship with Reign-Nan Sales Industry & General Contractor, Inc. (RSIGCI, 2025), where he was exposed to real-world engineering environments, documentation workflows, and basic construction and safety practices. He has also gained experience in the Marketing Department of RSIGCI as an on-site intern, and previously worked as a Freelance Video Editor (2022), refining his skills in storytelling, layout, and visual communication.

He has contributed to several academic and prototype projects—including a Point-of-Sale Cashier System, RSIGCI Website Development, a 6-DOF Robotic Arm, and an Automated Greenhouse Blinds and Irrigation System—primarily handling documentation, logistics, and hardware assistance to ensure that designs, procedures, and test results are clearly recorded and shared with the team. He pursues and builds his future career as an active member of ICpEP and ACoES.